# Intersection Theory

**— *EYNTKA* Series —**

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# *Introduction*

A reader who has worked through *Algebraic Geometry* [2] can already intersect. On a smooth projective surface two curves meet in a well-defined number of points, counted with multiplicity; that number depends only on the divisor classes, it is symmetric, and it computes things the naive count cannot, such as the self-intersection $E^2 = -1$ of an exceptional curve, which no set-theoretic count of points could produce. The construction was built by hand, and it was built for divisors on a surface. It says nothing about two surfaces inside a fourfold, nothing about a curve and a surface inside a threefold, and nothing about what happens when the two subvarieties share a component instead of meeting transversally.

Those are not exotic questions. They are the questions the subject was born from. How many lines meet four general lines in $\mathbb{P}^3$? How many conics are tangent to five general conics? Nineteenth century enumerative geometry answered such questions in quantity, by a principle of conservation of number: deform the configuration until the answer becomes visible, and trust that the count did not change along the way. The answers were, for the most part, right. The method was not a proof. Schubert's calculus of 1879 systematized it into a symbolic algebra of conditions so effective, and so unjustified, that Hilbert made its rigorous foundation the fifteenth of his 1900 problems.

The gap is precisely the one the surface theory papers over. Conservation of number needs an equivalence relation on cycles under which the count is invariant, and it needs the count to be defined even when the two cycles fail to meet properly. Rational equivalence supplies the first: two cycles are equivalent when one deforms to the other along a family parametrized by $\mathbb{P}^1$, and the quotient of the cycles by that relation is the Chow group. Deformation to the normal cone, introduced by Fulton and MacPherson in the 1970s, supplies the second: rather than moving the cycles until they meet properly, degenerate the ambient embedding to its normal cone, where the intersection can be read off the zero section of a bundle. That construction replaced the moving lemma, which had been the classical route and which is genuinely hard to prove in the generality one wants.

What comes out is more than a rigorous count. The Chow group is a functor, covariant for proper maps and contravariant for flat ones, and on a smooth variety it carries a ring structure, so questions about subvarieties become questions in a graded ring. Vector bundles acquire Chern classes valued in that ring, computed from Segre classes and the splitting principle. And Grothendieck's Riemann-Roch theorem relates the Chern character of a pushforward to the pushforward of a Chern character, subsuming Riemann-Roch for curves, which this reader has already seen [2, Riemann-Roch for curves], into a statement about arbitrary proper morphisms.

## Goal

The book opens with cycles and rational equivalence, and establishes the Chow group $A_*(X)$ together with the two functorialities that make it usable: proper pushforward and flat pullback, with the localization sequence and homotopy invariance that compute it in the first examples ($\mathbb{A}^n$, $\mathbb{P}^n$, curves). Divisors come next, in the form the rest of the book needs them: not as the divisor class group of a normal variety, but as an operator $c_1(L) \cap -$ on cycle classes, commutative and compatible with both functorialities. That operator is the first Chern class, and it is the germ of everything that follows.

Segre classes of cones generalize it. They give the Chern classes of a vector bundle, the Whitney sum formula, the splitting principle, and with them the projective bundle formula, which computes the Chow groups of $\mathbb{P}(E)$ and closes the one debt the opening chapter leaves open. Deformation to the normal cone then delivers the Gysin map for a regular embedding, and the intersection product follows by reduction to the diagonal: on a smooth variety $X$, the product of two classes is the Gysin pullback along $X \hookrightarrow X \times X$. This is the point at which $A^*(X)$ becomes a ring, Bézout's theorem becomes a corollary, and the surface pairing built by hand upstream is recovered as a special case rather than an analogy.

The last movement turns outward. Comparing rational equivalence with algebraic, homological and numerical equivalence puts the Chow ring next to cohomology through the cycle class map, which is what makes algebraic cycles a Hodge-theoretic object and what the standard conjectures are about; correspondences, cycles on $X \times Y$ acting as operators, are the entry point to motives. Riemann-Roch is proved in Grothendieck's form and specialized back down to Hirzebruch-Riemann-Roch and Noether's formula. The book closes where it began, on enumerative geometry: the Chow ring of the Grassmannian, Schubert calculus done properly, degeneracy loci, and the classical counting problems that motivated the whole apparatus, now with proofs.

**0.0.1 Prerequisites and Place in the Series** This volume builds on *Algebraic Geometry* [2]. No later volume currently lists it as a prerequisite. The reader has access to every completed volume of the series but is not assumed to have read all of them; prerequisites are cited where invoked.

## Sources

The treatment follows Fulton's *Intersection Theory* [7] as its main source, with Vakil's intersection-theory notes [10] for exposition, adapted to a reader who already has scheme theory and sheaf cohomology in full [2]. The standing conventions are collected at the start of Part I, where they first bind.

# Part I

# The Chow Group

Cycles are easy to write down and useless as they stand: the group of formal integer combinations of subvarieties is enormous, is not an invariant of anything, and admits no product. This part cuts it down to something that is. Rational equivalence identifies two cycles when one sweeps out to the other in a family over $\mathbb{P}^1$, and the quotient, the Chow group, turns out to carry exactly the structure the classical geometers were assuming when they deformed a configuration and trusted the count. Proper pushforward and flat pullback make it functorial, the localization sequence and homotopy invariance make it computable, and intersecting with a divisor gives the first operation that lowers dimension. That last operation, the first Chern class, is the seed of everything the rest of the book builds.

**0.0.2 Convention: Standing Hypotheses** Throughout the book a scheme is a *separated* scheme of finite type over a fixed algebraically closed field $\overline{k}$; such a scheme is called an *algebraic scheme*. A point is a closed point unless stated otherwise, a subvariety of a scheme is a closed integral subscheme, and a *variety* is a reduced and irreducible algebraic scheme, hence separated. Separatedness is folded in here rather than carried statement by statement because it is exactly the condition making the diagonal $\delta \colon X \to X \times X$ a closed immersion, and from chapter 5 onward every intersection product is built by pulling back along that diagonal. A non-separated example such as the line with a doubled origin is excluded by this convention, not by hypothesis.

**0.0.3 Convention: $\mathbb{P}(E)$ Is the Bundle of Lines** For a vector bundle $E$ on $X$, this book writes $\mathbb{P}(E)$ for the bundle of one-dimensional *subspaces* of $E$: the fibre over $x$ is the space of lines in $E_x$. This is Fulton's convention.
*Algebraic Geometry* [2] uses Grothendieck's opposite convention, under which $\mathbb{P}(\mathscr{E}) = \operatorname{Proj}(\operatorname{Sym}^\bullet \mathscr{E})$ parametrizes one-dimensional *quotients* of the fibres. The two are related by a dual: the bundle written $\mathbb{P}(E)$ here is the one written $\mathbb{P}(E^\vee)$ there, and a formula transported between the two must be dualized. The Segre classes of Chapter 1 onward are defined by pushing forward powers of $c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$, so reading such a formula in the wrong convention silently computes the classes of $E^\vee$ instead of $E$.

# 1

# *Rational Equivalence*

Recall from [6] that we had to define *projective plane curves* to be an equivalence relation on homogeneous polynomials of $k[x,y,z]$ up to a constant ($f \sim g$ if and only if $f = \alpha g$). This was specifically done keep track of multiplicity information. By the correspondence between homogeneous and affine curves, we can without loss of generality work with $[f(x,y)], [g(x,y)] \in k[x,y]$ and choose any representative; we shall simply call these *plane curves*. Given a choice of plane curves $f, g$ let $F, G$ denote the resulting variety. The intersection of two (affine or projective) plane curves $F, G$ at $P$ was defined as:

$$i(P, f \cdot g) = \dim_K \frac{\mathcal{O}\mathbb{A}^2_k, p}{(f,g)} = \dim_K \mathcal{O}_{Z,P}$$

where $Z = \mathcal{Z}(f,g)$. It was proven that:

1. (commutative on intersection): $i(P, F \cdot G) = i(P, G \cdot F)$
2. (linear on intersection): if $F_1 + F_2$ is the curve defined by $f_1 f_2$, then $i(P, (F_1 + F_2) \cdot G) = i(P, F_1 \cdot G) = i(P, F_2 \cdot G)$
3. (invariance of representation): if $F'$ is given by $f + gh$ for some $h \in k[x,y]$:
   $$i(P, F' \cdot G) = i(P, F \cdot G)$$
4. If $F, G$ have have a common component through $P$ (i.e. the dimension of the irreducible variety in $F \cap G$ containing $p$ is greater than 0), then $i(P, F \cdot G) = \infty$. Otherwise, $i(P, F \cdot G) \in \mathbb{N}_{\geq 0}$.
5. If $P \notin F \cap G$:
   $$i(P, F \cdot G) = 0$$
6. if $f(x,y) = x - a$ and $g(x,y) = y - b$ (more generally $\frac{\partial(f,g)}{\partial(x,y)} \neq 0$ at $p$) then:
   $$i(P, F \cdot G) = 1$$

7. If $F$ has no common component with $G$ and $H$ and $P$ is a simple point of $F$:

$$i(P, G \cdot H) \geq \min i(P, F \cdot G), i(P, F \cdot H)$$

## 1.1 Cycles and Rational Equivalence

The classical intersection number recalled above attaches an integer to a pair of plane curves meeting at a point. Turning that local count into a theory valid on any variety and in every dimension calls for a home for "formal combinations of subvarieties" together with an equivalence relation flexible enough to make intersection numbers well defined. That home is the Chow group, the higher-dimensional descendant of the divisor class group of [2, Divisor Class Group]: where the divisor class group records codimension-one subvarieties modulo the divisors of rational functions, the Chow group records subvarieties of *every* dimension modulo the same mechanism.

This section builds the three objects in turn, the group of cycles $Z_k(X)$, the subgroup $\mathrm{Rat}_k(X)$ of cycles rationally equivalent to zero, and the quotient $A_k(X)$. Throughout the chapter $X$ is an algebraic scheme (a scheme of finite type over the algebraically closed field $\overline{k}$), a *subvariety* means a closed integral subscheme, and a point means a closed point unless stated otherwise.

The construction rests on the order of vanishing of a rational function along a codimension-one subvariety, introduced for divisors in [2, Divisors: Geometric Representation of Line Bundles]. On a variety $W$ a prime divisor is a codimension-one subvariety $V$, and the local ring $\mathcal{O}_{W,V}$ is a one-dimensional local domain with fraction field the function field $k(W)$. When $W$ is regular in codimension one this ring is a discrete valuation ring and its valuation $\nu_V$ records orders of vanishing ([2, Principal Divisor]); the length-theoretic definition below removes the regularity hypothesis, which matters as soon as $W$ is an arbitrary subvariety of $X$.

**Definition 1.1.1: Order Along a Subvariety**

Let $W$ be a variety and $V \subseteq W$ a codimension-one subvariety, and set $A = \mathcal{O}_{W,V}$, a one-dimensional local domain with fraction field $k(W)$. For nonzero $a \in A$ the quotient $A/(a)$ has finite length, and the *order of* $a$ *along* $V$ is

$$\mathrm{ord}_V(a) = \mathrm{length}_A\left(A/(a)\right)$$

Because $\mathrm{ord}_V(ab) = \mathrm{ord}_V(a) + \mathrm{ord}_V(b)$, this extends to a homomorphism $\mathrm{ord}_V : k(W)^* \to \mathbb{Z}$ by $\mathrm{ord}_V(a/b) = \mathrm{ord}_V(a) - \mathrm{ord}_V(b)$.

When $\mathcal{O}_{W,V}$ is a discrete valuation ring, in particular whenever $W$ is normal, this length equals the valuation $\nu_V(a)$, so definition 1.1.1 agrees with the order function of [2, Divisors: Geometric Representation of Line Bundles]. For a fixed $r \in k(W)^*$ the integer $\mathrm{ord}_V(r)$ vanishes for all but finitely many $V$ ([2, Almost All Valuations Vanish]), so each sum $\sum_V \mathrm{ord}_V(r)[V]$ below is finite.

**Definition 1.1.2: Cycle**

Let $X$ be an algebraic scheme. A *$k$-cycle* on $X$ is a finite formal $\mathbb{Z}$-linear combination $\sum_i n_i[V_i]$ of $k$-dimensional subvarieties $V_i \subseteq X$ with $n_i \in \mathbb{Z}$. The $k$-cycles form the free abelian group

$$Z_k(X) = \bigoplus_{\dim V = k} \mathbb{Z}\,[V]$$

on the set of $k$-dimensional subvarieties of $X$, and the group of all cycles is $Z_*(X) = \bigoplus_{k\geq 0} Z_k(X)$.

A cycle remembers only which subvarieties occur and with what integer multiplicity; it detects neither embedded nor nonreduced structure, a point made precise by the equality $Z_k(X) = Z_k(X_{\text{red}})$. Two cases sit at the extremes. A 0-cycle is a finite formal sum of closed points. At the top, if $X$ has dimension $n$ with $n$-dimensional irreducible components $X_1, \ldots, X_r$, then $Z_n(X)$ is free on $[X_1], \ldots, [X_r]$.

The *fundamental cycle* of $X$ ranges over *all* irreducible components $X_1, \ldots, X_t$, of whatever dimension:

$$[X] = \sum_{i=1}^{t} \operatorname{length}_{\mathcal{O}_{X,X_i}}\left(\mathcal{O}_{X,X_i}\right)[X_i] \;\in\; Z_*(X)$$

weighting each component by the multiplicity with which $X$ carries it, and reducing to $\sum_i [X_i]$ when $X$ is reduced. When $X$ is equidimensional the sum is concentrated in degree $n$; in general it is not, and it must not be truncated to the top-dimensional components. A scheme such as the union of a plane and a line meeting it in a point carries a fundamental cycle with a 2-dimensional and a 1-dimensional term, and discarding the line would make the Segre classes of chapter 3 disagree with the standard ones on exactly such a base.

The one construction that produces relations among cycles is the divisor of a rational function. Given a $(k+1)$-dimensional subvariety $W \subseteq X$ and a nonzero $r \in k(W)^*$, its *divisor* is the $k$-cycle

$$[\operatorname{div}(r)] = \sum_V \operatorname{ord}_V(r)\,[V] \in Z_k(X)$$

the sum running over the codimension-one subvarieties $V$ of $W$, each of which is a $k$-dimensional subvariety of $X$. The sum is finite by [2, Almost All Valuations Vanish], and $[\operatorname{div}(rs)] = [\operatorname{div}(r)] + [\operatorname{div}(s)]$ because $\operatorname{ord}_V$ is a homomorphism.

**Definition 1.1.3: Rational Equivalence**

A $k$-cycle $\alpha \in Z_k(X)$ is *rationally equivalent to zero*, written $\alpha \sim 0$, if there are finitely many $(k+1)$-dimensional subvarieties $W_i \subseteq X$ and rational functions $r_i \in k(W_i)^*$ with

$$\alpha = \sum_i [\operatorname{div}(r_i)]$$

Two $k$-cycles are *rationally equivalent*, written $\alpha \sim \beta$, when $\alpha - \beta \sim 0$. The cycles rationally equivalent to zero form a subgroup $\operatorname{Rat}_k(X) \leq Z_k(X)$: it is closed under addition by definition and under negation because $[\operatorname{div}(r^{-1})] = -[\operatorname{div}(r)]$.

Geometrically, a subvariety $W$ together with a rational function $r$ presents $[\mathrm{div}(r)]$ as the difference between the zeros and the poles of $r$, that is, between the fibers over 0 and $\infty$ of the rational map $r\colon W \dashrightarrow \mathbb{P}^1$. Rational equivalence is thus the relation generated by sweeping one cycle to another through a family parametrized by the line. In the one codimension seen before, $k = n-1$ on an $n$-dimensional variety, a cycle is a Weil divisor and rational equivalence is exactly the linear equivalence of [2, Divisor Class Group]; the Chow group below carries $\mathrm{Cl}\, X$ to all dimensions at once.

**Definition 1.1.4: Chow Group**

The *k-th Chow group* of an algebraic scheme $X$ is the quotient

$$A_k(X) = Z_k(X)/\mathrm{Rat}_k(X)$$

Its elements are *k-cycle classes*, and the class of a cycle $\alpha$ is written $[\alpha]$, or simply $\alpha$ when no confusion results. The *total Chow group* is

$$A_*(X) = \bigoplus_{k \geq 0} A_k(X) = Z_*(X)/\mathrm{Rat}_*(X), \qquad \mathrm{Rat}_*(X) = \bigoplus_{k \geq 0} \mathrm{Rat}_k(X)$$

Following the cohomological convention, much of the literature grades by codimension, writing $A^p(X) = A_{n-p}(X)$ or $CH^p(X)$ when $X$ is equidimensional of dimension $n$; the two indexings carry the same information, and definition 1.1.4 is the promised general form of the construction previewed for surfaces in [2, Chow Rings and the General Intersection Product]. A cycle $\sum_i n_i[V_i]$ is *positive* (or *effective*) if it is nonzero with every $n_i \geq 0$, and a cycle class is positive if it has a positive representative. Positivity is the algebraic trace of an actual geometric subvariety, and deciding which classes are positive is among the central difficulties of the subject.

Passing to the quotient collapses exactly the ambiguity that a complete variety imposes on its points.

**Example 1.1.5: Zero-Cycles on the Line**

On $X = \mathbb{P}^1$ any two closed points are rationally equivalent. Write $t$ for the affine coordinate and $\infty$ for the point at infinity. Distinct points $p, q \in \mathbb{A}^1 \subseteq \mathbb{P}^1$ are cut out by $t-p$ and $t-q$, and the rational function $r = (t-p)/(t-q)$ has

$$\mathrm{ord}_p(r) = 1, \qquad \mathrm{ord}_q(r) = -1, \qquad \mathrm{ord}_x(r) = 0 \text{ for } x \neq p, q$$

since $r$ takes the finite nonzero value 1 at $\infty$. Hence $[\mathrm{div}(r)] = [p] - [q]$ and $[p] \sim [q]$; taking $r = t-p$ instead gives $[\mathrm{div}(t-p)] = [p] - [\infty]$, so $[\infty]$ lies in the same class. Every point of $\mathbb{P}^1$ therefore represents a single class $h$. A rational function on $\mathbb{P}^1$ has as many zeros as poles, so $\sum_V \mathrm{ord}_V(r) = 0$ for all $r$, and the total-coefficient map $\sum_i n_i[p_i] \mapsto \sum_i n_i$ descends to an isomorphism

$$A_0(\mathbb{P}^1) \xrightarrow{\sim} \mathbb{Z}, \qquad h \mapsto 1$$

The affine line behaves oppositely. On $X = \mathbb{A}^1$ the function $t-p$ has the single zero $p$ and no pole, so $[\mathrm{div}(t-p)] = [p]$ and every point satisfies $[p] \sim 0$; thus $A_0(\mathbb{A}^1) = 0$. On an affine line a point can be swept away to nothing, whereas on the complete line it cannot, and the surviving invariant of a class is its degree.

The contrast in example 1.1.5 is why completeness recurs throughout the theory: the degree of a zero-cycle is an invariant of its rational equivalence class only on a complete scheme, a point taken up in section 1.2. The remaining first properties of Chow groups are collected next; each is either immediate or follows from the localization sequence established later in the chapter.

**Example 1.1.6: First Properties of Chow Groups**
Let $X$ be an algebraic scheme.

1. *Reduction.* Since cycles see only subvarieties, $A_k(X) = A_k(X_{\text{red}})$ for every $k$.
2. *Disjoint unions.* If $X = X' \sqcup X''$ is a disjoint union of open-and-closed subschemes, then $A_k(X) = A_k(X') \oplus A_k(X'')$, as every subvariety lies in one piece.
3. *Top-dimensional cycles.* If $\dim X = n$, there is no subvariety of dimension $n+1$, so $\mathrm{Rat}_n(X) = 0$ and $A_n(X) = Z_n(X)$ is free on the $n$-dimensional components of $X$. In particular the coefficient of such a component in a class $\alpha \in A_n(X)$ is well defined.
4. *Right-exactness.* For closed subschemes $X_1, X_2 \subseteq X$ there is an exact sequence
$$A_k(X_1 \cap X_2) \to A_k(X_1) \oplus A_k(X_2) \to A_k(X_1 \cup X_2) \to 0$$
with no left-exactness in general. This is a consequence of the localization sequence developed later in the chapter.

### Exercise 1.1.1

1. Let $X$ be the reduced scheme consisting of two distinct points. Compute $Z_k(X)$ and $A_k(X)$ for all $k$.
2. Show that $A_0(\mathbb{A}^n) = 0$ for every $n \geq 1$.
3. Prove from the definitions that $Z_k(X) = Z_k(X_{\text{red}})$ and $\mathrm{Rat}_k(X) = \mathrm{Rat}_k(X_{\text{red}})$, and deduce $A_k(X) = A_k(X_{\text{red}})$.
4. Let $X$ have dimension $n$ with $n$-dimensional irreducible components $X_1, \ldots, X_r$. Show that $A_n(X) = \bigoplus_{i=1}^r \mathbb{Z}\,[X_i]$.
5. Let $C$ be a smooth projective curve over $\overline{k}$. Show that $A_0(C) \cong \mathrm{Cl}\, C = \mathrm{Pic}\, C$, and that the degree gives a short exact sequence $0 \to \mathrm{Pic}^0 C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$.

## 1.2 Proper Pushforward and the Degree

Cycles and their classes change under morphisms, and the first of these functorialities is covariant: a proper morphism pushes cycles forward, respects rational equivalence, and on a complete scheme produces the degree of a zero-cycle.

A proper morphism $f\colon X \to Y$ carries closed sets to closed sets, so the image of a subvariety $V \subseteq X$ is again a subvariety $W = f(V) \subseteq Y$: the image of an irreducible set is irreducible, and properness (universal closedness) makes it closed. Without properness the image can fail to be closed, as for the projection of [2, Non-Proper Morphisms], and pushforward is not defined. The inclusion of function fields $k(W) \hookrightarrow k(V)$ is a finite extension exactly when $\dim V = \dim W$, and its degree counts the points of $V$ over a general point of $W$.

**Definition 1.2.1: Proper Pushforward**

Let $f\colon X \to Y$ be a proper morphism and $V \subseteq X$ a subvariety with image $W = f(V)$. Set

$$\deg(V/W) = \begin{cases} [k(V) : k(W)] & \dim V = \dim W \\ 0 & \dim V > \dim W \end{cases}$$

(the extension $k(W) \hookrightarrow k(V)$ is finite in the first case), and put $f_*[V] = \deg(V/W)\,[W]$. Extending $\mathbb{Z}$-linearly gives the *pushforward* $f_*\colon Z_k(X) \to Z_k(Y)$.

Pushforward is covariant: for composable proper morphisms $(g \circ f)_* = g_* \circ f_*$, since degrees of function-field extensions multiply in towers. When $\dim V = \dim W$ the number $\deg(V/W)$ counts the sheets of the generically finite cover $V \to W$, matching the topological degree over $\mathbb{C}$. That $f_*$ descends to Chow groups is the content of this section, and rests on how it acts on the divisor of a rational function.

**Proposition 1.2.2: Pushforward of Divisor**

Let $f : X \to Y$ be a proper surjective morphisms between varieties, and let $r \in k(X)^*$. Then:

1. if $\dim(Y) < \dim(X)$, then
$$f_*[\mathrm{div}(r)] = 0$$
2. if $\dim(Y) = \dim(X)$, then
$$f_*[\mathrm{div}(r)] = [\mathrm{div}(N(r))]$$
where $N(r)$ is the *norm* of $r$ (computed as the determinant of the $R(Y)$-linear endomorphism of $R(X)$ given by multiplying by $r$).

***Proof*:**
The order function and the norm are multiplicative, so both sides of each identity are homomorphisms in $r$. Both parts reduce to two model computations; write $n = \dim X$.

*Step 1: the line over a point.* Let $Y = \operatorname{Spec} k$ and $X = \mathbb{P}^1$, with $f$ the structure map. Every closed point of $\mathbb{P}^1$ has residue field $k = \overline{k}$, so $f_*[p] = [Y]$ for each $p$ and $f_*[\mathrm{div}(r)] = \big(\sum_V \operatorname{ord}_V(r)\big)[Y]$. Writing $r = c\prod_j (t - a_j)^{m_j}$ in the affine coordinate $t$, the order of $r$ at $a_j$ is $m_j$ and its order at $\infty$ is $-\sum_j m_j$ (in $s = 1/t$ one has $r = s^{-\sum_j m_j} u$ with $u$ a unit at $s = 0$), so $\sum_V \operatorname{ord}_V(r) = 0$ and $f_*[\mathrm{div}(r)] = 0$. The same count shows that on any proper curve $C$ over a field $K$ a principal divisor has degree zero: normalize $C$ and choose a finite map $\widetilde{C} \to \mathbb{P}^1_K$, then apply Step 2 to that map and this computation to $\mathbb{P}^1_K \to \operatorname{Spec} K$.

*Step 2: finite morphisms of equal dimension.* Suppose $f$ is finite with $\dim Y = \dim X$, so that $K = k(Y) \subseteq L = k(X)$ is a finite extension. Fix a codimension-one subvariety $W \subseteq Y$ and let $A = \mathcal{O}_{Y,W}$, a one-dimensional local domain with fraction field $K$. Its integral closure $B$ in $L$ is a finite $A$-module whose maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_s$ correspond to the codimension-one subvarieties $V_i \subseteq X$ over $W$, with $B_{\mathfrak{m}_i} = \mathcal{O}_{X,V_i}$ and residue degree $[k(V_i) : k(W)] = [B/\mathfrak{m}_i : A/\mathfrak{m}]$. Only these $V_i$ contribute to the coefficient of $[W]$ in $f_*[\mathrm{div}(r)]$, which is therefore $\sum_i [k(V_i) : k(W)] \operatorname{ord}_{V_i}(r)$, while the coefficient of $[W]$ in $[\operatorname{div}(N(r))]$ is $\operatorname{ord}_W(N(r))$. For $r \in B$ a nonzerodivisor these agree:

$$\operatorname{ord}_W(N(r)) = \operatorname{length}_A\big(A/N(r)A\big) = \operatorname{length}_A\big(B/rB\big) = \sum_i [k(V_i) : k(W)] \operatorname{ord}_{V_i}(r)$$

the middle equality is the determinant-length formula, since $N(r)$ is the determinant of multiplication by $r$ on the finite $A$-module $B$, and the last is additivity of length across the semilocal decomposition of $B$, weighted by residue degrees (see [7, Appendix A] for this length bookkeeping). As $L = \operatorname{Frac}(B)$ and both sides are homomorphisms, the identity holds for all $r \in L^*$, which is part (2) for finite $f$.

*Step 3: the general case.* A proper surjective $f$ with $\dim Y = \dim X$ is generically finite, and the coefficient of a codimension-one $W \subseteq Y$ in $f_*[\operatorname{div}(r)]$ is computed over the generic point of $W$, where $f$ becomes finite; Step 2 gives (2). For (1), let $\dim Y < n$. Since $f_*[\operatorname{div}(r)] \in Z_{n-1}(Y)$ vanishes when $\dim Y < n-1$, only $\dim Y = n-1$ remains, and there $f_*[\operatorname{div}(r)] = c\,[Y]$ with $c$ its coefficient over the generic point of $Y$. Base-changing to $K = k(Y)$ replaces $f$ by a proper curve $C = X_K$ over the field $K$ and $r$ by a function in $k(C)^*$, and $c = \sum_{P \in C} \operatorname{ord}_P(r)\,[k(P) : K]$ is the degree of its principal divisor, which vanishes by Step 1. Hence $f_*[\operatorname{div}(r)] = 0$, completing the proof.

**Theorem 1.2.3: Pushforward Preserves Rational Equivalence**

Let $f \colon X \to Y$ be proper. If $\alpha \in Z_k(X)$ is rationally equivalent to zero, then so is $f_*\alpha$. Consequently $f_*$ descends to a homomorphism

$$f_* \colon A_k(X) \to A_k(Y)$$

and $A_*(-)$ equipped with proper pushforward is a covariant functor.

***Proof*:**
It suffices to push forward a generator $\alpha = [\operatorname{div}(r)]$, where $r \in k(W)^*$ for some $(k+1)$-dimensional subvariety $W \subseteq X$. Replacing $X$ by $W$ and $Y$ by $f(W)$ makes $f$ proper and surjective, and proposition 1.2.2 identifies $f_*[\operatorname{div}(r)]$ with $0$ (when $\dim f(W) < \dim W$) or with $[\operatorname{div}(N(r))]$ (when $\dim f(W) = \dim W$). Either value is a principal divisor, hence rationally equivalent to zero, so $f_*\alpha \sim 0$. Functoriality on classes then follows from $(g \circ f)_* = g_* \circ f_*$ on cycles, as we sought to show.

**Definition 1.2.4: Degree of a Zero-Cycle**

Let $X$ be complete (proper over $\operatorname{Spec} k$), with structure morphism $\pi\colon X \to \operatorname{Spec} k$, and let $\alpha = \sum_P n_P[P]$ be a zero-cycle on $X$. The *degree* of $\alpha$ is

$$\deg(\alpha) = \int_X \alpha = \sum_P n_P\,[k(P) : k]$$

Since $\operatorname{Spec} k$ is a single point with no lower-dimensional subvarieties, $A_0(\operatorname{Spec} k) = Z_0(\operatorname{Spec} k) = \mathbb{Z}\,[\operatorname{Spec} k] \cong \mathbb{Z}$, and under this identification $\deg(\alpha) = \pi_*\alpha$. When the scheme is clear one writes $\int \alpha$ for $\int_X \alpha$.

By theorem 1.2.3 rationally equivalent zero-cycles have the same degree, so $\deg\colon A_0(X) \to \mathbb{Z}$ is well defined on a complete scheme. Extending by $\int_X \alpha = 0$ for $\alpha \in A_k(X)$ with $k > 0$ and using $\pi_X = \pi_Y \circ f$, any morphism $f\colon X \to Y$ of complete schemes satisfies

$$\int_X \alpha = \int_Y f_*\alpha$$

for every $\alpha \in A_*(X)$, a special case of the functoriality of pushforward.

**1.2.5 Assuming Pushforward with subschemes** Let $Y_1, \ldots, Y_r$ be closed subschemes of a scheme $X$. Let $Y$ be a closed subscheme of $X$ which contains all the $Y_i$. Given $\alpha_i \in A_*Y_i$, $i = 1, \ldots, r$, and $\beta \in A_*Y$, the notation "$\beta = \sum_{i=1}^r \alpha_i$ in $A_*Y$" stands in for the precise equation $\beta = \sum_{i=1}^r \varphi_{i*}(\alpha_i)$ where $\varphi_i$ is the inclusion of $Y_i$ in $Y$.

**Example 1.2.6: Degrees, Bézout, and the Role of Properness**

1. *Bézout's theorem.* Two curves $C, D \subseteq \mathbb{P}^2$ of degrees $c, d$ with no common component meet in a zero-cycle $C \cdot D$ of degree $cd$. Pushing forward to a point, $\int_{\mathbb{P}^2}[C \cdot D] = cd$: the classical intersection number from the chapter opening is the degree of a zero-cycle, and its independence of the choice of $C$ and $D$ in their linear systems is rational equivalence together with theorem 1.2.3.

2. *Properness is necessary.* For the non-proper projection of [2, Non-Proper Morphisms] the image of a subvariety can fail to be closed, so $f_*$ is not even defined; and on $\mathbb{A}^1$ every point is rationally equivalent to zero (example 1.1.5), so no degree survives. The degree is an invariant of a rational equivalence class only on a complete scheme.

3. *A finite cover of curves.* A finite morphism $\varphi\colon C \to C'$ of smooth projective curves satisfies $\varphi_*[P] = [k(P) : k(\varphi(P))]\,[\varphi(P)]$. For a point $Q \in C'$ away from the branch locus, $\varphi^{-1}(Q)$ consists of $\deg\varphi = [k(C) : k(C')]$ reduced points, so $\varphi_*\big[\varphi^{-1}(Q)\big] = (\deg\varphi)\,[Q]$; since $\deg(\varphi_*\alpha) = \deg(\alpha)$ on the complete curves, the sheet count is exactly the degree recorded by $\varphi$.

## Exercise 1.2.1

1. For the finite morphism $\varphi\colon \mathbb{P}^1 \to \mathbb{P}^1$, $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (in characteristic $\neq 2$), compute $\varphi_*[p]$ for a closed point $p$ and $\varphi_*[\varphi^{-1}(q)]$ for a closed point $q$. Verify that $\deg(\varphi_*\alpha) = \deg(\alpha)$ for every zero-cycle $\alpha$, and explain where the degree $\deg\varphi = 2$ appears.
2. Let $\alpha = \sum_i n_i[P_i]$ be a zero-cycle on $\mathbb{P}^n$ over $\bar{k}$. Compute $\deg(\alpha)$ and prove it is invariant under rational equivalence.
3. Deduce from proposition 1.2.2 that a nonzero rational function on a smooth projective curve $C$ has principal divisor of degree zero, and show by example that this fails on $\mathbb{A}^1$.
4. For the structure map $\pi\colon \mathbb{P}^1 \to \operatorname{Spec}\bar{k}$ and a rational function $r \in k(\mathbb{P}^1)^*$, verify $\int_{\mathbb{P}^1}[\operatorname{div}(r)] = \int_{\operatorname{Spec}\bar{k}} \pi_*[\operatorname{div}(r)]$, and say which hypothesis of definition 1.2.4 would fail were $\mathbb{P}^1$ replaced by $\mathbb{A}^1$.

## 1.3 Flat Pullback

Proper pushforward of section 1.2 is covariant: it moves cycles along a morphism in the direction of the arrow, lowering nothing and raising nothing in dimension. The second functoriality of the Chow group runs the other way. A morphism $f\colon X \to Y$ pulls a subvariety $V \subseteq Y$ back to its preimage $f^{-1}(V) \subseteq X$, and one would like to read that preimage as a cycle on $X$. For an arbitrary $f$ this fails: the preimage can jump in dimension from fiber to fiber, can acquire embedded components, and carries no canonical multiplicities. The hypothesis that repairs every one of these defects at once is flatness. A flat morphism of relative dimension $n$ has equidimensional fibers, so the preimage of a $k$-dimensional subvariety is pure of dimension $k + n$, and the scheme-theoretic preimage supplies the multiplicities with which its components must be counted. The resulting map $f^*\colon Z_k(Y) \to Z_{k+n}(X)$ raises dimension by $n$, is contravariant, and descends to Chow groups.

This section constructs $f^*$, records its three structural properties (it is a graded homomorphism, it composes contravariantly, and it commutes with proper pushforward across a fiber square), and proves that it respects rational equivalence. The two guiding examples are the ones every later computation rests on: restriction to an open subscheme, and the projection $X \times \mathbb{A}^n \to X$ of a trivial affine bundle.

Recall that a morphism $f\colon X \to Y$ of algebraic schemes is flat if each local ring $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. Flatness is the algebraic incarnation of the geometric require that the fibers of $f$ vary without collapsing or jumping, and its precise dimensional consequence is what makes pullback of cycles possible.

**Definition 1.3.1: Flat Morphism of Relative Dimension $n$**

A morphism $f\colon X \to Y$ of algebraic schemes is *flat of relative dimension* $n$ if $f$ is flat and, for every subvariety $W \subseteq Y$ and every irreducible component $Z$ of $f^{-1}(W)$,

$$\dim Z = \dim W + n$$

Equivalently, $f$ is flat and every nonempty fiber of $f$ has pure dimension $n$.

The equivalence stated in the definition is the mechanism the whole construction turns on, so it is worth isolating. Flatness forces the fiber dimension to be locally constant, and it forces the preimage of a subvariety to inherit that constant as an additive shift; neither holds without it.

**Proposition 1.3.2: Flat Preimages Are Equidimensional**

Let $f\colon X \to Y$ be a flat morphism of algebraic schemes with $Y$ a variety, and suppose every nonempty fiber of $f$ has pure dimension $n$. Then for every subvariety $W \subseteq Y$ each irreducible component of $f^{-1}(W)$ has dimension $\dim W + n$. In particular $f^{-1}(W)$ is either empty or pure of dimension $\dim W + n$.

***Proof*:**
Write $g\colon f^{-1}(W) \to W$ for the restriction of $f$; it is the base change of $f$ along the closed immersion $W \hookrightarrow Y$, hence flat, and its fibers are fibers of $f$, so each nonempty fiber of $g$ has pure dimension $n$. Thus it suffices to prove the statement for $W = Y$: that a flat morphism $g\colon X' \to W$ onto a variety $W$, with all nonempty fibers of pure dimension $n$, has $\dim Z = \dim W + n$ for every irreducible component $Z$ of $X'$.

Let $Z \subseteq X'$ be an irreducible component, with generic point $\eta$. Flatness satisfies going-down (the going-down property of flat local homomorphisms, [8, p. III.9]), so the minimal prime $\eta$ of $X'$ contracts to the minimal prime of $W$; that is, $g(\eta)$ is the generic point of $W$, and $g$ restricts to a dominant morphism $g\big|_Z\colon Z \to W$ of varieties. Both are integral of finite type over $\overline{k}$, so their dimensions are transcendence degrees of function fields, and the dimension of a fiber of a dominant morphism of varieties can be read at a closed point.
The fiber-dimension theorem for the dominant morphism $g\big|_Z\colon Z \to W$ of varieties ([8, Exercise II.3.22]) provides a dense open $U \subseteq Z$ such that for every closed point $z \in U$, the fiber $(g\big|_Z)^{-1}(g(z))$ has dimension exactly $\dim Z - \dim W$ at $z$. The complement in $Z$ of the other irreducible components of $X'$ is also a dense open, so their intersection is nonempty and contains a closed point $z$; write $w = g(z) \in W$. Since $z$ lies off the other components, $X'$ coincides with $Z$ near $z$, so the fiber $X'_w = g^{-1}(w)$ coincides with $(g\big|_Z)^{-1}(w)$ near $z$, and its dimension at $z$ is $\dim Z - \dim W$. By hypothesis $X'_w$ is nonempty of pure dimension $n$, so this local fiber dimension equals $n$. Therefore

$$n = \dim Z - \dim W$$

hence $\dim Z = \dim W + n$, as we sought to show.

The proposition is the geometric license behind the definition of flat pullback. If $V \subseteq Y$ is a $k$-dimensional subvariety and $f$ is flat of relative dimension $n$, then $f^{-1}(V)$ is a closed subscheme of $X$, pure of dimension $k + n$, and one takes its associated cycle, weighting each component by the length that measures the multiplicity with which the scheme carries it. This is exactly the fundamental cycle of definition 1.1.2 applied to the scheme-theoretic preimage.

**Definition 1.3.3: Flat Pullback**

Let $f\colon X \to Y$ be a flat morphism of relative dimension $n$ and $V \subseteq Y$ a $k$-dimensional subvariety. The scheme-theoretic preimage $f^{-1}(V) = X \times_Y V$ is a closed subscheme of $X$, pure of dimension $k+n$, with irreducible components $Z_1, \ldots, Z_r$. The *flat pullback* of $[V]$ is the $(k+n)$-cycle

$$f^*[V] = \left[f^{-1}(V)\right] = \sum_{i=1}^{r} \operatorname{length}_{\mathcal{O}_{f^{-1}(V),Z_i}}\left(\mathcal{O}_{f^{-1}(V),Z_i}\right)[Z_i]$$

the length being taken in the local ring of the preimage scheme at the generic point of $Z_i$. Extending $\mathbb{Z}$-linearly gives the flat pullback

$$f^*\colon Z_k(Y) \to Z_{k+n}(X)$$

The multiplicities are the point of the construction. The preimage as a topological set records only the components $Z_i$; the scheme structure records how many times $f$ "covers" each one, and it is these lengths that make $f^*$ compatible with divisors and hence with rational equivalence. When $f$ is an open immersion the preimage is reduced and every length is 1, so $f^*[V] = [V \cap X]$; this is the source of the restriction map computed in the next example. The relative dimension $n$ is the amount by which $f^*$ raises the dimension of a cycle, so $f^*$ is graded of degree $+n$.

**Example 1.3.4: Open Restriction and the Projection $X \times \mathbb{A}^n \to X$**

Two flat morphisms recur throughout the theory, and their pullbacks can be read off the definition.

1. *Open immersion.* Let $j\colon U \hookrightarrow Y$ be the inclusion of an open subscheme. Such $j$ is flat of relative dimension 0: it is a localization at the level of local rings, and its fibers are single reduced points (over $u \in U$) or empty (over $y \notin U$). For a $k$-dimensional subvariety $V \subseteq Y$ the preimage $j^{-1}(V) = V \cap U$ is the open subscheme of $V$, which is reduced. If $V \cap U \neq \emptyset$ it is a $k$-dimensional subvariety of $U$ and $j^*[V] = [V \cap U]$; if $V \subseteq Y \setminus U$ the preimage is empty and $j^*[V] = 0$. Thus flat pullback along an open immersion is the *restriction of cycles* $Z_k(Y) \to Z_k(U)$ that forgets the part of a cycle supported off $U$, the map at the heart of the localization sequence of section 1.4.

2. *Trivial affine bundle.* Let $p\colon X \times \mathbb{A}^n \to X$ be the projection. It is flat, being a base change of the flat morphism $\mathbb{A}^n_{\overline{k}} \to \operatorname{Spec} \overline{k}$ along $X \to \operatorname{Spec} \overline{k}$, and every fiber is a copy of $\mathbb{A}^n$, pure of dimension $n$; so $p$ is flat of relative dimension $n$. For a $k$-dimensional subvariety $V \subseteq X$ the preimage is $V \times \mathbb{A}^n$, which is integral of dimension $k+n$, so

$$p^*[V] = [V \times \mathbb{A}^n]$$

the cylinder over $V$. No multiplicity appears because $V \times \mathbb{A}^n$ is a variety. This is the computation that will make $p^*$ an isomorphism on Chow groups (theorem 1.4.6).

Both examples exhibit the pattern the general theory generalizes: pullback restricts a cycle to an open part in the first case and thickens it uniformly along the fiber in the second. The structural properties below show that these behaviors are functorial.

The first property collects the ways in which $f^*$ is well behaved as a map of cycle groups. It is a graded homomorphism raising dimension by the relative dimension, and it composes contravariantly, so that a tower of flat morphisms pulls back through the composite exactly as one expects.

**Proposition 1.3.5: Basic Properties of Flat Pullback**

Let $f\colon X \to Y$ be flat of relative dimension $n$.

1. *Homomorphism.* $f^*\colon Z_k(Y) \to Z_{k+n}(X)$ is a homomorphism of abelian groups for each $k$, and assembles to a graded homomorphism $f^*\colon Z_*(Y) \to Z_*(X)$ of degree $n$.
2. *Fundamental cycle.* $f^*[Y] = [X]$ for any algebraic scheme $Y$: flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space.
3. *Composition.* If $g\colon Y \to Y'$ is flat of relative dimension $m$, then $g \circ f$ is flat of relative dimension $m + n$ and
$$(g \circ f)^* = f^* \circ g^*$$
as maps $Z_k(Y') \to Z_{k+m+n}(X)$.
4. *Identity.* $(\mathrm{id}_X)^* = \mathrm{id}$ on $Z_*(X)$.

***Proof*:**

1. *Homomorphism.* By construction $f^*$ is defined on generators $[V]$ and extended $\mathbb{Z}$-linearly, so it is a homomorphism $Z_k(Y) \to Z_{k+n}(X)$ by fiat; proposition 1.3.2 is what guarantees the target group is $Z_{k+n}(X)$, since $f^{-1}(V)$ is pure of dimension $k + n$. Summing over $k$ gives a degree-$n$ graded homomorphism.
2. *Fundamental cycle.* Write $[Y] = \sum_i m_i[Y_i]$ over the irreducible components of $Y$, with $m_i = \mathrm{length}_{\mathcal{O}_{Y,Y_i}}(\mathcal{O}_{Y,Y_i})$. If $Y$ is a variety this is the single generator $[Y]$ with multiplicity 1, and $f^{-1}(Y) = X$ gives $f^*[Y] = [X]$ at once.

   In general, $f^*[Y] = \sum_i m_i[f^{-1}(Y_i)]$ by definition of flat pullback, and since $Y = \bigcup_i Y_i$ we have $X = \bigcup_i f^{-1}(Y_i)$. The two sides are cycles, so it is enough to compare the coefficient of each subvariety of $X$, and the work is in matching up which subvarieties occur. Flatness enters through going-down: $\mathcal{O}_{Y,f(\eta)} \to \mathcal{O}_{X,\eta}$ is a flat local homomorphism for every $\eta \in X$, so every generization of $f(\eta)$ in $Y$ lifts to a generization of $\eta$ in $X$.

   *Every component of $X$ dominates a component of $Y$.* Let $X_j$ be a component of $X$, with generic point $\eta_j$, and put $y = f(\eta_j)$. A generization of $y$ lifts to a generization of $\eta_j$; but $\eta_j$ is a generic point of $X$, so its only generization in $X$ is itself, and therefore $y$ has no proper generization in $Y$. So $y$ is the generic point of some component $Y_i$, and $X_j$ is a component of $f^{-1}(Y_i)$. That $i$ is unique: for $i' \neq i$ the generic point of $Y_i$ does not lie in $Y_{i'}$, so $\eta_j \notin f^{-1}(Y_{i'})$ and only the $i$-th term of $\sum_i m_i[f^{-1}(Y_i)]$ contributes to the coefficient of $[X_j]$.

   *Conversely, every component of every $f^{-1}(Y_i)$ is a component of $X$.* Let $W$ be one, with generic point $\eta_W$. Going-down applied at $\eta_W$ lifts the generic point of $Y_i$ to a generization of $\eta_W$ inside $f^{-1}(Y_i)$, and $\eta_W$ is maximal there, so $W$ dominates $Y_i$, that is $f(\eta_W)$ is the generic point of $Y_i$. If $W$ were not a component of $X$ it would lie in some component $X_{j'}$, whose generic point generizes $\eta_W$ and therefore maps to a generization of the generic point of $Y_i$; the latter being a generic point of $Y$, that forces $X_{j'}$ to dominate $Y_i$ as well, so $\dim X_{j'} = \dim Y_i + n = \dim W$ by proposition 1.3.2, and $W = X_{j'}$. A component $Y_i$ with $f^{-1}(Y_i) = \emptyset$ contributes to neither side.

*Multiplicities.* It remains to see that the multiplicity of $X_j$ in $[X]$ equals $m_i$ times its multiplicity in $[f^{-1}(Y_i)]$. This is a length computation: $\mathcal{O}_{Y,Y_i} \to \mathcal{O}_{X,X_j}$ is a flat local homomorphism of Noetherian local rings, and [3, Length Along a Flat Local Homomorphism] gives $\operatorname{length}(\mathcal{O}_{X,X_j}) = \operatorname{length}(\mathcal{O}_{Y,Y_i}) \cdot \operatorname{length}\big(\mathcal{O}_{X,X_j}/\mathfrak{m}_{Y_i}\mathcal{O}_{X,X_j}\big)$. Because $Y_i$ is a subvariety and so reduced, $\mathcal{O}_{Y,Y_i}/\mathfrak{m}_{Y_i}$ is the function field of $Y_i$, whence $\mathcal{O}_{X,X_j}/\mathfrak{m}_{Y_i}\mathcal{O}_{X,X_j}$ is the local ring of $f^{-1}(Y_i)$ at $X_j$ and the second factor is exactly the multiplicity of $X_j$ in $[f^{-1}(Y_i)]$. Summing over $i$ and $j$ gives $f^*[Y] = [X]$, as we sought to show.

3. *Composition.* Composites of flat morphisms are flat, and relative dimensions add because fiber dimensions add along the tower: a fiber of $g \circ f$ over $y' \in Y'$ maps to the fiber $g^{-1}(y')$ of pure dimension $m$, with fibers of $f$ of pure dimension $n$, so it has pure dimension $m+n$. Hence $g \circ f$ is flat of relative dimension $m+n$. For the cycle identity it suffices to check on a generator $[V]$ with $V \subseteq Y'$ a $k$-dimensional subvariety. Both sides are cycles supported on $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$, so the claim is that the multiplicities match component by component. The scheme-theoretic preimage is computed by iterated fiber product,
$$(g \circ f)^{-1}(V) = X \times_{Y'} V = X \times_Y (Y \times_{Y'} V) = f^{-1}\big(g^{-1}(V)\big)$$
so the underlying scheme of $(g \circ f)^{-1}(V)$ equals the underlying scheme of $f^{-1}(g^{-1}(V))$. It remains to see that the length multiplicities agree, that is, that applying $f^*$ to the cycle $g^*[V] = \sum_i \ell_i [W_i]$ (where $W_i$ are the components of $g^{-1}(V)$ with lengths $\ell_i$) reproduces the multiplicities of $(g \circ f)^{-1}(V)$. Fix a component $Z$ of the common preimage, with generic point $z$ lying over the generic point $w$ of some $W_i$. Flatness of $f$ makes $\mathcal{O}_{X,z}$ flat over $\mathcal{O}_{Y,w}$, and length is additive and multiplies through a flat local homomorphism along the fiber: the length of $\mathcal{O}_{(g\circ f)^{-1}(V),z}$ equals $\ell_i$ times the length of $\mathcal{O}_{f^{-1}(W_i),z}$. Summing $z$ over the components of $f^{-1}(W_i)$ and $i$ over the components of $g^{-1}(V)$ gives $(g \circ f)^*[V] = f^*(g^*[V])$, as required.

4. *Identity.* The identity map is flat of relative dimension 0 with $(\mathrm{id})^{-1}(V) = V$ reduced, so $(\mathrm{id}_X)^*[V] = [V]$ on generators, completing the proof.

Contravariant functoriality (part 3) is what will let a Chow-group computation be assembled from simpler flat maps, exactly as covariant functoriality of pushforward let degrees be computed in towers. The fundamental-cycle identity (part 2) is the seed of every computation in section 1.5: it identifies the generator of the top Chow group of the total space of a flat family in terms of the base.

The remaining two properties are the interaction of flat pullback with the machinery already built. The first is its compatibility with proper pushforward. Pullback and pushforward move cycles in opposite directions, and they do not commute for a single morphism; but across a fiber square, where one has a flat map and a proper map meeting at right angles, the two operations do commute. This is the cycle-level form of the projection formula, and it is the technical engine behind essentially every later compatibility.

**Proposition 1.3.6: Compatibility of Flat Pullback and Proper Pushforward**

Consider a fiber square of algebraic schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

that is Cartesian, $X' = X \times_Y Y'$, with $f$ proper and $g$ flat of relative dimension $n$. Then $f'$ is proper, $g'$ is flat of relative dimension $n$, and

$$g^* \circ f_* = f'_* \circ g'^* \colon Z_k(X) \to Z_{k+n}(Y')$$

Consequently the identity holds on Chow groups.

***Proof*:**
*Step 1: the maps in the base-changed square.* Properness and flatness are stable under base change, so $f'$ is proper and $g'$ is flat; the fibers of $g'$ over a point of $X$ are fibers of $g$ over its image in $Y$, so $g'$ has the same pure relative dimension $n$ as $g$. Both composites $g^* \circ f_*$ and $f'_* \circ g'^*$ therefore land in $Z_{k+n}(Y')$, and both are homomorphisms, so it suffices to check the identity on a generator $[V]$ with $V \subseteq X$ a $k$-dimensional subvariety.

*Step 2: reduction to $X = V$.* Replacing $X$ by $V$, $Y$ by $W = f(V)$, and forming the induced fiber squares, the four schemes become $V$, $W$, $V' = V \times_W W'$, $W' = W \times_Y Y'$, with $f \colon V \to W$ proper surjective and $g \colon W' \to W$ flat of relative dimension $n$. It suffices to prove

$$g^*(f_*[V]) = f'_*\big(g'^*[V]\big)$$

where now $f_*[V] = \deg(V/W)[W]$ by definition 1.2.1 and $g'^*[V] = [g'^{-1}(V)] = [V']$ (with its multiplicities).

*Step 3: the two dimension regimes.* There are two cases according to whether $f$ drops dimension.

If $\dim V > \dim W$, then $\deg(V/W) = 0$, so the left side is $g^*(0) = 0$. On the right, $V' = V \times_W W'$ maps to $W'$ under $f'$ with fibers the fibers of $V \to W$, which are positive dimensional over the generic point; hence $\dim V' > \dim W'$ and every component of $V'$ has image of strictly smaller dimension in $W'$, so $f'_*[V'] = 0$ by the dimension clause of definition 1.2.1. Both sides vanish.

If $\dim V = \dim W$, write $d = \deg(V/W) = [k(V) : k(W)]$. The left side is $g^*(d\,[W]) = d\,[W']$, using $g^*[W] = [W'] = [g^{-1}(W)]$ (the base $W$ is a variety and $g$ is flat, so $g^*$ of its fundamental cycle is the fundamental cycle of $W'$ by proposition 1.3.5(2), which here is the total space $W' = g^{-1}(W)$). For the right side, $[V'] = g'^*[V]$ carries the pullback multiplicities flagged above, and $f'_*[V']$ must account for both those multiplicities and the residue-field degrees; flatness is what makes only the field degree survive. Fix a component $W'_j$ of $W'$, with generic point $\eta_j$. Localizing at $\eta_j$, the fiber of $f' \colon V' \to W'$ over $\eta_j$ is Spec of $k(V) \otimes_{k(W)} k(W'_j)$: the generic fiber of $f \colon V \to W$ is $\operatorname{Spec} k(V)$, a $k(W)$-algebra of dimension $d$ since $V, W$ are varieties, and flat base change along $k(W) \hookrightarrow k(W'_j)$ preserves that dimension, so $\dim_{k(W'_j)}\big(k(V) \otimes_{k(W)} k(W'_j)\big) = d$. This dimension is exactly the total multiplicity that $f'_*[V']$ records over $W'_j$: summing the pullback multiplicity of each component of $V'$ dominating $W'_j$ against its residue degree over $k(W'_j)$ recovers the length $d$ of the fiber algebra, because flatness of $g$ (hence of $g'$) forbids any length being created or lost in the base change. Thus $f'_*[V'] = \sum_j d\,[W'_j] = d\,[W']$, matching the left side. The two sides agree in both regimes, which establishes $g^* f_* = f'_* g'^*$ on cycles. Since each of $g^*$, $f_*$, $f'_*$, $g'^*$ descends to Chow groups by theorem 1.2.3 and proposition 1.3.7 below, the identity descends as well, completing the proof.

The compatibility should be read as the statement that pushing a cycle forward and then restricting to a flat base change gives the same class as first restricting and then pushing forward within the base change. In the special case where $g$ is an open immersion, it says that pushforward commutes with restriction to an open set, which is the reason the localization sequence of the next section is a sequence of $A_*$-functorial maps. In the case where $f$ is the projection to a point, it recovers the invariance of the degree of a zero-cycle under a flat base change of the ground scheme.

Everything so far concerns cycles. To obtain a map of Chow groups, flat pullback must carry $\mathrm{Rat}_k$ into $\mathrm{Rat}_{k+n}$. The mechanism is that flat pullback commutes with taking the divisor of a rational function, up to the multiplicities that flatness controls, so a cycle that is a sum of divisors pulls back to a sum of divisors.

**Proposition 1.3.7: Flat Pullback Preserves Rational Equivalence**

Let $f\colon X \to Y$ be flat of relative dimension $n$. If $\alpha \in Z_k(Y)$ is rationally equivalent to zero, then $f^*\alpha \in Z_{k+n}(X)$ is rationally equivalent to zero. Consequently $f^*$ descends to a homomorphism

$$f^*\colon A_k(Y) \to A_{k+n}(X)$$

and $A_*(-)$ equipped with flat pullback is a contravariant functor on flat morphisms of algebraic schemes, raising dimension by the relative dimension.

***Proof*:**
It suffices to treat a generator of $\mathrm{Rat}_k(Y)$, namely $\alpha = [\mathrm{div}(r)]$ for a $(k+1)$-dimensional subvariety $W \subseteq Y$ and a nonzero $r \in k(W)^*$. The claim is that $f^*[\mathrm{div}(r)]$ is a sum of divisors of rational functions on subvarieties of $X$.

*Step 1: reduction to the total space over $W$.* The cycle $[\mathrm{div}(r)]$ is supported on $W$, and $f^*[\mathrm{div}(r)]$ is supported on $f^{-1}(W)$. Let $W' = f^{-1}(W) = X \times_Y W$ with its scheme-theoretic (possibly non-reduced) structure, and let $f_W\colon W' \to W$ be the restriction of $f$, which is flat of relative dimension $n$ by base change. Because the components of $f^{-1}(W)$ are the components of $W'$ and the pullback multiplicities are computed inside $W'$, it is enough to prove that $f_W^*[\mathrm{div}(r)] \sim 0$ in $Z_{k+n}(W')$; the class then remains rationally equivalent to zero in $X$ under the proper (closed immersion) pushforward along $W' \hookrightarrow X$, which preserves rational equivalence by theorem 1.2.3. Thus one may assume $Y = W$ is a variety and $r \in k(Y)^*$.

*Step 2: the key identity on components.* With $Y$ a variety and $r \in k(Y)^*$, the divisor $[\mathrm{div}(r)] = \sum_V \mathrm{ord}_V(r)[V]$ runs over the codimension-one subvarieties $V \subseteq Y$. The assertion to prove is the flat-pullback divisor formula

$$f^*[\mathrm{div}(r)] = \left[\mathrm{div}(f^\# r)\right] \quad \text{on each component of } X \tag{1.1}$$

interpreted componentwise as follows. Let $X_1, \ldots, X_s$ be the irreducible components of $X$, each of dimension $n + \dim Y$ by proposition 1.3.2, with generic points dominating $Y$ (flatness sends generic points to the generic point of the base). On each $X_j$ the pullback $f^\# r$ of $r$ is a nonzero rational function, since $r \neq 0$ and $X_j$ dominates $Y$. The claim is

$$f^*[\operatorname{div}(r)] = \sum_{j=1}^{s} m_j \, [\operatorname{div}(f^{\#}r\big|_{X_j})]$$

where $m_j = \operatorname{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$ is the multiplicity of the component $X_j$ in the fundamental cycle $[X]$. Each term on the right is a divisor of a rational function on the subvariety $(X_j)_{\mathrm{red}}$, hence rationally equivalent to zero, so (1.1) gives $f^*[\operatorname{div}(r)] \sim 0$.

*Step 3: verifying the identity.* Fix a codimension-one subvariety $D \subseteq X_j$, a $(k+n)$-dimensional subvariety of $X$, and let $V = \overline{f(D)} \subseteq Y$. Two cases arise by the relative-dimension count of proposition 1.3.2 applied to $f\big|_{X_j}$.

If $\dim V = \dim Y$, then $V = Y$ and $D$ dominates $Y$; the coefficient of $[D]$ in $[\operatorname{div}(f^{\#}r)]$ is $\operatorname{ord}_D(f^{\#}r)$, computed in the discrete-valuation-style local ring $\mathcal{O}_{X_j,D}$, while $D$ does not appear in $f^*[\operatorname{div}(r)]$ because $[\operatorname{div}(r)]$ has no component equal to $Y$. But $\operatorname{ord}_D(f^{\#}r) = 0$. The point is that $D$ dominates $Y$: its generic point $\xi$ maps to the generic point of $Y$, at which $r$ is a nonzero rational function, that is, a unit of $k(Y)$. Since $f^{\#}r$ is the pullback of $r$ along the local homomorphism $\mathcal{O}_{Y,Y} = k(Y) \to \mathcal{O}_{X_j,D}$ induced on stalks at $\xi$, the image $f^{\#}r$ is a unit of $\mathcal{O}_{X_j,D}$, so $\operatorname{ord}_D(f^{\#}r) = 0$ and no such $D$ contributes to either side. (A nonzero rational function need not be a unit along a codimension-one subvariety in general; here it is one precisely because $D$ dominates the base and so sees $r$ only at the generic point of $Y$.)

If $\dim V = \dim Y - 1$, then $V$ is one of the codimension-one subvarieties appearing in $[\operatorname{div}(r)]$, and $D$ is a component of $f^{-1}(V)$, lying on one or more of the components $X_j$. Two kinds of local ring enter the count, and keeping them distinct is the crux. Write

$$A = \mathcal{O}_{Y,V}, \qquad \widetilde{B} = \mathcal{O}_{X,D}$$

Here $A$ is a one-dimensional local domain and $\operatorname{ord}_V(r) = \operatorname{length}_A(A/rA)$; the ring $\widetilde{B}$ is the local ring of the whole (possibly nonreduced) scheme $X$ at the generic point of $D$, and it is flat over $A$ because $f$ is flat. The minimal primes of the one-dimensional ring $\widetilde{B}$ correspond to the components $X_j$ passing through $D$; for such a component the quotient $B_j = \widetilde{B}/\mathfrak{p}_j = \mathcal{O}_{X_j,D}$ is the discrete-valuation-style local ring of the reduced component $X_j$ at $D$, in which $\operatorname{ord}_D(f^{\#}r\big|_{X_j}) = \operatorname{length}_{B_j}(B_j/(f^{\#}r)B_j)$ is computed. When $X$ is irreducible along $D$ there is a single such $j$, but in general several components may meet along $D$.
The coefficient of $[D]$ in $f^*[\operatorname{div}(r)]$ is $\operatorname{ord}_V(r) \cdot e_D$, where

$$e_D = \operatorname{length}_{\mathcal{O}_{f^{-1}(V),D}}\big(\mathcal{O}_{f^{-1}(V),D}\big) = \operatorname{length}_{\widetilde{B}}\big(\widetilde{B}/\mathfrak{m}_A\widetilde{B}\big)$$

is the multiplicity of $D$ in the pullback cycle $f^*[V]$, the second equality holding because $\mathcal{O}_{f^{-1}(V),D} = \widetilde{B} \otimes_A A/\mathfrak{m}_A = \widetilde{B}/\mathfrak{m}_A\widetilde{B}$. The coefficient of $[D]$ in $\sum_j m_j[\operatorname{div}(f^{\#}r\big|_{X_j})]$ is $\sum_{j\,:\,D \subseteq X_j} m_j \cdot \operatorname{ord}_D(f^{\#}r\big|_{X_j})$, the sum running over the components $X_j$ that pass through $D$. That the two coefficients agree is two multiplicative facts chained together. First, flatness of $\widetilde{B}$ over $A$ makes length multiply through the base change: filtering $A/rA$ by its $\operatorname{ord}_V(r)$ composition factors $A/\mathfrak{m}_A$ and applying the exact functor $- \otimes_A \widetilde{B}$ gives

$$\operatorname{length}_{\widetilde{B}}\big(\widetilde{B}/r\widetilde{B}\big) = \operatorname{length}_A(A/rA) \cdot \operatorname{length}_{\widetilde{B}}\big(\widetilde{B}/\mathfrak{m}_A\widetilde{B}\big) = \operatorname{ord}_V(r) \cdot e_D$$

Second, order in the one-dimensional local ring $\widetilde{B}$ distributes over its minimal primes with the generic multiplicities as weights: each minimal prime $\mathfrak{p}_j$ has residue ring $B_j = \mathcal{O}_{X_j,D}$ and localization $\widetilde{B}_{\mathfrak{p}_j} = \mathcal{O}_{X,X_j}$ of length $m_j = \operatorname{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$, so the additivity of order over the minimal primes of a one-dimensional local ring ([7, App. A.3]) gives, for the nonzerodivisor $f^\# r$,

$$\operatorname{length}_{\widetilde{B}}\big(\widetilde{B}/(f^\# r)\widetilde{B}\big) = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \operatorname{length}_{B_j}\big(B_j/(f^\# r)B_j\big) = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \operatorname{ord}_D(f^\# r|_{X_j})$$

Since $f^\# r$ is the image of $r$ in $\widetilde{B}$, the left sides of the two displays coincide, whence $\operatorname{ord}_V(r)\cdot e_D = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \operatorname{ord}_D(f^\# r|_{X_j})$: the two coefficients of $[D]$ match. When $X$ is irreducible along $D$ the sum has a single term and reduces to $e_D = m_j \cdot e_{D/V}$, where $e_{D/V} = \operatorname{length}_{B_j}(B_j/\mathfrak{m}_A B_j)$ is the ramification-type multiplicity of $D$ over $V$ computed in the reduced component ring $B_j$; the pullback multiplicity $e_D$ over the full preimage scheme then exceeds it by exactly the factor $m_j$. Summing over the divisors $D$ and, on the right, over the components $X_j$ reproduces both sides of (1.1) term by term. Hence $f^*[\operatorname{div}(r)] = \sum_j m_j[\operatorname{div}(f^\# r|_{X_j})]$ is a sum of divisors of rational functions, so $f^*[\operatorname{div}(r)] \sim 0$.

Descent to Chow groups follows: $f^*(\operatorname{Rat}_k(Y)) \subseteq \operatorname{Rat}_{k+n}(X)$, so $f^*$ induces $A_k(Y) \to A_{k+n}(X)$. Functoriality on classes is inherited from proposition 1.3.5(3),(4) on cycles, giving a contravariant functor as claimed, completing the proof.

The proposition completes the package: flat pullback is a well-defined operation on Chow groups, contravariant and dimension-raising, dual to the covariant dimension-preserving pushforward of section 1.2. The pair $(f_*, f^*)$ makes $A_*$ into a functor covariant for proper maps and contravariant for flat maps, with the compatibility of proposition 1.3.6 tying the two together across fiber squares. This is the structure the localization sequence and homotopy invariance of section 1.4 exploit, and it is why the affine-bundle projection $p\colon X \times \mathbb{A}^n \to X$, whose pullback was computed in example 1.3.4, will turn out to induce an isomorphism on Chow groups.

**Example 1.3.8: Pullback to a Product and to an Open Set**
The two flat morphisms of example 1.3.4 descend to Chow groups explicitly.

1. *Cylinder map.* For $p\colon X \times \mathbb{A}^1 \to X$ the induced map $p^*\colon A_k(X) \to A_{k+1}(X \times \mathbb{A}^1)$ sends the class of a subvariety $V$ to the class of the cylinder $V \times \mathbb{A}^1$. On $X = \operatorname{Spec} \overline{k}$ this reads $A_0(\operatorname{Spec} \overline{k}) = \mathbb{Z} \to A_1(\mathbb{A}^1)$, sending the class of the point to $[\mathbb{A}^1]$, the fundamental class of the affine line, consistent with $p^*[Y] = [X]$ from proposition 1.3.5(2).

2. *Restriction to an open set.* For an open immersion $j\colon U \hookrightarrow Y$ the map $j^*\colon A_k(Y) \to A_k(U)$ restricts a class to $U$, forgetting components supported on the complement $Y \setminus U$. On $Y = \mathbb{P}^1$ with $U = \mathbb{A}^1$ this is the surjection $A_0(\mathbb{P}^1) = \mathbb{Z} \twoheadrightarrow A_0(\mathbb{A}^1) = 0$ from example 1.1.5: the class of a point on $\mathbb{P}^1$ restricts to a point of $\mathbb{A}^1$, which is rationally trivial. The kernel is generated by the class of the point at infinity, the part of $\mathbb{P}^1$ removed. This surjectivity with the removed piece generating the kernel is the prototype of the localization sequence.

## Exercise 1.3.1

1. Let $f\colon X \to Y$ be flat of relative dimension 0 (for instance a finite flat morphism, or an open

immersion). Show directly from definition 1.3.3 that $f^*$ maps $Z_k(Y) \to Z_k(X)$ for every $k$, preserving dimension, and that $f^*[V] = \sum_i e_i[Z_i]$ where $Z_i$ are the components of $f^{-1}(V)$ and $e_i$ their multiplicities in the fiber over the generic point of $V$.

2. Let $\pi\colon \mathbb{A}^{n+1} \setminus \{0\} \to \mathbb{P}^n$ be the tautological $\mathbb{G}_m$-bundle. It is flat of relative dimension 1. Compute $\pi^*[\mathbb{P}^k]$ for a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$, identifying its class as a cycle on $\mathbb{A}^{n+1} \setminus \{0\}$.

3. Give an example of a flat morphism $f$ for which $f^*\colon A_*(Y) \to A_*(X)$ is not injective, and one for which it is not surjective. (Hint: the projection $\mathbb{P}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ and the open immersion $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$.)

4. Let $q\colon X \times \mathbb{A}^m \times \mathbb{A}^n \to X \times \mathbb{A}^m$ and $p\colon X \times \mathbb{A}^m \to X$ be the projections. Verify the composition law proposition 1.3.5(3) for $p \circ (\mathrm{id} \times \text{-})$ directly on the class of a subvariety $V \subseteq X$, confirming both routes give $[V \times \mathbb{A}^m \times \mathbb{A}^n]$.

5. Let $f\colon X \to Y$ be proper and $j\colon U \hookrightarrow Y$ an open immersion, with fiber square $X_U = f^{-1}(U) \to X$ over $j$. Using proposition 1.3.6, prove that $(f|_{X_U})_*(\alpha|_{X_U}) = (f_*\alpha)|_U$ in $A_*(U)$ for every $\alpha \in A_*(X)$, that is, proper pushforward commutes with restriction to an open subset.

6. Let $E \to X$ be a vector bundle of rank $r$ over a variety $X$ of dimension $d$, with projection $\pi$ flat of relative dimension $r$. Compute $\pi^*[X]$ and identify the top Chow group $A_{d+r}(E)$, first showing that $E$ is irreducible.

## 1.4 The Fundamental Sequences

The two functorialities already in hand, proper pushforward (section 1.2) and flat pullback (section 1.3), combine into the two structural results that make Chow groups computable. The first is a long-exact-sequence surrogate: cutting a scheme into a closed piece and its open complement relates the three Chow groups by a right-exact sequence, the algebraic analogue of the excision property of a homology theory. The second is homotopy invariance in its algebraic form: an affine bundle over $X$, and in particular the trivial bundle $X \times \mathbb{A}^n$, has the same Chow groups as $X$, shifted in degree by the fibre dimension. Neither result computes a single Chow group in isolation, but together they reduce the computation of $A_*(X)$ for the spaces of the next section to bookkeeping: stratify $X$ by locally closed pieces on which the answer is known, and assemble the pieces through the sequence.

Both proofs rest on a single principle. A cycle on $X$ is a formal combination of subvarieties; a subvariety either meets the open set $U$ or is contained in the closed complement $Z$, and rational equivalence is generated by divisors of rational functions, which live on subvarieties and so obey the same dichotomy. The localization sequence is the exact translation of this observation into a statement about the three groups, and affine-bundle invariance is its first substantial payoff, extracted by Noetherian induction.

Throughout, $Z \subseteq X$ is a closed subscheme with open complement $U = X \setminus Z$, and

$$i\colon Z \hookrightarrow X, \qquad j\colon U \hookrightarrow X$$

denote the inclusions. The closed immersion $i$ is proper, so it pushes cycles forward by definition 1.2.1: a subvariety $V \subseteq Z$ is also a subvariety of $X$, and $i_*[V] = [V]$ since $\deg(V/V) = 1$. The open immersion $j$ is flat of relative dimension 0, so it pulls cycles back by definition 1.3.3: for a subvariety

$V \subseteq X$ the restriction is $j^*[V] = [V \cap U]$ when $V$ meets $U$ and $j^*[V] = 0$ when $V \subseteq Z$. These two operations are what the sequence is built from.

**Theorem 1.4.1: Localization Sequence**

Let $X$ be an algebraic scheme, $Z \subseteq X$ a closed subscheme, and $U = X \setminus Z$ its open complement, with $i\colon Z \hookrightarrow X$ and $j\colon U \hookrightarrow X$ the inclusions. For every $k$ the sequence

$$A_k(Z) \xrightarrow{i_*} A_k(X) \xrightarrow{j^*} A_k(U) \longrightarrow 0$$

is exact: $j^*$ is surjective, and $\ker(j^*) = \operatorname{im}(i_*)$.

***Proof*:**
The map $i_*$ is proper pushforward along the closed immersion (theorem 1.2.3) and $j^*$ is flat pullback along the open immersion (proposition 1.3.7); both are defined on Chow groups, so the sequence is a sequence of homomorphisms. Exactness is checked at the two nonzero spots.

*Step 1: surjectivity of $j^*$.* A subvariety $V' \subseteq U$ is a $k$-dimensional closed integral subscheme of the open set $U$. Its closure $V = \overline{V'}$ in $X$ is a $k$-dimensional subvariety of $X$ with $V \cap U = V'$, since closure adds only points lying in the closed set $Z$ and does not raise the dimension. Thus $j^*[V] = [V']$, and every generator of $Z_k(U)$ is hit by a cycle on $X$. Hence $j^*\colon Z_k(X) \to Z_k(U)$ is already surjective at the level of cycles, and a fortiori $j^*\colon A_k(X) \to A_k(U)$ is surjective.

*Step 2: the inclusion* $\operatorname{im}(i_*) \subseteq \ker(j^*)$. For a subvariety $V \subseteq Z$ the intersection $V \cap U$ is empty, so $j^* i_*[V] = j^*[V] = 0$. By linearity $j^* \circ i_* = 0$ on $Z_k(Z)$, hence on $A_k(Z)$, which gives $\operatorname{im}(i_*) \subseteq \ker(j^*)$.

*Step 3: the inclusion* $\ker(j^*) \subseteq \operatorname{im}(i_*)$. This is the substance of the theorem. Let $\alpha \in Z_k(X)$ be a cycle whose class in $A_k(X)$ lies in $\ker(j^*)$, so $j^*\alpha \sim 0$ on $U$; the goal is to modify $\alpha$ by a cycle rationally equivalent to zero on $X$ until it is supported on $Z$, that is, until it lies in the image of $i_*\colon Z_k(Z) \to Z_k(X)$.

Split the cycle by whether its components meet $U$. Group the subvarieties occurring in $\alpha$ as those contained in $Z$ and those meeting $U$, and write

$$\alpha = \alpha_Z + \beta, \qquad \alpha_Z \in Z_k(Z), \quad \beta = \sum_i n_i [V_i]$$

where each $V_i$ meets $U$. Since $\alpha_Z = i_*\alpha_Z$ already lies in the image of $i_*$, it may be discarded, and it suffices to treat $\beta$. Applying $j^*$ gives $j^*\alpha = j^*\beta = \sum_i n_i [V_i']$ with $V_i' = V_i \cap U$ the distinct $k$-dimensional subvarieties of $U$, and by hypothesis this cycle is rationally equivalent to zero on $U$:

$$\sum_i n_i [V_i'] = \sum_j [\operatorname{div}(r_j')] \quad \text{in } Z_k(U)$$

for finitely many $(k+1)$-dimensional subvarieties $W_j' \subseteq U$ and rational functions $r_j' \in k(W_j')^*$.

The idea is now to lift each $W_j'$ and $r_j'$ from $U$ to $X$ and compare divisors. Let $W_j = \overline{W_j'} \subseteq X$ be the closure, a $(k+1)$-dimensional subvariety of $X$ with $W_j \cap U = W_j'$. Restriction of functions to the dense open $W_j' \subseteq W_j$ is an isomorphism of function fields $k(W_j) \xrightarrow{\sim} k(W_j')$, since a variety and a dense open subset have the same function field; let $r_j \in k(W_j)^*$ be the function corresponding to $r_j'$. Form the divisor $[\operatorname{div}(r_j)] \in Z_k(X)$. Its restriction to $U$ is governed by the flat-pullback compatibility of the divisor construction: for a codimension-one subvariety $V \subseteq W_j$ meeting $U$, the local ring $\mathcal{O}_{W_j,V} = \mathcal{O}_{W'_j,V\cap U}$ is unchanged by passing to the open set, so $\mathrm{ord}_V(r_j) = \mathrm{ord}_{V\cap U}(r'_j)$, while a codimension-one subvariety of $W_j$ contained in $Z$ contributes a component of $[\mathrm{div}(r_j)]$ that lies in $Z_k(Z)$. Splitting the sum over these two kinds of $V$,

$$[\mathrm{div}(r_j)] = [\mathrm{div}(r'_j)] + \gamma_j, \qquad \gamma_j \in Z_k(Z) \tag{1.2}$$

where the first term on the right is read inside $Z_k(X)$ via the identification $V \leftrightarrow \overline{V}$ of subvarieties of $U$ with their closures, and $\gamma_j$ collects the codimension-one subvarieties of $W_j$ that lie in $Z$.

*Step 3, conclusion.* Sum (1.2) over $j$ and use $\sum_j [\mathrm{div}(r'_j)] = \beta$ as cycles on $X$ under the closure identification (both sides are the cycle $\sum_i n_i[V_i]$, since $V_i = \overline{V'_i}$). This yields

$$\sum_j [\mathrm{div}(r_j)] = \beta + \sum_j \gamma_j$$

so that

$$\beta = \sum_j [\mathrm{div}(r_j)] - \sum_j \gamma_j$$

The first sum is a $\mathbb{Z}$-combination of divisors of rational functions on $X$, hence lies in $\mathrm{Rat}_k(X)$; the second sum $\gamma := \sum_j \gamma_j$ lies in $Z_k(Z)$. Therefore, in $A_k(X)$,

$$[\beta] = -[\gamma] = i_*(-[\gamma]) \in \mathrm{im}(i_*)$$

Adding back the discarded $\alpha_Z = i_*[\alpha_Z]$ gives $[\alpha] = i_*\big([\alpha_Z] - [\gamma]\big)$, exhibiting the class of $\alpha$ in the image of $i_*$. This proves $\ker(j^*) \subseteq \mathrm{im}(i_*)$, and with Steps 1 and 2 the sequence is exact, completing the proof.

The mechanism of the proof is worth isolating, because it recurs whenever Chow groups are computed by stratification. Surjectivity on the right is elementary: a subvariety of the open set extends to its closure, so no class on $U$ is unreachable from $X$. The content is in the middle, and it is entirely a statement about lifting rational equivalences. A relation $j^*\alpha \sim 0$ on $U$ is witnessed by finitely many rational functions on subvarieties of $U$; each such function extends across $Z$ to a rational function on a subvariety of $X$ (function fields do not see the boundary), and the extended divisor differs from the original only in components supported on $Z$. The failure of the sequence to be exact on the left, which is left as it stands, measures exactly how much rational equivalence is created by the extra room $X$ provides over $Z$: a cycle on $Z$ can become rationally equivalent to zero in $X$ without being so in $Z$, because $X$ contains $(k+1)$-dimensional subvarieties running out of $Z$ into $U$. The affine line makes this concrete below.

**1.4.2 Remark: No Left-Exactness** The map $i_* \colon A_k(Z) \to A_k(X)$ need not be injective. Take $X = \mathbb{A}^1$, $Z = \{0\}$, and $k = 0$. Then $A_0(Z) = \mathbb{Z}$, generated by $[0]$, but $[0] \sim 0$ in $A_0(\mathbb{A}^1)$ by example 1.1.5, so $i_*[0] = 0$ and $i_*$ is the zero map on a nonzero group. The kernel of $i_*$ records cycles on $Z$ that bound in $X$ but not in $Z$, and there is no functorial description of it. This is why the localization sequence is only a three-term right-exact sequence, not a long exact sequence: Bloch's higher Chow groups supply the missing terms, but that theory is outside the scope of this book.

The localization sequence is stated for one closed subscheme, but its most frequent use is for a pair. Two closed subschemes $X_1, X_2$ of an ambient $X$ can be compared through their union and intersection, and the resulting sequence is the Mayer-Vietoris right-exactness that was promised without proof in example 1.1.6. It is a companion to the localization sequence rather than a corollary of it: the derivation below runs directly at the level of cycles, using only that a subvariety of the union lies in one of the two pieces, so no appeal to theorem 1.4.1 is needed.

**Corollary 1.4.3: Right-Exactness for Two Closed Subschemes**

Let $X_1$ and $X_2$ be closed subschemes of an algebraic scheme, and set $X_1 \cup X_2$ and $X_1 \cap X_2$ (with reduced or any fixed scheme structure; Chow groups see only the reduced structure by example 1.1.6). For every $k$ the sequence

$$A_k(X_1 \cap X_2) \xrightarrow{(\iota_{1*}, -\iota_{2*})} A_k(X_1) \oplus A_k(X_2) \xrightarrow{j_{1*}+j_{2*}} A_k(X_1 \cup X_2) \longrightarrow 0$$

is exact, where $\iota_m \colon X_1 \cap X_2 \hookrightarrow X_m$ and $j_m \colon X_m \hookrightarrow X_1 \cup X_2$ are the inclusions. There is no exactness at $A_k(X_1 \cap X_2)$ in general.

***Proof*:**
Write $Y = X_1 \cup X_2$. The argument is cleanest at the level of cycles, using the one structural fact that an irreducible closed subset of $Y = X_1 \cup X_2$ is contained in $X_1$ or in $X_2$. Consequently a subvariety $V \subseteq Y$ lies in $X_1$ or in $X_2$ (or in both, in which case $V \subseteq X_1 \cap X_2$), and because $Z_k(Y)$ is free on subvarieties, a cycle expressible both by subvarieties of $X_1$ and by subvarieties of $X_2$ has all its components in $X_1 \cap X_2$.

*Step 1: surjectivity of $j_{1*} + j_{2*}$.* Every subvariety of $Y$ lies in $X_1$ or $X_2$, so at the level of cycles $Z_k(Y) = j_{1*}Z_k(X_1) + j_{2*}Z_k(X_2)$; the map $Z_k(X_1) \oplus Z_k(X_2) \to Z_k(Y)$, $(a_1, a_2) \mapsto a_1 + a_2$, is surjective. Passing to classes, $j_{1*} + j_{2*} \colon A_k(X_1) \oplus A_k(X_2) \to A_k(Y)$ is surjective.

*Step 2: the composite is zero.* For a subvariety $V \subseteq X_1 \cap X_2$ the two pushforwards $j_{1*}\iota_{1*}[V]$ and $j_{2*}\iota_{2*}[V]$ both equal $[V]$ in $Z_k(Y)$, since each inclusion sends $[V]$ to $[V]$. Hence $(j_{1*} + j_{2*}) \circ (\iota_{1*}, -\iota_{2*}) = j_{1*}\iota_{1*} - j_{2*}\iota_{2*} = 0$, giving $\operatorname{im}(\iota_{1*}, -\iota_{2*}) \subseteq \ker(j_{1*} + j_{2*})$.

*Step 3: exactness in the middle.* Let $(\eta_1, \eta_2) \in A_k(X_1) \oplus A_k(X_2)$ satisfy $j_{1*}\eta_1 + j_{2*}\eta_2 = 0$ in $A_k(Y)$. Represent $\eta_1$ and $\eta_2$ by cycles $a_1 \in Z_k(X_1)$ and $a_2 \in Z_k(X_2)$. The hypothesis says $a_1 + a_2 \sim 0$ in $Z_k(Y)$, so there are finitely many $(k+1)$-dimensional subvarieties $W_j \subseteq Y$ and $r_j \in k(W_j)^*$ with

$$a_1 + a_2 = \sum_j [\operatorname{div}(r_j)] \quad \text{in } Z_k(Y)$$

Each $W_j$ lies in $X_1$ or in $X_2$, and the divisor $[\operatorname{div}(r_j)]$ of a rational function on $W_j$ is a cycle supported on $W_j$, hence on $X_1$ or on $X_2$ accordingly. Collect the terms:

$$b_1 = \sum_{W_j \subseteq X_1} [\operatorname{div}(r_j)] \in \operatorname{Rat}_k(X_1), \qquad b_2 = \sum_{W_j \subseteq X_2,\, W_j \not\subseteq X_1} [\operatorname{div}(r_j)] \in \operatorname{Rat}_k(X_2)$$

where a $W_j$ lying in both is assigned to the first group; then $\sum_j [\operatorname{div}(r_j)] = b_1 + b_2$ and so $a_1 + a_2 = b_1 + b_2$ in $Z_k(Y)$. Rearranging,

$$c := a_1 - b_1 = b_2 - a_2$$

The left expression $a_1 - b_1$ is a cycle supported on $X_1$, and the right expression $b_2 - a_2$ is a cycle supported on $X_2$; being the same cycle in the free group $Z_k(Y)$, it is supported on $X_1 \cap X_2$, that is $c \in Z_k(X_1 \cap X_2)$. Now compute classes. In $A_k(X_1)$, $b_1 \in \mathrm{Rat}_k(X_1)$ is zero, so $\eta_1 = [a_1] = [a_1 - b_1] = \iota_{1*}[c]$; in $A_k(X_2)$, $b_2 \in \mathrm{Rat}_k(X_2)$ is zero, so $\eta_2 = [a_2] = [a_2 - b_2] = -\iota_{2*}[c]$. Therefore

$$(\eta_1, \eta_2) = (\iota_{1*}[c], -\iota_{2*}[c]) = (\iota_{1*}, -\iota_{2*})\big([c]\big)$$

lies in the image of $(\iota_{1*}, -\iota_{2*})$. This gives $\ker(j_{1*} + j_{2*}) \subseteq \mathrm{im}(\iota_{1*}, -\iota_{2*})$, and with Step 2 exactness in the middle.

*No left-exactness.* The map $(\iota_{1*}, -\iota_{2*})$ need not be injective. Take $X_1$ and $X_2$ two distinct lines through the origin in $\mathbb{A}^2$, so $X_1 \cap X_2 = \{0\}$, in degree $k = 0$: then $A_0(X_1 \cap X_2) = \mathbb{Z}$ is generated by $[0]$, while $\iota_{1*}[0] = [0] \sim 0$ in $A_0(X_1) = A_0(\mathbb{A}^1)$ and likewise $\iota_{2*}[0] = 0$ in $A_0(X_2)$, so $(\iota_{1*}, -\iota_{2*})[0] = 0$ on a nonzero group. Thus there is no exactness at $A_k(X_1 \cap X_2)$, as we sought to show.

The corollary is the exact sequence recorded, without proof, as item (4) of example 1.1.6; the present derivation discharges that promissory note. Its shape is the Chow-group form of the Mayer-Vietoris sequence familiar from topology, truncated on the left for the same reason the localization sequence is: the intersection $X_1 \cap X_2$ can carry cycles that become rationally equivalent to zero once they are free to move in the larger $X_1 \cup X_2$. Where it is used most is in dévissage arguments, where a scheme is written as a union of simpler closed pieces and its Chow groups are assembled from those of the pieces and their overlaps.

A single application of the localization sequence recovers, from scratch, the computation of $A_0(\mathbb{A}^1)$ recorded in example 1.1.5, this time without touching a rational function by hand. The point is that $\mathbb{A}^1$ is $\mathbb{P}^1$ with a point removed, and the localization sequence turns the known $A_0(\mathbb{P}^1) = \mathbb{Z}$ into the answer for the open complement.

**Example 1.4.4: Recomputing $A_0(\mathbb{A}^1)$ by Localization**

Take $X = \mathbb{P}^1$, $Z = \{\infty\}$ the point at infinity, and $U = \mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$. The localization sequence in degree 0 reads

$$A_0(\{\infty\}) \xrightarrow{i_*} A_0(\mathbb{P}^1) \xrightarrow{j^*} A_0(\mathbb{A}^1) \longrightarrow 0$$

The left group is $A_0(\{\infty\}) = \mathbb{Z}$, generated by the class of the single point $\infty$, and $i_*[\infty] = [\infty] \in A_0(\mathbb{P}^1)$. By example 1.1.5 the group $A_0(\mathbb{P}^1) \cong \mathbb{Z}$ is generated by the class $h$ of any point, and $[\infty] = h$ is a generator, so $i_*$ is surjective. By exactness $\mathrm{im}(i_*) = \ker(j^*)$, and since $i_*$ is onto, $\ker(j^*) = A_0(\mathbb{P}^1)$; the quotient by the whole group is trivial:

$$A_0(\mathbb{A}^1) = A_0(\mathbb{P}^1)/\ker(j^*) \cong \mathrm{coker}(i_*) = \mathbb{Z}/\mathbb{Z} = 0$$

Thus $A_0(\mathbb{A}^1) = 0$, in agreement with the direct computation. The single generator of $A_0(\mathbb{P}^1)$ is killed precisely because the point removed to form $\mathbb{A}^1$ was rationally equivalent to every remaining point; removing it collapses the whole group.

With the localization sequence available, the second structural result of the section follows by Noetherian induction. Homotopy invariance in algebraic geometry takes the form of affine-bundle invariance: pulling a cycle back along the projection of an affine bundle is an isomorphism on Chow groups. The motivating case is the trivial bundle $X \times \mathbb{A}^n \to X$, which asserts that appending an affine factor does not change the Chow group beyond the expected degree shift, the algebraic counterpart of the fact that $\mathbb{A}^n$ is contractible.

The result is stated for a general affine bundle, of which the trivial bundle is the leading case.

**Definition 1.4.5: Affine Bundle of Relative Dimension $n$**

An *affine bundle of relative dimension* $n$ over $X$ is a scheme $p\colon E \to X$ that is locally on $X$ a product with affine space: there is an open cover $\{U_\lambda\}$ of $X$ and isomorphisms $p^{-1}(U_\lambda) \cong U_\lambda \times \mathbb{A}^n$ over $U_\lambda$.

A vector bundle of rank $n$, with its transition maps in $GL_n$, is such a bundle, as is any torsor under a vector bundle; the trivial example is $X \times \mathbb{A}^n$ itself. In every case $p$ is flat of relative dimension $n$, so flat pullback $p^*\colon A_k(X) \to A_{k+n}(E)$ is defined by definition 1.3.3.

**Theorem 1.4.6: Affine-Bundle Invariance**

Let $p\colon E \to X$ be an affine bundle of relative dimension $n$ over an algebraic scheme $X$. Then the flat pullback

$$p^*\colon A_k(X) \xrightarrow{\sim} A_{k+n}(E)$$

is an isomorphism for every $k$. In particular, for the trivial bundle, $A_k(X) \xrightarrow{\sim} A_{k+n}(X \times \mathbb{A}^n)$.

***Proof*:**
The proof runs the argument directly at relative dimension $n$; the fibre dimension enters only through the trivial bundle, where a global tower is available. Note that a global tower of rank-one affine bundles is *not* claimed for a general bundle: the local identification $U \times \mathbb{A}^n = (U \times \mathbb{A}^{n-1}) \times \mathbb{A}^1$ does not glue to a factorization $E \to X$ over $X$, since a general affine (or vector) bundle carries no global flag of subbundles. The trivial bundle $X \times \mathbb{A}^n$, being a literal product, *is* such a tower, $X \times \mathbb{A}^n = (X \times \mathbb{A}^{n-1}) \times \mathbb{A}^1$; the construction below invokes the tower only there. Fix $n$ and prove that $p^*\colon A_k(X) \to A_{k+n}(E)$ is an isomorphism for all $k$, by Noetherian induction on the closed subschemes of $X$. For a closed subscheme $W \subseteq X$ write $E_W = p^{-1}(W)$ and $p_W\colon E_W \to W$ for the restricted bundle, again an affine bundle of relative dimension $n$.

*Step 1: reductions.* The scheme $X$ is Noetherian, so its closed subschemes satisfy the descending chain condition and it suffices to prove: if $p_W^*$ is an isomorphism for every proper closed subscheme $W \subsetneq X$, then $p^*$ is an isomorphism for $X$. Chow groups depend only on the reduced structure (example 1.1.6), so take $X$ reduced. If $X$ is reducible, write $X = X_1 \cup X_2$ with $X_1$ an irreducible component and $X_2$ the union of the remaining components, both proper closed subschemes; the pullbacks $p^*_{X_1}, p^*_{X_2}, p^*_{X_1 \cap X_2}$ are isomorphisms by hypothesis. The Mayer-Vietoris sequences of corollary 1.4.3 for $X = X_1 \cup X_2$ and for $E = E_{X_1} \cup E_{X_2}$ (whose intersection is $E_{X_1 \cap X_2}$) form a commutative ladder linked by the vertical pullbacks; a diagram chase using the three outer isomorphisms shows $p^*$ is an isomorphism for $X$. So assume $X$ is a variety (reduced and irreducible), and let $\eta = \operatorname{Spec} k(X)$ be its generic point, with generic fibre $E_\eta \cong \mathbb{A}^n_{k(X)}$.

*Step 2: surjectivity of $p^*$.* It suffices to realize the class of a single $(k+n)$-dimensional subvariety $V \subseteq E$ as a pullback modulo the image of $A_{k+n}(E_Z)$ for some proper closed $Z \subsetneq X$. Let $W = \overline{p(V)} \subseteq X$. If $W \subsetneq X$ is proper, then $V \subseteq E_W$ is supported over $Z = W$, and $[V] \in \operatorname{im}\big(A_{k+n}(E_W) \to A_{k+n}(E)\big)$; by the inductive hypothesis $p_W^*$ is onto, and compatibility of $p^*$ with the closed immersion (the fibre square for $p$ over $W \hookrightarrow X$ gives $p^* i_* = i_* p_W^*$ by proposition 1.3.6) writes $[V]$ as a pullback from $X$. If instead $W = X$, so $V$ dominates $X$, restrict to the generic fibre. There $V_\eta \subseteq \mathbb{A}^n_{k(X)}$ is a closed subvariety, and its class is controlled by the trivial-bundle case of the theorem over the field $K = k(X)$: the projection $\mathbb{A}^n_K \to \operatorname{Spec} K$ has surjective flat pullback. This last fact is elementary and needs no base induction, being the trivial bundle over a point: writing the tower $\mathbb{A}^n_K = (\mathbb{A}^{n-1}_K) \times \mathbb{A}^1$ and inducting on $n$, the base case $A_*(\mathbb{A}^1_K)$ is generated over $A_*(\operatorname{Spec} K)$ by the pullback (every closed point of $\mathbb{A}^1_K$ is the divisor of its minimal polynomial, so $A_0(\mathbb{A}^1_K) = 0$, exactly as in example 1.1.5 with $K$ in place of $\overline{k}$), and the tower propagates this up the coordinates. Thus $[V_\eta]$ is rationally equivalent on $\mathbb{A}^n_{k(X)}$ to a pullback $p^*_\eta(a_\eta)$ of a class $a_\eta$ on $\operatorname{Spec} k(X)$. Spreading out the finitely many rational functions that realize this equivalence, together with $a_\eta$, to an open $U \subseteq X$ shows $[V]|_{E_U} = p^*_U(\alpha_U)$ for a class $\alpha_U \in A_k(U)$ and hence $[V] - p^*(\alpha_V)$ (with $\alpha_V \in A_k(X)$ lifting $\alpha_U$ via surjectivity of $j^*$, theorem 1.4.1) restricts to 0 on $E_U$, so its support lies over the proper closed $Z = X \setminus U$. In every case

$$[V] = p^*(\alpha_V) + \gamma_V \quad \text{in } A_{k+n}(E), \qquad \gamma_V \in \operatorname{im}\big(A_{k+n}(E_Z) \to A_{k+n}(E)\big)$$

for some proper closed $Z \subsetneq X$ and $\alpha_V \in A_k(X)$. By the inductive hypothesis $p^*_Z$ is onto $A_*(E_Z)$, so $\gamma_V = i_* p^*_Z(\delta_V) = p^*(i_*\delta_V)$ for a class $\delta_V$ on $Z$, again by proposition 1.3.6. Hence $[V] = p^*(\alpha_V + i_*\delta_V)$ and $p^*$ is surjective.

Surjectivity of $p^*$ is now in hand for every $X$; injectivity is the remaining point, and it is not a formal consequence of the localization ladder, since the two rows are only right-exact and provide no left term to chase against. It is proved by exhibiting a left inverse to $p^*$ for the trivial bundle and reducing the general bundle to that case.

*Step 3: a left inverse for the trivial bundle.* The construction is carried out for a single affine coordinate over an arbitrary base $Y$; the trivial bundle $X \times \mathbb{A}^n$ is a tower of $n$ such single coordinates, and the rank-$n$ left inverse is the composite of the $n$ single-coordinate left inverses (Step 3d). Take $E = Y \times \mathbb{A}^1$, with coordinate $t$ on the second factor and projection $p\colon Y \times \mathbb{A}^1 \to Y$, and define a homomorphism $\sigma\colon Z_{k+1}(Y \times \mathbb{A}^1) \to Z_k(Y)$ on a generator $[V]$, $V \subseteq Y \times \mathbb{A}^1$ a $(k+1)$-dimensional subvariety, by

$$\sigma[V] = \begin{cases} p_*\big[\operatorname{div}_V(t|_V)\big] & t|_V \in k(V)^* \text{ nonconstant} \\ 0 & t|_V \text{ constant} \end{cases}$$

extended additively; here "$t|_V$ constant" is read to include the degenerate case $t|_V \equiv 0$ (when $V \subseteq Y \times \{0\}$), which likewise contributes 0, since then $t|_V \notin k(V)^*$ and no divisor $\operatorname{div}_V(t|_V)$ is formed. When $t|_V$ is nonconstant, $[\operatorname{div}_V(t|_V)]$ is a $k$-cycle on $V$; the pushforward $p_*$ of this particular cycle is well-defined even though $p|_V\colon V \to \overline{p(V)} \subseteq Y$ need not be proper. The point is that $t$ is a regular function on $Y \times \mathbb{A}^1$, so its restriction $t|_V$ is regular on $V$ and $\operatorname{div}_V(t|_V)$ is effective, with support contained in the zero locus $V \cap (Y \times \{0\}) \subseteq Y \times \{0\}$. On $Y \times \{0\}$ the projection $p$ is the closed immersion identifying $Y \times \{0\}$ with $Y$, in particular proper, so $p_*[\operatorname{div}_V(t|_V)]$ is a well-defined $k$-cycle on $Y$ by definition 1.2.1, computed component by component over the locus where $t|_V$ vanishes. Geometrically $\sigma[V]$ is the fibre of $V$ over $t = 0$, projected to $Y$: the specialization of $V$ to the zero section.

*Step 3a: $\sigma \circ p^* = \mathrm{id}$.* For a $k$-dimensional subvariety $W \subseteq Y$ the pullback is $p^*[W] = [W \times \mathbb{A}^1]$, and $t$ restricts to the coordinate on $W \times \mathbb{A}^1$, which is nonconstant with $[\operatorname{div}_{W\times\mathbb{A}^1}(t)] = [W \times \{0\}]$ (the coordinate $t$ has its single zero along $W \times \{0\}$ and no pole on $W \times \mathbb{A}^1$). Pushing forward, $p_*[W \times \{0\}] = [W]$, so $\sigma(p^*[W]) = [W]$. By additivity $\sigma \circ p^* = \mathrm{id}$ on $Z_k(Y)$, hence on $A_k(Y)$ once $\sigma$ is known to descend.

*Step 3b: $\sigma$ respects rational equivalence.* A generator of $\mathrm{Rat}_{k+1}(Y \times \mathbb{A}^1)$ is $[\mathrm{div}_S(g)]$ for a $(k+2)$-dimensional subvariety $S \subseteq Y \times \mathbb{A}^1$ and $g \in k(S)^*$, and unwinding the definitions gives $\sigma[\mathrm{div}_S(g)] = p_*\big(\sum_V \mathrm{ord}_V(g)\,[\mathrm{div}_V(t|_V)]\big)$, the outer sum over the codimension-one subvarieties $V \subseteq S$ on which $t|_V$ is nonconstant (a $V$ with $t|_V$ constant, including $V \subseteq Y \times \{0\}$ where $t|_V \equiv 0$, drops out, contributing 0). This iterated divisor is symmetric in the two functions $g$ and $t|_S$ modulo a principal divisor on $S$:

$$\sum_V \mathrm{ord}_V(g)\,[\mathrm{div}_V(t)] - \sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)] \in \mathrm{Rat}_k(S) \tag{1.3}$$

The relation (1.3) is the commutativity of two divisor operations on the $(k+2)$-dimensional $S$, equivalently the well-definedness of intersecting a cycle with a Cartier divisor; its proof requires the divisor-theoretic machinery of the next chapter, and is supplied there by theorem 2.3.2 (its surface case is exactly (1.3)), so no tool used here depends on it circularly. Granting (1.3), $\sigma[\mathrm{div}_S(g)]$ differs from $p_*\big(\sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)]\big)$ by the pushforward of a principal divisor, and $\sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)]$ is itself a $\mathbb{Z}$-combination of principal divisors on the subvarieties $V$, so its pushforward lies in $\mathrm{Rat}_k(Y)$ by theorem 1.2.3. Either way $\sigma[\mathrm{div}_S(g)] \in \mathrm{Rat}_k(Y)$, so $\sigma$ descends to

$$\sigma\colon A_{k+1}(Y \times \mathbb{A}^1) \to A_k(Y), \qquad \sigma \circ p^* = \mathrm{id}$$

In particular $p^*$ is injective for the trivial rank-one bundle over any base $Y$, and being surjective by Step 2, it is an isomorphism there. The left inverse $\sigma$ commutes with proper pushforward along closed immersions and with flat restriction to opens, since each of $p_*$ and $\mathrm{div}(t)$ does; this compatibility is what carries the trivial-bundle result across the localization sequence in Step 3c.

*Step 3d: the trivial rank-n bundle.* For $E = X \times \mathbb{A}^n$ with coordinates $t_1, \ldots, t_n$, the tower

$$X \times \mathbb{A}^n \xrightarrow{q_n} X \times \mathbb{A}^{n-1} \xrightarrow{q_{n-1}} \cdots \xrightarrow{q_1} X$$

factors $p = q_1 \circ \cdots \circ q_n$ into single-coordinate trivial rank-one bundles $q_m\colon (X \times \mathbb{A}^{m-1}) \times \mathbb{A}^1 \to X \times \mathbb{A}^{m-1}$, so $p^* = q_n^* \circ \cdots \circ q_1^*$ by functoriality of flat pullback (proposition 1.3.5). Applying Step 3b with $Y = X \times \mathbb{A}^{m-1}$ gives a left inverse $\sigma_m$ to each $q_m^*$; the composite $\sigma := \sigma_1 \circ \cdots \circ \sigma_n\colon A_{k+n}(X \times \mathbb{A}^n) \to A_k(X)$ satisfies $\sigma \circ p^* = \mathrm{id}$, so $p^*$ is injective, and with Step 2 it is an isomorphism for the trivial bundle. The compatibility of each $\sigma_m$ with proper pushforward along closed immersions and with flat restriction to opens passes to the composite, so $\sigma$ has the same compatibility used in the next step.

*Step 3c: injectivity for a general bundle.* Let $p\colon E \to X$ be an affine bundle of relative dimension $n$ over the variety $X$ and $\alpha \in A_k(X)$ with $p^*\alpha = 0$. Choose a dense open $U \subseteq X$ over which $E$ trivializes, $E_U \cong U \times \mathbb{A}^n$, and set $Z = X \setminus U$, a proper closed subscheme. Flat base change (proposition 1.3.5) gives $p_U^*(j^*\alpha) = j^*(p^*\alpha) = 0$, and $p_U^*$ is injective by Step 3d applied to the trivial bundle over $U$; hence $j^*\alpha = 0$. By the localization sequence for $Z \subseteq X$ (theorem 1.4.1), $\alpha = i_*\beta$ for some $\beta \in A_k(Z)$. Pushing $\beta$ into $E$ through the fibre square over $Z \hookrightarrow X$ (proposition 1.3.6) gives $p^*\alpha = p^*i_*\beta = i_*(p_Z^*\beta) = 0$, so $p_Z^*\beta$ lies in the kernel of $i_*\colon A_{k+n}(E_Z) \to A_{k+n}(E)$. Over the total space $E$ the left inverse of Step 3d, restricted to the trivializing locus and extended by $\sigma_Z := (p_Z^*)^{-1}$ over $E_Z$ (the inductive isomorphism), assembles into a retraction $r_E\colon A_{k+n}(E) \to A_k(X)$ of $p^*$; the compatibility noted at the end of Step 3d makes the square with $i_*$ commute, so that $r_E \circ i_* = i_* \circ \sigma_Z$ on $A_{k+n}(E_Z)$. Evaluating at $p_Z^*\beta$,

$$i_*\beta = i_*\big(\sigma_Z p_Z^*\beta\big) = r_E\big(i_* p_Z^*\beta\big) = r_E(0) = 0$$

so $\alpha = i_*\beta = 0$. Hence $p^*$ is injective.

> Steps 2 and 3 show $p^*$ is an isomorphism for every $X$ and every relative dimension $n$, completing the Noetherian induction; the tower entered only for the trivial bundle (Step 3d), where it is an actual global factorization, so no global tower was claimed for a general bundle, completing the proof.

The theorem is the algebraic-geometric form of a topological triviality: the projection $X \times \mathbb{A}^n \to X$ is a homotopy equivalence when $\mathbb{A}^n = \mathbb{C}^n$ is given the classical topology, so it induces isomorphisms on singular homology, and Chow groups behave the same way. The degree shift by $n$ is the algebraic bookkeeping for the real dimension shift $2n$ on the topological side, since a $k$-cycle (complex dimension $k$) has real dimension $2k$ and $A_{k+n}$ matches $H_{2(k+n)} = H_{2k+2n}$. The proof uses none of this: surjectivity of $p^*$ needs only that every zero-cycle on an affine line over a field is rationally trivial ($A_0(\mathbb{A}^1_K) = 0$), propagated up the base by localization, while injectivity needs a left inverse, the specialization to the zero section. The one step left open, the divisor symmetry (1.3), is exactly the input that makes intersecting with a Cartier divisor well defined on rational equivalence, the payload of the next chapter; the isomorphism here is the first place that well-definedness is needed, and the forward reference is deliberate rather than a gap. It is discharged in theorem 2.3.2. The reason the argument needs Noetherian induction rather than a direct computation is the same reason the localization sequence is not left-exact: one cannot control what rational equivalences a cycle acquires on $E$ without dismantling the base into pieces small enough that the bundle trivializes and the fibrewise vanishing takes over.

## Exercise 1.4.1

1. Let $Z \subseteq X$ be closed with complement $U$. Show directly from theorem 1.4.1 that $A_k(U) \cong A_k(X)/\operatorname{im}\big(A_k(Z) \to A_k(X)\big)$, and use this to prove that if $A_k(Z) = 0$ then $j^*\colon A_k(X) \to A_k(U)$ is an isomorphism.

2. Let $X = \mathbb{A}^2$ and let $Z = \{0\}$ be the origin, so $U = \mathbb{A}^2 \setminus \{0\}$ is the punctured plane. Using the localization sequence together with the values $A_k(\mathbb{A}^2)$ (namely $A_2(\mathbb{A}^2) = \mathbb{Z}$ and $A_k(\mathbb{A}^2) = 0$ for $k < 2$, which you may take from section 1.5), compute $A_k(U)$ for all $k$.

3. Let $X$ be a variety of dimension $n$ and $U \subseteq X$ a nonempty open subset. Show that $A_n(X) \to A_n(U)$ is an isomorphism, and that both are $\mathbb{Z}$, generated by the fundamental class.

4. Compute $A_*(\mathbb{P}^n \times \mathbb{A}^m)$ in terms of $A_*(\mathbb{P}^n)$, assuming the value $A_k(\mathbb{P}^n) = \mathbb{Z}$ for $0 \le k \le n$ from section 1.5. State the answer as a graded group.

5. Let $p\colon E \to X$ be a vector bundle of rank $r$ over an algebraic scheme $X$, with zero section $s\colon X \hookrightarrow E$. Affine-bundle invariance makes $p^*\colon A_k(X) \to A_{k+r}(E)$ an isomorphism. Explain why one cannot conclude $A_*(E) \cong A_*(X)$ by inverting $p^*$ with a proper pushforward $p_*$, identifying the precise obstruction, and describe the degree by which $p^*$ and the closed-immersion pushforward $s_*$ each shift a cycle class.

6. Take $X_1$ and $X_2$ to be two distinct lines through the origin in $\mathbb{A}^2$, meeting only at $\{0\}$. Write out the sequence of corollary 1.4.3 in degree 0 and verify its exactness by hand, given that $A_0$ of a line is 0 and $A_0$ of a point is $\mathbb{Z}$.

## 1.5 First Computations

The machinery of the preceding sections is now assembled: cycles, rational equivalence, proper pushforward, flat pullback, and the two fundamental sequences. This section puts it to work on the spaces one meets first. We compute the Chow groups of affine space, of projective space, and of a smooth projective curve, and record the shape the answer takes for a projective bundle. These are the building blocks from which nearly every later computation is assembled, and they already display the two extremes of the theory: affine space, whose cycles almost all sweep away to nothing, and projective space, whose cycles are pinned down by a single integer in each dimension.

The pattern to watch is how the localization sequence and affine-bundle invariance from section 1.4 do the actual work. Affine space is an affine bundle over a point, so its Chow groups collapse; projective space is built from affine space by adding a hyperplane at infinity, so an induction along the localization sequence peels off one $\mathbb{Z}$ per dimension. Every computation below is an instance of one of these two moves.

Affine space is the first computation, and $\mathbb{A}^n$ retracts, in the sense of rational equivalence, all the way down to a point: every subvariety of dimension less than $n$ can be swept off to zero, and only the fundamental class in top dimension survives. This is the higher-dimensional form of the observation $A_0(\mathbb{A}^1) = 0$ from example 1.1.5, and it follows in one line from affine-bundle invariance once one notices that $\mathbb{A}^n$ is an affine bundle over $\operatorname{Spec} \overline{k}$.

**Proposition 1.5.1: Chow Groups of Affine Space**

For every $n \geq 0$ the Chow groups of affine space are

$$A_k(\mathbb{A}^n) = \begin{cases} \mathbb{Z}\,[\mathbb{A}^n] & k = n \\ 0 & k \neq n \end{cases}$$

the group $A_n(\mathbb{A}^n)$ being free of rank one on the fundamental class $[\mathbb{A}^n]$.

***Proof*:**
The structure morphism $\pi\colon \mathbb{A}^n \to \operatorname{Spec} \overline{k}$ exhibits $\mathbb{A}^n$ as the trivial affine bundle of relative dimension $n$ over a point. Affine-bundle invariance (theorem 1.4.6) states that flat pullback

$$\pi^*\colon A_k(\operatorname{Spec} \overline{k}) \xrightarrow{\sim} A_{k+n}(\mathbb{A}^n)$$

is an isomorphism for every $k$. The point $\operatorname{Spec} \overline{k}$ has a single subvariety, itself, of dimension 0, so $A_0(\operatorname{Spec} \overline{k}) = Z_0(\operatorname{Spec} \overline{k}) = \mathbb{Z}\,[\operatorname{Spec} \overline{k}]$ and $A_k(\operatorname{Spec} \overline{k}) = 0$ for $k \neq 0$. Transporting these through $\pi^*$ gives $A_{k+n}(\mathbb{A}^n) = 0$ for $k \neq 0$, that is $A_j(\mathbb{A}^n) = 0$ for $j \neq n$, and $A_n(\mathbb{A}^n) = \pi^*\big(\mathbb{Z}\,[\operatorname{Spec} \overline{k}]\big)$. Flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space (definition 1.3.3), so $\pi^*[\operatorname{Spec} \overline{k}] = [\mathbb{A}^n]$, and $A_n(\mathbb{A}^n) = \mathbb{Z}\,[\mathbb{A}^n]$ is free of rank one, completing the proof.

Affine space thus has no cycles in low dimension worth remembering, a fact that reads two ways. From below it says that any $k$-dimensional subvariety $V \subseteq \mathbb{A}^n$ with $k < n$ bounds: there is a family of cycles, parametrized by the line, sweeping $[V]$ to zero. From above it isolates the one invariant that does survive, the coefficient of the fundamental class, which for a top cycle records the multiplicity with which $\mathbb{A}^n$ appears in it. The vanishing in low dimension is what makes affine space so poor a stage for enumerative geometry, and simultaneously what makes it the natural chart in which to localize a computation on a larger space. This second role is exactly how projective space is handled next.

The contrast to keep in mind is the affine line. A single point of $\mathbb{A}^1$ is a 0-cycle in dimension $0 < 1$, so proposition 1.5.1 predicts $A_0(\mathbb{A}^1) = 0$, recovering the computation of example 1.1.5: the coordinate function $t - p$ has $[\operatorname{div}(t-p)] = [p]$, presenting the point as a principal divisor and hence as zero in the Chow group. The proposition is the assertion that this collapse happens in every dimension below the top, uniformly and for the same reason.

Projective space is the opposite. Nothing sweeps to zero for free, because a rational function on a subvariety of $\mathbb{P}^n$ has as many zeros as poles, and the surviving invariant in each dimension is an integer. The computation is an induction that peels $\mathbb{P}^n$ apart along a hyperplane: the closed subscheme $\mathbb{P}^{n-1} \subseteq \mathbb{P}^n$ at infinity, with open complement the affine chart $\mathbb{A}^n$. Affine space contributes nothing below the top by proposition 1.5.1, so the localization sequence forces every class of $\mathbb{P}^n$ in dimension $k < n$ to come from $\mathbb{P}^{n-1}$, and the induction runs.

The motivating principle is that projective space is *cellular*: it is stratified by the chain

$$\mathbb{P}^0 \subset \mathbb{P}^1 \subset \cdots \subset \mathbb{P}^{n-1} \subset \mathbb{P}^n$$

of linear subspaces, each obtained from the previous by adding an affine cell $\mathbb{A}^k = \mathbb{P}^k \setminus \mathbb{P}^{k-1}$. The Chow groups turn out to be free on the classes of these cells, one generator in each dimension. This is the algebraic form of the cell decomposition that computes the singular homology of $\mathbb{C}\mathrm{P}^n$ in [9], where the same chain of linear subspaces yields $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}$ and $H_{\text{odd}} = 0$.

**Theorem 1.5.2: Chow Groups of Projective Space**

For every $n \geq 0$ and every $0 \leq k \leq n$, the Chow group $A_k(\mathbb{P}^n)$ is free abelian of rank one, generated by the class $[\mathbb{P}^k]$ of any linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$ of dimension $k$:

$$A_k(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^k], \qquad 0 \leq k \leq n$$

and $A_k(\mathbb{P}^n) = 0$ for $k > n$. Any two linear subspaces of the same dimension are rationally equivalent, so the generator does not depend on the choice.

***Proof*:**
The argument is an induction on $n$, run through projection from a point, which realizes the complement of a point in $\mathbb{P}^n$ as a line bundle over $\mathbb{P}^{n-1}$. Write $h_k$ for the class $[\mathbb{P}^k]$ of a linear $k$-plane; the claim is that $A_k(\mathbb{P}^n) = \mathbb{Z}\,h_k$ is free of rank one for $0 \leq k \leq n$, and 0 otherwise. Groups $A_k(\mathbb{P}^n)$ with $k > n = \dim \mathbb{P}^n$ vanish because $\mathbb{P}^n$ has no subvariety of that dimension, so only $0 \leq k \leq n$ is at issue.

*Step 1: base cases $n = 0$ and $k = 0$.* For $n = 0$ the space $\mathbb{P}^0 = \operatorname{Spec} \overline{k}$ is a point, with $A_0(\mathbb{P}^0) = \mathbb{Z}\,[\mathbb{P}^0]$ and $A_k(\mathbb{P}^0) = 0$ for $k \neq 0$ by proposition 1.5.1, which is the claim. For general $n$ and $k = 0$, every closed point of $\mathbb{P}^n$ is rationally equivalent to every other: two points $p, q$ both lie on a line $\ell \cong \mathbb{P}^1 \subseteq \mathbb{P}^n$, and on $\ell$ they are rationally equivalent by example 1.1.5, so $[p] \sim [q]$ in $\mathbb{P}^n$. Thus $A_0(\mathbb{P}^n)$ is cyclic, generated by $h_0 = [p]$. Since $\mathbb{P}^n$ is complete, the degree map $\deg\colon A_0(\mathbb{P}^n) \to \mathbb{Z}$ of definition 1.2.4 is well defined and sends $h_0 = [p] \mapsto [k(p) : \overline{k}] = 1$, so $h_0$ has infinite order and $A_0(\mathbb{P}^n) = \mathbb{Z}\,h_0$ is free of rank one.

*Step 2: projection from a point as a line bundle.* Fix $n \geq 1$ and a closed point $P \in \mathbb{P}^n$. Choosing coordinates with $P = [0 : \cdots : 0 : 1]$, linear projection away from $P$ is a morphism

$$q \colon \mathbb{P}^n \setminus \{P\} \to \mathbb{P}^{n-1}, \qquad [x_0 : \cdots : x_n] \mapsto [x_0 : \cdots : x_{n-1}]$$

whose fiber over a point $[a] \in \mathbb{P}^{n-1}$ is the line through $P$ and $[a]$ with $P$ removed, an affine line $\mathbb{A}^1$. This exhibits $\mathbb{P}^n \setminus \{P\}$ as a line bundle over $\mathbb{P}^{n-1}$, the total space of $\mathcal{O}_{\mathbb{P}^{n-1}}(1)$; in particular it is an affine bundle of relative dimension 1. Only the last clause is used below. Affine-bundle invariance (theorem 1.4.6) makes flat pullback

$$q^* \colon A_{k-1}(\mathbb{P}^{n-1}) \xrightarrow{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \tag{1.4}$$

an isomorphism for every $k$.

*Step 3: removing the point via localization.* The point $\{P\}$ is a closed subscheme of $\mathbb{P}^n$ with open complement $\mathbb{P}^n \setminus \{P\}$. The localization sequence (theorem 1.4.1) for this pair reads

$$A_k(\{P\}) \xrightarrow{i_*} A_k(\mathbb{P}^n) \xrightarrow{j^*} A_k(\mathbb{P}^n \setminus \{P\}) \to 0 \tag{1.5}$$

exact at $A_k(\mathbb{P}^n)$ and $A_k(\mathbb{P}^n \setminus \{P\})$. The point contributes $A_k(\{P\}) = 0$ for every $k \geq 1$ and $A_0(\{P\}) = \mathbb{Z}[P]$. Hence for $k \geq 1$ the left term vanishes and $j^*$ is an isomorphism, which combined with (1.4) gives, for $k \geq 1$,

$$A_k(\mathbb{P}^n) \xrightarrow[j^*]{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \xleftarrow[q^*]{\sim} A_{k-1}(\mathbb{P}^{n-1}) \tag{1.6}$$

so $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$ as abelian groups.

*Step 4: the induction.* Fix $k$ with $1 \leq k \leq n$. Iterating (1.6) $k$ times descends both indices by $k$:

$$A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1}) \cong \cdots \cong A_0(\mathbb{P}^{n-k})$$

and $A_0(\mathbb{P}^{n-k}) = \mathbb{Z}$ is free of rank one by Step 1. Together with the case $k = 0$ of Step 1, this proves $A_k(\mathbb{P}^n) \cong \mathbb{Z}$ is free of rank one for every $0 \leq k \leq n$. For $k > n$ the group vanishes as noted at the outset.

*Step 5: the generator is the class of a linear subspace.* It remains to identify the free generator with $h_k = [\mathbb{P}^k]$. Trace a linear $\mathbb{P}^k \subseteq \mathbb{P}^n$ through (1.6). Choose the point $P \in \mathbb{P}^k$; projection of the linear subspace $\mathbb{P}^k$ from the point $P \in \mathbb{P}^k$ is again linear, of one lower dimension, so $q(\mathbb{P}^k \setminus \{P\})$ is a linear $\mathbb{P}^{k-1} \subseteq \mathbb{P}^{n-1}$. Restricting the bundle $q \colon \mathbb{P}^n \setminus \{P\} \to \mathbb{P}^{n-1}$ of Step 2 over this $\mathbb{P}^{k-1}$ realizes $\mathbb{P}^k \setminus \{P\}$ as the affine bundle over $\mathbb{P}^{k-1}$; concretely $q^{-1}(\mathbb{P}^{k-1}) = \mathbb{P}^k \setminus \{P\}$ as a reduced irreducible subvariety, so flat pullback gives $q^*[\mathbb{P}^{k-1}] = [q^{-1}(\mathbb{P}^{k-1})] = [\mathbb{P}^k \setminus \{P\}]$ with multiplicity 1. Hence by construction of (1.6) the class $h_k = [\mathbb{P}^k]$ satisfies $j^* h_k = [\mathbb{P}^k \setminus \{P\}] = q^*[\mathbb{P}^{k-1}]$, so $h_k$ corresponds to $h_{k-1}$ under the isomorphism $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$. Descending to $A_0(\mathbb{P}^{n-k})$, the class $h_k$ maps to $h_0 = [\mathrm{pt}]$, which is the free generator by Step 1. Thus $h_k$ is a generator of $A_k(\mathbb{P}^n) = \mathbb{Z}$. The top case $k = n$ is consistent with the fundamental-class computation $A_n(\mathbb{P}^n) = Z_n(\mathbb{P}^n) = \mathbb{Z}[\mathbb{P}^n]$ of example 1.1.6, since $\mathbb{P}^n$ is irreducible and $h_n = [\mathbb{P}^n]$.

*Step 6: independence of the linear subspace.* Any two linear $k$-planes $\Lambda, \Lambda' \subseteq \mathbb{P}^n$ have classes differing by an integer multiple of the generator; the previous step assigns each of them the generator $h_0$ under the descent to $A_0(\mathbb{P}^{n-k})$, so $[\Lambda] = [\Lambda']$ in $A_k(\mathbb{P}^n)$. Concretely a projective linear transformation carrying $\Lambda$ to $\Lambda'$ sweeps out a rational equivalence between them, so the generator $h_k$ is independent of the chosen $k$-plane, completing the induction and the proof.

The theorem says that the Chow groups of $\mathbb{P}^n$ take the simplest possible form: one copy of $\mathbb{Z}$ in each dimension from 0 to $n$, with a canonical generator, the class of a linear subspace. Writing the total group additively,

$$A_*(\mathbb{P}^n) = \bigoplus_{k=0}^{n} \mathbb{Z}\, h_k, \qquad h_k = [\mathbb{P}^k]$$

so $A_*(\mathbb{P}^n)$ is free of rank $n+1$. In codimension indexing the generator $h_k = [\mathbb{P}^k]$ has codimension $n-k$, and it is common to write $h = [\mathbb{P}^{n-1}]$ for the hyperplane class and record $h_k$ as the intersection $h^{n-k}$ of $n-k$ hyperplanes; the multiplicative structure that makes this literal is the intersection product developed in a later chapter. For now the additive statement is what matters: on projective space, a $k$-cycle carries exactly one integer of information, its degree, the coefficient of $h_k$.

This is the precise sense in which projective space is rigid where affine space is soft. A subvariety $V \subseteq \mathbb{P}^n$ of dimension $k$ and degree $d$ (its degree as a projective variety, the number of points in which it meets a general linear $\mathbb{P}^{n-k}$) has class $[V] = d\, h_k$, and no amount of rational equivalence can change $d$. The degree is a complete invariant of the class, and rational equivalence on $\mathbb{P}^n$ is, for irreducible cycles, exactly the relation "same dimension and same degree" (and for arbitrary cycles, "same degree in each dimension"). The comparison with singular homology is exact: under the cycle class map to $H_{2k}(\mathbb{C}\mathrm{P}^n)$ studied later, $h_k$ maps to the generator, and the freeness of $A_k(\mathbb{P}^n)$ mirrors the freeness of $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ ([9]).

The one-dimensional case deserves its own statement, both because curves are where the theory began and because on a curve the Chow group is an object the reader already knows under another name. On a smooth projective curve $C$ the only cycles are 0-cycles and the fundamental class, and the 0-cycles modulo rational equivalence are precisely the divisor classes. This ties the general construction back to the divisor class group $\mathrm{Cl}\, C$ and the Picard group $\mathrm{Pic}\, C$ of [2, Divisors: Geometric Representation of Line Bundles].

**Proposition 1.5.3: Chow Groups of a Smooth Projective Curve**

Let $C$ be a smooth projective curve over $\overline{k}$. Then

$$A_1(C) = \mathbb{Z}\,[C], \qquad A_0(C) \cong \mathrm{Cl}\, C = \mathrm{Pic}\, C$$

and $A_k(C) = 0$ for $k \neq 0, 1$. Under the second isomorphism the degree of a zero-cycle corresponds to the degree of a divisor, giving a short exact sequence

$$0 \to \mathrm{Pic}^0 C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$$

where $\mathrm{Pic}^0 C$ is the group of divisor classes of degree zero.

***Proof*:**
Since $\dim C = 1$, the only nonzero groups are $A_0(C)$ and $A_1(C)$, and $A_k(C) = 0$ for $k \neq 0, 1$. The curve $C$ is irreducible of dimension 1, so by example 1.1.6 its top group is $A_1(C) = Z_1(C) = \mathbb{Z}\,[C]$, free on the fundamental class.

*The zero-cycles.* A 0-cycle on $C$ is a finite $\mathbb{Z}$-combination of closed points, so $Z_0(C)$ is the group $\mathrm{WDiv}\, C$ of Weil divisors on $C$. The subgroup $\mathrm{Rat}_0(C)$ of cycles rationally equivalent to zero is generated by the cycles $[\mathrm{div}(r)]$ for $r \in k(W)^*$ ranging over 1-dimensional subvarieties $W \subseteq C$; the only such $W$ is $C$ itself, so $\mathrm{Rat}_0(C)$ is generated by the principal divisors $[\mathrm{div}(r)]$ with

$r \in k(C)^*$. Thus $\mathrm{Rat}_0(C)$ is exactly the group $\mathrm{Prin}\, C$ of principal divisors in the sense of [2, Principal Divisor], and

$$A_0(C) = Z_0(C)/\mathrm{Rat}_0(C) = \mathrm{WDiv}\, C/\,\mathrm{Prin}\, C = \mathrm{Cl}\, C$$

is the divisor class group of [2, Divisor Class Group]. Because $C$ is smooth, the divisor-to-line-bundle correspondence of [2, Divisors: Geometric Representation of Line Bundles] identifies $\mathrm{Cl}\, C$ with $\mathrm{Pic}\, C$.

*The degree sequence.* On a smooth projective curve linearly equivalent divisors have equal degree: if $D \sim D'$ then $\deg D = \deg D'$, since the Riemann-Roch theorem ([2, Riemann-Roch for curves]) gives $\chi(\mathcal{L}(D)) = \deg D + 1 - g$, a quantity depending only on the class of $D$, so $\deg D$ is determined by it. Hence deg descends to a homomorphism $\deg\colon A_0(C) \to \mathbb{Z}$, agreeing under the identification $A_0(C) = \mathrm{Cl}\, C$ with the degree of a divisor class. It is surjective because a single closed point has degree $[k(p) : \overline{k}] = 1$, and its kernel is the group $\mathrm{Pic}^0 C$ of degree-zero classes by definition. This gives the stated short exact sequence, completing the proof.

The proposition recovers, in the language of Chow groups, the entire divisor theory of a curve. The degree map splits the class group into a discrete part, the copy of $\mathbb{Z}$ recording degree, and a connected part $\mathrm{Pic}^0 C$, the Jacobian, carrying the continuous moduli of divisor classes of a fixed degree. For $C = \mathbb{P}^1$ the Jacobian is trivial and $A_0(\mathbb{P}^1) = \mathbb{Z}$, in agreement with example 1.1.5 and with the case $n = 1$ of theorem 1.5.2; for a curve of genus $g \geq 1$ the group $\mathrm{Pic}^0 C$ is a $g$-dimensional abelian variety and $A_0(C)$ is strictly larger than $\mathbb{Z}$, so the degree is no longer a complete invariant of a class. The Chow group of a curve is therefore the first place in the theory where an interesting continuous invariant appears, and it is the model on which the higher-dimensional theory is built.

The Chow group $A_0(C)$ also illustrates the failure of the total-degree map to see everything. On $\mathbb{P}^n$ the degree was a complete invariant of a $k$-cycle class; on a curve of positive genus it is not, because two points $p, q \in C$ need not be rationally equivalent even though $[p]$ and $[q]$ have the same degree. Their difference $[p] - [q]$ is a nonzero class in $\mathrm{Pic}^0 C$ precisely when $p$ and $q$ are not linearly equivalent, which for $g \geq 1$ is the generic situation. This is the geometric content of the statement that a curve of positive genus admits no nonconstant rational function of degree one.

The last computation of the section is stated but not proved, because its proof needs Segre classes, which are developed later. It generalizes the projective-space computation from the trivial bundle to an arbitrary vector bundle: the Chow groups of a projectivized vector bundle are a free module over the Chow groups of the base, on the powers of the tautological class. This is the projective-bundle formula, and it is the engine that will later define Chern classes.

Given a vector bundle $E$ of rank $r+1$ on a scheme $X$, write $\mathbb{P}(E) \to X$ for the associated projective bundle, whose fiber over a point $x$ is the projective space $\mathbb{P}(E_x) \cong \mathbb{P}^r$ of one-dimensional subspaces of the fiber $E_x$. Let $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1)) \in A^1(\mathbb{P}(E))$ be the first Chern class of the Serre-twist line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ (the dual of the tautological subbundle, with the "one-dimensional subspaces" convention above), viewed as an operator on cycles. The projective-bundle formula states that the flat pullback $\pi^*$ together with capping against powers of $\xi$ makes $A_*(\mathbb{P}(E))$ a free module over $A_*(X)$ on the classes $1, \xi, \ldots, \xi^r$:

$$A_*(\mathbb{P}(E)) = \bigoplus_{i=0}^{r} \xi^i \cap \pi^* A_*(X)$$

Concretely, every class on $\mathbb{P}(E)$ is uniquely $\sum_{i=0}^r \xi^i \cap \pi^* \alpha_i$ with $\alpha_i \in A_*(X)$. Taking $X = \mathrm{Spec}\, \overline{k}$ and $E = \overline{k}^{\,n+1}$ recovers $\mathbb{P}(E) = \mathbb{P}^n$ and the free-module statement collapses to $A_*(\mathbb{P}^n) = \bigoplus_{i=0}^n \mathbb{Z}\, \xi^i$

of theorem 1.5.2, with $\xi^i$ the class of a linear subspace of codimension $i$. The full statement, and the definition of the Chern class operators it rests on, are theorem 3.1.9 in section 3.1; the formula is recorded here because it is the natural common generalization of the two computations above, and because it is the shape every later Chow-group computation of a bundle takes.

Before the exercises, the three worked instances of the theory are collected side by side, so the common mechanism is visible in one place.

**Example 1.5.4: The Chow Groups of $\mathbb{P}^1$, $\mathbb{P}^2$, and a Plane Curve**

1. *The projective line.* For $n = 1$, theorem 1.5.2 gives $A_0(\mathbb{P}^1) = \mathbb{Z}\,[\mathrm{pt}]$ and $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$, so
$$A_*(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1] \oplus \mathbb{Z}\,[\mathrm{pt}]$$
The degree-0 part $A_0(\mathbb{P}^1) = \mathbb{Z}$ agrees with example 1.1.5, and with proposition 1.5.3 for the genus-zero curve $\mathbb{P}^1$, whose Jacobian $\operatorname{Pic}^0\mathbb{P}^1$ is trivial.
2. *The projective plane.* For $n = 2$,
$$A_*(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2] \oplus \mathbb{Z}\,[\mathbb{P}^1] \oplus \mathbb{Z}\,[\mathrm{pt}]$$
with $A_2(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^1]$ generated by the class of a line, and $A_0(\mathbb{P}^2) = \mathbb{Z}\,[\mathrm{pt}]$. A plane curve $D \subseteq \mathbb{P}^2$ of degree $d$ has class $[D] = d\,[\mathbb{P}^1] = d\,h_1$ in $A_1(\mathbb{P}^2)$, since $D$ is cut out by a form of degree $d$ and hence is rationally equivalent to $d$ lines (the pencil spanned by the form and $\ell^d$ for a linear form $\ell$ sweeps one to the other). Two plane curves of degrees $c$ and $d$ therefore have classes meeting, after the intersection product is available, in $cd$ points, which is Bézout's theorem ([6]) read off from $h_1 \cdot h_1 = h_0$.
3. *A plane curve as a curve in its own right.* Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree $d$, so $C$ is a smooth projective curve of genus $g = \binom{d-1}{2}$. Two computations of its Chow groups coexist. Viewed inside $\mathbb{P}^2$, the class $[C] = d\,h_1 \in A_1(\mathbb{P}^2)$ records only its degree. Viewed intrinsically, proposition 1.5.3 gives $A_0(C) \cong \operatorname{Cl} C = \operatorname{Pic} C$ with its degree sequence $0 \to \operatorname{Pic}^0 C \to A_0(C) \to \mathbb{Z} \to 0$, and for $d \geq 3$ the genus is positive, so $\operatorname{Pic}^0 C$ is nontrivial and $A_0(C)$ is strictly larger than $\mathbb{Z}$. The pushforward $A_0(C) \to A_0(\mathbb{P}^2) = \mathbb{Z}$ along the inclusion is the degree map, collapsing all of $\operatorname{Pic}^0 C$ to zero: the intrinsic Chow group of the curve sees the moduli that the ambient projective plane cannot.

## Exercise 1.5.1

1. Compute $A_*(\mathbb{A}^2 \setminus \{p\})$, the Chow groups of the affine plane with one closed point removed. (Use the localization sequence for the closed point inside $\mathbb{A}^2$ together with proposition 1.5.1.)
2. Let $H \cong \mathbb{P}^{n-1}$ be a hyperplane in $\mathbb{P}^n$ and $U = \mathbb{P}^n \setminus H \cong \mathbb{A}^n$ its complement. Using the localization sequence, show directly that $j^*\colon A_k(\mathbb{P}^n) \to A_k(\mathbb{A}^n)$ is an isomorphism for $k = n$ and the zero map for $k < n$, and reconcile this with theorem 1.5.2.
3. Let $Q \subseteq \mathbb{P}^3$ be a smooth quadric surface. Assuming the isomorphism $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ and that $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ is free on the classes of a point, the two rulings, and the fundamental class, determine the rank of each $A_k(Q)$ and describe generators. Compare with $A_*(\mathbb{P}^2)$ and explain why $A_1(Q)$ has rank 2 rather than 1.

4. For $\mathbb{P}^m$ and $\mathbb{P}^n$, use theorem 1.5.2 and the projective-bundle formula (with $\mathbb{P}^m \times \mathbb{P}^n = \mathbb{P}(\mathcal{O}^{\oplus(m+1)})$ over $\mathbb{P}^n$) to show that $A_k(\mathbb{P}^m \times \mathbb{P}^n)$ is free of rank equal to the number of pairs $(i, j)$ with $0 \leq i \leq m$, $0 \leq j \leq n$, and $i + j = k$.

5. Let $C$ be a smooth projective curve of genus 1 over $\overline{k}$. Show that $A_0(C)$ is not finitely generated, and identify the classes $[p] - [q]$ of differences of points inside $\operatorname{Pic}^0 C$. Contrast with $A_0(\mathbb{P}^1)$.

6. Let $X \subseteq \mathbb{A}^3$ be the affine cone $x^2 + y^2 = z^2$ over a conic, a surface singular at the origin. Using $A_k(X) = A_k(X_{\text{red}})$ and the vanishing of $A_k(\mathbb{A}^n)$ in low degree as a guide, determine $A_2(X)$ and explain why the singular point does not contribute a class in $A_0(X)$ that survives rational equivalence.

# 2

# Divisors, Line Bundles, and the First Chern Class

Chapter 1 built the Chow groups and their two functorialities, proper pushforward and flat pullback, but produced no way to multiply cycles: nothing so far intersects a $k$-cycle with anything to lower its dimension. The first and most basic such operation is intersection with a divisor. A Cartier divisor $D$ on $X$, equivalently a line bundle together with a rational section, cuts each subvariety $V$ not contained in its support down to a $(k-1)$-cycle $D \cdot [V]$, the divisor of the restricted section. This chapter makes that operation precise, proves it descends to rational equivalence, and establishes its one indispensable property, commutativity: intersecting with $D$ and then with $D'$ agrees with intersecting in the other order.

Commutativity is not a formality. It is the identity whose proof, resting on the vanishing degree of a principal divisor on a complete curve, supplies the well-definedness that the affine-bundle-invariance theorem of section 1.4 took on faith. Packaging intersection with a divisor as the *first Chern class* $c_1(L) \cap -$ of the associated line bundle turns these results into the seed of the entire theory of Chern classes, developed for bundles of higher rank in the chapter that follows.

## 2.1 Cartier Divisors, Line Bundles, and the Class Map

The Chow group of section 1.1 is graded by dimension, and every construction of sections 1.2 to 1.4 either preserves the grading (proper pushforward, flat pullback of the fundamental class) or shifts it by the fixed relative dimension of a flat map. What is still missing is an operation that *lowers* dimension by one, cutting a $k$-cycle down to a $(k-1)$-cycle by intersecting it with a hypersurface. The classical prototype is Bézout's theorem: a curve of degree $d$ and a curve of degree $e$ in $\mathbb{P}^2$ meet in $de$ points. The intersection is a codimension-one operation, and the object one intersects with is a divisor. This section sets up the divisor side of that operation. It recalls, from *Algebraic Geometry* [2], the two faces of a divisor (the geometric Weil divisor and the sheaf-theoretic Cartier divisor) and the line bundle that packages the Cartier data, then constructs the first bridge from divisor theory to the Chow group: the class map $\operatorname{Pic}(X) \to A_{n-1}(X)$ sending a line bundle to the cycle of its rational sections. The intersection operation itself, and the proof that it is well defined on rational equivalence, occupy section 2.2; here the goal is to fix the input and record how it sits inside the Chow group of a fixed variety.

Throughout, $X$ is an algebraic scheme over $\overline{k}$ and $V$ ranges over subvarieties (closed integral subschemes); a point is a closed point, and $n = \dim X$.

*Algebraic Geometry* [2] developed two theories of divisors, and the reader who has that material in hand can move quickly here. A *Weil divisor* ([2, Weil Divisor]) on a Noetherian integral separated scheme regular in codimension one is a finite formal integer combination $\sum_i n_i[Y_i]$ of prime divisors, where a prime divisor is a codimension-one subvariety. This is precisely a codimension-one algebraic cycle in the sense of section 1.1: when $\dim X = n$, the group $\operatorname{WDiv} X$ of Weil divisors is exactly the cycle group $Z_{n-1}(X)$. A *principal divisor* $\operatorname{div}(f) = \sum_Y \nu_Y(f)[Y]$ ([2, Principal Divisor]) records the zeros and poles of a rational function $f \in K(X)^*$, weighted by the order $\nu_Y$ measured in the discrete valuation ring $\mathcal{O}_{X,\eta_Y}$; this is the same order-along-a-subvariety $\operatorname{ord}_Y(f)$ that defines rational equivalence in definition 1.1.1. The divisor class group $\operatorname{Cl} X = \operatorname{WDiv} X/\operatorname{PDiv}(X)$ of [2, Divisor Class Group] is therefore literally the top Chow group $A_{n-1}(X)$ in the case where the only cycles rationally equivalent to zero come from functions on $X$ itself, that is, on a variety.

A *Cartier divisor* ([2, Cartier Divisor]) encodes the same data locally and sheaf-theoretically: it is a global section of $\mathscr{K}_X^*/\mathcal{O}_X^*$, represented by an open cover $\{U_i\}$ with rational functions $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ whose ratios $f_i/f_j$ are units on overlaps. A Cartier divisor is *principal* when it comes from a single global rational function; two Cartier divisors are linearly equivalent when they differ by a principal one. The gluing data $\{(U_i, f_i)\}$ builds a line bundle $\mathcal{O}_X(D)$, the sub-$\mathcal{O}_X$-module of $\mathscr{K}_X$ generated locally by $f_i^{-1}$, and this assignment is the heart of the theory: $D \mapsto \mathcal{O}_X(D)$ is a homomorphism $\operatorname{CDiv} X \to \operatorname{Pic} X$ that carries addition of divisors to tensor product of line bundles, sends $-D$ to the dual, and vanishes exactly on principal divisors. On an integral scheme it induces an isomorphism from the Cartier divisor class group $\operatorname{CaCl} X$ of [2, Cartier Divisor Class Group] onto the whole Picard group (all of this is developed in [2, Divisors: Geometric Representation of Line Bundles]),

$$\operatorname{CaCl} X \xrightarrow{\sim} \operatorname{Pic} X$$

so a line bundle is the same thing as a Cartier divisor up to linear equivalence. This is the sense in which the section title lists “Cartier divisors” and “line bundles” as two names for one object.

The reader should keep both faces in view. The Cartier divisor is what one intersects with, because its local equations $f_i$ give the equations to cut against; the line bundle $\mathcal{O}_X(D)$ is the coordinate-free invariant, the thing that survives when the choice of local equations is forgotten. The class map built below is precisely the passage from the line bundle back to a well-defined cycle class.

The two theories agree under a regularity hypothesis, and pinning down that hypothesis is what makes the class map land in the Chow group cleanly. The point is one of local principality: a Weil divisor is Cartier exactly when each of its prime divisors is locally cut out by a single equation. Regularity in codimension one is enough to send a Cartier divisor to a Weil divisor; local factoriality is what makes the map an isomorphism.

**Proposition 2.1.1: Weil-Cartier Comparison**

Let $X$ be a variety of dimension $n$ over $\overline{k}$.

1. If $X$ is regular in codimension one (for instance $X$ normal), there is a homomorphism

$$\operatorname{CDiv} X \longrightarrow \operatorname{WDiv} X = Z_{n-1}(X), \qquad D \longmapsto [D]$$

carrying principal Cartier divisors to principal Weil divisors. In local data $D = \{(U_i, f_i)\}$ the associated Weil cycle is $[D] = \sum_Y \nu_Y(f_i)[Y]$, the coefficient of a prime divisor $Y$ being $\nu_Y(f_i)$ for any index $i$ with $\eta_Y \in U_i$.

2. If $X$ is locally factorial (every local ring $\mathcal{O}_{X,x}$ is a unique factorization domain; smooth varieties qualify), this map is an isomorphism $\operatorname{CDiv} X \xrightarrow{\sim} \operatorname{WDiv} X$ carrying principal divisors to principal divisors, hence an isomorphism of class groups

$$\operatorname{Pic} X \;\cong\; \operatorname{CaCl} X \;\cong\; \operatorname{Cl} X \;=\; A_{n-1}(X)$$

***Proof*:**
Both parts are proved in full in [2, Divisors: Geometric Representation of Line Bundles] and only recalled here. The homomorphism of part (1) is [2, Cartier Divisors to Weil Divisors], and its upgrade to an isomorphism under local factoriality (part (2)) is [2, Weil and Cartier Divisors Under Locally Factorial Hypotheses]; the resulting chain of class-group isomorphisms $\operatorname{Pic} X \cong \operatorname{CaCl} X \cong \operatorname{Cl} X$ is [2, Divisor Class Group and Picard Group]. Only the final equality $\operatorname{Cl} X = A_{n-1}(X)$ is new to this Part, and it is the identification made in the paragraph below: on a variety the codimension-one cycles modulo rational equivalence are exactly the Weil divisors modulo principal divisors, both quotients of $Z_{n-1}(X)$ by $\operatorname{PDiv}(X)$, as we sought to show.

The equality $\operatorname{Cl} X = A_{n-1}(X)$ deserves a moment's reflection, because it is easy to mistake for a definition. Rational equivalence on $Z_{n-1}(X)$ is generated by cycles $[\operatorname{div}(r)]$ where $r$ is a rational function on a subvariety $W \subseteq X$ of dimension $n$. On a variety $X$, the only $n$-dimensional subvariety is $X$ itself, so those generators are exactly the principal Weil divisors $\operatorname{div}(r)$ with $r \in K(X)^*$. Hence $A_{n-1}(X) = Z_{n-1}(X)/\operatorname{Rat}_{n-1}(X)$ and $\operatorname{Cl} X = \operatorname{WDiv} X/\operatorname{PDiv}(X)$ are the same quotient of the same group. Linear equivalence of divisors *is* rational equivalence of codimension-one cycles. This is what will let the class map below produce Chow classes, not just divisor classes.

**Example 2.1.2: The Quadric Cone: Cartier Strictly Inside Weil**
The comparison map need not be surjective when factoriality fails. On the affine quadric cone $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy - z^2)$, a normal surface singular at the vertex, [2, Cartier Divisors] computes that the ruling $Y = V(y, z)$ is a Weil divisor which is not Cartier at the vertex, while its double $2[Y] = \operatorname{div}(y)$ is, so

$$\operatorname{CaCl} X = 0 \subsetneq \operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z} = A_1(X)$$

the final equality being the identification $\operatorname{Cl} X = A_{n-1}(X)$ of proposition 2.1.1. Normality supplies part (1), but the vertex obstructs local principality, so part (2) fails; on a smooth variety no such gap exists and Weil, Cartier, Picard, and $A_{n-1}$ all coincide.

On a smooth variety the isomorphism $\operatorname{Pic} X \cong A_{n-1}(X)$ of proposition 2.1.1 already provides a bridge from line bundles to Chow classes. To intersect and to compute Chern classes on *arbitrary* algebraic schemes, one needs this bridge without any regularity hypothesis, and one needs it stated at the level of line bundles rather than Cartier divisors, so that the input is a coordinate-free invariant. The construction is direct: a line bundle $L$ has rational sections, a rational section has a divisor of zeros and poles, and that divisor is a codimension-one cycle whose class in $A_{n-1}(X)$ turns out not to depend on the section chosen.

Fix a variety $X$ of dimension $n$ with function field $K(X)$, and a line bundle $L$ on $X$. A *rational section* of $L$ is a nonzero section of $L \otimes_{\mathcal{O}_X} K(X)$ over $X$, equivalently a section of $L$ over some nonempty open subset. Since $X$ is a variety, $L \otimes K(X)$ is a one-dimensional $K(X)$-vector space, so rational sections exist and any two differ by multiplication by an element of $K(X)^*$. On the open set where $s$ is defined and nonvanishing, $s$ trivializes $L$; along each prime divisor $Y$ one may compare $s$ with a local trivialization of $L$ near $\eta_Y$ and read off an integer $\operatorname{ord}_Y(s)$, the order of vanishing (or of the pole) of $s$ along $Y$. Concretely, if $t$ is a local generator of $L$ near $\eta_Y$ and $s = g \cdot t$ with $g \in K(X)^*$, set $\operatorname{ord}_Y(s) := \nu_Y(g)$; a different generator $t' = ut$ with $u \in \mathcal{O}^*_{X,\eta_Y}$ changes $g$ to $u^{-1}g$ and leaves $\nu_Y(g)$ unchanged, so $\operatorname{ord}_Y(s)$ is well defined. The *divisor of* $s$ is the Weil divisor

$$\operatorname{div}(s) \;=\; \sum_Y \operatorname{ord}_Y(s)\,[Y] \;\in\; Z_{n-1}(X)$$

finite because $s$ is a regular nonvanishing section over some nonempty open $U \subseteq X$, where it trivializes $L$ and so $\operatorname{ord}_Y(s) = 0$ for every prime divisor $Y$ meeting $U$; the only possible contributors are the finitely many prime divisors in the proper closed complement $X \setminus U$. This is the same finiteness that makes the principal divisor $\operatorname{div}(f)$ of [2, Principal Divisor] a well-defined cycle. If $L = \mathcal{O}_X(D)$ for a Cartier divisor $D$ and $s$ is the canonical rational section (the constant $1 \in \mathscr{K}_X$, which against the local generator $f_i^{-1}$ of $\mathcal{O}_X(D)|_{U_i}$ is written $1 = f_i \cdot f_i^{-1}$, giving $\operatorname{ord}_Y(s) = \nu_Y(f_i)$), then $\operatorname{div}(s)$ is exactly the Weil cycle $[D]$ of proposition 2.1.1.

**Proposition 2.1.3: The Class Map**

Let $X$ be a variety of dimension $n$. For a line bundle $L$ on $X$, choose any nonzero rational section $s$ of $L$ and set

$$c(L) \;:=\; [\operatorname{div}(s)] \;\in\; A_{n-1}(X)$$

Then $c(L)$ is independent of the choice of $s$, and the resulting assignment

$$c\colon \operatorname{Pic} X \longrightarrow A_{n-1}(X), \qquad L \longmapsto c(L)$$

is a group homomorphism. When $X$ is locally factorial (in particular smooth), $c$ is the isomorphism $\operatorname{Pic} X \cong A_{n-1}(X)$ of proposition 2.1.1; on a general variety $c$ need be neither injective nor surjective.

***Proof*:**
*Step 1: independence of the section.* Let $s, s'$ be two nonzero rational sections of $L$. Both lie in the one-dimensional $K(X)$-vector space $L \otimes K(X)$, so $s' = gs$ for a unique $g \in K(X)^*$. Along each prime divisor $Y$, choose a local generator $t$ of $L$ near $\eta_Y$ and write $s = at$, $s' = a't$ with $a, a' \in K(X)^*$; then $a' = ga$ and

$$\operatorname{ord}_Y(s') = \nu_Y(a') = \nu_Y(g) + \nu_Y(a) = \nu_Y(g) + \operatorname{ord}_Y(s)$$

because $\nu_Y$ is a valuation. Summing over $Y$ gives $\operatorname{div}(s') = \operatorname{div}(g) + \operatorname{div}(s)$ in $Z_{n-1}(X)$, where $\operatorname{div}(g) = \sum_Y \nu_Y(g)[Y]$ is the principal Weil divisor of $g \in K(X)^*$. Since $\operatorname{div}(g)$ is a principal divisor on the variety $X$, it is rationally equivalent to zero in $A_{n-1}(X)$ by the identification $\mathrm{Rat}_{n-1}(X) = \mathrm{PDiv}(X)$ recorded after proposition 2.1.1. Hence $[\mathrm{div}(s')] = [\mathrm{div}(s)]$ in $A_{n-1}(X)$, and $c(L)$ is well defined.

*Step 2: homomorphism property.* Let $L, L'$ be line bundles with rational sections $s, s'$. Then $s \otimes s'$ is a nonzero rational section of $L \otimes L'$, and along each prime divisor $Y$, choosing local generators $t$ of $L$ and $t'$ of $L'$ near $\eta_Y$ and writing $s = at$, $s' = a't'$, the product $s \otimes s' = (aa')(t \otimes t')$ is expressed against the local generator $t \otimes t'$ of $L \otimes L'$. Thus

$$\mathrm{ord}_Y(s \otimes s') = \nu_Y(aa') = \nu_Y(a) + \nu_Y(a') = \mathrm{ord}_Y(s) + \mathrm{ord}_Y(s')$$

so $\mathrm{div}(s \otimes s') = \mathrm{div}(s) + \mathrm{div}(s')$, and passing to classes gives $c(L \otimes L') = c(L) + c(L')$. The trivial bundle $\mathcal{O}_X$ has the rational section 1 with $\mathrm{div}(1) = 0$, so $c(\mathcal{O}_X) = 0$, and $c(L^{-1}) = -c(L)$ follows from $L \otimes L^{-1} = \mathcal{O}_X$. Therefore $c$ is a homomorphism.

*Step 3: comparison with the smooth case.* When $X$ is locally factorial, every line bundle is $\mathcal{O}_X(D)$ for a Cartier divisor $D$, and the canonical rational section of $\mathcal{O}_X(D)$ has $\mathrm{div}(s) = [D]$, the Weil cycle of $D$. Under $\mathrm{Pic}\, X \cong \mathrm{CaCl}\, X$ ([2, Divisors: Geometric Representation of Line Bundles]) and $\mathrm{CaCl}\, X \cong \mathrm{Cl}\, X = A_{n-1}(X)$ (proposition 2.1.1), the assignment $L = \mathcal{O}_X(D) \mapsto [D]$ is exactly that isomorphism, so $c$ is an isomorphism. On a variety where these hypotheses fail, $c$ can lose injectivity or surjectivity, as example 2.1.2 shows for the class-group comparison. This completes the proof.

The class map is the first of the two ways line bundles act on the Chow group. It records *where* a line bundle is, as a codimension-one class $c(L) \in A_{n-1}(X)$, but it does not yet say how $L$ intersects cycles of other dimensions. That is the job of the operator $c_1(L) \cap -$ built in section 2.4, whose value on the fundamental class $[X]$ recovers $c(L)$: writing ahead, $c_1(L) \cap [X] = c(L)$. The class map is thus the top-dimensional case of the full first Chern class, and everything in this chapter can be read as promoting $c$ from a map on line bundles to an operator on all of $A_*(X)$.

The value of stating $c$ at the level of line bundles, rather than resting on the isomorphism $\mathrm{Pic} \cong A_{n-1}$, is that $c$ makes sense with no regularity hypothesis on $X$: a rational section and its order along a prime divisor require only that $X$ be a variety, and the group $A_{n-1}(X)$ is defined for every algebraic scheme. The intersection theory of section 2.2 is built for arbitrary $X$, and it needs $c$ in this general form.

**Example 2.1.4: The Class Map on Projective Space**

Projective space is where every ingredient can be computed by hand. Since $\mathbb{P}^n$ has all local rings unique factorization domains, it is locally factorial, so $\mathrm{Pic}\, \mathbb{P}^n \cong \mathrm{Cl}\, \mathbb{P}^n \cong \mathbb{Z}$, generated by $\mathcal{O}_{\mathbb{P}^n}(1)$, and by theorem 1.5.2 the top Chow group is $A_{n-1}(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^{n-1}]$, generated by the class of a hyperplane. The class map identifies the two.

A rational section of $\mathcal{O}_{\mathbb{P}^n}(1)$ is a nonzero linear form. Take $s = x_0$, the section whose zero scheme is the hyperplane $H = V_+(x_0)$, whose Cartier data $\{(U_i, x_0/x_i)\}$ was recorded in [2, Cartier Divisors]. On the standard chart $U_i = D_+(x_i) \cong \mathbb{A}^n$ the sheaf $\mathcal{O}_{\mathbb{P}^n}(1)$ is trivialized by $x_i$, and $s = x_0 = (x_0/x_i)\, x_i$, so $s$ corresponds to the rational function $x_0/x_i \in K(\mathbb{P}^n)$ against the generator $x_i$. Along the prime divisor $H$, the generic point lies in charts $U_i$ with $i \neq 0$, where $x_0/x_i$ is a local equation cutting out $H$ with order one, so $\mathrm{ord}_H(s) = 1$; along every other prime divisor $x_0/x_i$ is a unit near the generic point, contributing zero. Hence

$$\mathrm{div}(x_0) = [H] = [\mathbb{P}^{n-1}], \qquad c\big(\mathcal{O}_{\mathbb{P}^n}(1)\big) = [\mathbb{P}^{n-1}] \in A_{n-1}(\mathbb{P}^n)$$

Choosing instead $s = x_1$ gives the hyperplane $V_+(x_1)$, a different divisor but the same class: $x_1 = (x_1/x_0)\,x_0$ differs from $x_0$ by the rational function $x_1/x_0$, and $\operatorname{div}(x_1) - \operatorname{div}(x_0) = \operatorname{div}(x_1/x_0) \sim 0$, confirming the independence of proposition 2.1.3 in this case. More generally, $\mathcal{O}_{\mathbb{P}^n}(d)$ has rational sections given by degree-$d$ forms; the section $x_0^d$ has divisor $d[H]$, so $c(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,[\mathbb{P}^{n-1}]$, and $c\colon \mathbb{Z} \to \mathbb{Z}$ is the identity. The generator $\mathcal{O}(1)$ of Pic maps to the generator [hyperplane] of $A_{n-1}$, and the class map is the announced isomorphism $\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$.

There is one further refinement to record before intersecting, and it addresses a bookkeeping problem rather than a new geometric idea. When a Cartier divisor $D$ is intersected with a subvariety $V$, one wants the resulting cycle to live on $V \cap |D|$, the part of $V$ where $D$ cuts, not just somewhere on $V$. Localizing the answer to the support is what makes the intersection product refined enough to carry information (for instance, an excess-intersection or a self-intersection formula) and what makes it functorial under the localization sequence of theorem 1.4.1. The device that carries this localized data is the *pseudo-divisor*: a line bundle, a chosen closed support, and a trivialization of the bundle away from that support.

**Definition 2.1.5: Pseudo-Divisor**

Let $X$ be an algebraic scheme. A *pseudo-divisor* on $X$ is a triple

$$\mathfrak{D} = (L, Z, s)$$

consisting of a line bundle $L$ on $X$, a closed subset $Z \subseteq X$ (the *support*, written $|\mathfrak{D}|$), and a trivialization $s\colon \mathcal{O}_{X\setminus Z} \xrightarrow{\sim} L|_{X\setminus Z}$ of $L$ over the complement of $Z$. The line bundle $L$ is denoted $\mathcal{O}_X(\mathfrak{D})$.

A Cartier divisor $D$ on $X$ determines a pseudo-divisor $(\mathcal{O}_X(D), |D|, s_D)$ whose support is the support $|D|$ of $D$ and whose trivialization $s_D$ is the canonical rational section of $\mathcal{O}_X(D)$, which is a nonvanishing regular section away from $|D|$. Two Cartier divisors $D, D'$ with the same support and the same associated line bundle give the same pseudo-divisor when their canonical sections agree off the support.

The three pieces of data are exactly what a divisor supplies for cutting. The line bundle $L$ is the coordinate-free invariant whose class map value is $c(L)$; the support $Z$ pins down where the cutting happens, so the product lands in $A_*(Z)$ rather than $A_*(X)$; and the trivialization $s$ off $Z$ is what lets one measure orders of vanishing, because on the locus where $L$ is trivialized by $s$ there is nothing to cut, and all the divisorial content is concentrated on $Z$. A Cartier divisor is the special case where $Z = |D|$ is as small as possible, but allowing $Z$ to be larger is what gives the flexibility to restrict, pull back, and add pseudo-divisors while keeping supports under control. The next section takes a subvariety $V$ not contained in the support and defines $\mathfrak{D} \cdot [V]$ as a class in $A_{k-1}(V \cap Z)$; when $V \subseteq Z$, the restricted line bundle $L|_V$ supplies the answer through its own class map. Here the pseudo-divisor is introduced only to fix the input; its use is the content of section 2.2.

**Example 2.1.6: The Pseudo-Divisor of a Hyperplane**

Let $X = \mathbb{P}^n$ and $H = V_+(x_0)$ the hyperplane at $x_0 = 0$. The associated pseudo-divisor is

$$\mathfrak{H} = \bigl(\mathcal{O}_{\mathbb{P}^n}(1),\ H,\ s\bigr)$$

with support the hyperplane $H$ and trivialization $s$ given by the section $x_0$ over the complement $\mathbb{P}^n \setminus H = D_+(x_0) \cong \mathbb{A}^n$, where $x_0$ is nonvanishing and so trivializes $\mathcal{O}(1)$. The line bundle $\mathcal{O}_{\mathbb{P}^n}(1)$ carries the class $c(\mathcal{O}(1)) = [H] \in A_{n-1}(\mathbb{P}^n)$ computed in example 2.1.4; the support restricts any intersection against $\mathfrak{H}$ to $H$; and the trivialization records that off $H$ the hyperplane cuts nothing. Intersecting $\mathfrak{H}$ with a subvariety $V \not\subseteq H$ will, in section 2.2, produce the cycle $[V \cap H]$ with its multiplicities, supported on $V \cap H$, which is the algebraic incarnation of Bézout's degree count.

The class map and the pseudo-divisor together fix the two ways a line bundle enters intersection theory: as a class $c(L) \in A_{n-1}(X)$, which is its total content, and as a localized cutting datum $(L, Z, s)$, which is how it acts. With the input settled, the operation $\mathfrak{D} \cdot -$ can be defined and shown well defined on rational equivalence, which is the subject of section 2.2.

### Exercise 2.1.1

1. Let $X = \mathbb{P}^n$ and let $Q = V_+(q)$ be a smooth quadric hypersurface, cut out by a nondegenerate quadratic form $q$. Compute $c(\mathcal{O}_{\mathbb{P}^n}(Q))$, the class map value of the line bundle of $Q$, in $A_{n-1}(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^{n-1}]$, and explain the answer in terms of the degree of $Q$.
2. Let $X$ be a variety and $r \in K(X)^*$ a nonzero rational function. Show that the line bundle $\mathcal{O}_X(\operatorname{div}(r))$ associated to the principal Cartier divisor $\operatorname{div}(r)$ is trivial, and deduce that $c(\mathcal{O}_X(\operatorname{div}(r))) = 0$ directly from proposition 2.1.3. Which step of the class-map construction is this computing?
3. On the affine quadric cone $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy - z^2)$ of example 2.1.2, the class map $c\colon \operatorname{Pic} X \to A_1(X)$ is a homomorphism $\operatorname{CaCl} X \to \operatorname{Cl} X$. Using the values $\operatorname{CaCl} X = 0$ and $\operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z}$ from that example, describe $c$ explicitly and state whether it is injective, surjective, or neither. What does this say about the necessity of the factoriality hypothesis in proposition 2.1.1(2)?
4. Pseudo-divisors add: given $\mathfrak{D} = (L, Z, s)$ and $\mathfrak{D}' = (L', Z', s')$, define $\mathfrak{D} + \mathfrak{D}' := (L \otimes L',\ Z \cup Z',\ s \otimes s')$. Verify that this is a well-defined pseudo-divisor, that $\mathcal{O}_X(\mathfrak{D} + \mathfrak{D}') = \mathcal{O}_X(\mathfrak{D}) \otimes \mathcal{O}_X(\mathfrak{D}')$, and that on the line-bundle level this addition is compatible with the homomorphism property of the class map in proposition 2.1.3.
5. Let $L$ be a line bundle on a variety $X$, $s$ a nonzero rational section, and $Y$ a prime divisor. Show that the order $\operatorname{ord}_Y(s) := \nu_Y(a)$, defined by writing $s = a\,t$ against a local generator $t$ of $L$ near $\eta_Y$ with $a \in K(X)^*$, does not depend on the choice of generator $t$. Then show that if $s$ is a regular nonvanishing section of $L$ near $\eta_Y$, then $\operatorname{ord}_Y(s) = 0$.

## 2.2 Intersecting with a Divisor

The class map of section 2.1 sends a line bundle to a cycle class in codimension one, but it says nothing about how that cycle meets the other cycles already living on $X$. This section supplies the missing operation. A Cartier divisor $D$ acts on cycles by cutting each subvariety down along the zeros and poles of a rational section, an operation that lowers dimension by one and records where the cut takes place. The construction is the first instance of a general principle of the subject, that a line bundle acts on the Chow group as an operator, and it is the operation whose associativity and commutativity the rest of the chapter is built to establish.

The geometry is the one already visible in codimension one. A Cartier divisor is locally the divisor of a single rational function $f_i$ on an open $U_i$, and the transition ratios $f_i/f_j$ are units, so the zeros and poles of the local equations patch to a well-defined codimension-one cycle on $X$. To intersect $D$ with a $k$-dimensional subvariety $V$, restrict this local data to $V$ and read off the resulting $(k-1)$-cycle on $V$. Two features force themselves on the construction. First, the answer is supported on the intersection $V \cap |D|$, since away from the support of $D$ the local equations are units and contribute nothing. Second, the recipe stumbles when $V$ lies inside $|D|$, where the local equations $f_i|_V$ may vanish identically and $\operatorname{div}(f_i|_V)$ is undefined; the trivialization off $|D|$ is exactly what is lost. The definition below treats the two cases separately and unifies them through the restricted line bundle.

**Definition 2.2.1: Intersection with a Cartier Divisor**

Let $D$ be a Cartier divisor on $X$ with support $|D|$ and associated line bundle $\mathcal{O}_X(D)$, and let $V \subseteq X$ be a $k$-dimensional subvariety. The *intersection class* $D \cdot [V] \in A_{k-1}(V \cap |D|)$ is defined as follows.

1. If $V \not\subseteq |D|$, then the local equations of $D$ restrict to a nonzero rational function $s_V$ on $V$: choosing a representative $\{(U_i, f_i)\}$ of $D$, the ratios $f_i|_V$ agree up to units on overlaps and glue to a rational section $s_V$ of $\mathcal{O}_X(D)|_V$ that is nonzero on the dense open $V \setminus |D|$. Its divisor is a $(k-1)$-cycle supported on $V \cap |D|$, and one sets
$$D \cdot [V] = \big[\operatorname{div}(s_V)\big] \in A_{k-1}(V \cap |D|)$$
2. If $V \subseteq |D|$, then no such nonzero section exists, and one instead uses the restricted line bundle. The class map of proposition 2.1.3 applied to $\mathcal{O}_X(D)|_V$ produces a class in $A_{k-1}(V)$; since $V \subseteq |D|$ gives $V \cap |D| = V$, one sets
$$D \cdot [V] = c\big(\mathcal{O}_X(D)|_V\big) \in A_{k-1}(V) = A_{k-1}(V \cap |D|)$$

Extending $\mathbb{Z}$-linearly over the subvarieties appearing in a cycle $\alpha = \sum_i n_i [V_i]$ gives the *intersection cycle* $D \cdot \alpha = \sum_i n_i\,(D \cdot [V_i])$, a class in $A_{k-1}(|D| \cap \operatorname{Supp}\alpha)$, where $\operatorname{Supp}\alpha = \bigcup_i V_i$.

Two conventions in the definition deserve unpacking, since both recur throughout the theory. The first is that the intersection class is recorded on the intersection $V \cap |D|$, not on all of $X$. This is a refinement one does not want to discard prematurely: the intersection of a curve with a divisor is a set of points on the curve, and remembering that the points lie in $V \cap |D|$ is exactly what makes localized intersection numbers and excess-intersection formulas possible later. When only the class in $A_{k-1}(X)$ is wanted, push forward along the inclusion $V \cap |D| \hookrightarrow X$, a proper map. The second convention is the treatment of $V \subseteq |D|$. Here the naive recipe fails because the local equations restrict to zero, so there is no rational section to take the divisor of; but $\mathcal{O}_X(D)|_V$ is still a line bundle on $V$, and its class under the class map $c\colon \operatorname{Pic}(V) \to A_{k-1}(V)$ of proposition 2.1.3 is a $(k-1)$-cycle class. Both clauses are the same construction, taking the class of the restricted bundle $\mathcal{O}_X(D)|_V$. When $V \not\subseteq |D|$ the bundle comes with a preferred rational section $s_V$ and the class is computed as $[\operatorname{div}(s_V)]$; when $V \subseteq |D|$ that section degenerates and only the bundle survives. The packaging of the whole operation as the first Chern class $c_1(\mathcal{O}_X(D)) \cap -$ is the subject of section 2.4.

That the operation lowers dimension by exactly one is built into the definition: the divisor of a rational section on a $k$-dimensional variety is a codimension-one cycle, hence $(k-1)$-dimensional. Geometrically, $D \cdot [V]$ is the divisor cut out on $V$ by the equation defining $D$, the higher-dimensional form of substituting a curve into the equation of a hypersurface and reading off the intersection points with multiplicity. The next example makes this concrete in the case a reader already knows from the classical degree of a plane curve.

**Example 2.2.2: A Hyperplane Meeting a Plane Curve**

Let $X = \mathbb{P}^2$ with homogeneous coordinates $[x_0 : x_1 : x_2]$, let $H = \{x_0 = 0\}$ be the line at infinity, a Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(H) = \mathcal{O}_{\mathbb{P}^2}(1)$, and let $V \subseteq \mathbb{P}^2$ be a smooth conic, say the closure of $\{x_1^2 = x_0x_2\}$, a subvariety of dimension 1 not contained in $H$. The unique intersection point turns out to be $p = [0 : 0 : 1]$, which lies in the chart $U_2 = \{x_2 \neq 0\}$; work there, with affine coordinates $u = x_0/x_2$ and $v = x_1/x_2$. On $U_2$ the conic is $V = \{u = v^2\}$ (from $x_1^2 = x_0x_2$), the divisor $H = \{x_0 = 0\}$ is $\{u = 0\}$, and against the local generator $x_2$ of $\mathcal{O}_{\mathbb{P}^2}(1)$ the section $x_0$ representing $H$ is the function $x_0/x_2 = u$. Its restriction to $V$ is $s_V = u|_V = v^2$, which vanishes exactly where $V$ meets $H$: $u = 0$ forces $v = 0$, so $p$ is the only intersection point, and there $V$ is tangent to $H$. In the local coordinate $v$ on $V$ near $p$, the function $s_V = v^2$ has a double zero (the conic meets its tangent line to order two), so

$$H \cdot [V] = \big[\mathrm{div}(s_V)\big] = 2\,[p] \in A_0(V \cap H)$$

Pushing forward to $\mathbb{P}^2$ and taking degrees gives $\deg(H \cdot [V]) = 2$, the degree of the conic. The intersection class remembers more than the number 2: it records that both units of intersection are concentrated at the single point $p$, information the bare degree discards.

The example illustrates the two things the intersection class is meant to capture, the total count and its distribution, and shows how the classical intersection number of two plane curves reappears as the degree of $D \cdot [V]$. For a curve $V$ of degree $d$ and a line $H$ in general position the same computation gives $d$ distinct simple points, and $\deg(H \cdot [V]) = d$ in every case; this is the statement, once the well-definedness below is in hand, that intersecting with a hyperplane computes the degree. Before turning to that, the second worked example records the complementary phenomenon, what happens when $V$ lies inside the support of $D$, so that the second clause of definition 2.2.1 is in force.

**Example 2.2.3: A Divisor Restricted to a Component of Its Support**

Let $X = \mathbb{P}^2$ again and let $D = L$ be a line, say $L = \{x_0 = 0\}$, so $|D| = L$ and $\mathcal{O}_{\mathbb{P}^2}(L) = \mathcal{O}_{\mathbb{P}^2}(1)$. Take $V = L$ itself, a 1-dimensional subvariety contained in $|D|$. The first clause does not apply, since $x_0$ restricts to zero on $L$; the second clause instructs one to compute the class of the restricted line bundle $\mathcal{O}_{\mathbb{P}^2}(1)|_L$. Now $L \cong \mathbb{P}^1$, and $\mathcal{O}_{\mathbb{P}^2}(1)|_L = \mathcal{O}_{\mathbb{P}^1}(1)$ by the adjunction of the hyperplane bundle to a linear subspace. The class map $\mathrm{Pic}\,(\mathbb{P}^1) = \mathbb{Z} \to A_0(\mathbb{P}^1) = \mathbb{Z}$ of proposition 2.1.3 carries $\mathcal{O}_{\mathbb{P}^1}(1)$ to the class of a single point $[q]$, so

$$L \cdot [L] = c\big(\mathcal{O}_{\mathbb{P}^2}(1)|_L\big) = [q] \in A_0(L)$$

the class of one point on $L$. This is the self-intersection $L \cdot L = 1$ of a line in the plane, the positive-degree form of the fact that two distinct lines meet in one point: moving $L$ to a nearby line $L'$ and intersecting $L \cdot [L'] = [L \cap L']$ recovers the same class, now visibly one transverse point. The second clause of the definition is what lets self-intersection make sense at all, since a divisor cannot literally be substituted into a variety it defines.

The two examples exhibit the operation on both sides of the containment condition and already hint at the compatibility the operation must have with rational equivalence. In example 2.2.3 the value $L \cdot [L]$ was computed by restricting the bundle, but it was also claimed to agree with $L \cdot [L']$ for a nearby line $L'$, obtained by the first clause. Since $L \sim L'$ as divisors, these two computations must give the same class if the operation is to depend only on the rational-equivalence class of the cycle, and only then does the self-intersection number deserve its name. Establishing that agreement in general is the content of the section, and it is the property that will let the intersection class be defined on the Chow group rather than on cycles alone.

The stake is higher than notational convenience. Intersection with a divisor is meant to be an operator on $A_*(X)$, so that $D \cdot -$ carries $A_k(X)$ to $A_{k-1}(X)$ and can be iterated; without invariance under rational equivalence the operation lives only on cycles, where iteration is not controlled and no ring structure can form. The following proposition is the foundation of everything downstream, and it is one half of the identity (1.3) that the affine-bundle-invariance theorem of section 1.4 deferred to this chapter, the half asserting that $D \cdot -$ is well defined; the commutativity half is section 2.3.

**Proposition 2.2.4: Rational Invariance of Divisor Intersection**

Let $D$ be a Cartier divisor on $X$. If $\alpha \in Z_k(X)$ is rationally equivalent to zero and no component of $\alpha$ lies in $|D|$, then $D \cdot \alpha \sim 0$ in $A_{k-1}(|D|)$. Consequently, for any class $a \in A_k(X)$ the intersection $D \cdot a \in A_{k-1}(|D|)$ is well defined by choosing a representative cycle none of whose components lies in $|D|$, and $D \cdot -$ descends to a homomorphism

$$D \cdot - \colon A_k(X) \to A_{k-1}(|D|)$$

independent of all choices.

The proof is the technical heart of the divisor calculus, and it is worth naming the obstacle before entering it. A cycle rationally equivalent to zero is a sum of divisors $[\operatorname{div}(r)]$ of rational functions $r$ on $(k+1)$-dimensional subvarieties $W$. Intersecting with $D$ replaces each such $W$-divisor by its trace under the section $s_W$ of $\mathcal{O}_X(D)|_W$, and the claim is that this trace is again a boundary. On the surface $W$ there are now two rational functions in play, the given $r$ and the local equation of $D$, and the argument turns on the symmetry between them: the order of $r$ measured along the zeros of $D$ matches the order of the equation of $D$ measured along the zeros of $r$, up to a principal divisor. That symmetry is a statement about a two-dimensional surface, and it is proved by pushing everything down to a complete curve, where a principal divisor has degree zero. Three inputs carry the argument: this degree-zero fact, proposition 1.2.2; the reciprocity lemma on the auxiliary curve that turns it into the two-function symmetry; and the localization sequence theorem 1.4.1 reducing the boundary strata to lower dimension. The degree-zero fact is why the argument could not be given in section 1.4 before the divisor machinery was in place.

***Proof*:**
By linearity it suffices to treat a single generator of $\operatorname{Rat}_k(X)$, so let $W \subseteq X$ be a $(k+1)$-dimensional subvariety and $r \in k(W)^*$, and set $\alpha = [\operatorname{div}(r)]$. The hypothesis that no component of $\alpha$ lies in $|D|$ is used only to make each $D \cdot [V]$ fall under the first clause of definition 2.2.1; the reduction below produces $D \cdot \alpha$ as the divisor of a rational function, and the containment cases are absorbed at the end.

*Step 1: Reduction to the surface $W$.* Replacing $X$ by $W$ and $D$ by its restriction $D|_W$ changes neither side: the components $V$ of $[\operatorname{div}(r)]$ are codimension-one subvarieties of $W$, the section $s_V$ used to form $D \cdot [V]$ is the restriction to $V$ of the section $s_W$ of $\mathcal{O}_X(D)|_W$, and the inclusion $W \hookrightarrow X$ carries $A_{k-1}(|D|_W|) \to A_{k-1}(|D|)$ compatibly. It therefore suffices to prove the statement with $X = W$, so that $D$ is a Cartier divisor on the $(k+1)$-dimensional variety $W$, $r \in k(W)^*$, and $D \cdot [\operatorname{div}(r)]$ is to be shown rationally equivalent to zero in $A_{k-1}(W)$. Restrict further to the dense open $W^\circ \subseteq W$ where $D$ is principal, given by a single rational function $g \in k(W)^*$ with $\mathcal{O}_W(D)|_{W^\circ} = \mathcal{O}_{W^\circ} \cdot g^{-1}$; the codimension-one subvarieties of $W$ meeting $W^\circ$

are exactly those on which the computation is nontrivial, and those contained in the complement $W \setminus W^\circ$ contribute to both a principal divisor and the divisor of $D$ and are handled by the localization sequence in Step 4. On $W^\circ$ the section $s_V$ of clause (1) is $g|_V$, so

$$D \cdot [\operatorname{div}(r)] = \sum_V \operatorname{ord}_V(r)\, \big[\operatorname{div}_V(g|_V)\big]$$

the outer sum over the codimension-one subvarieties $V \subseteq W$ with $r$ having a zero or pole along $V$, and the inner divisor formed on each such $V$.

*Step 2: The two-function symmetry.* The right-hand side is not visibly symmetric in $r$ and $g$, but its symmetrization is the object that vanishes. Consider the two iterated sums

$$\Phi(r,g) = \sum_V \operatorname{ord}_V(r)\, \big[\operatorname{div}_V(g|_V)\big], \qquad \Phi(g,r) = \sum_V \operatorname{ord}_V(g)\, \big[\operatorname{div}_V(r|_V)\big]$$

both $(k-1)$-cycles on $W$, where in each the outer sum runs over codimension-one subvarieties $V \subseteq W$ and the inner divisor is formed on $V$ of the restricted function. The claim of this step is the tame symmetry

$$\Phi(r,g) - \Phi(g,r) \sim 0 \quad \text{in } A_{k-1}(W)$$

Granting it, the proof is nearly complete: $\Phi(g,r) = \sum_V \operatorname{ord}_V(g)\, [\operatorname{div}_V(r|_V)]$ is a $\mathbb{Z}$-linear combination of divisors of the rational functions $r|_V$ on the $k$-dimensional subvarieties $V$, hence lies in $\operatorname{Rat}_{k-1}(W)$ by definition 1.1.3; combined with the symmetry, $\Phi(r,g) = D \cdot [\operatorname{div}(r)]$ is rationally equivalent to $\Phi(g,r) \sim 0$, which is the assertion.

*Step 3a: Localizing the symmetry at a codimension-two locus.* The symmetry is a statement whose content is concentrated in codimension two on $W$, so fix a codimension-two subvariety $Z \subseteq W$, a $(k-1)$-dimensional variety, and compare the coefficients of $[Z]$ on the two sides. A codimension-one $V \subseteq W$ contributes to the coefficient of $[Z]$ in $\Phi(r,g)$ only when $Z \subseteq V$, that is when $Z$ is a prime divisor of $V$; write the coefficient of $[Z]$ in $\Phi(r,g)$ as

$$c_Z(r,g) = \sum_{Z \subseteq V} \operatorname{ord}_V(r) \operatorname{ord}_{Z,V}(g|_V)$$

the sum over codimension-one $V$ with $Z \subseteq V \subseteq W$, and $\operatorname{ord}_{Z,V}$ the order along $Z$ computed on $V$. The task is to show $c_Z(r,g) = c_Z(g,r)$ modulo the coefficient of $[Z]$ in a principal divisor, and this is a computation on the two-dimensional local geometry of $W$ along $Z$. Passing to the local ring $\mathcal{O}_{W,Z}$, a two-dimensional local domain with fraction field $L = k(W)$ and residue field $K = k(Z)$, the codimension-one $V$ containing $Z$ correspond to the height-one primes $\mathfrak{p} \subseteq \mathcal{O}_{W,Z}$; each carries a discrete valuation $\operatorname{ord}_{\mathfrak{p}} = \operatorname{ord}_V$ on $L$, and the residue field $k(\mathfrak{p})$ of $\mathfrak{p}$ is the function field $k(V)$ of the divisor $V$. In these terms

$$c_Z(r,g) = \sum_{\mathfrak{p}} \operatorname{ord}_{\mathfrak{p}}(r) \operatorname{ord}_Z(g \bmod \mathfrak{p})$$

where $g \bmod \mathfrak{p}$ is the image of $g$ in $k(\mathfrak{p}) = k(V)^*$ and $\operatorname{ord}_Z$ on $k(V)$ is the order along $Z$ regarded as a prime divisor of $V$. The two factors play asymmetric roles: $\operatorname{ord}_{\mathfrak{p}}(r)$ is the transverse order of $r$ along the entire divisor $V$, a single integer, while $\operatorname{ord}_Z(g \bmod \mathfrak{p})$ is the order of the residue $g|_V$ along the codimension-one subvariety $Z \subseteq V$. Swapping $r$ and $g$ interchanges these two roles rather than permuting equal factors, so $c_Z(r,g) = c_Z(g,r)$ is a genuine reciprocity between a transverse divisorial order and a residual order, not the commutativity of a product.

*Step 3b: The exceptional curve and the tame symbol.* The two integers in each term of $c_Z(r,g)$, the transverse order $\operatorname{ord}_{\mathfrak{p}}(r)$ and the residual order $\operatorname{ord}_Z(g \bmod \mathfrak{p})$, are the two ends of a single valuation datum, and completeness enters by resolving the two-dimensional geometry of $\mathcal{O}_{W,Z}$ into a curve. Choose a proper birational morphism $\pi\colon S \to \operatorname{Spec}\mathcal{O}_{W,Z}$ with $S$ a regular two-dimensional scheme (a resolution of the localized surface, by normalization and blow-up), and let $E = \pi^{-1}(\mathfrak{m})$ be its exceptional fibre over the closed point, a complete curve over $K$ whose integral components are $E_1, \dots, E_m$, each a complete curve over $K$ with function field $k(E_i)$. The coefficient $c_Z(r,g)$ is invariant under $\pi$: since $\pi$ is proper and birational, proposition 1.2.2 identifies the coefficient of $[Z]$ downstairs with the coefficient of the exceptional fibre upstairs, and among the prime divisors of $S$ only the exceptional $E_i$ map into $Z$, so the strict transforms of the $V$ drop out and the coefficient is carried by $E_1, \dots, E_m$ alone. Along each exceptional component $E_i$ the function $r$ has a transverse order $a_i = \operatorname{ord}_{E_i}(r)$ and $g$ a transverse order $b_i = \operatorname{ord}_{E_i}(g)$, both single integers, and once these orders are cleared $r$ and $g$ restrict to rational functions on $E_i$; their zeros and poles on $E_i$ are exactly the points where the strict transforms $\widetilde{V}$ meet $E_i$. Attach to $E_i$ the *tame symbol* of $r$ and $g$, the residue element

$$\tau_i(r,g) = (-1)^{a_i b_i} \left(\frac{r^{\,b_i}}{g^{\,a_i}}\right)\Big|_{E_i} \in k(E_i)^*$$

which is a well-defined nonzero rational function on $E_i$ because $r^{b_i}/g^{a_i}$ has order zero along $E_i$. Reading off its order at a closed point $P \in E_i$ gives $\operatorname{ord}_P(\tau_i(r,g)) = b_i \operatorname{ord}_P(r|_{E_i}) - a_i \operatorname{ord}_P(g|_{E_i})$. The contribution of $(i,P)$ to $c_Z(r,g)$ is the transverse order of $r$ against the residual order of $g$, namely $a_i \operatorname{ord}_P(g|_{E_i})$, and its contribution to $c_Z(g,r)$ is $b_i \operatorname{ord}_P(r|_{E_i})$; their difference is thus $-\operatorname{ord}_P(\tau_i(r,g))$. Summing over $S$ and using the $\pi$-invariance above, the two coefficients organize as

$$c_Z(r,g) - c_Z(g,r) = -\sum_{i=1}^{m} \sum_{P \in E_i} [k(P):K] \operatorname{ord}_P\bigl(\tau_i(r,g)\bigr)$$

the double sum over the components $E_i$ of the exceptional curve and their closed points $P$. The pairing is asymmetric: the summand pairs the transverse order of one function along $E_i$ against the residual order of the other at $P$, and swapping $r$ and $g$ replaces $\tau_i(r,g)$ by $\tau_i(r,g)^{-1}$ up to sign, so the difference is the sum of the point orders of the tame symbols, not a rearrangement of a symmetric product $\operatorname{ord}_P(r)\operatorname{ord}_P(g)$.

*Step 3c: Assembling by degree zero.* Each tame symbol $\tau_i(r,g)$ is a rational function on the complete curve $E_i$ over $K$, so its divisor $\operatorname{div}_{E_i}(\tau_i(r,g))$ is a principal divisor on a complete curve. By proposition 1.2.2(1) applied to the proper structure map $E_i \to \operatorname{Spec} K$, its degree over $K$ vanishes: $\deg_K \operatorname{div}_{E_i}(\tau_i(r,g)) = \sum_{P \in E_i} [k(P):K] \operatorname{ord}_P(\tau_i(r,g)) = 0$ for every $i$. Summing over the finitely many components,

$$c_Z(r,g) - c_Z(g,r) = -\sum_{i=1}^{m} \deg_K \operatorname{div}_{E_i}\bigl(\tau_i(r,g)\bigr) = 0$$

so $c_Z(r,g) = c_Z(g,r)$ for every codimension-two $Z$. This is the reciprocity that a naive term-by-term rearrangement would miss: it pairs the transverse order of one function against the residual order of the other, and its vanishing rests on completeness of each exceptional component $E_i$ through the degree-zero property alone, with no external tame-symbol machinery imported. Consequently $\Phi(r,g) = \Phi(g,r)$ coefficient by coefficient on the locus where $D$ is principal, any residual difference being a principal divisor supported off $W^\circ$. This is the symmetry claimed in Step 2, as needed.

*Step 4: The boundary components.* It remains to account for the codimension-one subvarieties $V \subseteq W \setminus W^\circ$, where $D$ was not represented by the single function $g$. Let $U = W^\circ$ and $Y = W \setminus W^\circ$ be its closed complement, of codimension at least one in $W$. The localization sequence (theorem 1.4.1) for $Y \subseteq W$,

$$A_{k-1}(Y) \to A_{k-1}(W) \to A_{k-1}(U) \to 0$$

is right exact, and Steps 1 through 3 established $D \cdot [\mathrm{div}(r)] \sim \Phi(g, r) \sim 0$ after restriction to $U$, that is, the class of $D \cdot [\mathrm{div}(r)]$ dies in $A_{k-1}(U)$. By exactness its class in $A_{k-1}(W)$ comes from $A_{k-1}(Y)$; but $Y$ has dimension at most $k$, and the restriction $D|_Y$ is again a Cartier divisor, so the boundary class is computed by the same recipe on the lower-dimensional $Y$. To pin the inductive object down, decompose $Y = \bigcup_j Y_j$ into irreducible components, each of dimension at most $k$, with inclusions $\iota_j \colon Y_j \hookrightarrow W$. The class in $A_{k-1}(Y)$ produced by the boundary of the localization sequence is represented by $\sum_j (\iota_j)_*\big(D|_{Y_j} \cdot [\mathrm{div}(r_j)]\big)$, where $r_j := r|_{Y_j} \in k(Y_j)^*$ is the restriction of $r$ to the generic point of $Y_j$, a nonzero rational function since the zero and pole components of $r$ were split off in Steps 1 through 3, and $[\mathrm{div}(r_j)]$ is the principal cycle it cuts on the variety $Y_j$ of dimension at most $k$. Each summand is thus $D|_{Y_j}$ intersected with a principal cycle on $Y_j$, the exact object to which the proposition applies one dimension lower. A Noetherian induction on $\dim W$ closes the argument: the base case $\dim W = 0$ is vacuous, and for the inductive step each summand $D|_{Y_j} \cdot [\mathrm{div}(r_j)]$ is rationally trivial on $Y_j$ by the inductive hypothesis applied to the Cartier divisor $D|_{Y_j}$ and the principal cycle $[\mathrm{div}(r_j)]$ on the lower-dimensional $Y_j$; pushing forward along $\iota_j$ preserves rational equivalence, so the whole boundary class in $A_{k-1}(Y)$ vanishes. Hence $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(W)$, and pushing forward along $W \hookrightarrow X$ gives $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(|D|)$.

*Step 5: Well-definedness of $D \cdot a$.* Fix a class $a \in A_k(X)$. Every rational-equivalence class has a representative cycle none of whose finitely many components lies in $|D|$, since a component inside $|D|$ can be replaced by a rationally equivalent cycle whose components meet $|D|$ properly; on such a representative every $D \cdot [V]$ falls under the first clause of definition 2.2.1. Let $\alpha \sim \alpha'$ be two such representatives, so their difference lies in $\mathrm{Rat}_k(X)$, a finite $\mathbb{Z}$-combination of generators $[\mathrm{div}(r)]$ with $r \in k(W)^*$ on $(k+1)$-dimensional $W$. Steps 1 through 4 give $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(|D|)$ for each generator, so summing gives $D \cdot (\alpha - \alpha') \sim 0$, and $D \cdot a := [D \cdot \alpha]$ is independent of the chosen representative. Additivity in $a$ is immediate from additivity of $D \cdot -$ on cycles, so $D \cdot - \colon A_k(X) \to A_{k-1}(|D|)$ is a homomorphism, completing the proof.

The proposition is the license to speak of $D \cdot a$ for a class $a$, and with it the intersection with a divisor becomes an operator on the Chow group. Read backward, the argument explains why the well-definedness was out of reach in section 1.4: the entire content is the tame symmetry of Step 2, and that symmetry rests on the degree-zero property of a principal divisor on a complete curve, a fact about the divisor calculus that section did not yet have. The specialization map used there to invert affine-bundle pullback was built from exactly the operation $\mathrm{div}(t) \cdot -$ for the coordinate $t$, and its descent to Chow groups is this proposition applied to $D = \{t = 0\}$; the forward reference in that proof pointed here.

Two features of the argument recur and are worth isolating. The reduction of Step 1, replacing $X$ by the $(k+1)$-dimensional $W$ on which $r$ lives, is the standard move that turns a statement about intersecting a principal cycle into a statement about a single surface with two functions on it, and it will reappear when commutativity is proved in section 2.3. The symmetry of Step 3, that pairing the zeros of $r$ against $g$ equals pairing the zeros of $g$ against $r$, is the algebraic form of Weil reciprocity, here proved from the ground up rather than cited: the tame symbol is constructed by hand along each component of the exceptional curve, and its reciprocity is reduced to the vanishing degree of a principal divisor on that complete curve, so no external tame-symbol machinery is imported. The same reciprocity, promoted one dimension higher, is what makes $D \cdot D' = D' \cdot D$, and the two proofs share their engine.

The operator now available, $D \cdot -\colon A_k(X) \to A_{k-1}(|D|)$, invites the computation that motivated the whole construction. Intersecting repeatedly with a hyperplane on projective space should descend through the dimensions and terminate in a point, recovering the degree; the next example carries this out and previews the Chern-class computation of section 2.4.

**Example 2.2.5: Iterated Hyperplane Sections on Projective Space**

Let $X = \mathbb{P}^n$ and $H = \{x_0 = 0\}$ a hyperplane, so $\mathcal{O}_{\mathbb{P}^n}(H) = \mathcal{O}_{\mathbb{P}^n}(1)$ and $|H| \cong \mathbb{P}^{n-1}$. By theorem 1.5.2, $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ with $h_k$ the class of a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$. To compute $H \cdot h_k$, represent $h_k$ by a linear $\mathbb{P}^k$ meeting $H$ transversally, that is by an $L \cong \mathbb{P}^k$ not contained in $H$; then $L \cap H$ is a linear $\mathbb{P}^{k-1}$, and the restricted section $x_0/x_j$ on $L$ has its zero divisor equal to $L \cap H$ with multiplicity one. Hence

$$H \cdot h_k = [L \cap H] = h_{k-1} \in A_{k-1}(\mathbb{P}^{n-1})$$

pushed into $\mathbb{P}^n$, the class of a linear $\mathbb{P}^{k-1}$. Intersecting $n$ times starting from the fundamental class $h_n = [\mathbb{P}^n]$,

$$\underbrace{H \cdot H \cdots H}_{k} \cdot [\mathbb{P}^n] = h_{n-k}$$

and in particular $H^n \cdot [\mathbb{P}^n] = h_0 = [\mathrm{pt}]$, a single reduced point of degree one. The iterated intersection walks down the tower of linear subspaces one step at a time, and its terminal value is the class of a point, whose degree 1 is the degree of $\mathbb{P}^n$ under the embedding $\mathcal{O}(1)$. The choice of a transversal representative $L$ at each stage is legitimated precisely by proposition 2.2.4: the answer $h_{k-1}$ does not depend on which $\mathbb{P}^k$ in the class $h_k$ is used, so long as it meets $H$ properly.

The computation is the prototype of every enumerative count the theory produces. Reinterpreted through the first Chern class in section 2.4, it reads $c_1(\mathcal{O}(1))^k \cap [\mathbb{P}^n] = h_{n-k}$, and its terminal instance $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathrm{pt}]$ is the statement that the top self- intersection of the hyperplane class on $\mathbb{P}^n$ is one, the algebraic form of the assertion that $n$ general hyperplanes meet in a single point. That the intermediate steps are independent of the representatives chosen is the payoff of this section, and it is what turns a naive count of intersection points into an invariant.

## Exercise 2.2.1

1. On $X = \mathbb{P}^2$ let $D = \{x_0x_1 = 0\}$ be the union of two lines, a Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(D) = \mathcal{O}_{\mathbb{P}^2}(2)$, and let $V$ be the conic $\{x_2^2 = x_0x_1\}$. Compute $D \cdot [V] \in A_0(V \cap |D|)$ explicitly as a sum of points with multiplicities, and verify that its degree equals 4. Explain how the multiplicities distribute among the intersection points.

2. Let $V \subseteq \mathbb{P}^2$ be a smooth curve of degree $d$ and $H$ a line meeting $V$ transversally in $d$ distinct points. Using definition 2.2.1, show that $H \cdot [V]$ is the sum of those $d$ points each with multiplicity one, and deduce $\deg(H \cdot [V]) = d$. Then explain why the value $\deg(H \cdot [V]) = d$ holds for every line $H$, transversal or not, without recomputing, citing proposition 2.2.4.

3. Let $D = \mathrm{div}(f)$ be a principal Cartier divisor on a variety $X$, so $\mathcal{O}_X(D) \cong \mathcal{O}_X$ is trivial. Show that $D \cdot a = 0$ in $A_{k-1}(X)$ for every class $a \in A_k(X)$. (Push $D \cdot a$ forward from $|D|$ into $X$;

the point is that a global trivialization of $\mathcal{O}_X(D)$ makes the restricted section extend to all of $V$.)

4. On $\mathbb{P}^3$ let $H$ be a hyperplane and let $V$ be a smooth cubic surface, regarded through its inclusion as a subvariety, with the induced very ample bundle $\mathcal{O}_V(1) = \mathcal{O}_{\mathbb{P}^3}(1)|_V$. Compute $H \cdot [V]$ as a class in $A_1(V \cap H)$, identify $V \cap H$ geometrically, and compute the degree of the further intersection $H \cdot (H \cdot [V])$.

5. Let $Q = \mathbb{P}^1 \times \mathbb{P}^1$ and let $D = f_1 = \{a\} \times \mathbb{P}^1$ be a ruling of the first kind, with $\mathcal{O}_Q(D) = \mathrm{pr}_1^*\mathcal{O}_{\mathbb{P}^1}(1)$. Compute $D \cdot [f_1]$ and $D \cdot [f_2]$, where $f_2 = \mathbb{P}^1 \times \{b\}$ is a ruling of the second kind, and interpret the results as the self-intersection $f_1^2$ and the cross-intersection $f_1 \cdot f_2$. Which case uses the second clause of definition 2.2.1?

6. Give an explicit example on $X = \mathbb{P}^1$ of a Cartier divisor $D$ and two rationally equivalent zero-cycles $\alpha \sim \alpha'$ for which the naive formula would assign different answers if one ignored the requirement that components avoid $|D|$, and explain how definition 2.2.1 and proposition 2.2.4 together resolve the apparent discrepancy. (Here $k = 0$, so $D \cdot \alpha$ lands in $A_{-1} = 0$; use this to frame what the proposition is and is not asserting in the lowest dimension.)

## 2.3 Commutativity and the Projection Formula

Intersection with a divisor, made precise in section 2.2, is an operation $D \cdot -$ that lowers dimension by one and, by proposition 2.2.4, descends to rational equivalence. Nothing so far says these operations may be applied in either order. Given two Cartier divisors $D$ and $D'$ on $X$ and a $k$-cycle $\alpha$, one can form $D \cdot (D' \cdot \alpha)$ and $D' \cdot (D \cdot \alpha)$, each a class in $A_{k-2}(X)$; that they agree is the single algebraic fact on which the entire intersection calculus rests. Its proof does not follow from the definitions by formal manipulation, because the two orders of intersection produce different cycles before passing to rational equivalence. The content is that the difference is always a boundary, and the reason it is a boundary is the vanishing of the degree of a principal divisor on a complete curve, established back in proposition 1.2.2.

This is also the point at which a debt from chapter 1 comes due. The proof of affine-bundle invariance (theorem 1.4.6) deferred one identity, the divisor symmetry (1.3), on the grounds that the tools were not yet in place; that identity is exactly the commutativity proved here. This section discharges it in full and then draws the two functorial consequences, compatibility of $D \cdot -$ with proper pushforward and with flat pullback, that make the operation part of the same covariant/contravariant package as pushforward and pullback themselves.

The obstruction to commutativity lives on a surface, and the mechanism that removes it is a symmetry between two rational functions. Before the theorem, the mechanism deserves to be isolated as a lemma. Let $S$ be a complete variety of dimension two and let $f, g \in k(S)^*$ be nonzero rational functions. Each has a divisor $[\mathrm{div}(f)] = \sum_W \mathrm{ord}_W(f)\,[W]$, the sum over the codimension-one subvarieties (the curves) $W \subseteq S$, and on each such curve $W$ the other function $g$ restricts, wherever it is neither zero nor a pole identically along $W$, to a rational function $g|_W$ with its own divisor of points. The two-function symmetry lemma compares the two zero-cycles obtained by cutting first with $f$ and then with $g$, against the same two cuts in the opposite order.

The statement needs the order of one function measured against the divisor of another. For a curve $W \subseteq S$ with $\mathrm{ord}_W(g) = 0$, so that $g$ is a unit at the generic point of $W$ and restricts to a nonzero rational function $g|_W \in k(W)^*$, write $[\mathrm{div}_W(g)] = \sum_{P\in W} \mathrm{ord}_P(g|_W)\,[P]$ for the divisor of that restriction, a zero-cycle on $W$ and hence on $S$. The lemma is a reciprocity between the iterated cuts $\sum_W \mathrm{ord}_W(f)\,[\mathrm{div}_W(g)]$ and $\sum_W \mathrm{ord}_W(g)\,[\mathrm{div}_W(f)]$.

**Lemma 2.3.1: Two-Function Reciprocity on a Surface**

Let $S$ be a complete variety of dimension two and let $f, g \in k(S)^*$. Then

$$\sum_W \mathrm{ord}_W(f)\,[\mathrm{div}_W(g)] \;\sim\; \sum_W \mathrm{ord}_W(g)\,[\mathrm{div}_W(f)] \quad \text{in } A_0(S)$$

where each sum runs over the curves $W \subseteq S$ on which the relevant restriction is defined, and the divisors $[\mathrm{div}_W(g)]$, $[\mathrm{div}_W(f)]$ are the zero-cycles of the restricted functions. In fact both sides have the same degree, and their difference is a sum of principal divisors of rational functions on the curves $W \subseteq S$, hence rationally trivial.

The lemma is the two-dimensional heart of commutativity, and its proof is the one place where the degree-zero property of principal divisors on complete curves does the decisive work. The strategy is to reduce the equality of two zero-cycles to an equality of degrees along each curve, and then to recognize each such degree as the value at a point of the *tame symbol*, a bimultiplicative pairing on pairs of rational functions whose reciprocity across a complete curve is precisely the statement that a principal divisor has degree zero.

***Proof*:**
*Step 1: the tame symbol at a curve.* Fix a curve $W \subseteq S$ and let $A = \mathcal{O}_{S,W}$ be the local ring of $S$ at the generic point of $W$. Because $S$ is a variety and $W$ has codimension one, $A$ is a one-dimensional local domain with fraction field $k(S)$ and residue field $k(W)$; its normalization is a discrete valuation ring, and after replacing $S$ by its normalization along $W$ (which changes neither side of the asserted identity, both being computed by orders and residues that are birational invariants) one may assume $A$ is itself a discrete valuation ring with valuation $v_W = \mathrm{ord}_W$ and uniformizer $\pi$. Define the *tame symbol* of $f, g$ at $W$ to be the residue class

$$\partial_W(f,g) \;=\; (-1)^{v_W(f)\,v_W(g)}\,\frac{f^{\,v_W(g)}}{g^{\,v_W(f)}}\bigg|_W \;\in\; k(W)^*$$

the restriction to $W$ of the indicated unit, which lies in $A$ and is a unit there because its valuation

$$v_W\Big((-1)^{v_W(f)v_W(g)}\,f^{v_W(g)}g^{-v_W(f)}\Big) = v_W(g)\,v_W(f) - v_W(f)\,v_W(g) = 0$$

vanishes. The construction is bimultiplicative in $f$ and $g$ and antisymmetric, $\partial_W(g,f) = \partial_W(f,g)^{-1}$, both directly from the formula. The point of the sign is that this antisymmetry holds on the nose, not just up to squares.

*Step 2: the tame symbol computes the difference of iterated divisors, pointwise.* Fix a closed point $P \in S$ and a curve $W$ through $P$. When $\mathrm{ord}_W(g) = 0$ the restriction $g|_W$ is a nonzero function on $W$ and the coefficient of $[P]$ in $\mathrm{ord}_W(f)\,[\mathrm{div}_W(g)]$ is $\mathrm{ord}_W(f)\,\mathrm{ord}_P(g|_W)$; symmetrically for the other cut. Comparing the two iterated cuts at $P$ amounts to comparing, over all curves $W \ni P$,

$$\sum_{W\ni P} \mathrm{ord}_W(f)\,\mathrm{ord}_P(g|_W) \quad \text{against} \quad \sum_{W\ni P} \mathrm{ord}_W(g)\,\mathrm{ord}_P(f|_W)$$

The tame symbol reconciles them, but the reconciliation is delicate at the curves where both functions have nonzero order, and getting it right is the whole content of the step. For a single curve $W$ with $\operatorname{ord}_W(f) = a$ and $\operatorname{ord}_W(g) = b$, write $f = \pi^a u$ and $g = \pi^b w$ with $u, w$ units at $W$. The tame symbol of *Step 1* then simplifies to

$$\partial_W(f, g) = (-1)^{ab}\, \bar{u}^{\,b}\, \bar{w}^{\,-a} \;\in\; k(W)^*$$

the powers of $\pi$ cancelling, so its order at $P$ is always

$$\operatorname{ord}_P\big(\partial_W(f, g)\big) = b \operatorname{ord}_P(\bar{u}) - a \operatorname{ord}_P(\bar{w}) \tag{2.1}$$

regardless of whether $a$ or $b$ vanishes. The iterated cuts, on the other hand, see only the restricted functions, and $f|_W = \bar{u}$ is defined precisely when $a = 0$ while $g|_W = \bar{w}$ is defined precisely when $b = 0$. Compare the two curve-by-curve.

*If $a = 0$ and $b \neq 0$:* then $\operatorname{ord}_W(g)\,[\operatorname{div}_W(f)]$ contributes $b \operatorname{ord}_P(\bar{u})$ at $P$ while $\operatorname{ord}_W(f)\,[\operatorname{div}_W(g)] = 0$, so the iterated-cut difference is $b \operatorname{ord}_P(\bar{u})$, which is exactly (2.1) since $a = 0$ kills the second term. *If $a \neq 0$ and $b = 0$:* symmetrically the difference is $-a \operatorname{ord}_P(\bar{w})$, again (2.1). *If $a = b = 0$:* both restrictions are units, both sides vanish. In all three of these *non-mixed* cases,

$$\Big(\operatorname{ord}_W(g) \operatorname{ord}_P(f|_W) - \operatorname{ord}_W(f) \operatorname{ord}_P(g|_W)\Big) = \operatorname{ord}_P\big(\partial_W(f, g)\big)$$

the left side being the contribution of $W$ to the iterated-cut difference at $P$, where a term whose restricted function is undefined is read as zero because its coefficient $\operatorname{ord}_W(g)$ or $\operatorname{ord}_W(f)$ then vanishes. The remaining case is the *mixed* one, $a \neq 0$ and $b \neq 0$: here *neither* $f|_W$ nor $g|_W$ is defined, so $W$ contributes nothing to either iterated cut, yet $\partial_W(f, g) = (-1)^{ab}\bar{u}^{\,b}\bar{w}^{\,-a}$ is a genuine nonzero rational function on $W$ whose divisor is generally nonzero. Thus on a mixed curve the iterated-cut difference and $\operatorname{ord}_P(\partial_W)$ do *not* agree; the tame symbol overcounts by exactly the mixed contribution. Summing the non-mixed agreement over the curves through $P$,

$$\Big(\sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)]\Big)_P = \sum_{\substack{W \ni P \\ \text{non-mixed}}} \operatorname{ord}_P\big(\partial_W(f, g)\big) \tag{2.2}$$

the sum on the right running only over curves $W \ni P$ on which at least one of $f, g$ has order zero, and the left side denoting the coefficient of $[P]$ in the difference of the two iterated cuts. The mixed curves are deliberately excluded from the right side; they carry no iterated-cut contribution, and Step 3 will show their tame symbols are individually harmless.

*Step 3: reciprocity along each curve kills the total.* Fix a curve $W \subseteq S$. The tame symbol $\partial_W(f, g)$ is a nonzero rational function on the complete curve $W$ (it is complete because $S$ is complete and $W$ is closed in $S$). Its divisor $[\operatorname{div}(\partial_W(f, g))] = \sum_P \operatorname{ord}_P(\partial_W(f, g))\,[P]$ is a principal divisor on $W$, so by proposition 1.2.2(1) applied to the structure map $W \to \operatorname{Spec} \overline{k}$, which is proper surjective with target of strictly smaller dimension, its pushforward vanishes: $\deg[\operatorname{div}(\partial_W(f, g))] = 0$. This holds for *every* curve $W$, mixed or not. Summing (2.2) over all closed points $P$ and reorganizing the double sum by curves,

$$\deg\Big(\sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)]\Big) = \sum_{\text{non-mixed } W} \deg[\operatorname{div}(\partial_W(f, g))] = 0$$

each summand vanishing individually. Thus the difference of the two iterated zero-cycles has degree zero on $S$.

*Step 4: from degree zero to rational equivalence.* A zero-cycle of degree zero on the complete variety $S$ need not be rationally equivalent to zero in general, but the difference here is more than a degree-zero cycle: (2.2), summed over all closed points $P$ and reorganized by curves, exhibits it as a sum of principal divisors,

$$\sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)] = \sum_{\text{non-mixed } W} [\operatorname{div}(\partial_W(f,g))]$$

the right side running over the curves on which one of $f, g$ has order zero, each term the divisor of the rational function $\partial_W(f,g)$ on the complete curve $W \subseteq S$. Each $[\operatorname{div}(\partial_W(f,g))]$ is by definition rationally equivalent to zero on $W$, hence on $S$ under the proper pushforward along $W \hookrightarrow S$, which preserves rational equivalence by theorem 1.2.3. Therefore the difference is a finite sum of cycles each $\sim 0$, so it is itself $\sim 0$ in $A_0(S)$, which is the asserted reciprocity, as we sought to show.

The lemma is the algebraic incarnation of a classical reciprocity, but the mechanism it uses is elementary: on each curve $W$ the tame symbol $\partial_W(f,g)$ is a single rational function, its divisor on the complete curve $W$ is principal and so rationally trivial, and the difference of the two iterated cuts is a finite sum of such divisors. No appeal to Weil reciprocity or to the point-level tame symbol $\prod_P \partial_P(f,g) = 1$ as an external input is needed, and indeed none is available in this book; the entire force of the lemma comes from the single fact that a rational function on a complete curve has as many zeros as poles, counted with multiplicity and residue degree, applied one curve at a time. That fact is proposition 1.2.2(1), proved in chapter 1 by the model computation on $\mathbb{P}^1$. The reader should notice that the surface $S$ never needed to be smooth; normalizing along each curve reduced every local computation to a discrete valuation ring, which is all the tame symbol requires.

With the reciprocity lemma in hand, commutativity is a matter of reducing the general assertion to the surface case the lemma governs. The reduction is entirely formal, following the pattern used throughout chapter 1: the operation $D \cdot -$ is a homomorphism, so it suffices to check the identity on the class of a single subvariety, and the intersection with $D$ and $D'$ is supported where the two divisors meet the subvariety, so one may replace the ambient scheme by the subvariety itself.

**Theorem 2.3.2: Commutativity of Divisor Intersection**

Let $D$ and $D'$ be Cartier divisors on an algebraic scheme $X$, and let $\alpha \in A_k(X)$. Then

$$D \cdot (D' \cdot \alpha) = D' \cdot (D \cdot \alpha) \quad \text{in } A_{k-2}(X)$$

More precisely, both sides are represented by well-defined classes in $A_{k-2}\big(|D| \cap |D'| \cap \operatorname{supp}\alpha\big)$, and they coincide there. This identity is the divisor symmetry (1.3) deferred in the proof of theorem 1.4.6.

***Proof*:**
*Step 1: reduction to a variety.* Both $D \cdot -$ and $D' \cdot -$ are homomorphisms $A_k \to A_{k-1}$ by definition 2.2.1 and proposition 2.2.4, so both iterated operations are homomorphisms $A_k(X) \to A_{k-2}(X)$. It therefore suffices to verify the identity on a generator $\alpha = [V]$, the class of a $k$-dimensional subvariety $V \subseteq X$. Restricting the line bundles $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ and their rational sections to $V$, the entire computation of $D \cdot (D' \cdot [V])$ and $D' \cdot (D \cdot [V])$ takes place

on $V$ using the restricted divisors $D|_V$ and $D'|_V$. Replacing $X$ by $V$, one is reduced to proving

$$D \cdot (D' \cdot [X]) = D' \cdot (D \cdot [X])$$

for $X$ itself a $k$-dimensional variety and $D, D'$ Cartier divisors on it, with the common value a class in $A_{k-2}(|D| \cap |D'|)$.

*Step 2: choosing sections and unwinding the two orders.* Let $s$ and $s'$ be the rational sections of $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ cutting out $D$ and $D'$, so that on the variety $X$,

$$D \cdot [X] = [\mathrm{div}(s)] = \sum_W \mathrm{ord}_W(s)\,[W], \qquad D' \cdot [X] = [\mathrm{div}(s')] = \sum_{W'} \mathrm{ord}_{W'}(s')\,[W']$$

the sums over the codimension-one subvarieties $W, W' \subseteq X$. To compute the double intersection one then intersects each $[W']$ with $D$ and each $[W]$ with $D'$. On a codimension-one subvariety $W$ not contained in $|D'|$, the section $s'$ restricts to a rational section $s'|_W$ of $\mathcal{O}_X(D')|_W$, whose divisor is $[\mathrm{div}_W(s')]$, and by definition 2.2.1,

$$D' \cdot (D \cdot [X]) = \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')], \qquad D \cdot (D' \cdot [X]) = \sum_{W'} \mathrm{ord}_{W'}(s')\,[\mathrm{div}_{W'}(s)]$$

as cycles supported on $|D| \cap |D'|$. The two expressions are the two iterated cuts of the reciprocity lemma, with the roles of the two divisors interchanged.

*Step 3: the difference as a sum of tame symbols.* The difference

$$\Delta \;=\; \sum_{W'} \mathrm{ord}_{W'}(s')\,[\mathrm{div}_{W'}(s)] \;-\; \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')]$$

is a $(k-2)$-cycle supported on $|D| \cap |D'|$, and the claim is $\Delta \sim 0$ in $A_{k-2}(|D| \cap |D'|)$. The strategy is to write $\Delta$ as a $\mathbb{Z}$-combination of divisors of rational functions on $(k-1)$-dimensional subvarieties of $X$, using the tame symbol of lemma 2.3.1. For a $(k-1)$-dimensional subvariety $W \subseteq X$ on which one of the two sections has nonzero order, choose a common local trivialization of $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ near the generic point of $W$ (a line bundle is locally trivial, and one open set trivializes both), turning $s, s'$ into rational functions in the local ring $\mathcal{O}_{X,W}$, a two-dimensional local domain. The tame symbol

$$\partial_W(s, s') = (-1)^{\mathrm{ord}_W(s)\mathrm{ord}_W(s')}\,\frac{s^{\mathrm{ord}_W(s')}}{s'^{\,\mathrm{ord}_W(s)}}\bigg|_W \;\in\; k(W)^*$$

is then a nonzero rational function on the $(k-1)$-dimensional variety $W$, defined exactly as in the lemma. Localizing $X$ along $W$ removes the trivialization ambiguity: two trivializations differ by a unit, which does not change $\partial_W(s, s')$ up to a rational function that is regular and nonvanishing at the generic point of $W$, so its divisor on $W$ is well defined.

*Step 4: applying reciprocity, one codimension-two point at a time.* Fix a $(k-2)$-dimensional subvariety $Z \subseteq |D| \cap |D'|$ and compute the coefficient of $[Z]$ on both sides of the asserted identity $\Delta = \sum_{\text{non-mixed } W} [\mathrm{div}(\partial_W(s, s'))]$, the sum running over $(k-1)$-dimensional $W$ on which one of $s, s'$ has order zero. That coefficient is determined at the generic point $\zeta$ of $Z$, where $\mathcal{O}_{X,\zeta}$ has dimension two; the two-dimensional local situation there is governed by lemma 2.3.1, whose (2.2) reads, at $\zeta$,

$$\Big(\sum_W \mathrm{ord}_W(s')\,[\mathrm{div}_W(s)] - \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')]\Big)_\zeta \;=\; \sum_{\substack{W \ni \zeta \\ \text{non-mixed}}} \mathrm{ord}_\zeta\big(\partial_W(s, s')\big)$$

the right sum over the $(k-1)$-dimensional subvarieties $W$ through $\zeta$ on which one of $s, s'$ has order zero, exactly as in the lemma; the mixed $W$, on which both sections have nonzero order, are excluded because they contribute to neither iterated cut. This is the coefficient of $[Z]$ in $\Delta$ set equal to the coefficient of $[Z]$ in $\sum_{\text{non-mixed } W}[\operatorname{div}(\partial_W(s,s'))]$. Since the two sides agree at every such $Z$, they agree as cycles:

$$\Delta = \sum_{\text{non-mixed } W} [\operatorname{div}(\partial_W(s,s'))]$$

Each term $[\operatorname{div}(\partial_W(s,s'))]$ is the divisor of a rational function on the $(k-1)$-dimensional subvariety $W$, hence rationally equivalent to zero on $W$, and its proper pushforward to $|D| \cap |D'|$ along $W \hookrightarrow |D| \cap |D'|$ preserves that equivalence by theorem 1.2.3. As a finite sum of cycles each $\sim 0$, therefore $\Delta \sim 0$, so

$$D \cdot (D' \cdot [X]) = D' \cdot (D \cdot [X]) \quad \text{in } A_{k-2}(|D| \cap |D'|)$$

Pushing forward to $X$ along the inclusion, the identity holds in $A_{k-2}(X)$ for $\alpha = [X]$, and by Step 1 for every $\alpha \in A_k(X)$. The surface case $X = S$, in which the difference of the two orders is exhibited as a sum of principal cycles, is precisely the deferred divisor symmetry (1.3), with the two rational functions $s|_S$ and $s'|_S$ here playing the roles of $g$ and the coordinate $t$ there, completing the proof.

Commutativity is the property that makes the phrase "intersection number" unambiguous. Without it, the number attached to a chain of divisors would depend on the order in which they are imposed, and no well-defined product on cycles could exist. With it, one may write $D_1 \cdot D_2 \cdots D_r \cdot \alpha$ for an iterated intersection and speak of it without specifying an order, and the class of a complete intersection $D_1 \cap \cdots \cap D_r$ in $A_*$ is symmetric in the defining divisors. The proof also settles the debt from chapter 1: the identity (1.3) that theorem 1.4.6 took on faith is the surface case here, so affine-bundle invariance is now unconditional. The tame symbol was the only tool required beyond the definitions, and its reciprocity was nothing more than the degree-zero property of principal divisors, propagated from curves to surfaces.

A first reading of the theorem should note what it does *not* require. The divisors need not be effective, the sections need not be regular, and $X$ need not be smooth or even normal; the entire argument localizes to one-dimensional local rings, where a divisor is controlled by a single valuation. This is why intersection with a Cartier divisor is available on every algebraic scheme, in contrast to the intersection product of arbitrary cycles, which will require the deformation-to-the-normal-cone machinery of a later chapter and works cleanly only on smooth varieties.

The second half of the section records how $D \cdot -$ interacts with the two functorialities of the Chow group. Both statements are what one would guess from the topological analogy, in which cap product with a cohomology class is natural for maps: pushforward absorbs the pullback of a divisor, and flat pullback commutes with intersection outright. The proofs run through the same reduction to a single subvariety and the same input, the pushforward of a principal divisor, that powered commutativity.

For a proper morphism $f\colon X \to Y$ and a Cartier divisor $D$ on $Y$, the pullback $f^*D$ is defined whenever no component of $X$ maps into $|D|$, by pulling back the line bundle $\mathcal{O}_Y(D)$ and its rational section; it is the Cartier divisor $\operatorname{div}(f^\# s)$ of the pulled-back section. The projection formula compares intersecting upstairs and pushing down against pushing down and intersecting downstairs.

**Proposition 2.3.3: Projection Formula for Divisors**

Let $D$ be a Cartier divisor on $Y$.

1. *Proper pushforward.* If $f\colon X \to Y$ is proper and $\alpha \in A_k(X)$ is a cycle no component of whose support maps into $|D|$, so that $f^*D \cdot \alpha$ is defined, then
$$f_*\big(f^*D \cdot \alpha\big) = D \cdot f_*\alpha \quad \text{in } A_{k-1}(Y)$$
and, more precisely, both sides are classes in $A_{k-1}\big(|D| \cap f(\mathrm{supp}\alpha)\big)$ and agree there, $f$ being proper so that it carries $f^{-1}(|D|) \cap \mathrm{supp}\alpha$ onto $|D| \cap f(\mathrm{supp}\alpha)$.

2. *Flat pullback.* If $g\colon X' \to Y$ is flat of relative dimension $n$ and $\beta \in A_k(Y)$, then
$$g^*\big(D \cdot \beta\big) = g^*D \cdot g^*\beta \quad \text{in } A_{k+n-1}(X')$$
where $g^*D$ is the pullback Cartier divisor $\mathrm{div}(g^{\#}s)$; more precisely both sides are classes in $A_{k+n-1}\big(g^{-1}(|D| \cap \mathrm{supp}\beta)\big)$ and agree there.

***Proof*:**
Both statements are linear in $\alpha$ and $\beta$, so it suffices to verify each on the class of a subvariety, and both reduce, by the argument that opened the commutativity proof, to a computation with the divisor of a rational function that is settled by proposition 1.2.2 and proposition 1.3.6. The refined statements come for free: definition 2.2.1 records $D \cdot [V]$ on $V \cap |D|$ to begin with, and each identity below is an equality of cycles supported there, so no step passes to the ambient group. This refinement is what chapter 4 consumes.

1. *Proper pushforward.* Take $\alpha = [V]$ for a $k$-dimensional subvariety $V \subseteq X$, and set $W = f(V) \subseteq Y$, so $f_*[V] = \deg(V/W)\,[W]$ if $\dim W = k$ and $f_*[V] = 0$ if $\dim W < k$, by definition 1.2.1. Restricting to $V$ and $W$, one may assume $X = V$, $Y = W$, and $f\colon V \to W$ proper surjective with $V$ a variety. Let $s$ be the rational section of $\mathcal{O}_Y(D)$ cutting out $D$; its pullback $f^{\#}s$ is a rational section of $f^*\mathcal{O}_Y(D)$ cutting out $f^*D$, and $f^*D \cdot [V] = [\mathrm{div}(f^{\#}s)]$ while $D \cdot [W] = [\mathrm{div}(s)]$, both by definition 2.2.1. There are two cases.

   If $\dim W < k$, then $f_*[V] = 0$, so the right side $D \cdot f_*[V]$ vanishes. On the left, $[\mathrm{div}(f^{\#}s)]$ is a $(k-1)$-cycle on $V$, and $f$ pushes it into $Y$; but $f(V) = W$ has dimension $< k$, so $f_*[\mathrm{div}(f^{\#}s)]$ is a cycle of dimension $k-1$ supported on the $(< k)$-dimensional $W$, and proposition 1.2.2(1) applied to $f\colon V \to W$ (proper surjective, target of strictly smaller dimension) gives $f_*[\mathrm{div}(f^{\#}s)] = 0$. Both sides vanish.

   If $\dim W = k$, then $f\colon V \to W$ is proper surjective of equal dimension, generically finite of degree $d = \deg(V/W) = [k(V) : k(W)]$. Now $f^{\#}s$ is a rational section whose local expression, in a trivialization of $\mathcal{O}_Y(D)$ near a curve of $W$, is a rational function $r \in k(V)^*$ pulled back from $k(W)$, and proposition 1.2.2(2) gives $f_*[\mathrm{div}(f^{\#}s)] = [\mathrm{div}(N(f^{\#}s))]$ with $N$ the norm along $k(V)/k(W)$. Because $f^{\#}s$ is pulled back from $s$, its norm is $N(f^{\#}s) = s^d$ up to the local trivialization, so $[\mathrm{div}(N(f^{\#}s))] = d\,[\mathrm{div}(s)] = d\,(D \cdot [W])$. Hence
$$f_*\big(f^*D \cdot [V]\big) = f_*[\mathrm{div}(f^{\#}s)] = d\,(D \cdot [W]) = D \cdot (d\,[W]) = D \cdot f_*[V]$$

as required.

2. *Flat pullback.* Take $\beta = [V]$ for a $k$-dimensional subvariety $V \subseteq Y$. Both sides are supported over $V \cap |D|$, so restricting to $V$ one may assume $Y = V$ is a variety and $D$ a Cartier divisor on it with cutting section $s$, so $D \cdot [V] = [\mathrm{div}(s)]$. Then $g^*(D \cdot [V]) = g^*[\mathrm{div}(s)]$, and by the flat-pullback divisor formula established in proposition 1.3.7, flat pullback commutes with taking the divisor of a rational function: $g^*[\mathrm{div}(s)] = [\mathrm{div}(g^\# s)]$ componentwise, with the multiplicities of the components of $g^{-1}(V)$ accounted by flatness. On the other side, $g^*D$ is the Cartier divisor $\mathrm{div}(g^\# s)$ on $X'$ by definition, and $g^*[V] = [g^{-1}(V)] = \sum_j m_j[X'_j]$ over the components $X'_j$ of the preimage with their flat multiplicities $m_j$ by proposition 1.3.5. Intersecting,

$$g^*D \cdot g^*[V] = g^*D \cdot \sum_j m_j[X'_j] = \sum_j m_j\,[\mathrm{div}((g^\# s)|_{X'_j})] = [\mathrm{div}(g^\# s)] = g^*(D \cdot [V])$$

the middle equality being definition 2.2.1 applied on each component and the last the same componentwise divisor formula. This is the claim; the compatibility of these classes with the base change is proposition 1.3.6, which guarantees the two routes land in the same class of $A_{k+n-1}(X')$. This establishes both parts, completing the proof.

The projection formula is the mechanism by which a computation on a resolved or covering space transfers to the base. In its pushforward form, intersecting the pullback of $D$ upstairs and pushing the answer down is the same as pushing the cycle down and intersecting with $D$ there, which is exactly what one does when computing an intersection number on $Y$ by passing to a modification $X \to Y$ where the geometry is transparent. The scalar $d = \deg(V/W)$ in the equal-dimensional case is the reason degrees multiply under finite covers, and it is the algebraic source of formulas like $\deg(f^*D) = (\deg f)(\deg D)$ for a finite map of curves. In its flat form, the formula says intersection with a divisor is a local construction, insensitive to flat base change: pulling back the divisor and pulling back the cycle and then intersecting reproduces the pullback of the intersection. Together the two halves make $D \cdot -$ compatible with the covariant/contravariant package $(f_*, g^*)$ of chapter 1, the bimodule structure that all of intersection theory is built to respect.

The abstractions above are best seen at work on the projective plane, where every class is a known multiple of a linear space and the intersection numbers can be read off by hand. The following example verifies commutativity concretely and recovers the intersection number of two lines.

**Example 2.3.4: Two Lines on the Projective Plane**

Let $X = \mathbb{P}^2$ over $\overline{k}$, with homogeneous coordinates $[x_0 : x_1 : x_2]$. By theorem 1.5.2, $A_2(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}\,[L]$ generated by the class of a line $L$, and $A_0(\mathbb{P}^2) = \mathbb{Z}\,[P]$ generated by the class of a point. Take the two Cartier divisors $D = \{x_0 = 0\}$ and $D' = \{x_1 = 0\}$, each a line, with $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(D') \cong \mathcal{O}(1)$ and cutting sections $s = x_0$, $s' = x_1$ (rational sections of $\mathcal{O}(1)$ in the trivialization on the chart $U_2 = \{x_2 \neq 0\}$, where the affine coordinates are $u = x_0/x_2$, $v = x_1/x_2$).

*First order.* Intersecting $[\mathbb{P}^2]$ with $D$ gives $D \cdot [\mathbb{P}^2] = [\mathrm{div}(s)] = [D] = [L]$, the line $\{x_0 = 0\}$. Now intersect with $D'$. Since $D = \{x_0 = 0\}$ is not contained in $|D'| = \{x_1 = 0\}$, the section $s'|_D = x_1|_D$ is a rational section of $\mathcal{O}(1)|_D$; on the line $D \cong \mathbb{P}^1$ with coordinate $[x_1 : x_2]$ it vanishes to order one at the single point $x_1 = 0$, namely $[0 : 0 : 1]$. Thus

$$D' \cdot (D \cdot [\mathbb{P}^2]) = D' \cdot [L] = [\mathrm{div}_D(s')] = \big[\,[0 : 0 : 1]\,\big]$$

the class of the point where the two lines meet.

*Reverse order.* Intersecting $[\mathbb{P}^2]$ with $D'$ first gives $D' \cdot [\mathbb{P}^2] = [\mathrm{div}(s')] = [D'] = [L]$, the line $\{x_1 = 0\}$. Intersecting with $D$, the section $s|_{D'} = x_0|_{D'}$ on $D' \cong \mathbb{P}^1$ with coordinate $[x_0 : x_2]$ vanishes to order one at $x_0 = 0$, again $[0 : 0 : 1]$, so

$$D \cdot (D' \cdot [\mathbb{P}^2]) = D \cdot [L] = [\mathrm{div}_{D'}(s)] = \big[\,[0:0:1]\,\big]$$

The two orders produce the same zero-cycle, the reduced point $[0 : 0 : 1]$, on the nose, which is theorem 2.3.2 made visible: here no rational equivalence was even needed, because the two lines are transverse and the tame symbol $\partial_W(x_0, x_1)$ has no zeros or poles. Taking degrees on the complete $\mathbb{P}^2$,

$$\deg\big(D \cdot D' \cdot [\mathbb{P}^2]\big) = 1$$

the intersection number of two distinct lines: they meet in exactly one point. This is the case $c = d = 1$ of Bézout's theorem ([6]), recovered from the divisor calculus.

*Where reciprocity would enter.* Replacing $D'$ by $D$ itself, one must intersect $[L]$ with the same line, which requires moving $D$ within its linear system: since $\mathcal{O}(1)$ has many sections, choose instead $s'' = x_1$, whose divisor $\{x_1 = 0\}$ is linearly equivalent to $\{x_0 = 0\}$, and the previous computation gives $D \cdot D \cdot [\mathbb{P}^2] = [\,[0 : 0 : 1]\,]$ of degree 1, so the self-intersection $L^2 = 1$ on $\mathbb{P}^2$. The passage from $s = x_0$ to the linearly equivalent $s'' = x_1$ changes the cutting section by the rational function $x_1/x_0$, and that the answer is unchanged is precisely proposition 2.2.4, the well-definedness whose commutativity companion is proved in this section.

The example is the enumerative payoff in miniature. The number 1 is the degree of a zero-cycle, the two lines meet where their defining sections simultaneously vanish, and the independence of the answer from the order of intersection and from the choice of representative in each linear system is exactly what theorem 2.3.2 and proposition 2.2.4 guarantee. Iterating, $r$ general hyperplanes in $\mathbb{P}^n$ intersect in $c_1(\mathcal{O}(1))^r \cap [\mathbb{P}^n] = [\mathbb{P}^{n-r}]$, and taking $r = n$ recovers a single reduced point of degree one, the statement that $n$ general hyperplanes meet in one point. This is the computation the next section packages as the first Chern class.

## Exercise 2.3.1

1. On $\mathbb{P}^2$, let $D = \{x_0x_1 = 0\}$ be the union of two coordinate lines (a divisor in the class $2L$) and let $D' = \{x_2 = 0\}$ be a third line. Compute $D\cdot(D'\cdot[\mathbb{P}^2])$ and $D'\cdot(D\cdot[\mathbb{P}^2])$ as zero-cycles, verify they agree, and compute the degree of the product. Confirm the answer is $2 = \deg D \cdot \deg D'$, consistent with Bézout.

2. Let $S = \mathbb{P}^2$ with affine coordinates $u = x_0/x_2$, $v = x_1/x_2$ on the chart $x_2 \neq 0$. For the rational functions $f = u$ and $g = v$, compute the tame symbol $\partial_W(f, g)$ at each coordinate curve $W \in \{u = 0\}, \{v = 0\}$ and at the line at infinity, and verify directly that $\sum_W \mathrm{ord}_W(f)\,[\mathrm{div}_W(g)] = \sum_W \mathrm{ord}_W(g)\,[\mathrm{div}_W(f)]$ as zero-cycles.

3. Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be the degree-two map $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (characteristic $\neq 2$) and let $D = [\infty]$ be the divisor of the point at infinity on the target. Using proposition 2.3.3(1), compute both $f_*(f^*D \cdot [\mathbb{P}^1])$ and $D \cdot f_*[\mathbb{P}^1]$, identify the scalar $d = \deg f$, and confirm they agree.

4. Let $g\colon \mathbb{A}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ be the projection, flat of relative dimension one, and let $D = [0]$ be the origin on $\mathbb{P}^1$. Compute $g^*(D \cdot [\mathbb{P}^1])$ and $g^*D \cdot g^*[\mathbb{P}^1]$ and verify proposition 2.3.3(2) directly.

5. Deduce from theorem 2.3.2 that for any Cartier divisor $D$ on a complete variety $X$ of dimension

$n$, the iterated intersection $D^n \cdot [X] \in A_0(X)$ is symmetric under any reordering of the $n$ copies of $D$ (trivially, but state precisely why the number $\deg(D^n \cdot [X])$ is well defined), and use theorem 1.5.2 to compute $H^n \cdot [\mathbb{P}^n]$ for $H$ a hyperplane.

6. Explain why the proof of theorem 2.3.2 does not require $X$ to be normal, identifying the exact step where normalization is used and why it changes neither side of the identity. (One sentence on each.)

## 2.4 The First Chern Class of a Line Bundle

The three previous sections built a single operation and studied it thoroughly: a Cartier divisor $D$ on $X$ acts on cycles by $\alpha \mapsto D \cdot \alpha$, this action descends to rational equivalence (proposition 2.2.4), and two such actions commute (theorem 2.3.2). The divisor $D$, however, is not the most invariant object in sight. What acts on $A_*(X)$ is not $D$ itself but its linear-equivalence class, since $D$ and $D + \operatorname{div}(r)$ produce the same operator, and a linear-equivalence class of Cartier divisors is exactly a line bundle. The right name for the operator is therefore the *first Chern class* of the associated line bundle $\mathcal{O}_X(D)$, written $c_1(\mathcal{O}_X(D)) \cap -$.

The renaming records that the operator depends only on $L = \mathcal{O}_X(D)$, so it is defined for a line bundle whether or not a divisor is at hand; it makes the additive structure transparent, because tensoring line bundles adds their Chern classes; and it is the first instance of a construction that in the next chapter will attach to a vector bundle $E$ of rank $r$ a whole family of operators $c_1(E), \dots, c_r(E)$, the higher Chern classes. Everything the first Chern class does is already contained in the divisor calculus of the preceding sections; this section repackages that calculus into its final, functorial form and reads off the consequences.

The starting point is that the operator $D \cdot -$ sees only the line bundle $\mathcal{O}_X(D)$. Two Cartier divisors with isomorphic associated bundles are linearly equivalent, that is, they differ by the divisor of a rational function, and a principal divisor intersects every subvariety in a principal divisor, hence acts as zero on $A_*(X)$; proposition 2.4.2 below records this and is the independence needed to base the definition on $L$ rather than on $D$.

**Definition 2.4.1: The First Chern Class of a Line Bundle**

Let $L$ be a line bundle on an algebraic scheme $X$, and suppose $L \cong \mathcal{O}_X(D)$ for some Cartier divisor $D$ on $X$. The *first Chern class* of $L$ is the operator on Chow groups

$$c_1(L) \cap - \colon A_k(X) \longrightarrow A_{k-1}(X), \qquad c_1(L) \cap \alpha := D \cdot \alpha$$

defined for every $k$, where $D \cdot \alpha$ is the intersection with the Cartier divisor $D$ of definition 2.2.1. The class $c_1(L)$ is the operator itself; the notation $c_1(L) \cap \alpha$ reads "$c_1(L)$ capped with $\alpha$".

The definition names $D$ but must not depend on it: a line bundle admits many divisors of sections, and the Chern class has to be the same for all of them. The following proposition is the price of the definition, and its proof is short because the work was done in section 2.2.

**Proposition 2.4.2: Independence of the Divisor**

Let $L$ be a line bundle on $X$ and let $D, D'$ be Cartier divisors with $\mathcal{O}_X(D) \cong L \cong \mathcal{O}_X(D')$. Then $D \cdot \alpha = D' \cdot \alpha$ in $A_{k-1}(X)$ for every $\alpha \in A_k(X)$. Consequently $c_1(L) \cap -$ of definition 2.4.1 is well defined, independent of the choice of $D$.

***Proof*:**
An isomorphism $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ of line bundles is exactly a linear equivalence $D \sim D'$ of Cartier divisors: the tautological rational section of $\mathcal{O}_X(D)$ and that of $\mathcal{O}_X(D')$, compared through the isomorphism, differ by a global invertible section $r$ of the sheaf of total quotient rings $\mathcal{K}_X^*$, so $D - D' = \operatorname{div}(r)$ is principal (on a variety $\mathcal{K}_X^* = k(X)^*$ and $r$ is a rational function; on a general algebraic scheme $r$ is a section of the total-quotient sheaf, which is all the definition of a principal Cartier divisor requires). It therefore suffices to show that a principal Cartier divisor $\operatorname{div}(r)$ acts as zero on rational equivalence classes.

Fix $\alpha \in A_k(X)$ and represent it by a cycle $\sum_i n_i[V_i]$. Since $\operatorname{div}(r) \cdot -$ is defined component by component and extended $\mathbb{Z}$-linearly (definition 2.2.1), it suffices to show $\operatorname{div}(r) \cdot [V] = 0$ in $A_{k-1}(X)$ for a single $k$-dimensional subvariety $V$. The associated bundle of the principal divisor $\operatorname{div}(r)$ is $\mathcal{O}_X$, and the rational section of $\mathcal{O}_X$ that represents $\operatorname{div}(r)$ is $r$ itself; unwinding definition 2.2.1 for this section gives

$$\operatorname{div}(r) \cdot [V] = \big[\operatorname{div}\big(r|_V\big)\big] \in A_{k-1}(V)$$

where $r|_V \in k(V)^*$ is the restriction of $r$ to $V$ (a nonzero rational function when $V \not\subseteq |\operatorname{div}(r)|$, and otherwise computed through the restricted trivial bundle $\mathcal{O}_X|_V \cong \mathcal{O}_V$, whose section is again a rational function on $V$). In either case $\operatorname{div}(r|_V)$ is the divisor of a rational function on the variety $V$, hence a *principal* Weil divisor on $V$, which is rationally equivalent to zero by the very definition of rational equivalence (definition 1.1.3). Its class in $A_{k-1}(V)$, and so its image in $A_{k-1}(X)$, is therefore 0: intersecting a subvariety with a principal divisor produces a principal divisor, which dies in the Chow group. This is the same computation that drives the well-definedness of proposition 2.2.4, read now with the roles reversed, the divisor rather than the cycle being principal. Summing over the components of $\alpha$ gives $\operatorname{div}(r) \cdot \alpha = 0$, and so

$$D \cdot \alpha - D' \cdot \alpha = (D - D') \cdot \alpha = \operatorname{div}(r) \cdot \alpha = 0$$

giving $D \cdot \alpha = D' \cdot \alpha$, as we sought to show.

The proposition is the reason the notation $c_1(L)$ is legitimate at all. It shows the divisor served only to build the operator: once the operator exists, it is recorded by the line bundle alone. From here on a divisor is used to compute $c_1(L)$, never to define it, and the choice of divisor is a matter of convenience. On a variety, every line bundle admits a Cartier divisor, so the definition applies to all line bundles; on a general scheme one takes $L$ to be of the form $\mathcal{O}_X(D)$, which covers every line bundle whose class lies in the image of the Cartier divisor class group. The definition and the properties below are stated for a general algebraic scheme, but every worked example and exercise that follows runs on a variety, where a rational section and its function field $k(V)$ are always at hand; the general-scheme statements specialize to these illustrations by taking $X$ integral.

A first sanity check grounds the definition before its formal properties: the Chern class of the trivial bundle is the zero operator, and the two constructions available for a hyperplane bundle agree.

**Example 2.4.3: The Trivial Bundle and the Hyperplane Bundle**

1. *The trivial bundle.* For $L = \mathcal{O}_X$ one may take $D = 0$, the empty divisor. Then $c_1(\mathcal{O}_X) \cap \alpha = 0 \cdot \alpha = 0$ for every $\alpha$, so $c_1(\mathcal{O}_X)$ is the zero operator. This matches the reading of $c_1$ as a measure of the twisting of $L$: a trivial bundle has no twisting to record.

2. *The hyperplane bundle on $\mathbb{P}^n$.* Let $L = \mathcal{O}_{\mathbb{P}^n}(1)$ and let $H \subseteq \mathbb{P}^n$ be a hyperplane, cut out by a linear form; then $\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1)$. The class map of proposition 2.1.3 sends $\mathcal{O}_{\mathbb{P}^n}(1) \mapsto [H] \in A_{n-1}(\mathbb{P}^n)$, the generator of theorem 1.5.2 in codimension one. Capping with $[\mathbb{P}^n]$ recovers this class,

$$c_1(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n] = H \cdot [\mathbb{P}^n] = [H]$$

since $\mathbb{P}^n \nsubseteq H$ and the restricted section of $\mathcal{O}_{\mathbb{P}^n}(1)$ to $\mathbb{P}^n$ has divisor exactly $H$. The first Chern class of $\mathcal{O}(1)$, capped against the fundamental class, is the hyperplane class.

With the operator anchored, its formal properties follow. Each is a translation of a divisor identity already proved, and the point of collecting them is to see the first Chern class as a well-behaved endomorphism of the graded group $A_*(X)$: additive in $\alpha$, commuting with its own kind, additive in $L$, compatible with pushforward and pullback, and eventually nilpotent. The statement below is the working summary of the theory of $c_1$, and every clause is used repeatedly in the chapters that follow.

**Proposition 2.4.4: Properties of the First Chern Class**

Let $X$ be an algebraic scheme, $L, L'$ line bundles on $X$, and $\alpha \in A_k(X)$.

1. *Homomorphism.* $c_1(L) \cap -\colon A_k(X) \to A_{k-1}(X)$ is a group homomorphism: $c_1(L) \cap (\alpha + \beta) = c_1(L) \cap \alpha + c_1(L) \cap \beta$.

2. *Commutativity.* For line bundles $L, L'$,
$$c_1(L) \cap \big(c_1(L') \cap \alpha\big) = c_1(L') \cap \big(c_1(L) \cap \alpha\big)$$
in $A_{k-2}(X)$.

3. *Additivity.* $c_1(L \otimes L') = c_1(L) + c_1(L')$ as operators; that is, $c_1(L \otimes L') \cap \alpha = c_1(L) \cap \alpha + c_1(L') \cap \alpha$. In particular $c_1(L^\vee) = -c_1(L)$.

4. *Projection formula.* For a proper morphism $f\colon X' \to X$, a line bundle $L$ on $X$, and $\alpha' \in A_k(X')$,
$$f_*\big(c_1(f^*L) \cap \alpha'\big) = c_1(L) \cap f_*\alpha'$$
and for a flat morphism $g\colon X'' \to X$ of relative dimension $m$ and $\beta \in A_k(X)$,
$$c_1(g^*L) \cap g^*\beta = g^*\big(c_1(L) \cap \beta\big)$$
in $A_{k+m-1}(X'')$.

5. *Nilpotence.* If $X$ is a variety of dimension $n$, then for any line bundle $L$,
$$c_1(L)^{n+1} \cap [X] = 0$$
where $c_1(L)^{n+1} \cap [X]$ means $c_1(L)$ capped $n+1$ times with the fundamental class $[X]$.

***Proof*:**
Choose Cartier divisors $D, D'$ with $\mathcal{O}_X(D) \cong L$ and $\mathcal{O}_X(D') \cong L'$; by proposition 2.4.2 the choices do not affect any of the claims. Each part translates a divisor identity from section 2.2 or section 2.3.

1. *Homomorphism.* The intersection $D \cdot -$ is defined on cycles component by component and extended $\mathbb{Z}$-linearly (definition 2.2.1), so it is additive on $Z_k(X)$, and proposition 2.2.4 makes it descend to a well-defined additive map on $A_k(X)$. Hence $c_1(L) \cap - = D \cdot -$ is a homomorphism $A_k(X) \to A_{k-1}(X)$.

2. *Commutativity.* By definition $c_1(L) \cap (c_1(L') \cap \alpha) = D \cdot (D' \cdot \alpha)$ and $c_1(L') \cap (c_1(L) \cap \alpha) = D' \cdot (D \cdot \alpha)$. These are equal in $A_{k-2}(X)$ by theorem 2.3.2, the commutativity of intersection with two Cartier divisors.

3. *Additivity.* The associated-bundle construction of [2, Divisors: Geometric Representation of Line Bundles] converts the sum of Cartier divisors into the tensor product of line bundles, $\mathcal{O}_X(D+D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$, so $D+D'$ is a Cartier divisor with $\mathcal{O}_X(D+D') \cong L \otimes L'$. Using this divisor to compute the Chern class of $L \otimes L'$,
$$c_1(L \otimes L') \cap \alpha = (D + D') \cdot \alpha = D \cdot \alpha + D' \cdot \alpha = c_1(L) \cap \alpha + c_1(L') \cap \alpha$$

where the middle equality is the additivity of $- \cdot \alpha$ in the divisor, and it holds already at the level of cycles, on *every* subvariety $V$, with no repositioning of $\alpha$ required. Since $D \cdot -$ is defined component by component and extended $\mathbb{Z}$-linearly (definition 2.2.1), it suffices to check $(D+D')\cdot[V] = D\cdot[V]+D'\cdot[V]$ for a single $k$-dimensional subvariety $V$. Restricting the tensor-product isomorphism above to $V$ gives $\mathcal{O}_X(D+D')|_V \cong \mathcal{O}_X(D)|_V \otimes_{\mathcal{O}_V} \mathcal{O}_X(D')|_V$, so a rational section $s_V$ of $\mathcal{O}_X(D+D')|_V$ representing $(D+D')|_V$ is the product $s_V = s_V^D \cdot s_V^{D'}$ of representing rational sections of $\mathcal{O}_X(D)|_V$ and $\mathcal{O}_X(D')|_V$. By definition 2.2.1 the cap is the class of the divisor of the restricted section computed inside $V$, and the order function along a codimension-one subvariety $W$ of $V$ is additive on a product of rational sections, $\operatorname{ord}_W(s_V^D \cdot s_V^{D'}) = \operatorname{ord}_W(s_V^D) + \operatorname{ord}_W(s_V^{D'})$, since $\operatorname{ord}_W$ is a valuation. Summing over such $W$ gives $[\operatorname{div}(s_V)] = [\operatorname{div}(s_V^D)] + [\operatorname{div}(s_V^{D'})]$ in $A_{k-1}(V)$, that is $(D+D')\cdot[V] = D\cdot[V]+D'\cdot[V]$; pushing forward to $X$ and summing over the components of $\alpha$ gives the middle equality. This is a cycle-level identity holding for each $V$, so no moving lemma repositioning $\alpha$ off $|D|$ and $|D'|$ is needed, and none exists in general. Taking $L' = L^\vee$, so that $\mathcal{O}_X \cong L \otimes L^\vee$ and $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, gives $c_1(L) + c_1(L^\vee) = 0$, hence $c_1(L^\vee) = -c_1(L)$.

4. *Projection formula.* The pullback along $f$ of the Cartier divisor $D$, in the pseudo-divisor formalism of definition 2.1.5, is a pseudo-divisor $f^*D$ with $f^*\mathcal{O}_X(D) \cong \mathcal{O}_{X'}(f^*D)$; so $c_1(f^*L) \cap \alpha' = f^*D \cdot \alpha'$. The proper case is the divisor projection formula $f_*(f^*D \cdot \alpha') = D \cdot f_*\alpha'$ of proposition 2.3.3, which rewritten with Chern-class notation is exactly $f_*(c_1(f^*L) \cap \alpha') = c_1(L) \cap f_*\alpha'$. The flat case is the compatibility $g^*(D \cdot \beta) = g^*D \cdot g^*\beta$ of the same proposition, and since $c_1(g^*L) \cap g^*\beta = g^*D \cdot g^*\beta$ this reads $c_1(g^*L) \cap g^*\beta = g^*(c_1(L) \cap \beta)$.

5. *Nilpotence.* Each cap with $c_1(L)$ lowers dimension by one, since $c_1(L) \cap - \colon A_j(X) \to A_{j-1}(X)$ by definition 2.4.1, so repeatedly capping $[X] \in A_n(X)$ gives

$$c_1(L)^{n+1} \cap [X] \in A_{n-(n+1)}(X) = A_{-1}(X)$$

and $A_j(X) = 0$ for $j < 0$, there being no subvarieties of negative dimension. Hence the class is zero.

Each clause has been reduced to a result of the preceding sections, completing the proof.

The well-definedness proved above lets $D \cdot -$ be evaluated on a representative chosen to avoid $|D|$, which is all the divisor calculus of this chapter needs. Later constructions cannot make that choice: in chapter 4 a cycle is intersected with the fibre at infinity of a degeneration, and *every* component of the relevant representative lies in that fibre. The following strengthening removes the avoidance hypothesis, and it is what makes those constructions legitimate.

**Corollary 2.4.5: Divisor Intersection on a Closed Subscheme**

Let $D$ be a Cartier divisor on $X$ and $T \subseteq X$ a closed subscheme, with no hypothesis relating $T$ to $|D|$ and none making $D|_T$ a Cartier divisor on $T$. Then intersection with $D$ descends to a homomorphism

$$D \cdot - \ : \ A_k(T) \longrightarrow A_{k-1}\big(T \cap |D|\big)$$

computed on any representative cycle, and compatible with the inclusion of $T$ in $X$: for a cycle on $T$ the two readings of definition 2.2.1, one in $T$ and one in $X$, agree.

***Proof*:**
At the level of cycles there is nothing to prove: definition 2.2.1 computes $D \cdot [V]$ for a subvariety $V$ from the restricted line bundle $\mathcal{O}_X(D)|_V$ and from whether $V \subseteq |D|$, and neither datum changes when $V$ is regarded inside $T$ rather than $X$. This is the reason the definition was framed through the pseudo-divisor $(\mathcal{O}_X(D), |D|, s_D)$ of definition 2.1.5: a pseudo-divisor restricts to any subscheme, whereas a Cartier divisor need not, its local equations being free to become zero divisors.

What needs an argument is that the operation kills cycles rationally equivalent to zero *on* $T$, and lands in the smaller group. Such a cycle is a sum of $\mathrm{div}(r)$ for $(k+1)$-dimensional subvarieties $W \subseteq T$ and $r \in k(W)^*$ (definition 1.1.3). The proof of proposition 2.4.6 treats exactly these generators, and in both of its cases the vanishing it produces is already recorded on $W$: in the case $W \subseteq |D|$ the class is computed in $A_{k-1}(W)$ before being pushed forward, and in the case $W \not\subseteq |D|$ the appeal to theorem 2.3.2 places the identity in $A_{k-1}(|D| \cap |\mathrm{div}(r)| \cap W)$. Both groups map to $A_{k-1}(T \cap |D|)$, so $D \cdot \mathrm{div}(r) = 0$ there, as we sought to show.

**Proposition 2.4.6: Divisor Intersection on an Arbitrary Representative**

Let $D$ be a Cartier divisor on an algebraic scheme $X$ and let $\alpha \in Z_k(X)$ be a cycle rationally equivalent to zero, with *no* hypothesis on its components. Then

$$D \cdot \alpha = 0 \quad \text{in } A_{k-1}(|D|)$$

Consequently $D \cdot -$ may be computed from any representative of a class in $A_k(X)$, including one all of whose components lie in $|D|$, and every representative gives the same class in $A_{k-1}(|D|)$.

***Proof*:**
Both sides are additive, so it suffices to treat $\alpha = [\mathrm{div}(r)]$ for a $(k+1)$-dimensional subvariety $W \subseteq X$ and a rational function $r \in k(W)^*$, these being the generators of the subgroup of cycles rationally equivalent to zero (definition 1.1.3). Write $\iota\colon W \hookrightarrow X$ for the inclusion.

*Case 1:* $W \subseteq |D|$. Every component of $\mathrm{div}(r)$ lies in $W$, hence in $|D|$, so clause (2) of definition 2.2.1 computes $D \cdot [\mathrm{div}(r)]$ from the restricted line bundle. By definition 2.4.1 that value is $c_1\big(\mathcal{O}_X(D)|_W\big) \cap [\mathrm{div}(r)]$, computed in $A_{k-1}(W)$. Now $[\mathrm{div}(r)] = 0$ in $A_k(W)$ by the definition of rational equivalence, and $c_1(\mathcal{O}_X(D)|_W) \cap -$ is a homomorphism on Chow *classes* (proposition 2.4.4(1)), so it sends 0 to 0. Pushing forward along the closed immersion $W \hookrightarrow |D|$ gives $D \cdot [\mathrm{div}(r)] = 0$ in $A_{k-1}(|D|)$.

*Case 2:* $W \not\subseteq |D|$. Then $D$ restricts to a Cartier divisor $D|_W$ on $W$, and $[\mathrm{div}(r)] = \mathrm{div}(r) \cdot [W]$ exhibits $\alpha$ as the result of intersecting the fundamental class of $W$ with the principal Cartier divisor $\mathrm{div}(r)$ on $W$. Commutativity of divisor intersection (theorem 2.3.2), applied on $W$ to the pair $D|_W$ and $\mathrm{div}(r)$, gives

$$D|_W \cdot \big(\mathrm{div}(r) \cdot [W]\big) \;=\; \mathrm{div}(r) \cdot \big(D|_W \cdot [W]\big)$$

and that theorem records the identity in the refined group $A_{k-1}\big(|D| \cap |\mathrm{div}(r)| \cap W\big)$, so it may be read there. The right-hand side vanishes: the line bundle of a principal divisor is trivial, $\mathcal{O}_W(\mathrm{div}(r)) \cong \mathcal{O}_W$, so $\mathrm{div}(r) \cdot -$ is $c_1(\mathcal{O}_W) \cap -$, and $c_1$ of a trivial bundle is the zero operator by proposition 2.4.4(3) applied to $L = L' = \mathcal{O}_W$. Hence $D \cdot [\mathrm{div}(r)] = 0$ in $A_{k-1}(|D|)$, completing the proof.

Nothing in the argument is circular: theorem 2.3.2 is proved from the avoidance-hypothesis form proposition 2.2.4 alone, so it may be used here to remove that hypothesis.

Parts (1) through (4) say that $c_1$ is a homomorphism of graded groups compatible with all the structure already on $A_*$, and that $L \mapsto c_1(L)$ turns tensor product into addition, so $c_1 \colon \mathrm{Pic}\,(X) \to \mathrm{End}(A_*(X))$ is a homomorphism from the Picard group into the operators lowering degree by one. Part (2), the commutativity, is the fact that makes these operators generate a commutative subring of $\mathrm{End}(A_*(X))$; it is the discharged form of the divisor symmetry that theorem 1.4.6 in section 1.4 took on faith, now proved through theorem 2.3.2. Part (5), the nilpotence, is a dimension count: it says a single line bundle can only cut down the fundamental class $n$ times before the result must vanish, and it is what makes the top self-intersection number $c_1(L)^n \cap [X]$, a zero-cycle, the natural degree-type invariant of $L$ on a complete variety.

The nilpotence has a converse worth flagging: while $c_1(L)^{n+1} \cap [X]$ always vanishes, the intermediate power $c_1(L)^n \cap [X] \in A_0(X)$ need not, and its degree $\int_X c_1(L)^n$ is the numerical invariant of a line bundle on an $n$-dimensional complete variety. For $n = 1$ this is the degree of a line bundle on a curve; for a surface it is the self-intersection $L^2$. These invariants are the raw material of the intersection numbers computed throughout this book.

The first substantial computation is the self-intersection of $\mathcal{O}(1)$ on projective space, which reproduces the entire linear-subspace basis of theorem 1.5.2 from a single line bundle. It shows the first Chern class already generates the Chow ring of $\mathbb{P}^n$, and it is the model computation the higher Chern classes generalize.

**Example 2.4.7: Powers of the Hyperplane Class on $\mathbb{P}^n$**

Let $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$ act on $A_*(\mathbb{P}^n)$. By theorem 1.5.2 the Chow group $A_*(\mathbb{P}^n) = \bigoplus_{j=0}^{n} \mathbb{Z}\,[\mathbb{P}^j]$ is free on the classes of linear subspaces, with $[\mathbb{P}^j]$ the class of a $j$-plane. The claim is

$$c_1(\mathcal{O}_{\mathbb{P}^n}(1))^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}], \qquad 0 \le k \le n$$

so the $k$-th self-cap of the hyperplane bundle is the class of a linear subspace of codimension $k$.

The proof is induction on $k$. For $k = 0$ the left side is $[\mathbb{P}^n]$, the fundamental class, and $[\mathbb{P}^{n-0}] = [\mathbb{P}^n]$. Assume $c_1(\mathcal{O}(1))^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ for some $k < n$, so the class is represented by the subvariety $\mathbb{P}^{n-k} \subseteq \mathbb{P}^n$, a linear subspace of dimension $n-k$. Cap once more with $H$. There is a hyperplane $H_0$ with $\mathbb{P}^{n-k} \not\subseteq H_0$, because some linear form does not vanish identically on the nonempty $\mathbb{P}^{n-k}$ (equivalently, $\mathbb{P}^{n-k}$ is not contained in every hyperplane); its zero locus is such an $H_0$, and $H_0 \cap \mathbb{P}^{n-k}$ is then a hyperplane inside $\mathbb{P}^{n-k}$, of dimension $n-k-1$. Thus $\mathbb{P}^{n-k} \not\subseteq |H_0|$, and by definition 2.2.1 the cap $H \cdot [\mathbb{P}^{n-k}]$ is the divisor of the restricted linear form on $\mathbb{P}^{n-k}$, namely the hyperplane section $H_0 \cap \mathbb{P}^{n-k}$, a linear subspace $\mathbb{P}^{n-k-1}$ of dimension $n-k-1$:

$$c_1(\mathcal{O}(1))^{k+1} \cap [\mathbb{P}^n] = H \cdot [\mathbb{P}^{n-k}] = [H_0 \cap \mathbb{P}^{n-k}] = [\mathbb{P}^{n-k-1}]$$

completing the induction. In particular $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathbb{P}^0] = [\mathrm{pt}]$ is the class of a single point, and its degree is

$$\int_{\mathbb{P}^n} c_1(\mathcal{O}_{\mathbb{P}^n}(1))^n = 1$$

the statement that $n$ general hyperplanes in $\mathbb{P}^n$ meet in a single reduced point. Capping once more, $c_1(\mathcal{O}(1))^{n+1} \cap [\mathbb{P}^n] = H \cdot [\mathrm{pt}] = 0$, in agreement with the nilpotence of proposition 2.4.4(5). Every generator $[\mathbb{P}^j]$ of $A_*(\mathbb{P}^n)$ is thus a power of $c_1(\mathcal{O}(1))$ applied to $[\mathbb{P}^n]$, so the first Chern class of the hyperplane bundle generates the Chow group of projective space as a module over the operators.

The computation is the first appearance of the ring structure that intersection theory is heading toward: on $\mathbb{P}^n$ the operators $H^k$ multiply the way linear subspaces intersect, $[\mathbb{P}^a] \cdot [\mathbb{P}^b] = [\mathbb{P}^{a+b-n}]$ when $a + b \geq n$ and zero otherwise, and $\int_{\mathbb{P}^n} H^n = 1$ is the normalization. The Chow group $A_*(\mathbb{P}^n)$ is the truncated polynomial ring $\mathbb{Z}[H]/(H^{n+1})$ with $H = c_1(\mathcal{O}(1))$, and the nilpotence relation $H^{n+1} = 0$ is exactly proposition 2.4.4(5). This ring will be built in general later in this Part, once the intersection product is available; here it is visible for projective space through the single line bundle $\mathcal{O}(1)$.

The second computation identifies $c_1$ on a curve with the classical degree, tying the abstract operator to the invariant every reader already knows. On a smooth projective curve the first Chern class is the degree map, and capping with $c_1(L)$ is subtraction of that degree from the point count.

**Example 2.4.8: The First Chern Class on a Curve is the Degree**

Let $C$ be a smooth projective curve over $\overline{k}$ and $L$ a line bundle on $C$. Since $\dim C = 1$, the only nonzero cap is on $A_1(C) = \mathbb{Z}\,[C]$, and it lands in $A_0(C)$. Write $L \cong \mathcal{O}_C(D)$ for a Cartier divisor $D = \sum_P n_P\,[P]$, which on a smooth curve is the same as a Weil divisor, a finite $\mathbb{Z}$-combination of closed points. Then

$$c_1(L) \cap [C] = D \cdot [C] = \sum_P n_P\,[P] \in A_0(C)$$

by definition 2.2.1, the section of $\mathcal{O}_C(D)$ having divisor $D$ on the curve $C$ itself. Pushing this zero-cycle to a point through $\pi\colon C \to \operatorname{Spec} \overline{k}$ and taking degrees (definition 1.2.4),

$$\int_C c_1(L) \cap [C] = \sum_P n_P\,[\overline{k}(P) : \overline{k}] = \sum_P n_P = \deg D = \deg L$$

since every closed point of a variety over the algebraically closed field $\overline{k}$ has residue field $\overline{k}$. The number $\int_C c_1(L)$ is the degree of the line bundle. Under the identification $A_0(C) \cong \operatorname{Cl} C = \operatorname{Pic} C$ of proposition 1.5.3, capping with $c_1(L)$ against $[C]$ is the class map $L \mapsto [D]$, and its composite with the degree $A_0(C) \xrightarrow{\deg} \mathbb{Z}$ recovers the degree of the divisor class, fitting into the sequence $0 \to \operatorname{Pic}^0 C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$ of proposition 1.5.3.

The two computations frame the whole theory. On projective space $c_1(\mathcal{O}(1))$ generates the Chow ring and its top power counts a single point; on a curve $c_1(L)$ is the degree, the invariant that governs Riemann-Roch through $\chi(L) = \deg L + 1 - g$ ([2, Riemann-Roch for curves]). Both come from one construction: the first Chern class turns a line bundle into an operator on cycles, and its powers are the numerical invariants that intersection theory computes. The chapter that follows generalizes the construction from line bundles to vector bundles of arbitrary rank: a rank-$r$ bundle $E$ carries Chern classes $c_1(E), \ldots, c_r(E)$, built so that $c_1$ of a line bundle is the operator of this section, and so that the total Chern class is multiplicative in short exact sequences. The machine that produces them, Segre classes of the projectivized bundle $\mathbb{P}(E)$, is the subject of the next chapter, and the divisor calculus of this chapter is the rank-one base case on which it is built.

## Exercise 2.4.1

1. Compute $\int_{\mathbb{P}^1} c_1(\mathcal{O}_{\mathbb{P}^1}(d))$ for every integer $d$, both by the curve-degree formula of example 2.4.8 and directly from a divisor of a section of $\mathcal{O}_{\mathbb{P}^1}(d)$. Confirm that $c_1\colon \operatorname{Pic}(\mathbb{P}^1) \to A_0(\mathbb{P}^1) = \mathbb{Z}$ is an isomorphism.

2. Let $S$ be a smooth projective surface and $L = \mathcal{O}_S(C)$ for a curve $C \subseteq S$. Express $c_1(L)^2 \cap [S] \in A_0(S)$ and its degree $\int_S c_1(L)^2$ in terms of the divisor $C$, and identify $\int_S c_1(L)^2$ with the self-intersection number $C \cdot C$ of the surface theory of [2, Intersection Theory on Surfaces]. Why does proposition 2.4.4(5) say $c_1(L)^3 \cap [S] = 0$?

3. Using proposition 2.4.4(3), show that $c_1 \colon \mathrm{Pic}\,(X) \to \mathrm{End}(A_*(X))$ is a group homomorphism from the Picard group to the additive group of degree-lowering operators. Deduce that $c_1(L^{\otimes m}) = m\, c_1(L)$ for every integer $m$, and that $c_1$ factors through the numerical class of $L$ in the sense that linearly equivalent line bundles have equal first Chern class.

4. On $\mathbb{P}^1 \times \mathbb{P}^1$, let $p_1, p_2$ be the two projections and $L = p_1^*\mathcal{O}(a) \otimes p_2^*\mathcal{O}(b)$ the line bundle of bidegree $(a, b)$. Using the projection formula proposition 2.4.4(4) and additivity, compute $\int_{\mathbb{P}^1 \times \mathbb{P}^1} c_1(L)^2$ in terms of $a$ and $b$. (You may use $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ free on the classes of a point, the two rulings, and the fundamental class, with each ruling of self-intersection zero and the two rulings meeting in one point.)

5. Let $f \colon X \to Y$ be a proper surjective morphism of varieties with $\dim X = \dim Y = n$ and generic degree $\delta$, and let $L$ be a line bundle on $Y$. Show that $\int_X c_1(f^*L)^n = \delta \int_Y c_1(L)^n$. (Cap the projection formula $n$ times and use $f_*[X] = \delta\,[Y]$.) Then explain why $c_1(f^*L)^{n+1} \cap [X] = 0$ holds regardless of $\delta$.

# Part II

# Chern Classes and the Intersection Product

Everything so far has been rank one. The Chow group was cut out of the cycle group by rational equivalence, made functorial by proper pushforward and flat pullback, and then acted on by a single operator: intersecting with a Cartier divisor, which is to say capping with the first Chern class of a line bundle. That operator lowers dimension by one, commutes with itself, and satisfies a projection formula. It is also the only operator available, and a line bundle is the only bundle it sees.

This part removes both restrictions. A vector bundle of rank $r+1$ carries $r+1$ operators rather than one, and the projective bundle of its lines is the space on which they can be constructed: pushing powers of the tautological class down to the base produces the Segre classes, inverting the resulting series produces the Chern classes, and the geometry of the bundle forces the relations among them. With operators of every codimension in hand, the remaining question is whether two cycles can be multiplied rather than operated on. They can, and the mechanism is deformation to the normal cone, which replaces the classical manoeuvre of moving cycles into general position with a degeneration of the ambient embedding. The product it yields makes $A^*(X)$ a ring for smooth $X$, recovers Bézout's theorem, and subsumes the surface intersection pairing that *Algebraic Geometry* [2] built by hand.

# 3

# Cones, Segre Classes, and Chern Classes of Vector Bundles

The construction of this chapter rests on one geometric observation. A vector bundle $E$ on $X$ has no interesting cycles of its own, being an affine bundle and so contributing nothing beyond $A_*(X)$ by theorem 1.4.6. Its projectivization $\mathbb{P}(E)$ does: it is proper over $X$, it carries the tautological line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$, and the first Chern class of that line bundle is an operator of the kind Chapter 2 already supplies. Pushing its powers back down to $X$ therefore produces operators on $A_*(X)$ out of nothing but the rank-one theory, and those operators are the Segre classes.

The chapter develops this in four movements. The Segre classes come first, along with the projective-bundle formula that computes $A_*(\mathbb{P}(E))$ completely and which was promised without proof in chapter 1. Chern classes follow by inverting the Segre series, and the splitting principle, made legitimate by the flag bundle, reduces their identities to the rank-one case already in hand. The Chern character and Todd class are then assembled as polynomial expressions in these operators, with the rational coefficients their denominators force. The chapter closes by extending Segre classes from vector bundles to cones, and so to arbitrary closed subschemes, which is the input chapter 4 needs.

## 3.1 Segre Classes and the Projective Bundle

A vector bundle $E$ on $X$ determines a projective bundle $\pi\colon \mathbb{P}(E) \to X$ carrying a tautological line bundle. Pushing powers of that line bundle's first Chern class down to $X$ produces a family of operators on $A_*(X)$, the Segre classes, and the structural theorem behind them computes $A_*(\mathbb{P}(E))$ from $A_*(X)$ together with the powers of the tautological class. Both are built here, the second in full, discharging the formula recorded without proof in section 1.5.

The projective bundle itself is not built here. *Algebraic Geometry* [2] constructs Proj of a graded quasi-coherent algebra ([2, graded quasi-coherent algebra]) and its twisting sheaf ([2, Relative Proj]), specializes the construction to a locally free sheaf ([2, the projective bundle]), and proves there that the resulting morphism is proper and flat ([2, A Projective Bundle is Proper and Flat]) and locally trivial over a trivializing cover ([2, Local Triviality of a Projective Bundle]). One translation is needed at the point of import. What that volume writes $\mathbb{P}(\mathscr{E})$ parametrizes one-dimensional *quotients* of the fibres, while this book's $\mathbb{P}(E)$ parametrizes one-dimensional *subspaces* (box 0.0.3). Writing $\mathscr{E}$ for the sheaf of sections of $E$, the two are related by a dual:

$$\mathbb{P}(E) \;=\; \operatorname{Proj}\left(\operatorname{Sym}^{\bullet}\mathscr{E}^{\vee}\right)$$

so the bundle this book calls $\mathbb{P}(E)$ is the one [2] calls $\mathbb{P}(\mathscr{E}^{\vee})$. Under that identification the twisting sheaf $\mathcal{O}(1)$ of the relative Proj is the universal rank-one quotient of $\pi^*\mathscr{E}^{\vee}$ ([2, The Tautological Quotient]); dualizing that surjection exhibits $\mathcal{O}(1)^{\vee}$ as the universal rank-one *subsheaf* of $\pi^*\mathscr{E}$, which is the tautological line. So $\mathcal{O}_{\mathbb{P}(E)}(1)$ is the dual of the tautological subbundle, which is what the subspace convention requires. Every formula below is read in that convention.

Throughout, $E$ has rank $r+1$, so the fibre $\mathbb{P}(E_x)$ is a projective space of dimension $r$. All fibres of $\pi$ are pure of dimension $r$, and $\pi$ is flat, so $\pi$ is flat of relative dimension $r$ in the sense of definition 1.3.1; flat pullback $\pi^*\colon A_k(X) \to A_{k+r}(\mathbb{P}(E))$ is therefore defined by definition 1.3.3, and $\pi$ being proper, pushforward $\pi_*\colon A_k(\mathbb{P}(E)) \to A_k(X)$ is defined by definition 1.2.1. These two maps and the divisor calculus of chapter 2 are all this section uses.

**Definition 3.1.1: The Tautological Class of a Projective Bundle**

Let $E$ be a vector bundle of rank $r+1$ on an algebraic scheme $X$, let $\pi\colon \mathbb{P}(E) \to X$ be the associated bundle of lines in $E$, and let $\mathcal{O}_{\mathbb{P}(E)}(1)$ be the dual of the tautological subbundle. The *tautological class* of $E$ is the first Chern class operator

$$\xi \;=\; c_1\left(\mathcal{O}_{\mathbb{P}(E)}(1)\right)$$

of definition 2.4.1, so that $\xi \cap -\colon A_k(\mathbb{P}(E)) \to A_{k-1}(\mathbb{P}(E))$ for every $k$. Its powers $\xi^i \cap -$ are the $i$-fold iterates of this operator, with $\xi^0 \cap -$ the identity.

The class $\xi$ is an operator on the Chow groups of the total space, not an element of a ring: there is no product on $A_*(\mathbb{P}(E))$ until the intersection product of a later chapter supplies one. What makes $\xi$ usable is that the operators $\xi \cap -$ commute among themselves and with every other first Chern class, by proposition 2.4.4(2), so an expression such as $\xi^i \cap \pi^*\alpha$ is unambiguous however it is bracketed.

**Example 3.1.2: The Tautological Class on $\mathbb{P}^n$ and on a Line Bundle**

Two extreme cases fix the meaning of $\xi$.

1. *A trivial bundle over a point.* Let $X = \operatorname{Spec}\overline{k}$ and $E = \overline{k}^{n+1}$, so $\mathbb{P}(E) = \mathbb{P}^n$ is the space of lines through the origin in $\overline{k}^{n+1}$ and $\mathcal{O}_{\mathbb{P}(E)}(1) = \mathcal{O}_{\mathbb{P}^n}(1)$ is the Serre twist. Here $\xi = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, and example 2.4.7 computes its powers outright:

$$\xi^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}], \qquad 0 \le k \le n$$

with $\xi^{n+1} \cap [\mathbb{P}^n] = 0$. The tautological class of the trivial bundle over a point is the hyperplane class.

2. *A line bundle.* Let $E = L$ have rank one, so $r = 0$. A one-dimensional subspace of a one-dimensional fibre is the whole fibre, so $\mathbb{P}(L) = X$ and $\pi = \mathrm{id}$; in the Proj description, $\mathrm{Sym}^\bullet$ of an invertible sheaf has $\mathrm{Proj}\,(\mathrm{Sym}^\bullet \mathscr{E}^\vee) = X$ and $\mathcal{O}(1) = \mathscr{E}^\vee$, where $\mathscr{E}$ is the invertible sheaf of sections of $L$. The tautological subbundle is $L$ itself, hence $\mathcal{O}_{\mathbb{P}(L)}(1) = L^\vee$ and, by proposition 2.4.4(3),
$$\xi = c_1(L^\vee) = -c_1(L)$$
The sign is forced by the convention: the tautological object is a *sub*bundle, and $\mathcal{O}(1)$ is its dual.

The tautological class lives on $\mathbb{P}(E)$, and the Chow groups one wants to compute with live on $X$. Transporting a class $\alpha$ on $X$ up to $\mathbb{P}(E)$, cutting it with powers of $\xi$, and pushing the result back down produces an operator on $A_*(X)$ built from $E$ alone. The bookkeeping is forced: $\pi^*$ raises dimension by $r$, capping with $\xi^m$ lowers it by $m$, and $\pi_*$ preserves it, so the composite lowers dimension by $m - r$. Writing $m = r + i$ makes $i$ the amount of dimension lost, and these composites are the Segre classes.

**Definition 3.1.3: Segre Class Operators of a Vector Bundle**

Let $E$ be a vector bundle of rank $r + 1$ on an algebraic scheme $X$, with $\pi\colon \mathbb{P}(E) \to X$ and tautological class $\xi$. For an integer $i \geq -r$ the *$i$-th Segre class* of $E$ is the operator defined on $\alpha \in A_k(X)$ by
$$s_i(E) \cap \alpha \;=\; \pi_*\big(\xi^{\,r+i} \cap \pi^*\alpha\big)$$
and $s_i(E) \cap \alpha = 0$ for $i < -r$. Each is a homomorphism $s_i(E) \cap -\colon A_k(X) \to A_{k-i}(X)$. The *total Segre class* of $E$ is the sum
$$s(E) \;=\; \sum_{i \geq -r} s_i(E)$$
which is a finite sum of operators on any fixed class, since $s_i(E)$ lowers dimension by $i$ and $A_j(X) = 0$ for $j < 0$.

That $s_i(E) \cap -$ is a homomorphism is inherited from its three constituents: $\pi^*$ is additive by proposition 1.3.5(1), capping with $\xi$ is additive by proposition 2.4.4(1), and $\pi_*$ is additive by definition 1.2.1. Each of the three also descends to rational equivalence, by proposition 1.3.7, proposition 2.2.4 and theorem 1.2.3 respectively, so $s_i(E) \cap -$ is well defined on Chow groups without further argument. The content of the definition is not that these operators exist but that they are computable and that they encode $E$; the rank-one case already shows what they look like.

**Example 3.1.4: The Segre Classes of a Line Bundle**

Let $L$ be a line bundle on $X$, so $r = 0$ and, by example 3.1.2(2), $\mathbb{P}(L) = X$ with $\pi = \mathrm{id}$ and $\xi = -c_1(L)$. The definition reads
$$s_i(L) \cap \alpha = \xi^{\,i} \cap \alpha = (-1)^i\, c_1(L)^i \cap \alpha, \qquad i \geq 0$$
there being no indices below $-r = 0$. The total Segre class is therefore the alternating series
$$s(L) = 1 - c_1(L) + c_1(L)^2 - c_1(L)^3 + \cdots$$
which on any fixed class terminates, since $c_1(L)^m \cap \alpha$ lies in $A_{k-m}(X)$ and this group vanishes once $m > k$. The series is the inverse of $1 + c_1(L)$: capping $s(L) \cap \alpha$ with $1 + c_1(L)$ telescopes,

every term cancelling against its successor, and leaves $\alpha$.
For a concrete instance take $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(1)$. Then $c_1(L)^i \cap [\mathbb{P}^n] = [\mathbb{P}^{n-i}]$ by example 2.4.7, so

$$s_i\big(\mathcal{O}_{\mathbb{P}^n}(1)\big) \cap [\mathbb{P}^n] = (-1)^i\,[\mathbb{P}^{n-i}], \qquad 0 \le i \le n$$

and $s_i(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n] = 0$ for $i > n$. The Segre classes of a line bundle carry exactly the information in $c_1$, packaged with alternating signs.

The alternating signs in example 3.1.4 indicate that $s(E)$ is the wrong normalization for a class one wants to be additive on direct sums; the repair, carried out in section 3.2, is to invert the series. Before that, the operators need their formal properties: which of them vanish, which one is the identity, whether they commute, and how they behave under the two functorialities of the Chow group. All four are consequences of the corresponding properties of $c_1$ and of the flat-proper calculus of chapter 1.

**Proposition 3.1.5: Basic Properties of Segre Classes**

Let $E$ be a vector bundle of rank $r+1$ on an algebraic scheme $X$.

1. *Vanishing.* $s_i(E) \cap \alpha = 0$ for every $i < 0$ and every $\alpha \in A_*(X)$.
2. *Normalization.* $s_0(E) \cap \alpha = \alpha$ for every $\alpha \in A_*(X)$; that is, $s_0(E)$ is the identity operator. Consequently $s(E) = 1 + s_1(E) + s_2(E) + \cdots$.
3. *Commutativity.* If $F$ is a second vector bundle on $X$, then for all $i, j$

$$s_i(E) \cap \big(s_j(F) \cap \alpha\big) = s_j(F) \cap \big(s_i(E) \cap \alpha\big)$$

4. *Functoriality.* If $f\colon X' \to X$ is proper and $\alpha' \in A_k(X')$, then

$$f_*\big(s_i(f^*E) \cap \alpha'\big) = s_i(E) \cap f_*\alpha'$$

   and if $g\colon X'' \to X$ is flat of relative dimension $m$ and $\beta \in A_k(X)$, then

$$s_i(g^*E) \cap g^*\beta = g^*\big(s_i(E) \cap \beta\big)$$

***Proof*:**
Formation of $\mathbb{P}(E)$ and of $\mathcal{O}_{\mathbb{P}(E)}(1)$ from $\mathrm{Sym}^\bullet \mathscr{E}^\vee$ commutes with base change along an arbitrary morphism ([2, Relative Proj Commutes with Base Change]), so for any morphism $h\colon Y \to X$ there is a fibre square

$$\begin{array}{ccc} \mathbb{P}(h^*E) & \xrightarrow{h'} & \mathbb{P}(E) \\ \downarrow{\scriptstyle \pi_Y} & & \downarrow{\scriptstyle \pi} \\ Y & \xrightarrow{h} & X \end{array}$$

with $\mathbb{P}(h^*E) = \mathbb{P}(E) \times_X Y$ and $\mathcal{O}_{\mathbb{P}(h^*E)}(1) = h'^*\mathcal{O}_{\mathbb{P}(E)}(1)$, hence $\xi_Y = h'^*\xi$ for the tautological classes. This square is used in every part.

1. *Vanishing.* By additivity it suffices to treat $\alpha = [V]$ for a $k$-dimensional subvariety $V \subseteq X$, with closed immersion $h\colon V \hookrightarrow X$. Since $h$ is proper and $\pi$ is flat, proposition 1.3.6 applied

to the square above gives $\pi^* h_* = h'_* \pi_V^*$, so

$$\pi^*[V] = h'_* \, \pi_V^*[V]$$

and proposition 2.4.4(4) for the proper $h'$, applied $r+i$ times with $\xi_V = h'^*\xi$, gives

$$\xi^{\,r+i} \cap \pi^*[V] = h'_* \big(\xi_V^{r+i} \cap \pi_V^*[V]\big)$$

Pushing forward by $\pi$ and using $\pi \circ h' = h \circ \pi_V$ with functoriality of proper pushforward,

$$s_i(E) \cap [V] = h_* \, \pi_{V*} \big(\xi_V^{r+i} \cap \pi_V^*[V]\big) \tag{3.1}$$

The class inside $h_*$ lies in $A_{k-i}(V)$, and for $i < 0$ one has $k - i > k = \dim V$, so that group is zero: $V$ has no subvariety of dimension exceeding its own. Hence $s_i(E) \cap [V] = 0$.

2. *Normalization.* The inner class in (3.1) is $s_i(E|_V) \cap [V]$ computed on $V$, so that identity reduces the claim to the case $\alpha = [X]$ with $X$ a variety of dimension $k$, where it reads $\pi_*(\xi^{\,r} \cap [\mathbb{P}(E)]) = [X]$; note that $\pi^*[X] = [\mathbb{P}(E)]$ by proposition 1.3.5(2). The total space $\mathbb{P}(E)$ is a variety: it is covered by open sets isomorphic to $U_\lambda \times \mathbb{P}^r$ for a trivializing cover $\{U_\lambda\}$ of $X$ ([2, Local Triviality of a Projective Bundle]), each of which is irreducible and reduced, and any two of them meet because their images in the irreducible $X$ do. So $\dim \mathbb{P}(E) = k + r$ and $\pi_*(\xi^r \cap [\mathbb{P}(E)])$ lies in $A_k(X) = Z_k(X) = \mathbb{Z}\,[X]$ by example 1.1.6(3); write it as $m\,[X]$ with $m \in \mathbb{Z}$.

   The integer $m$ is computed after restriction to a trivializing open. Choose a nonempty open $U \subseteq X$ with $E|_U$ trivial, and let $j\colon U \hookrightarrow X$ and $j'\colon \mathbb{P}(E|_U) \hookrightarrow \mathbb{P}(E)$ be the inclusions, both flat of relative dimension zero. Flat pullback commutes with $\pi_*$ across the fibre square by proposition 1.3.6 and with capping by $\xi$ by proposition 2.4.4(4), and $j'^*[\mathbb{P}(E)] = [\mathbb{P}(E|_U)]$ because restriction along an open immersion carries a fundamental cycle to a fundamental cycle (example 1.3.4(1)), so

$$j^*\big(m\,[X]\big) = \pi_{U*}\big(\xi_U^r \cap [\mathbb{P}(E|_U)]\big)$$

   Under a trivialization $\mathbb{P}(E|_U) \cong U \times \mathbb{P}^r$ the twisting sheaf becomes $\rho^*\mathcal{O}_{\mathbb{P}^r}(1)$ for the second projection $\rho\colon U \times \mathbb{P}^r \to \mathbb{P}^r$ ([2, Twisting Sheaf]), and $\rho$ is flat of relative dimension $\dim U$, so $\rho^*[\mathbb{P}^r] = [U \times \mathbb{P}^r]$ by proposition 1.3.5(2). Hence, by the flat case of proposition 2.4.4(4) and by example 2.4.7,

$$\xi_U^r \cap [U \times \mathbb{P}^r] = \rho^*\big(c_1(\mathcal{O}_{\mathbb{P}^r}(1))^r \cap [\mathbb{P}^r]\big) = \rho^*[\mathrm{pt}] = [U \times \{\mathrm{pt}\}]$$

   The projection $\pi_U$ restricts to an isomorphism $U \times \{\mathrm{pt}\} \to U$, so $\pi_{U*}[U \times \{\mathrm{pt}\}] = [U]$ by definition 1.2.1. Thus $m\,[U] = [U]$ in $A_k(U) = \mathbb{Z}\,[U]$, which is free on the fundamental class by example 1.1.6(3), and therefore $m = 1$. This gives $s_0(E) \cap [X] = [X]$, and with part (1) the total class has no terms in negative degree.

3. *Commutativity.* Let $F$ have rank $f+1$, with $q\colon \mathbb{P}(F) \to X$ and tautological class $\eta$, and form the fibre product $W = \mathbb{P}(E) \times_X \mathbb{P}(F)$ with its two projections $q'\colon W \to \mathbb{P}(E)$ and $p'\colon W \to \mathbb{P}(F)$. Since $q$ is proper and $\pi$ is flat, $q'$ is proper and $p'$ is flat of relative dimension $r$; write $\rho = \pi \circ q' = q \circ p'\colon W \to X$, which is proper, and flat of relative

dimension $r+f$ by proposition 1.3.5(3). Put $\tilde{\xi} = q'^*\xi$ and $\tilde{\eta} = p'^*\eta$. Then

$$\begin{aligned} s_i(E) \cap \big(s_j(F) \cap \alpha\big) &= \pi_*\Big(\xi^{\,r+i} \cap \pi^* q_*\big(\eta^{\,f+j} \cap q^*\alpha\big)\Big) \\ &= \pi_*\Big(\xi^{\,r+i} \cap q'_* p'^*\big(\eta^{\,f+j} \cap q^*\alpha\big)\Big) \\ &= \pi_*\Big(\xi^{\,r+i} \cap q'_*\big(\tilde{\eta}^{\,f+j} \cap \rho^*\alpha\big)\Big) \\ &= \pi_* q'_*\Big(\tilde{\xi}^{\,r+i} \cap \tilde{\eta}^{\,f+j} \cap \rho^*\alpha\Big) = \rho_*\Big(\tilde{\xi}^{\,r+i} \cap \tilde{\eta}^{\,f+j} \cap \rho^*\alpha\Big) \end{aligned}$$

the second line by proposition 1.3.6 for the fibre square defining $W$, the third by the flat case of proposition 2.4.4(4) together with $p'^*q^* = \rho^*$ (proposition 1.3.5(3)), and the fourth by the proper case of proposition 2.4.4(4) for $q'$. Running the same computation with the roles of $E$ and $F$ exchanged uses the same scheme $W$ with its two projections interchanged, and expresses $s_j(F) \cap (s_i(E) \cap \alpha)$ as $\rho_*(\tilde{\eta}^{\,f+j} \cap \tilde{\xi}^{\,r+i} \cap \rho^*\alpha)$. The two expressions differ only in the order of two first Chern class cap operators, which commute by proposition 2.4.4(2), so they agree; this is the asserted commutativity.

4. *Functoriality.* For the proper case take $h = f$ in the fibre square above, so $f' : \mathbb{P}(f^*E) \to \mathbb{P}(E)$ is proper and $\pi_{X'}$ is proper and flat of relative dimension $r$. Then

$$f_*\big(s_i(f^*E) \cap \alpha'\big) = f_*\, \pi_{X'*}\big(\xi_{X'}^{r+i} \cap \pi_{X'}^*\alpha'\big) = \pi_*\, f'_*\big(f'^*\xi^{\,r+i} \cap \pi_{X'}^*\alpha'\big)$$

using $f \circ \pi_{X'} = \pi \circ f'$ and $\xi_{X'} = f'^*\xi$. The proper case of proposition 2.4.4(4) moves the cap outside, and proposition 1.3.6 for the same square gives $f'_*\pi_{X'}^* = \pi^* f_*$, so the right side is $\pi_*(\xi^{\,r+i} \cap \pi^* f_*\alpha') = s_i(E) \cap f_*\alpha'$.

For the flat case take $h = g$, so $g' : \mathbb{P}(g^*E) \to \mathbb{P}(E)$ is flat of relative dimension $m$. By proposition 1.3.5(3), $\pi_{X''}^* g^* = (g \circ \pi_{X''})^* = (\pi \circ g')^* = g'^*\pi^*$, and by the flat case of proposition 2.4.4(4) with $\xi_{X''} = g'^*\xi$,

$$\xi_{X''}^{r+i} \cap \pi_{X''}^* g^*\beta = g'^*\big(\xi^{\,r+i} \cap \pi^*\beta\big)$$

Applying $\pi_{X''*}$ and using $\pi_{X''*}g'^* = g^*\pi_*$, again proposition 1.3.6, gives $s_i(g^*E) \cap g^*\beta = g^*(s_i(E) \cap \beta)$, completing the proof.

Parts (1) and (2) say that the total Segre class is an operator of the form 1+(dimension-lowering terms), hence invertible: a series $1 + s_1 + s_2 + \cdots$ whose higher terms strictly lower dimension has a two-sided inverse of the same shape, computed by the usual recursion, and section 3.2 names that inverse the total Chern class. Part (3) is what makes any such formal manipulation legitimate, since inverting a series requires its terms to commute. Part (4) says the Segre classes are natural: they are computed in whichever space is most convenient and transported by the functorialities already available. The normalization in part (2) has an immediate structural consequence for $\pi^*$ itself.

**Corollary 3.1.6: Flat Pullback along a Projective Bundle is Split Injective**

Let $E$ be a vector bundle of rank $r+1$ on algebraic scheme $X$, with $\pi : \mathbb{P}(E) \to X$ and tautological class $\xi$. Then for every $k$ the map $\pi^* : A_k(X) \to A_{k+r}(\mathbb{P}(E))$ is injective, and $\beta \mapsto \pi_*(\xi^{\,r} \cap \beta)$ is a left inverse for it.

***Proof*:**
For $\alpha \in A_k(X)$ the composite is $\pi_*(\xi^r \cap \pi^*\alpha) = s_0(E) \cap \alpha = \alpha$ by definition 3.1.3 and proposition 3.1.5(2). A map admitting a left inverse is injective, as we sought to show.

No cycle class on the base is lost on passing to the projective bundle, and the loss can be undone by an explicit formula rather than an abstract argument. This is the mechanism that will make the flag bundle a legitimate device for proving identities among Chern classes rather than a heuristic; the iteration is carried out in section 3.2. A second consequence of proposition 3.1.5 is a computation, and it is the one that calibrates the whole normalization.

**Example 3.1.7: The Segre Classes of a Trivial Bundle**

Let $E = \mathcal{O}_X^{\oplus(r+1)}$ be trivial of rank $r+1$, so $\mathbb{P}(E) = X \times \mathbb{P}^r$ with $\pi$ the first projection, $\rho$ the second, and $\mathcal{O}_{\mathbb{P}(E)}(1) = \rho^*\mathcal{O}_{\mathbb{P}^r}(1)$, whence $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$. Then

$$s_0(E) = 1, \qquad s_i(E) = 0 \ \text{ for } i \neq 0$$

so $s(E) = 1$.

The case $i < 0$ is proposition 3.1.5(1) and the case $i = 0$ is proposition 3.1.5(2), so only $i \geq 1$ needs an argument. By (3.1) it suffices to take $X = V$ a variety of dimension $k$ and $\alpha = [V]$. Then $\pi^*[V] = [V \times \mathbb{P}^r]$, and $[V \times \mathbb{P}^r] = \rho^*[\mathbb{P}^r]$ because $\rho$ is flat of relative dimension $k$ over the variety $\mathbb{P}^r$ (proposition 1.3.5(2)). The flat case of proposition 2.4.4(4) then moves the whole cap into the base:

$$\xi^{r+i} \cap \pi^*[V] = \rho^*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^r}(1)\big)^{r+i} \cap [\mathbb{P}^r]\Big)$$

For $i \geq 1$ the exponent exceeds $r = \dim \mathbb{P}^r$, so the class in brackets vanishes by proposition 2.4.4(5), and therefore $s_i(E) \cap [V] = \pi_*(0) = 0$. A trivial bundle has trivial total Segre class, which is the normalization that makes $s(E)$, and with it the Chern classes built from it, an invariant of the twisting of $E$ rather than of its rank.

The computation in example 3.1.7 used only the structure of $\mathbb{P}^r$ over a point, transported along a flat projection. The same transport, applied to a hyperplane rather than to the whole space, produces the identity that drives the main theorem: on a trivial projective bundle, the tautological class is the class of a sub-bundle of hyperplanes, so capping with $\xi$ is the same as pushing forward from a smaller projective bundle.

**Lemma 3.1.8: The Hyperplane Divisor on a Trivial Projective Bundle**

Let $X$ be an algebraic scheme, let $r \geq 1$, and let $P = X \times \mathbb{P}^r$ with projections $\pi \colon P \to X$ and $\rho \colon P \to \mathbb{P}^r$, and $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$. Fix a hyperplane $H \subseteq \mathbb{P}^r$, put $Q = X \times H$ with its closed immersion $\iota \colon Q \hookrightarrow P$, and let $\pi_Q \colon Q \to X$ be the projection. Then for every $\alpha \in A_k(X)$,

$$\xi \cap \pi^*\alpha = \iota_*\, \pi_Q^*\alpha \quad \text{in } A_{k+r-1}(P)$$

***Proof*:**
Since $H$ is cut out by a linear form, $\mathcal{O}_{\mathbb{P}^r}(H) \cong \mathcal{O}_{\mathbb{P}^r}(1)$ by example 2.4.3(2). No component of $P$ maps into $H$, so the local equations of $H$ pull back to local equations of a Cartier divisor $Q$ on $P$, and $\mathcal{O}_P(Q) \cong \rho^*\mathcal{O}_{\mathbb{P}^r}(1)$ because $\mathcal{O}(D)$ is formed from those same local equations. (This is

the pullback of a Cartier divisor along an arbitrary morphism meeting the support properly, not the flat pullback of proposition 2.3.3, whose relative-dimension hypothesis $\rho$ need not satisfy when $X$ has components of different dimensions.) By definition 2.4.1 this identifies the operator: $\xi \cap \pi^*\alpha = Q \cdot \pi^*\alpha$.

Both sides of the asserted identity are additive homomorphisms $A_k(X) \to A_{k+r-1}(P)$, the left by proposition 1.3.7 and proposition 2.2.4, the right by proposition 1.3.7 and theorem 1.2.3, so it suffices to check them on the class of a $k$-dimensional subvariety $W \subseteq X$. The morphism $\pi$ is flat of relative dimension $r$, being the base change of $\mathbb{P}^r \to \operatorname{Spec} \overline{k}$, so $\pi^*[W] = [W \times \mathbb{P}^r]$ by definition 1.3.3, the preimage $W \times \mathbb{P}^r$ being integral of dimension $k + r$ and hence carrying multiplicity one. This subvariety is not contained in $|Q| = X \times H$, so the first clause of definition 2.2.1 applies: writing $s$ for the rational section of $\mathcal{O}_P(Q)$ cutting out $Q$, which is the pullback along $\rho$ of the linear form defining $H$, its restriction to $W \times \mathbb{P}^r$ has divisor

$$\big[\operatorname{div}\big(s|_{W\times\mathbb{P}^r}\big)\big] = [W \times H]$$

with multiplicity one, since a linear form vanishes to order one along the hyperplane it defines. Hence $Q \cdot \pi^*[W] = [W \times H]$. On the other side $\pi_Q$ is flat of relative dimension $r - 1$, so $\pi_Q^*[W] = [W \times H]$ by the same reasoning, and $\iota_*$ carries this class to $[W \times H]$ in $Z_{k+r-1}(P)$, the closed immersion having degree one on each subvariety (definition 1.2.1). The two sides therefore agree on every generator, completing the proof.

The lemma converts a divisor operation into a pushforward, and pushforward is what the localization sequence controls. That is the whole strategy for the theorem below: peel $\mathbb{P}^r$ apart into a hyperplane and its affine complement, use theorem 1.4.6 on the complement, use the lemma to recognize the hyperplane contribution as one more power of $\xi$, and induct. The general bundle is then reduced to the trivial one by a second induction, over the closed subschemes of the base, since a bundle is trivial over some dense open set. Injectivity, by contrast, needs no induction at all: the Segre classes already invert the construction.

**Theorem 3.1.9: The Projective Bundle Formula**

Let $E$ be a vector bundle of rank $r + 1$ on an algebraic scheme $X$, with $\pi\colon \mathbb{P}(E) \to X$ and tautological class $\xi$. For every $k$ the homomorphism

$$\Psi\colon \bigoplus_{i=0}^{r} A_{k-r+i}(X) \longrightarrow A_k\big(\mathbb{P}(E)\big), \qquad (\alpha_0, \dots, \alpha_r) \longmapsto \sum_{i=0}^{r} \xi^{\,i} \cap \pi^*\alpha_i$$

is an isomorphism. Every class $\beta \in A_k(\mathbb{P}(E))$ is thus uniquely of the form $\beta = \sum_{i=0}^{r} \xi^{\,i} \cap \pi^*\alpha_i$ with $\alpha_i \in A_{k-r+i}(X)$, and the components are recovered from $\beta$ by the triangular system

$$\pi_*\big(\xi^{\,m} \cap \beta\big) = \alpha_{r-m} + \sum_{j=1}^{m} s_j(E) \cap \alpha_{r-m+j}, \qquad 0 \le m \le r \tag{3.2}$$

***Proof*:**
Each summand of $\Psi$ lands in $A_k(\mathbb{P}(E))$: for $\alpha_i \in A_{k-r+i}(X)$ the class $\pi^*\alpha_i$ lies in $A_{k+i}(\mathbb{P}(E))$ and capping with $\xi^i$ lowers dimension by $i$. The map is additive because $\pi^*$ and $\xi \cap -$ are.

*Step 1: the triangular system, and injectivity.* Let $\beta = \sum_{i=0}^{r} \xi^i \cap \pi^*\alpha_i$ and fix $m \geq 0$. Then

$$\pi_*(\xi^m \cap \beta) = \sum_{i=0}^{r} \pi_*(\xi^{m+i} \cap \pi^*\alpha_i) = \sum_{i=0}^{r} s_{m+i-r}(E) \cap \alpha_i$$

directly from definition 3.1.3. By proposition 3.1.5(1) the terms with $m+i-r<0$ vanish, and by proposition 3.1.5(2) the term with $m+i-r=0$ is $\alpha_{r-m}$; writing $j = m+i-r$ for the surviving terms, which satisfy $1 \leq j \leq m$ since $i \leq r$, gives exactly (3.2).

Suppose now $\Psi(\alpha_0, \ldots, \alpha_r) = 0$. Taking $m = 0$ in (3.2) gives $\alpha_r = 0$. Assume inductively that $\alpha_r = \cdots = \alpha_{r-m+1} = 0$ for some $m \leq r$; then every correction term in (3.2) involves one of these vanishing classes, so the identity reduces to $\alpha_{r-m} = 0$. Descending from $m=0$ to $m=r$ kills every component, so $\Psi$ is injective. This argument used no hypothesis on $X$ beyond those standing.

*Step 2: surjectivity for a trivial bundle.* Let $E = \mathcal{O}_X^{\oplus(r+1)}$, so $\mathbb{P}(E) = X \times \mathbb{P}^r$ and $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$ for the second projection $\rho$. The claim is proved by induction on $r$, uniformly in the base $X$.

For $r=0$ one has $\mathbb{P}(E) = X$ with $\pi = \mathrm{id}$, so $\Psi(\alpha_0) = \alpha_0$ is the identity map of $A_k(X)$, which is surjective. Let $r \geq 1$ and assume the claim for $r-1$ over every base. Fix a hyperplane $H \subseteq \mathbb{P}^r$, put $Q = X \times H$ with closed immersion $\iota\colon Q \hookrightarrow P = X \times \mathbb{P}^r$, and let $j\colon X \times \mathbb{A}^r \hookrightarrow P$ be the inclusion of the open complement, $\mathbb{A}^r = \mathbb{P}^r \setminus H$. The localization sequence (theorem 1.4.1) reads

$$A_k(Q) \xrightarrow{\iota_*} A_k(P) \xrightarrow{j^*} A_k(X \times \mathbb{A}^r) \longrightarrow 0$$

Let $q = \pi \circ j\colon X \times \mathbb{A}^r \to X$ be the projection. By affine-bundle invariance (theorem 1.4.6) the pullback $q^*\colon A_{k-r}(X) \to A_k(X \times \mathbb{A}^r)$ is an isomorphism, and $q^* = j^*\pi^*$ by proposition 1.3.5(3). Given $\beta \in A_k(P)$, there is therefore $\alpha_0 \in A_{k-r}(X)$ with $j^*\beta = q^*\alpha_0 = j^*(\pi^*\alpha_0)$, whence $\beta - \pi^*\alpha_0$ lies in $\ker j^* = \operatorname{im} \iota_*$ by exactness: say

$$\beta = \pi^*\alpha_0 + \iota_*\gamma, \qquad \gamma \in A_k(Q)$$

Now $Q = X \times H \cong X \times \mathbb{P}^{r-1}$ is the trivial projective bundle of one lower fibre dimension, and its tautological class is $\xi_Q = \iota^*\xi$: choosing a second hyperplane $H' \neq H$, the restriction $\mathcal{O}_{\mathbb{P}^r}(1)|_H \cong \mathcal{O}_{\mathbb{P}^r}(H')|_H$ is the line bundle of the hyperplane $H' \cap H$ of $H$, which is $\mathcal{O}_{\mathbb{P}^{r-1}}(1)$ under the identification $H \cong \mathbb{P}^{r-1}$. By the inductive hypothesis,

$$\gamma = \sum_{i=0}^{r-1} \xi_Q^i \cap \pi_Q^*\alpha_i', \qquad \alpha_i' \in A_{k-(r-1)+i}(X)$$

Pushing forward, the proper case of proposition 2.4.4(4) applied $i$ times with $\xi_Q = \iota^*\xi$ gives $\iota_*(\xi_Q^i \cap \pi_Q^*\alpha_i') = \xi^i \cap \iota_*\pi_Q^*\alpha_i'$, and lemma 3.1.8 identifies $\iota_*\pi_Q^*\alpha_i' = \xi \cap \pi^*\alpha_i'$. Hence

$$\iota_*\gamma = \sum_{i=0}^{r-1} \xi^{i+1} \cap \pi^*\alpha_i'$$

and setting $\alpha_{i+1} = \alpha_i' \in A_{k-r+(i+1)}(X)$ for $0 \leq i \leq r-1$ exhibits $\beta = \sum_{i=0}^{r} \xi^i \cap \pi^*\alpha_i$. This completes the induction on $r$, so $\Psi$ is surjective for every trivial bundle.

*Step 3: surjectivity in general.* The scheme $X$ is of finite type over $\overline{k}$, hence Noetherian, so its closed subschemes satisfy the descending chain condition and the following suffices: if $\Psi_{E|_W}$ is surjective in every degree for every proper closed subscheme $W \subsetneq X$, then $\Psi_E$ is surjective. The empty scheme starts the induction.

First produce a dense trivializing open. The scheme $X$ has finitely many irreducible components, with generic points $\eta_1, \ldots, \eta_s$. For each $t$ choose an open $V_t \ni \eta_t$ over which $E$ is trivial, and set $U_t = V_t \setminus \bigcup_{u \neq t} \overline{\{\eta_u\}}$, still an open neighbourhood of $\eta_t$ since $\eta_t$ lies on no other component. The $U_t$ are pairwise disjoint: each $U_u$ is contained in $\overline{\{\eta_u\}}$, because the components cover $X$, while $U_t$ misses $\overline{\{\eta_u\}}$ for $u \neq t$. Their union $U = \bigsqcup_t U_t$ is an open set containing every generic point, hence dense, and $E|_U$ is trivial of rank $r+1$, being trivial on each of the disjoint pieces. If $U = X$ then $E$ is trivial and Step 2 applies; otherwise let $Z = X \setminus U$ with its reduced structure, a proper closed subscheme.

Write $j\colon U \hookrightarrow X$ and $i_Z\colon Z \hookrightarrow X$ for the inclusions, and $j'\colon \mathbb{P}(E|_U) \hookrightarrow \mathbb{P}(E)$, $\iota\colon \mathbb{P}(E|_Z) \hookrightarrow \mathbb{P}(E)$ for the induced ones; these are the open and closed pieces of $\mathbb{P}(E)$, since forming $\mathbb{P}(E)$ commutes with base change. Let $\beta \in A_k(\mathbb{P}(E))$. Restricting to the trivializing part and applying Step 2 over $U$,

$$j'^*\beta = \sum_{i=0}^{r} \xi_U^{\,i} \cap \pi_U^* \alpha_i^U, \qquad \alpha_i^U \in A_{k-r+i}(U)$$

By theorem 1.4.1 the restriction $j^*\colon A_{k-r+i}(X) \to A_{k-r+i}(U)$ is surjective, so each $\alpha_i^U$ lifts to some $\alpha_i \in A_{k-r+i}(X)$. Flat pullback along $j'$ commutes with $\pi^*$, since $\pi \circ j' = j \circ \pi_U$ gives $j'^*\pi^* = \pi_U^* j^*$ by proposition 1.3.5(3), and with capping by $\xi$ by the flat case of proposition 2.4.4(4), using $\xi_U = j'^*\xi$. Hence $j'^*\big(\beta - \Psi_E(\alpha_0, \ldots, \alpha_r)\big) = 0$, and the localization sequence for the closed $\mathbb{P}(E|_Z) \subseteq \mathbb{P}(E)$ with open complement $\mathbb{P}(E|_U)$ produces $\delta \in A_k(\mathbb{P}(E|_Z))$ with

$$\beta - \Psi_E(\alpha_0, \ldots, \alpha_r) = \iota_*\delta$$

Since $Z \subsetneq X$ is a proper closed subscheme, the inductive hypothesis writes $\delta = \sum_{i=0}^{r} \xi_Z^{\,i} \cap \pi_Z^* \epsilon_i$ with $\epsilon_i \in A_{k-r+i}(Z)$. The proper case of proposition 2.4.4(4) with $\xi_Z = \iota^*\xi$, followed by proposition 1.3.6 for the fibre square over $i_Z$ (whose horizontal map is proper and whose vertical map $\pi$ is flat), gives

$$\iota_*\big(\xi_Z^{\,i} \cap \pi_Z^*\epsilon_i\big) = \xi^{\,i} \cap \iota_*\pi_Z^*\epsilon_i = \xi^{\,i} \cap \pi^*\big(i_{Z*}\epsilon_i\big)$$

Therefore $\beta = \Psi_E(\alpha_0 + i_{Z*}\epsilon_0, \ldots, \alpha_r + i_{Z*}\epsilon_r)$ lies in the image of $\Psi_E$. This closes the Noetherian induction, so $\Psi$ is surjective for every vector bundle, and with Step 1 it is an isomorphism, completing the proof.

The theorem computes the Chow groups of a projective bundle from those of its base, and it does so with an explicit basis and an explicit inversion. The classes $1, \xi, \ldots, \xi^r$ behave like a basis of $A_*(\mathbb{P}(E))$ over $A_*(X)$, and it is standard to say that $A_*(\mathbb{P}(E))$ is a free $A_*(X)$-module on them; that phrasing anticipates a ring structure on $A_*(X)$ that does not exist yet, so the content here is the isomorphism $\Psi$ of graded groups together with the fact that its components are the operators $\xi^i \cap \pi^*(-)$. The inversion (3.2) is what makes the statement usable in practice: to identify the components of a given class one pushes forward its caps with successive powers of $\xi$, and the Segre classes appear as the correction terms in a triangular system that can be solved from the top down. This is also the precise sense in which the Segre classes measure the failure of $\xi$ to behave like the hyperplane class on a product, since a trivial bundle has all corrections zero by example 3.1.7.

Taking $X = \operatorname{Spec} \overline{k}$ and $E = \overline{k}^{\,n+1}$ recovers $A_k(\mathbb{P}^n) = \mathbb{Z}\left(\xi^{\,n-k} \cap [\mathbb{P}^n]\right)$, the statement of theorem 1.5.2: the group $A_j(\operatorname{Spec} \overline{k})$ is $\mathbb{Z}$ for $j = 0$ and zero otherwise, so only the index $i = n - k$ survives in the sum, and $\pi^*[\operatorname{Spec} \overline{k}] = [\mathbb{P}^n]$. The theorem is thus the exact generalization of the projective-space computation from a point to an arbitrary base, which is how it was announced in section 1.5, and its proof runs by reducing to that computation twice: once through example 2.4.7 in the normalization $s_0(E) = 1$, and once through affine-bundle invariance in Step 2. The corresponding statement in singular cohomology, that $H^*(\mathbb{P}(E))$ is free over $H^*(X)$ on the powers of the class of $\mathcal{O}(1)$, holds as well, but it enters the theory at a different point: in [9] the topological Chern classes are defined as pullbacks of the polynomial generators of $H^*(BU(n))$, and the projective-bundle statement is a consequence, proved once they are in hand. Here the dependence runs the other way, the formula above coming first and the Chern operators of section 3.2 being defined by inverting the total Segre class; and the argument owes nothing to topology, being valid over any algebraically closed field and on singular schemes.

### Exercise 3.1.1

1. Let $L$ be a line bundle on an algebraic scheme $X$. Using example 3.1.2(2), show that $s_i(L) \cap \alpha = (-1)^i c_1(L)^i \cap \alpha$ for every $i \geq 0$ and every $\alpha \in A_k(X)$, and deduce that the operators $s(L) \cap -$ and $(1 + c_1(L)) \cap -$ are mutually inverse on $A_k(X)$. Explain why the series terminates on any fixed class.

2. Let $E$ have rank $r+1$ on $X$, with $\pi\colon \mathbb{P}(E) \to X$. Show that $\pi_*\pi^*\alpha = 0$ for every $\alpha \in A_*(X)$ when $r \geq 1$, while $\pi_*\pi^* = \mathrm{id}$ when $r = 0$. Show also that $\beta \mapsto \pi_*(\xi^r \cap \beta)$ is a left inverse of $\pi^*$ for every $r$, and reconcile the two computations.

3. Take $X = \mathbb{P}^n$ and $E = \mathcal{O}_{\mathbb{P}^n}^{\oplus(m+1)}$, so that $\mathbb{P}(E) = \mathbb{P}^n \times \mathbb{P}^m$ with $\pi$ the first projection. Use theorem 3.1.9 and theorem 1.5.2 to show that $A_k(\mathbb{P}^n \times \mathbb{P}^m)$ is free abelian of rank equal to the number of pairs $(a, b)$ with $0 \leq a \leq m$, $0 \leq b \leq n$ and $a + b = k$, and identify a basis geometrically.

4. Let $X$ be a complete variety of dimension $n$ and $E$ a vector bundle of rank $r+1$ on $X$. Show that
$$\int_X s_n(E) \cap [X] = \int_{\mathbb{P}(E)} \xi^{n+r} \cap [\mathbb{P}(E)]$$
and verify that both sides vanish for $E$ trivial and $n \geq 1$.

5. Let $E$ have rank $r+1$ on $X$ and let $\alpha \in A_k(X)$. By theorem 3.1.9 there are unique classes $\beta_i \in A_{k-1-r+i}(X)$ with $\xi^{r+1} \cap \pi^*\alpha = \sum_{i=0}^{r} \xi^i \cap \pi^*\beta_i$. Determine $\beta_r$ and $\beta_{r-1}$ in terms of the Segre operators of $E$ applied to $\alpha$, and describe the recursion that determines the remaining $\beta_i$.

## 3.2 Chern Classes

The Segre operators of section 3.1 record everything a vector bundle contributes to the Chow group, but in a form nothing can be computed with. The list $s_0(E), s_1(E), s_2(E), \ldots$ does not terminate, so no finite collection of them is an invariant attached to $E$; already for a line bundle every $s_i(L)$ is nonzero as long as $c_1(L)^i$ is. Worse, the operators attached to a direct sum are not the product of those attached to the summands, so a bundle assembled from simpler pieces has Segre operators that cannot be read off the pieces. Inverting the total Segre operator repairs both defects at once, and the inverse is what the rest of the book uses.

The section establishes four properties of the resulting Chern operators: they vanish above the rank of the bundle, they are natural for flat pullback and satisfy a projection formula, they are multiplicative on short exact sequences, and in rank one they reduce to the operator $c_1(L) \cap -$ of section 2.4. The last two are proved by the splitting principle, which replaces $X$ by a tower of projective bundles over it on which every bundle in sight is filtered by line bundles. That replacement is legitimate rather than heuristic because the projective-bundle formula makes $\pi^*$ injective, so an identity verified upstairs descends.

Throughout, $E$ is a vector bundle of rank $e \geq 1$ on an algebraic scheme $X$, the projection $\pi \colon \mathbb{P}(E) \to X$ has relative dimension $r = e - 1$, and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$ is the tautological class of section 3.1, so that

$$s_i(E) \cap \alpha = \pi_*\big(\xi^{r+i} \cap \pi^*\alpha\big)$$

for $\alpha \in A_*(X)$, with $s_0(E)$ the identity and $s_i(E) = 0$ for $i < 0$. Every vector bundle named below is assumed of positive rank: the operators are manufactured from $\mathbb{P}(E)$, which is empty for the zero bundle, so neither $s(0)$ nor $c(0)$ is defined.

The first point to settle is that the inversion makes sense. A Segre operator lowers the dimension of a cycle class, and on a scheme of finite dimension an operator that lowers dimension by too much has nowhere to land; that is the whole reason the total operator is invertible.

**Lemma 3.2.1: Dimension-Lowering Operators Vanish Above the Dimension**

Let $X$ be an algebraic scheme of dimension $n$. Call an operator on $A_*(X)$ *homogeneous of degree* $d$ if it carries $A_k(X)$ into $A_{k-d}(X)$ for every $k$. Then every operator on $A_*(X)$ homogeneous of degree $d > n$ is the zero operator.

***Proof*:**
A $k$-dimensional cycle class satisfies $0 \leq k \leq n$, because $X$ has no subvarieties of negative dimension and none of dimension exceeding $n$; hence $A_j(X) = 0$ for $j < 0$ and for $j > n$. An operator homogeneous of degree $d > n$ carries every $A_k(X)$ with $0 \leq k \leq n$ into $A_{k-d}(X)$ with $k - d < 0$, whose target vanishes, completing the proof.

Each $s_i(E)$ is homogeneous of degree $i$ (section 3.1): $\pi^*$ raises dimension by $r$ and capping with $\xi$ lowers it by one (definition 2.4.1). The lemma therefore applies to them and to every composite of them.

**Lemma 3.2.2: Invertibility of the Total Segre Operator**

Let $X$ be an algebraic scheme of dimension $n$ and $E$ a vector bundle on $X$. For $\alpha \in A_k(X)$ only finitely many of the classes $s_i(E) \cap \alpha$ are nonzero, so

$$s(E) := \sum_{i \geq 0} s_i(E)$$

is a well-defined operator on $A_*(X)$. It is invertible: there is a unique operator $\mu$ on $A_*(X)$ with

$$\mu \circ s(E) = s(E) \circ \mu = \mathrm{id}$$

and it is given by the finite sum $\mu = \sum_{j=0}^{n} (-1)^j \nu^j$, where $\nu = s(E) - \mathrm{id} = \sum_{i \geq 1} s_i(E)$.

***Proof*:**
By lemma 3.2.1, $s_i(E) = 0$ for $i > n$, so the sum defining $s(E)$ has at most $n+1$ nonzero terms and is a well-defined operator.

The same lemma makes $\nu = \sum_{i\geq 1} s_i(E)$ nilpotent. Expanding $\nu^{n+1}$ produces a sum of composites $s_{i_1}(E) \circ \cdots \circ s_{i_{n+1}}(E)$ with every $i_t \geq 1$, each homogeneous of degree $i_1 + \cdots + i_{n+1} \geq n+1 > n$, hence each zero by lemma 3.2.1. Thus $\nu^{n+1} = 0$, and the finite sum $\mu = \sum_{j=0}^{n} (-1)^j \nu^j$ satisfies

$$\mu\,(\mathrm{id} + \nu) = (\mathrm{id} + \nu)\,\mu = \sum_{j=0}^{n} (-1)^j \nu^j + \sum_{j=0}^{n} (-1)^j \nu^{j+1} = \mathrm{id} + (-1)^n \nu^{n+1} = \mathrm{id}$$

the two sums telescoping. Since $s(E) = \mathrm{id} + \nu$, the operator $\mu$ is a two-sided inverse. Uniqueness is the usual argument for inverses in a ring: if $\mu'$ is another, then $\mu' = \mu' \circ (s(E) \circ \mu) = (\mu' \circ s(E)) \circ \mu = \mu$, completing the proof.

Nilpotence, not any positivity or finiteness of $E$, is what powers the inversion. The bundle may have arbitrary rank and $X$ may be singular, reducible, or non-reduced; all that is used is that a cycle class cannot be cut down more times than the dimension of the ambient scheme allows. The inverse is therefore available on every algebraic scheme, which is the reason Chern classes in the Chow setting need no smoothness hypothesis.

**Definition 3.2.3: Chern Classes of a Vector Bundle**

Let $E$ be a vector bundle on an algebraic scheme $X$. The *total Chern class* of $E$ is the operator

$$c(E) := s(E)^{-1}$$

on $A_*(X)$ furnished by lemma 3.2.2, and the *i-th Chern class* $c_i(E)$ is its homogeneous component of degree $i$, an operator

$$c_i(E) \cap - \colon A_k(X) \longrightarrow A_{k-i}(X)$$

for every $k$. Equivalently, the $c_i(E)$ are determined by $c_0(E) = \mathrm{id}$ together with

$$\sum_{i+j=m} c_i(E) \circ s_j(E) = 0 \qquad \text{for every } m \geq 1$$

Unwinding the recursion $c_m(E) = -\sum_{j=1}^{m} c_{m-j}(E) \circ s_j(E)$ expresses each Chern class as an integer polynomial in the Segre classes:

$$\begin{aligned} &c_0(E) = \mathrm{id}, \qquad c_1(E) = -s_1(E), \qquad c_2(E) = s_1(E)^2 - s_2(E), \\ &c_3(E) = -s_1(E)^3 + 2\,s_1(E)s_2(E) - s_3(E) \end{aligned}$$

where a product of operators means composition. Each $c_m(E)$ is thus a $\mathbb{Z}$-combination of composites of $s_1(E), \ldots, s_m(E)$, a fact used below to transfer identities from Segre classes to Chern classes.

**3.2.4 Note: Chern Classes Are Operators, Not Elements** A Chern class here is an endomorphism of the graded group $A_*(X)$, not a member of it. There is no ring $A^*(X)$ yet: the intersection product that turns cycle classes into a ring for smooth $X$ rests on the deformation construction of chapter 4, and until it is available the only multiplication in sight is composition of operators. Expressions such as $c_1(E)^2$ or $c(E)c(F)$ always mean composites, and an equation between them is an equation of maps $A_*(X) \to A_*(X)$, asserted for every class $\alpha$. Once the product exists, capping against the fundamental class converts each operator into an element of $A^*(X)$ and the identities below become identities in a ring.

Every computation in this section runs through one imported object. The projective bundle carries a tautological line inside the pullback of $E$, and because this book and the scheme-theory volume use opposite conventions for $\mathbb{P}(-)$, that import has to be dualized in the open before it is used.

**3.2.5 Note: The Tautological Subbundle and Its Quotient** By box 0.0.3, the bundle of lines $\mathbb{P}(E)$ used here is $\operatorname{Proj}\left(\operatorname{Sym}^\bullet E^\vee\right)$, which is the projective bundle [2, the projective bundle $\mathbb{P}(\mathcal{E})$] of the *dual* sheaf. The universal surjection $\pi^*(E^\vee) \twoheadrightarrow \mathcal{O}_{\mathbb{P}(E)}(1)$ supplied there dualizes to an inclusion of vector bundles

$$\mathcal{O}_{\mathbb{P}(E)}(-1) \hookrightarrow \pi^* E$$

whose fibre over a point of $\mathbb{P}(E)$ lying above $x \in X$ and corresponding to a line $\ell \subseteq E_x$ is that line. Its cokernel $Q$ is locally free of rank $\operatorname{rank} E - 1$, since the inclusion is a subbundle inclusion: over an open $U \subseteq X$ trivializing $E$ the total space is $\mathbb{P}^r \times U$ and the inclusion is the tautological one on the first factor. Failing to dualize here replaces $E$ by $E^\vee$ throughout and negates every odd Chern class, which is why the convention is restated at each import.

The definition inverts a series and so says nothing recognizable on its face. The rank-one case makes it concrete and settles the compatibility the notation presumes: the symbol $c_1$ has already been used in section 2.4 for the divisor operator of a line bundle, and the two meanings must agree.

**Proposition 3.2.6: Chern Classes of a Line Bundle**

Let $L$ be a line bundle on an algebraic scheme $X$. Then

$$c(L) = \mathrm{id} + c_1(L)$$

where $c_1(L)$ on the right is the first Chern class operator of definition 2.4.1. In particular $c_1(L)$ in the sense of definition 3.2.3 is that operator, and $c_i(L) = 0$ for $i \geq 2$.

***Proof*:**
Both inputs are already computed in section 3.1: example 3.1.2(2) identifies $\mathbb{P}(L) = X$ with $\pi$ the identity and $\mathcal{O}_{\mathbb{P}(L)}(1) = L^\vee$, whence $\xi = c_1(L^\vee) = -c_1(L)$, and example 3.1.4 evaluates the resulting Segre operators as $s_i(L) \cap \alpha = (-1)^i c_1(L)^i \cap \alpha$, so $s(L) = \sum_{i\geq 0}(-1)^i c_1(L)^i$. The operator $c_1(L)$ is nilpotent by the lemma 3.2.1, so this sum is the inverse of $\mathrm{id} + c_1(L)$: indeed

$$\Big(\sum_{i\geq 0}(-1)^i c_1(L)^i\Big)\big(\mathrm{id} + c_1(L)\big) = \mathrm{id}$$

the terms cancelling in pairs and the tail vanishing by nilpotence. Inverses being unique (lemma 3.2.2), $c(L) = s(L)^{-1} = \mathrm{id} + c_1(L)$. Reading off homogeneous components gives $c_1(L)$ in degree one and zero in every degree $i \geq 2$, as we sought to show.

The agreement is what licenses a single symbol for both constructions, and it fixes the sign convention of the whole theory: had $\mathbb{P}(E)$ been taken as the bundle of hyperplanes rather than of lines, the tautological class would be $c_1(L)$ instead of $-c_1(L)$ and the inversion would produce $c_1(L^\vee)$ where $c_1(L)$ is wanted. Box 0.0.3 therefore cannot be skipped. The rank-one case also shows that Chern classes terminate while Segre classes do not: $c_i(L) = 0$ beyond $i = 1$, while $s_i(L)$ is nonzero for every $i$ with $c_1(L)^i \neq 0$.

**Example 3.2.7: The Twisting Sheaves on Projective Space**

Let $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(d)$ for an integer $d$. Write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, the hyperplane operator of example 2.4.7. Additivity of the first Chern class (proposition 2.4.4(3)) gives $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,H$, so by proposition 3.2.6

$$c(\mathcal{O}_{\mathbb{P}^n}(d)) = \mathrm{id} + d\,H$$

Capping against the fundamental class and using $H^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ from example 2.4.7,

$$c(\mathcal{O}_{\mathbb{P}^n}(d)) \cap [\mathbb{P}^n] = [\mathbb{P}^n] + d\,[\mathbb{P}^{n-1}]$$

so the total Chern class of $\mathcal{O}(d)$ records the single integer $d$ as the multiple of the hyperplane class it produces. The corresponding Segre operator is the full geometric series $s(\mathcal{O}_{\mathbb{P}^n}(d)) = \sum_{i\geq 0}(-d)^i H^i$, whose cap against $[\mathbb{P}^n]$ is $\sum_{i=0}^{n}(-d)^i[\mathbb{P}^{n-i}]$: every dimension carries a nonzero coefficient when $d \neq 0$, while the Chern class stops after one step. This is the terminating behaviour that theorem 3.2.9 establishes in general.

Termination in rank one came from an explicit computation. In general it comes from the projective-bundle formula, which pins down exactly one relation among the powers of $\xi$ and identifies its coefficients as the Chern classes. The following criterion extracts from that formula the tool used twice below: a class on $\mathbb{P}(E)$ is detected by finitely many pushforwards.

**Lemma 3.2.8: A Vanishing Criterion on the Projective Bundle**

Let $E$ be a vector bundle of rank $e = r + 1$ on $X$, with $\pi \colon \mathbb{P}(E) \to X$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$. A class $\beta \in A_*(\mathbb{P}(E))$ vanishes if and only if

$$\pi_*\big(\xi^m \cap \beta\big) = 0 \qquad \text{for } 0 \leq m \leq r$$

***Proof*:**
If $\beta = 0$ then every $\pi_*(\xi^m \cap \beta)$ vanishes, since capping and pushforward are homomorphisms.

Conversely, suppose the $r+1$ pushforwards vanish. By the projective-bundle formula (theorem 3.1.9) there are unique classes $\alpha_0, \ldots, \alpha_r \in A_*(X)$ with

$$\beta = \sum_{j=0}^{r} \xi^j \cap \pi^*\alpha_j$$

Capping with $\xi^m$ and pushing forward, the definition of the Segre operators gives

$$\pi_*\big(\xi^m \cap \beta\big) = \sum_{j=0}^{r} \pi_*\big(\xi^{m+j} \cap \pi^*\alpha_j\big) = \sum_{j=0}^{r} s_{m+j-r}(E) \cap \alpha_j$$

Take $m = 0$. Every index $j < r$ contributes $s_{j-r}(E) = 0$, since $j - r < 0$, so the sum reduces to $s_0(E) \cap \alpha_r = \alpha_r$, and the hypothesis forces $\alpha_r = 0$. Proceed by descending induction: suppose $\alpha_r = \cdots = \alpha_{r-m+1} = 0$ for some $1 \le m \le r$. In the displayed sum for that $m$, the indices $j < r - m$ contribute zero because $m + j - r < 0$, the indices $j > r - m$ contribute zero because $\alpha_j = 0$ by the inductive hypothesis, and the single index $j = r - m$ contributes $s_0(E) \cap \alpha_{r-m} = \alpha_{r-m}$. The hypothesis forces $\alpha_{r-m} = 0$, and the induction runs down to $m = r$, giving $\alpha_0 = 0$. All coefficients vanish, so $\beta = 0$, completing the proof.

With a criterion for vanishing in hand, the relation defining the Chern classes can be produced. The statement below is the one that makes them computable: on $\mathbb{P}(E)$ the power $\xi^e$ is not independent of the lower powers, and the coefficients expressing it in terms of them are exactly the Chern classes of $E$. Everything else in this section is extracted from it.

**Theorem 3.2.9: The Chern Relation on the Projective Bundle**

Let $E$ be a vector bundle of rank $e = r + 1$ on an algebraic scheme $X$, with $\pi\colon \mathbb{P}(E) \to X$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$.

1. For every $\alpha \in A_*(X)$,

$$\sum_{i=0}^{e} \xi^{e-i} \cap \pi^*\big(c_i(E) \cap \alpha\big) = 0 \quad \text{in } A_*(\mathbb{P}(E))$$

2. $c_i(E) = 0$ for every $i > e$; that is, the Chern classes of $E$ terminate at its rank.
3. The operators $c_0(E) = \mathrm{id}, c_1(E), \ldots, c_e(E)$ are the unique operators on $A_*(X)$, homogeneous of degrees $0, 1, \ldots, e$ respectively, whose degree-zero member is the identity and which satisfy the relation in (1).

***Proof*:**
Write $\gamma_\alpha = \sum_{i=0}^{e} \xi^{e-i} \cap \pi^*(c_i(E) \cap \alpha)$ for the class in question. Capping with $\xi^m$ and pushing forward, the definition of the Segre operators gives, for every $m \ge 0$,

$$\pi_*\big(\xi^m \cap \gamma_\alpha\big) = \sum_{i=0}^{e} \pi_*\Big(\xi^{m+e-i} \cap \pi^*\big(c_i(E) \cap \alpha\big)\Big) = \sum_{i=0}^{e} s_{m+1-i}(E) \cap \big(c_i(E) \cap \alpha\big) \tag{3.3}$$

using $m + e - i = r + (m + 1 - i)$, valid because $m + e - i \ge 0$ for $0 \le i \le e$ and $m \ge 0$, with the convention $s_j(E) = 0$ for $j < 0$ absorbing the terms where $i > m + 1$.

1. Fix $m$ with $0 \le m \le r$ and set $n = m + 1$, so $1 \le n \le e$. In (3.3) the terms with $i > n$ vanish, so the right side is $\sum_{i=0}^{n} s_{n-i}(E) c_i(E) \cap \alpha$, which is the degree-$n$ homogeneous component of $s(E) \circ c(E) = \mathrm{id}$ applied to $\alpha$. Since $n \ge 1$ that component is zero. Thus $\pi_*(\xi^m \cap \gamma_\alpha) = 0$ for $0 \le m \le r$, and lemma 3.2.8 gives $\gamma_\alpha = 0$.

2. By (1) the class $\gamma_\alpha$ vanishes, hence $\pi_*(\xi^m \cap \gamma_\alpha) = 0$ for *every* $m \geq 0$, not only for $m \leq r$. Reading (3.3) in that range and writing $n = m + 1$, this says
$$\sum_{i=0}^{e} s_{n-i}(E) \circ c_i(E) = 0 \qquad \text{for every } n \geq 1$$
since $\alpha$ was arbitrary. Set $\widetilde{c} := \sum_{i=0}^{e} c_i(E)$, the truncation of the total Chern class at degree $e$. The displayed identities, together with $s_0(E) \circ c_0(E) = \mathrm{id}$ in degree zero, say precisely that $s(E) \circ \widetilde{c} = \mathrm{id}$. The inverse of $s(E)$ is two-sided and unique (lemma 3.2.2), so $\widetilde{c} = c(E)$, and comparing homogeneous components gives $c_i(E) = 0$ for $i > e$.

3. Let $c_0' = \mathrm{id}, c_1', \ldots, c_e'$ be operators, homogeneous of the stated degrees, satisfying the relation of (1) for every $\alpha$. Subtracting the two relations, the terms with $i = 0$ cancel and
$$\sum_{i=1}^{e} \xi^{\,e-i} \cap \pi^*\Big(\big(c_i(E) - c_i'\big) \cap \alpha\Big) = 0$$
The exponents $e - i$ run over $0, 1, \ldots, e - 1 = r$, so this is a representation of the zero class in the form supplied by theorem 3.1.9. Uniqueness of that representation forces $(c_i(E) - c_i') \cap \alpha = 0$ for each $i$, and since $\alpha$ was arbitrary, $c_i' = c_i(E)$ for $1 \leq i \leq e$, completing the proof.

Part (2) is what gives a vector bundle a finite amount of Chern data: a rank-$e$ bundle has exactly $e$ potentially nonzero Chern classes, $c_1(E), \ldots, c_e(E)$, and the top one $c_e(E)$ carries the obstruction to finding a nowhere-vanishing section, as exercise 3.2.1 (3) below makes precise. Part (3) is the more useful of the three in practice. It says the Chern classes need not be computed by inverting a Segre series at all: any operators satisfying the relation on $\mathbb{P}(E)$ *are* the Chern classes. Every identity proved below is obtained by producing the relation from geometry and reading off its coefficients, and the Segre definition is never returned to.

The functoriality of the Segre operators transfers to the Chern classes for the same reason it held for them: the projective bundle of a pulled-back bundle is the pullback of the projective bundle. Two bundles on the same base are compared on the fibre product of their projective bundles, which is what makes their Chern classes commute.

**Proposition 3.2.10: Properties of the Chern Class Operators**

Let $X$ be an algebraic scheme and $E, F$ vector bundles on $X$.

1. *Commutativity.* For all $i, j \geq 0$ and $\alpha \in A_*(X)$,
$$c_i(E) \cap \big(c_j(F) \cap \alpha\big) = c_j(F) \cap \big(c_i(E) \cap \alpha\big)$$
2. *Projection formula.* If $f \colon X' \to X$ is proper and $\alpha' \in A_*(X')$, then
$$f_*\big(c_i(f^*E) \cap \alpha'\big) = c_i(E) \cap f_*\alpha'$$
3. *Flat pullback.* If $g \colon X'' \to X$ is flat of relative dimension $m$ and $\alpha \in A_*(X)$, then
$$c_i(g^*E) \cap g^*\alpha = g^*\big(c_i(E) \cap \alpha\big)$$

***Proof*:**
Each $c_i(E)$ is a $\mathbb{Z}$-combination of composites of $s_1(E), \ldots, s_i(E)$ by the recursion following definition 3.2.3, so each of the three statements follows from the corresponding statement for Segre classes: expand the polynomials and apply the Segre statement to each factor of each monomial in turn. All three Segre statements are already proved in section 3.1.

1. *Commutativity.* proposition 3.1.5(3) gives $s_i(E) \cap (s_j(F) \cap \alpha) = s_j(F) \cap (s_i(E) \cap \alpha)$. Polynomials in commuting operators commute, so $c_i(E)$ and $c_j(F)$ commute as well.
2. *Proper pushforward.* This is the first identity of proposition 3.1.5(4), $f_*(s_i(f^*E) \cap \alpha') = s_i(E) \cap f_*\alpha'$, together with the fact that $f_*$ is additive, so the identity passes through any $\mathbb{Z}$-combination of composites.
3. *Flat pullback.* This is the second identity of proposition 3.1.5(4), $s_i(g^*E) \cap g^*\beta = g^*(s_i(E) \cap \beta)$, and the same additivity argument, completing the proof.

Commutativity is what allows the Chern classes of all bundles on $X$ to be handled as though they were elements of a commutative ring, long before any such ring exists; every polynomial identity written below is an identity among commuting operators, and its meaning is unambiguous. The other two parts say that the Chern classes are attached to the bundle and not to the base: they survive restriction to an open set, pullback along a projective bundle, and pushforward along a proper map, which is exactly what a proof carried out after a base change needs in order to descend.

The identities that remain, multiplicativity on exact sequences chief among them, are proved by making $E$ as simple as possible, and the simplest a bundle can be is filtered by line bundles. For such a bundle the relation of theorem 3.2.9 can be written down by hand, because the tautological line at a point of $\mathbb{P}(E)$ must have a nonzero image in one of the successive quotients. That observation is the whole proof of the next lemma, and through it of the Whitney formula.

**Lemma 3.2.11: Chern Classes of a Filtered Bundle**

Let $E$ be a vector bundle of rank $e$ on an algebraic scheme $X$, and suppose $E$ admits a filtration by subbundles

$$0 = E_0 \subseteq E_1 \subseteq \cdots \subseteq E_e = E$$

with each quotient $L_i = E_i/E_{i-1}$ a line bundle. Then

$$c(E) = \prod_{i=1}^{e} \big(\mathrm{id} + c_1(L_i)\big)$$

as operators on $A_*(X)$; equivalently $c_m(E)$ is the $m$-th elementary symmetric expression

$$\sigma_m = \sum_{i_1<\cdots<i_m} c_1(L_{i_1})\cdots c_1(L_{i_m})$$

in the commuting operators $c_1(L_1), \ldots, c_1(L_e)$, for $0 \leq m \leq e$.

***Proof*:**
Write $P = \mathbb{P}(E)$ with projection $\pi$ and tautological class $\xi$, and for $1 \leq i \leq e$ let $P_i = \mathbb{P}(E_i) \subseteq P$ with projection $\pi_i$ and inclusion $\iota_i \colon P_i \hookrightarrow P$; set $P_0 = \mathbb{P}(E_0) = \emptyset$, so $P_e = P$. Each $P_i$ is a closed subscheme of $P$, because in a local trivialization of the filtration over $U \subseteq X$ the inclusion $P_i|_U \subseteq P|_U$ is the linear inclusion $\mathbb{P}^{i-1} \times U \subseteq \mathbb{P}^{e-1} \times U$, and these glue. Put

$$\theta_i = c_1\big(\mathcal{O}_P(1) \otimes \pi^* L_i\big) = \xi + c_1(\pi^* L_i)$$

the equality by additivity of the first Chern class (proposition 2.4.4(3)); the $\theta_i$ commute with each other by proposition 2.4.4(2).

*Step 1: each $\theta_i$ pushes a class on $P_i$ down onto $P_{i-1}$.* The tautological subbundle of box 3.2.5 restricts along $\iota_i$ to the tautological subbundle of $P_i$, since a point of $P_i$ is a line $\ell \subseteq (E_i)_x \subseteq E_x$ and the line attached to it is the same either way; thus $\iota_i^*\mathcal{O}_P(1) = \mathcal{O}_{P_i}(1)$ and $\mathcal{O}_{P_i}(-1) \subseteq \pi_i^* E_i$. Composing that inclusion with the quotient map $\pi_i^* E_i \to \pi_i^* L_i$ gives a homomorphism $\mathcal{O}_{P_i}(-1) \to \pi_i^* L_i$, that is, a global section $\tau_i$ of the line bundle

$$M_i := \mathcal{O}_{P_i}(1) \otimes \pi_i^* L_i = \iota_i^*\big(\mathcal{O}_P(1) \otimes \pi^* L_i\big)$$

At a point of $P_i$ above $x$ corresponding to a line $\ell \subseteq (E_i)_x$, the section $\tau_i$ vanishes exactly when $\ell$ maps to zero in $(L_i)_x$, that is when $\ell \subseteq (E_{i-1})_x$. Its zero locus is therefore $P_{i-1}$. In a local trivialization as above, with homogeneous fibre coordinates $y_1, \ldots, y_i$ chosen so that $E_{i-1}$ is cut out by $y_i = 0$, the section $\tau_i$ is the linear form $y_i$, a nonzerodivisor; hence $\mathrm{div}(\tau_i)$ is an effective Cartier divisor on $P_i$ with support $P_{i-1}$ and with $\mathcal{O}_{P_i}(\mathrm{div}\tau_i) \cong M_i$.

Now let $\beta = \iota_{i*}\beta_i$ for some $\beta_i \in A_*(P_i)$. The map $\iota_i$ is a closed immersion, hence proper, so the projection formula (proposition 2.4.4(4)) gives

$$\theta_i \cap \iota_{i*}\beta_i = \iota_{i*}\big(c_1(M_i) \cap \beta_i\big)$$

using $\iota_i^*(\mathcal{O}_P(1) \otimes \pi^* L_i) = M_i$. By definition 2.4.1 the class $c_1(M_i) \cap \beta_i$ is $\mathrm{div}(\tau_i) \cdot \beta_i$, which by proposition 2.2.4 is a well-defined class in $A_*\big(|\mathrm{div}\tau_i|\big) = A_*(P_{i-1})$. Pushing forward along $P_{i-1} \hookrightarrow P_i \hookrightarrow P$, the class $\theta_i \cap \beta$ is the image of a class on $P_{i-1}$, which is the assertion of this step. For $i = 1$ the filtration gives $E_1 = L_1$, the section $\tau_1$ is the identity of $L_1$ under $\mathcal{O}_{P_1}(-1) = L_1 = \pi_1^* L_1$, its divisor is empty, and $A_*(P_0) = A_*(\emptyset) = 0$, so $\theta_1$ annihilates every class coming from $P_1$.

*Step 2: the product of the $\theta_i$ is zero.* Let $\beta \in A_*(P) = A_*(P_e)$. Applying Step 1 with $i = e$ exhibits $\theta_e \cap \beta$ as coming from $A_*(P_{e-1})$; applying it again with $i = e-1$ exhibits $\theta_{e-1} \cap \theta_e \cap \beta$ as coming from $A_*(P_{e-2})$; after $e$ steps the class $\theta_1 \cap \cdots \cap \theta_e \cap \beta$ comes from $A_*(P_0) = 0$ and so vanishes. The $\theta_i$ commute, so the product may be taken in any order:

$$\Big(\prod_{i=1}^{e} \theta_i\Big) \cap \beta = 0 \qquad \text{for every } \beta \in A_*(P)$$

*Step 3: reading off the coefficients.* Take $\beta = \pi^*\alpha$ for $\alpha \in A_*(X)$ and expand the product, which is legitimate because the factors commute:

$$0 = \Big(\prod_{i=1}^{e} \big(\xi + c_1(\pi^* L_i)\big)\Big) \cap \pi^*\alpha = \sum_{m=0}^{e} \xi^{\,e-m} \cap \Big(\sigma_m\big(c_1(\pi^*L_1), \ldots, c_1(\pi^*L_e)\big) \cap \pi^*\alpha\Big)$$

The projection $\pi$ is flat, so the flat compatibility of proposition 2.4.4(4) gives $c_1(\pi^*L) \cap \pi^*\alpha = \pi^*(c_1(L) \cap \alpha)$ for each $L$; iterating over the factors of each monomial,

$$\sigma_m\big(c_1(\pi^*L_1), \ldots, c_1(\pi^*L_e)\big) \cap \pi^*\alpha = \pi^*\big(\sigma_m \cap \alpha\big)$$

with $\sigma_m = \sigma_m(c_1(L_1), \ldots, c_1(L_e))$ formed on $X$. The displayed vanishing is therefore

$$\sum_{m=0}^{e} \xi^{\,e-m} \cap \pi^*\big(\sigma_m \cap \alpha\big) = 0$$

for every $\alpha$, and $\sigma_0 = \mathrm{id}$. This is the relation of theorem 3.2.9(1) with $\sigma_m$ in place of $c_m(E)$, so the uniqueness statement theorem 3.2.9(3) gives $c_m(E) = \sigma_m$ for $0 \le m \le e$, while $c_m(E) = 0$ for $m > e$ by part (2). Summing over $m$ gives $c(E) = \prod_{i=1}^{e}(\mathrm{id} + c_1(L_i))$, as we sought to show.

The proof is geometric where it matters. A line inside $E$ cannot map to zero in every quotient $L_i$ of the filtration, and each factor $\theta_i$ cuts the support of a class down one step of the filtration, so after $e$ factors nothing is left; the algebra afterwards only names the coefficients. Note also what the lemma does not require: no hypothesis on $X$, and no splitting of the filtration. A filtration with line-bundle quotients is enough, and an extension that does not split has the same Chern classes as the direct sum of its pieces.

**Example 3.2.12: Split Bundles on Projective Space**

1. *The trivial bundle.* The bundle $\mathcal{O}_X^{\oplus e}$ is filtered by the coordinate subbundles $\mathcal{O}_X^{\oplus i}$, with all quotients equal to $\mathcal{O}_X$. Each factor of lemma 3.2.11 is then $\mathrm{id} + c_1(\mathcal{O}_X) = \mathrm{id}$, since $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, so
$$c\big(\mathcal{O}_X^{\oplus e}\big) = \mathrm{id}$$
and every trivial bundle has trivial total Chern class, of any rank and on any $X$.

2. *A sum of twists on $\mathbb{P}^n$.* Let $E = \mathcal{O}_{\mathbb{P}^n}(a) \oplus \mathcal{O}_{\mathbb{P}^n}(b)$ and write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$. The summand

$\mathcal{O}_{\mathbb{P}^n}(a)$ is a subbundle with quotient $\mathcal{O}_{\mathbb{P}^n}(b)$, so the lemma gives

$$c(E) = \big(\mathrm{id} + aH\big)\big(\mathrm{id} + bH\big) = \mathrm{id} + (a+b)H + ab\,H^2$$

and hence $c_1(E) = (a+b)H$ and $c_2(E) = ab\,H^2$. Capping with the fundamental class and using example 2.4.7,

$$c_1(E) \cap [\mathbb{P}^n] = (a+b)\,[\mathbb{P}^{n-1}], \qquad c_2(E) \cap [\mathbb{P}^n] = ab\,[\mathbb{P}^{n-2}]$$

for $n \geq 2$. On $\mathbb{P}^1$ the second class vanishes for a reason unrelated to the rank: $H^2$ lowers dimension by two on a one-dimensional scheme, so it is the zero operator by the degree count in lemma 3.2.2, and $c_2(E) = 0$ even though $\operatorname{rank} E = 2$.

Most bundles carry no filtration by line bundles. The projective bundle manufactures one, at the cost of replacing $X$ by a larger scheme, and iterating the construction strips off one line bundle at a time until nothing is left. The scheme at the end of the tower is the following.

**Definition 3.2.13: The Flag Bundle**

Let $E$ be a vector bundle of rank $e$ on an algebraic scheme $X$. The *flag bundle* of $E$ is the scheme $\mathrm{Fl}(E)$ over $X$ defined by induction on $e$: for $e = 1$ it is $X$ itself, and for $e > 1$ it is the flag bundle, formed over $\mathbb{P}(E)$, of the rank-$(e-1)$ quotient $Q = \pi^*E/\mathcal{O}_{\mathbb{P}(E)}(-1)$ of box 3.2.5. Its structure morphism $\mathrm{Fl}(E) \to X$ is the composite of the $e-1$ projective bundle projections in the tower, and its fibre over $x \in X$ is the variety of complete flags of subspaces of the fibre $E_x$. For a bundle of rank two the tower has one stage, so $\mathrm{Fl}(E) = \mathbb{P}(E)$.

The projective-bundle formula guarantees that nothing is lost in passing to the flag bundle: the pullback map on Chow groups is injective at each stage, so an identity of operators verified upstairs is an identity downstairs. That is the content of the splitting principle, and it is a theorem rather than a slogan precisely because of the injectivity.

**Theorem 3.2.14: The Splitting Principle**

Let $X$ be an algebraic scheme and $E_1, \ldots, E_t$ vector bundles on $X$. There is a scheme $X'$ and a proper, flat morphism $f\colon X' \to X$ such that

1. $f^*\colon A_*(X) \to A_*(X')$ is injective, and
2. each $f^*E_j$ admits a filtration by subbundles whose successive quotients are line bundles.

***Proof*:**
Suppose first that $t = 1$, and write $E = E_1$ of rank $e$; the morphism produced will be the structure morphism of the flag bundle $\mathrm{Fl}(E)$ of definition 3.2.13. Induct on $e$. If $e = 1$ the bundle is already a line bundle and $f = \mathrm{id}_X$ serves. Let $e > 1$ and let $\pi\colon \mathbb{P}(E) \to X$ be the projective bundle, proper and flat by [2, A Projective Bundle is Proper and Flat]. The Flat pullback along it is injective by corollary 3.1.6, which also exhibits the left inverse $\beta \mapsto \pi_*(\xi^{\,e-1} \cap \beta)$. On $\mathbb{P}(E)$ the tautological subbundle of box 3.2.5 gives an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}(E)}(-1) \longrightarrow \pi^*E \longrightarrow Q \longrightarrow 0$$

with $Q$ locally free of rank $e-1$. By the inductive hypothesis there is a proper flat $g\colon X' \to \mathbb{P}(E)$ with $g^*$ injective and $g^*Q$ filtered by subbundles with line-bundle quotients $N_2, \ldots, N_e$. Set $f = \pi \circ g$, proper and flat as a composite of such, with $f^* = g^* \circ \pi^*$ injective as a composite of injections (proposition 1.3.5(3) identifies the composite pullback with the pullback of the composite). Pulling the displayed sequence back along $g$, which is exact because pullback of locally free sheaves is exact, gives a line subbundle $F_1 = g^*\mathcal{O}_{\mathbb{P}(E)}(-1) \subseteq f^*E$ with quotient $g^*Q$. Taking $F_{j+1}$ to be the preimage in $f^*E$ of the $j$-th step of the filtration of $g^*Q$ produces

$$0 = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_e = f^*E$$

with $F_1$ of rank one and $F_{j+1}/F_j$ isomorphic to the $j$-th quotient of the filtration of $g^*Q$, hence a line bundle. This proves the case $t = 1$.

For general $t$, apply the case $t = 1$ to $E_1$, obtaining $f_1\colon X_1 \to X$; then to $f_1^*E_2$ on $X_1$, obtaining $f_2\colon X_2 \to X_1$; and so on, and take $f = f_1 \circ \cdots \circ f_t$. Each stage is proper and flat with injective pullback, so the composite is too. A filtration by subbundles with line-bundle quotients pulls back to a filtration of the same shape, again because pullback is exact on locally free sheaves, so the filtration of $f_j^*E_j$ constructed at stage $j$ survives all later stages. Hence every $f^*E_j$ is filtered with line-bundle quotients, completing the proof.

Two features of the statement carry the weight. Injectivity of $f^*$ is what turns the construction into a proof technique, since an identity of Chern operators is a family of identities of classes, and each descends. Flatness is what makes $f^*$ available at all at this stage of the theory, there being no pullback for a general morphism until chapter 4. Both come from the projective-bundle formula, which is why that formula had to be proved before any of this.

The reduction the splitting principle enables is the multiplicativity of Chern classes on exact sequences. This is the property that makes them computable in practice: bundles are almost always presented by exact sequences, and the formula converts such a presentation directly into Chern data.

**Theorem 3.2.15: The Whitney Sum Formula**

Let

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

be an exact sequence of vector bundles on an algebraic scheme $X$. Then

$$c(E) = c(E')\, c(E'')$$

as operators on $A_*(X)$; that is, for every $m \geq 0$ and every $\alpha \in A_*(X)$,

$$c_m(E) \cap \alpha = \sum_{i+j=m} c_i(E') \cap \big(c_j(E'') \cap \alpha\big)$$

***Proof*:**
Let $e' = \operatorname{rank} E'$ and $e'' = \operatorname{rank} E''$, so $\operatorname{rank} E = e' + e''$.

*Step 1: passing to a splitting cover.* By theorem 3.2.14 applied to the pair $E', E''$ there is a proper flat $f\colon X' \to X$ with $f^*$ injective, a filtration of $f^*E'$ with line-bundle quotients $L'_1, \ldots, L'_{e'}$, and a filtration of $f^*E''$ with line-bundle quotients $L''_1, \ldots, L''_{e''}$. Pulling the given sequence back along $f$ keeps it exact, pullback being exact on locally free sheaves, so

$$0 \longrightarrow f^*E' \longrightarrow f^*E \longrightarrow f^*E'' \longrightarrow 0$$

is exact on $X'$. Splice the two filtrations: take the filtration of $f^*E'$ as the bottom $e'$ steps of a filtration of $f^*E$, and above it take the preimages in $f^*E$ of the steps of the filtration of $f^*E''$. The successive quotients are $L'_1, \ldots, L'_{e'}$ followed by $L''_1, \ldots, L''_{e''}$, all line bundles, so $f^*E$ is filtered with line-bundle quotients.

*Step 2: the identity upstairs.* Lemma 3.2.11 applied to the three filtered bundles on $X'$ gives

$$c(f^*E) = \prod_{i=1}^{e'} \big(\mathrm{id} + c_1(L'_i)\big) \cdot \prod_{j=1}^{e''} \big(\mathrm{id} + c_1(L''_j)\big) = c(f^*E')\, c(f^*E'')$$

the middle expression being the product over the quotients of the spliced filtration, regrouped using commutativity of first Chern class operators (proposition 2.4.4(2)). Comparing homogeneous components of degree $m$,

$$c_m(f^*E) = \sum_{i+j=m} c_i(f^*E')\, c_j(f^*E'')$$

*Step 3: descending to $X$.* Fix $m$ and $\alpha \in A_*(X)$. Flat pullback naturality (proposition 3.2.10(3)) gives $f^*(c_m(E) \cap \alpha) = c_m(f^*E) \cap f^*\alpha$, and applying it twice,

$$f^*\Big(c_i(E') \cap \big(c_j(E'') \cap \alpha\big)\Big) = c_i(f^*E') \cap \big(c_j(f^*E'') \cap f^*\alpha\big)$$

Subtracting and using Step 2,

$$f^*\Big(c_m(E) \cap \alpha - \sum_{i+j=m} c_i(E') \cap \big(c_j(E'') \cap \alpha\big)\Big) = \Big(c_m(f^*E) - \sum_{i+j=m} c_i(f^*E')\, c_j(f^*E'')\Big) \cap f^*\alpha = 0$$

Since $f^*$ is injective, the class inside vanishes. As $\alpha$ and $m$ were arbitrary, the two operators agree, completing the proof.

The formula makes the total Chern class a homomorphism from short exact sequences to products, and this is what the Grothendieck group $K^\circ(X)$ of vector bundles will formalize when Riemann-Roch is taken up: the total Chern class is insensitive to the extension class, seeing only the sub and the quotient. Two immediate consequences deserve recording. First, $c(E)$ is invertible for every bundle $E$, with inverse $s(E)$: by definition 3.2.3 it is the inverse of $s(E)$, and that inverse is two-sided by lemma 3.2.2. So an exact sequence determines the Chern classes of any one of its three terms from the other two: $c(E'') = c(E)c(E')^{-1}$. Second, a trivial sub or quotient bundle contributes nothing, since $c(\mathcal{O}_X^{\oplus t}) = \mathrm{id}$ by example 3.2.12. Both are used repeatedly below.

**Corollary 3.2.16: Chern Classes of a Dual Bundle**

Let $E$ be a vector bundle on an algebraic scheme $X$. Then

$$c_i\big(E^\vee\big) = (-1)^i\, c_i(E)$$

as operators on $A_*(X)$, for every $i \geq 0$.

***Proof*:**
Write $e = \operatorname{rank} E$ and let $f\colon X' \to X$ be as in theorem 3.2.14 for the single bundle $E$, with $f^*E$ filtered by subbundles $0 = F_0 \subseteq \cdots \subseteq F_e = f^*E$ having line-bundle quotients $L_i = F_i/F_{i-1}$. Dualizing the filtration gives subbundles $(f^*E/F_i)^\vee \subseteq (f^*E)^\vee = f^*(E^\vee)$, increasing as $i$ decreases, with successive quotients

$$\big(f^*E/F_{i-1}\big)^\vee / \big(f^*E/F_i\big)^\vee \cong \big(F_i/F_{i-1}\big)^\vee = L_i^\vee$$

each a line bundle, so $f^*(E^\vee)$ is filtered with line-bundle quotients $L_e^\vee, \ldots, L_1^\vee$. By lemma 3.2.11 and $c_1(L^\vee) = -c_1(L)$ (proposition 2.4.4(3)),

$$c\big(f^*(E^\vee)\big) = \prod_{i=1}^{e} \big(\mathrm{id} - c_1(L_i)\big)$$

whose degree-$i$ component is $(-1)^i \sigma_i(c_1(L_1), \ldots, c_1(L_e)) = (-1)^i c_i(f^*E)$, again by lemma 3.2.11. Naturality (proposition 3.2.10(3)) rewrites both sides as pullbacks of operators on $X$, and injectivity of $f^*$ gives $c_i(E^\vee) = (-1)^i c_i(E)$, as we sought to show.

The remaining ingredient of a working calculus is a supply of computed examples. The tangent bundle of projective space is the one every later computation leans on, and it is produced from the Euler sequence by a single application of the Whitney formula.

**Example 3.2.17: The Tangent Bundle of Projective Space**

The Euler sequence [2, The Euler Sequence] relates the sheaf of differentials of $\mathbb{P}^n$ to the twisting sheaves through the exact sequence

$$0 \longrightarrow \Omega_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow 0$$

Projective space is smooth, so $\Omega_{\mathbb{P}^n}$ is locally free of rank $n$ and the sequence is a sequence of vector bundles; dualizing it, which preserves exactness, gives the sequence in the form used here,

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

with $T_{\mathbb{P}^n} = \Omega_{\mathbb{P}^n}^\vee$ the tangent bundle. Write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$. The Whitney formula (theorem 3.2.15) applied to this sequence reads

$$c\big(\mathcal{O}_{\mathbb{P}^n}\big)\, c\big(T_{\mathbb{P}^n}\big) = c\big(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}\big)$$

The left factor is the identity by example 3.2.12(1), and the right side is $(\mathrm{id} + H)^{n+1}$, the direct sum of $n+1$ copies of $\mathcal{O}(1)$ being filtered with all quotients $\mathcal{O}_{\mathbb{P}^n}(1)$ and each factor contributing

$\mathrm{id} + H$ by lemma 3.2.11. Hence

$$c(T_{\mathbb{P}^n}) = (\mathrm{id} + H)^{n+1}, \qquad c_i(T_{\mathbb{P}^n}) = \binom{n+1}{i} H^i$$

for $0 \le i \le n$, with $c_i(T_{\mathbb{P}^n}) = 0$ for $i > n$ in accordance with theorem 3.2.9(2). The binomial coefficient $\binom{n+1}{n+1} = 1$ in degree $n+1$ is not a contradiction: $H^{n+1}$ is the zero operator on $A_*(\mathbb{P}^n)$ by the degree count of lemma 3.2.2, matching the nilpotence recorded in proposition 2.4.4(5).

Capping the top class against the fundamental class and using $H^n \cap [\mathbb{P}^n] = [\mathbb{P}^0]$ from example 2.4.7,

$$c_n(T_{\mathbb{P}^n}) \cap [\mathbb{P}^n] = (n+1)\,[\mathbb{P}^0], \qquad \int_{\mathbb{P}^n} c_n(T_{\mathbb{P}^n}) = n+1$$

the degree being taken with definition 1.2.4. For $n = 1$ this gives $\int_{\mathbb{P}^1} c_1(T_{\mathbb{P}^1}) = 2$, the degree of the tangent bundle of a rational curve, consistent with $\deg T_C = 2 - 2g$ for $g = 0$ through example 2.4.8. For $n = 2$ it gives $\int_{\mathbb{P}^2} c_2(T_{\mathbb{P}^2}) = 3$.

The integers $n+1$ produced by the top Chern class are the Euler characteristics of the underlying spaces, which is the first sign that $c_n(T_X)$ counts something. The general statement, that the degree of the top Chern class of the tangent bundle of a smooth complete variety is its topological Euler characteristic, needs the comparison between Chow groups and cohomology through the cycle class map and is not available yet; what is available is the computation itself, and it is already enough to run enumerative arguments once the intersection product exists.

**3.2.18 Remark: The Topological Chern Classes** A reader who has met Chern classes topologically, as the characteristic classes of a complex vector bundle in *Algebraic Topology* [9], will recognize the shape of this section: the same axioms, the same Whitney formula, the same splitting principle through a flag bundle. The two theories are not the same construction. The classes there live in singular cohomology and are built for complex bundles on topological spaces; the classes here live in the Chow group of an algebraic scheme over an arbitrary algebraically closed field, and the scheme is allowed to be singular, reducible, and non-reduced. Neither proof above uses the other theory, and none could: no topology is available over a field of positive characteristic. The two are compared, for a smooth complex variety, by the cycle class map built later in this book, which carries the algebraic classes to the topological ones.

## Exercise 3.2.1

1. Using only definition 3.2.3, express $s_1(E)$, $s_2(E)$ and $s_3(E)$ in terms of $c_1(E), c_2(E), c_3(E)$, and check that the resulting formulas are obtained from the expressions for $c_1, c_2, c_3$ in terms of $s_1, s_2, s_3$ by exchanging the letters $c$ and $s$. Explain in one sentence why that symmetry is forced by the definition.
2. Let $E = \mathcal{O}_{\mathbb{P}^n}(a) \oplus \mathcal{O}_{\mathbb{P}^n}(b)$ with $n \ge 2$. Compute $c(E)$, $c(E^\vee)$ and $c(E \otimes \mathcal{O}_{\mathbb{P}^n}(1))$ as operators on $A_*(\mathbb{P}^n)$, and compute $c_2(E) \cap [\mathbb{P}^n]$. Then take $n = 1$ and explain why $c_2(E) = 0$ there, identifying which of the two possible reasons (the rank of $E$ or the dimension of the base) is responsible.
3. Let $E$ be a vector bundle of rank $e \ge 2$ on an algebraic scheme $X$ admitting a nowhere-

vanishing global section $\tau$, so that $\tau\colon \mathcal{O}_X \to E$ is an inclusion of a subbundle with locally free quotient $Q$ of rank $e-1 \geq 1$. Show that $c(E) = c(Q)$ and deduce $c_e(E) = 0$. Show separately that a line bundle with a nowhere-vanishing section is trivial, so that the conclusion holds in the excluded case $e = 1$ as well. Conversely, explain why the computation $\int_{\mathbb{P}^n} c_n(T_{\mathbb{P}^n}) = n+1 \neq 0$ of example 3.2.17 shows that $T_{\mathbb{P}^n}$ has no nowhere-vanishing global section.

4. Let $E$ be a vector bundle of rank two on an algebraic scheme $X$ and $L$ a line bundle on $X$. Using theorem 3.2.14 and lemma 3.2.11, prove

$$c_1(E \otimes L) = c_1(E) + 2\,c_1(L), \qquad c_2(E \otimes L) = c_2(E) + c_1(E)\,c_1(L) + c_1(L)^2$$

as operators on $A_*(X)$. (You may use that tensoring a filtration by a line bundle again gives a filtration, with each quotient tensored by that line bundle.)

5. Compute $c(\Omega_{\mathbb{P}^n})$ as an operator on $A_*(\mathbb{P}^n)$ in closed form, and evaluate $\int_{\mathbb{P}^2} c_2(\Omega_{\mathbb{P}^2})$ and $\int_{\mathbb{P}^2} c_1(\Omega_{\mathbb{P}^2})^2$. Compare the first with $\int_{\mathbb{P}^2} c_2(T_{\mathbb{P}^2})$ and account for the agreement.

6. Let $C$ be a smooth projective curve and $E$ a vector bundle of rank two on $C$. Show that $c_2(E) = 0$ as an operator on $A_*(C)$, and that $c(E)$ is determined by the single class $c_1(E) \cap [C] \in A_0(C)$. Show further that if $E$ sits in an exact sequence $0 \to L_1 \to E \to L_2 \to 0$ of bundles on $C$, then $\int_C c_1(E) = \deg L_1 + \deg L_2$.

## 3.3 The Chern Character and the Todd Class

The total Chern class of section 3.2 is multiplicative on short exact sequences: the Whitney formula turns an extension into a product. Riemann-Roch needs the opposite bookkeeping. It computes the Euler characteristic of a coherent sheaf, and Euler characteristics are *additive* on short exact sequences, so the characteristic class that Riemann-Roch pairs against must be additive too. This section builds that class, the Chern character, and alongside it the Todd class, the multiplicative correction factor the theorem attaches to the tangent bundle.

Both are universal power series in the Chern classes, and both differ from anything in section 3.2 in two ways. Their coefficients are rational, forced by denominators that no normalization removes, and they are infinite series rather than finite sums, which is legitimate only because a composite of more than $\dim X$ dimension-lowering operators is zero. Those two points come first, then the formal calculus of Chern roots that makes symmetric expressions in $r$ formal symbols into operators, and then the two classes and the identities they satisfy.

The first point is a change of coefficients. Nothing in section 3.2 required it: Segre and Chern classes are operators on $A_*(X)$ with integer coefficients. The Chern character does require it, because its degree-two component is $\frac{1}{2}(c_1^2 - 2c_2)$, and no rescaling removes that $\frac{1}{2}$ while keeping additivity.

**Definition 3.3.1: Chow Groups with Rational Coefficients**

For an algebraic scheme $X$, set

$$A_k(X)_{\mathbb{Q}} := A_k(X) \otimes_{\mathbb{Z}} \mathbb{Q}, \qquad A_*(X)_{\mathbb{Q}} := \bigoplus_k A_k(X)_{\mathbb{Q}}$$

A class $\alpha \in A_k(X)$ is identified with $\alpha \otimes 1$, and every operator $A_k(X) \to A_{k-i}(X)$ extends uniquely to a $\mathbb{Q}$-linear operator $A_k(X)_{\mathbb{Q}} \to A_{k-i}(X)_{\mathbb{Q}}$. The degree map of definition 1.2.4 on a complete $X$ is extended $\mathbb{Q}$-linearly and still written $\int_X : A_0(X)_{\mathbb{Q}} \to \mathbb{Q}$.

Since $\mathbb{Q}$ is flat over $\mathbb{Z}$, an injective map of Chow groups stays injective after this change of coefficients, so the injectivity statements of section 3.2 are available rationally without further comment. What is lost is torsion: the natural map $A_*(X) \to A_*(X)_{\mathbb{Q}}$ has kernel the torsion subgroup, so any torsion class becomes invisible. On $\mathbb{P}^n$ nothing is lost, since $A_*(\mathbb{P}^n) = \bigoplus_j \mathbb{Z}\,[\mathbb{P}^j]$ is free (theorem 1.5.2) and $A_*(\mathbb{P}^n)_{\mathbb{Q}} = \bigoplus_j \mathbb{Q}\,[\mathbb{P}^j]$; the class $\frac{3}{2}\,[\mathrm{pt}] \in A_0(\mathbb{P}^2)_{\mathbb{Q}}$, which will appear below as a component of $\mathrm{ch}(T_{\mathbb{P}^2})$, is a legitimate element of the rational Chow group and of nothing smaller.

The second point is that infinite series of operators are already finite sums. An operator lowering dimension can only be applied $\dim X$ times before the target group vanishes, which is the nilpotence recorded for a single line bundle in proposition 2.4.4(5), stated here for an arbitrary commuting family.

**Lemma 3.3.2: Power Series in Dimension-Lowering Operators**

Let $X$ be an algebraic scheme of dimension $n$ and let $\theta_1, \dots, \theta_m$ be pairwise commuting $\mathbb{Q}$-linear endomorphisms of $A_*(X)_{\mathbb{Q}}$ with $\theta_j\big(A_k(X)_{\mathbb{Q}}\big) \subseteq A_{k-d_j}(X)_{\mathbb{Q}}$ for integers $d_j \geq 1$. Write $\theta^\nu = \theta_1^{\nu_1} \cdots \theta_m^{\nu_m}$ and $|\nu| = \nu_1 + \cdots + \nu_m$.

1. $\theta^\nu = 0$ whenever $|\nu| > n$.
2. The assignment
$$\sum_\nu \lambda_\nu\, a^\nu \longmapsto \sum_{|\nu| \leq n} \lambda_\nu\, \theta^\nu$$
is a homomorphism of $\mathbb{Q}$-algebras $\mathbb{Q}[[a_1, \dots, a_m]] \to \mathrm{End}\big(A_*(X)_{\mathbb{Q}}\big)$ carrying $a_j$ to $\theta_j$.

***Proof*:**

1. A $k$-cycle is a combination of $k$-dimensional subvarieties, so $A_k(X)_{\mathbb{Q}} = 0$ unless $0 \leq k \leq n$. The operator $\theta^\nu$ carries $A_k(X)_{\mathbb{Q}}$ into $A_{k-e}(X)_{\mathbb{Q}}$ where $e = \sum_j \nu_j d_j \geq |\nu|$. If $|\nu| > n$ then for every $k$ with $A_k(X)_{\mathbb{Q}} \neq 0$ one has $k - e \leq n - |\nu| < 0$, so the target vanishes and $\theta^\nu = 0$.
2. Write $T$ for the assignment. It is $\mathbb{Q}$-linear by construction, and $T(1) = \mathrm{id}$. For multiplicativity, let $\Phi = \sum_\mu \lambda_\mu a^\mu$ and $\Psi = \sum_\rho \kappa_\rho a^\rho$. The product $\Phi\Psi$ has coefficients $\sum_{\mu+\rho=\nu} \lambda_\mu \kappa_\rho$, so
$$T(\Phi\Psi) = \sum_{|\nu| \leq n} \Big( \sum_{\mu+\rho=\nu} \lambda_\mu \kappa_\rho \Big) \theta^\nu$$

On the other side, since the $\theta_j$ commute, $T(\Phi)T(\Psi) = \sum_{|\mu|\le n,\, |\rho|\le n} \lambda_\mu \kappa_\rho\, \theta^{\mu+\rho}$. Every term with $|\mu+\rho| > n$ vanishes by part (1), and every pair $(\mu,\rho)$ with $|\mu+\rho| \le n$ automatically satisfies $|\mu| \le n$ and $|\rho| \le n$, so the two expressions have the same nonzero terms, completing the proof.

The Chern classes $c_1(E),\dots,c_r(E)$ of a bundle of rank $r$ are exactly such a family: they commute with one another by section 3.2, ultimately because two divisor operators commute (theorem 2.3.2), and $c_i(E)$ lowers dimension by $i$. A formal power series in $r$ variables therefore evaluates on them to a well-defined operator, given by a finite sum. This is the mechanism behind every definition below.

What remains is to say which power series. Both the Chern character and the Todd class are described most efficiently as symmetric expressions in $r$ formal symbols $a_1,\dots,a_r$, the Chern roots, which behave as though $E$ were a direct sum of line bundles with those first Chern classes. The splitting principle of section 3.2 is what makes the symbols legitimate.

**Definition 3.3.3: Chern Roots and Splitting Maps**

Let $E$ be a vector bundle of rank $r$ on an algebraic scheme $X$. A *splitting map* for $E$ is a flat proper morphism $f\colon X' \to X$ such that

1. $f^*\colon A_*(X) \to A_*(X')$ is injective, and
2. $f^*E$ admits a filtration by subbundles $0 = E_0 \subset E_1 \subset \cdots \subset E_r = f^*E$ whose successive quotients $L_p = E_p/E_{p-1}$ are line bundles.

The operators $a_p := c_1(L_p)$ on $A_*(X')_{\mathbb{Q}}$, for $p = 1,\dots,r$, are the *Chern roots* of $E$ relative to $f$.

The splitting principle of section 3.2 produces a splitting map for any bundle: the flag bundle of $E$ is a tower of projective bundles, proper and flat over $X$ by [2, A Projective Bundle is Proper and Flat], and the projective-bundle formula of section 3.1 makes each stage of the tower inject on Chow groups. By flatness of $\mathbb{Q}$ over $\mathbb{Z}$, condition (1) also gives injectivity of $f^*\colon A_*(X)_{\mathbb{Q}} \to A_*(X')_{\mathbb{Q}}$. Writing $a_p = c_1(L_p)$ for the Chern roots, lemma 3.2.11 gives

$$c(f^*E) = \prod_{p=1}^{r} \big(\mathrm{id} + a_p\big), \qquad \text{that is} \qquad c_i(f^*E) = e_i(a_1,\dots,a_r) \tag{3.4}$$

where $e_i$ is the $i$-th elementary symmetric polynomial. The roots are not canonical, and they are not operators on $X$ at all, only on $X'$. What descends to $X$ is any symmetric expression in them, and the reason is the classical structure theorem for symmetric functions.

Any symmetric formal power series in the Chern roots is a power series in the elementary symmetric polynomials, hence a genuine operator: this is [5, Symmetric Power Series in the Chern Roots], the formal-power-series refinement of the classical fundamental theorem on symmetric functions proved there.

The two lemmas combine into the calculus this section runs on: a symmetric power series in $r$ variables defines an operator attached to any rank-$r$ bundle, and that operator may be computed as though the bundle were split.

**Proposition 3.3.4: The Chern Root Calculus**

Let $E$ be a vector bundle of rank $r$ on an algebraic scheme $X$ and let $\Phi \in \mathbb{Q}[[a_1, \ldots, a_r]]$ be symmetric, with $\widetilde{\Phi}$ as in [5, Symmetric Power Series in the Chern Roots]. Define

$$\Phi(E) := \widetilde{\Phi}\big(c_1(E), \ldots, c_r(E)\big)$$

Then:

1. $\Phi(E)$ is a well-defined operator on $A_*(X)_\mathbb{Q}$, equal to a finite sum of composites of Chern classes of $E$, and it commutes with $\Psi(F)$ for every bundle $F$ and symmetric $\Psi$.
2. For a flat morphism $g\colon X'' \to X$ and $\alpha \in A_*(X)_\mathbb{Q}$,

$$g^*\big(\Phi(E) \cap \alpha\big) = \Phi(g^*E) \cap g^*\alpha$$

3. If $E$ itself admits a filtration by subbundles with line-bundle quotients $L_1, \ldots, L_r$, then $\Phi(E) = \Phi\big(c_1(L_1), \ldots, c_1(L_r)\big)$, the right-hand side being the evaluation of lemma 3.3.2.
4. If $f\colon X' \to X$ is a splitting map for $E$ with Chern roots $a_1, \ldots, a_r$, then $f^*\big(\Phi(E) \cap \alpha\big) = \Phi(a_1, \ldots, a_r) \cap f^*\alpha$ for all $\alpha \in A_*(X)_\mathbb{Q}$.

***Proof*:**

1. The classes $c_1(E), \ldots, c_r(E)$ commute pairwise and $c_i(E)$ lowers dimension by $i$, so lemma 3.3.2 applies and $\widetilde{\Phi}$ evaluates to a well-defined operator, equal to the sum of the terms of $\widetilde{\Phi}$ of weight at most $\dim X$. Uniqueness of $\widetilde{\Phi}$ in [5, Symmetric Power Series in the Chern Roots] is what makes the value depend on $\Phi$ alone. Chern classes of any two bundles commute by section 3.2, so the images of the two evaluation homomorphisms commute.
2. Chern classes are natural under flat pullback, $c_i(g^*E) \cap g^*\beta = g^*(c_i(E) \cap \beta)$ by section 3.2, generalizing proposition 2.4.4(4). Applying this once for each factor of a monomial in the $c_i(E)$, and summing over the finitely many monomials of $\widetilde{\Phi}$ that survive, gives the claim.
3. By (3.4) applied to the filtration of $E$ itself, $c_i(E) = e_i(a_1, \ldots, a_r)$ with $a_p = c_1(L_p)$. The evaluation homomorphism $\mathbb{Q}[[a_1, \ldots, a_r]] \to \mathrm{End}(A_*(X)_\mathbb{Q})$ of lemma 3.3.2 for the commuting family $a_1, \ldots, a_r$ is a ring homomorphism, so it may be applied to the identity $\Phi = \widetilde{\Phi}(e_1, \ldots, e_r)$ of [5, Symmetric Power Series in the Chern Roots] term by term, giving

$$\Phi(a_1, \ldots, a_r) = \widetilde{\Phi}\big(e_1(a), \ldots, e_r(a)\big) = \widetilde{\Phi}\big(c_1(E), \ldots, c_r(E)\big) = \Phi(E)$$

   as operators.
4. Combining parts (2) and (3): the morphism $f$ is flat, so $f^*(\Phi(E) \cap \alpha) = \Phi(f^*E) \cap f^*\alpha$, and $f^*E$ carries a filtration with line-bundle quotients $L_p$, so $\Phi(f^*E) = \Phi(a_1, \ldots, a_r)$ by part (3). This completes the proof.

Part (4) is the working form. To prove an identity between operators built from bundles on $X$, pass to a splitting map, verify the corresponding identity of symmetric power series in the roots, and descend by injectivity of $f^*$. The roots never appear in a statement, only in a proof.

The first series to feed into this machine is the exponential.

**Definition 3.3.5: The Chern Character**

Let $E$ be a vector bundle of rank $r$ on an algebraic scheme $X$. The *Chern character* of $E$ is the operator on $A_*(X)_{\mathbb{Q}}$ obtained from the symmetric power series

$$\mathrm{ch}(a_1,\dots,a_r) = \sum_{p=1}^{r} e^{a_p} = \sum_{p=1}^{r}\sum_{k\geq 0}\frac{a_p^k}{k!}$$

by proposition 3.3.4. It is written $\mathrm{ch}(E)$, and its component lowering dimension by $k$ is written $\mathrm{ch}_k(E)$, so that $\mathrm{ch}(E) = \sum_{k\geq 0}\mathrm{ch}_k(E)$ with the sum finite by lemma 3.3.2.

The definition is stated through the roots for brevity, but no roots are used: what proposition 3.3.4 returns is a specific polynomial in $c_1(E),\dots,c_r(E)$ with rational coefficients, and the next proposition writes it out. In every positive degree that polynomial is the same for all ranks, once the convention $c_i(E) = 0$ for $i > \mathrm{rank}E$ from section 3.2 is in force; the rank enters only in degree zero, where the constant term is the rank itself. That is what allows bundles of different ranks to be compared in a single identity: the positive-degree components match term by term and the ranks do their own bookkeeping in degree zero.

**Proposition 3.3.6: Components of the Chern Character**

Let $E$ be a vector bundle of rank $r$ on $X$, and write $c_i = c_i(E)$. Then $\mathrm{ch}_k(E) = \frac{1}{k!}\,p_k$, where $p_0 = r$ and the operators $p_k$ are determined by Newton's recursion

$$p_k - c_1p_{k-1} + c_2p_{k-2} - \cdots + (-1)^{k-1}c_{k-1}p_1 + (-1)^k k\,c_k = 0 \qquad (k\geq 1) \tag{3.5}$$

In particular

$$\mathrm{ch}(E) = r + c_1 + \tfrac{1}{2}\big(c_1^2 - 2c_2\big) + \tfrac{1}{6}\big(c_1^3 - 3c_1c_2 + 3c_3\big) + \cdots$$

so $\mathrm{ch}_0(E) = \mathrm{rank}E$ and $\mathrm{ch}_1(E) = c_1(E)$. For a line bundle $L$, $\mathrm{ch}(L) = e^{c_1(L)}$.

***Proof*:**
Work first with formal symbols. In $\mathbb{Q}[a_1,\dots,a_r][[t]]$ put $E(t) = \prod_{p=1}^{r}(1+a_pt) = \sum_{j\geq 0} e_jt^j$, where $e_j = e_j(a)$ and $e_j = 0$ for $j > r$. Each factor $1 + a_pt$ has constant term 1, hence is invertible, and the product rule gives

$$E'(t) = \sum_{p=1}^{r} a_p\prod_{q\neq p}(1+a_qt) = E(t)\sum_{p=1}^{r}\frac{a_p}{1+a_pt}$$

Expanding $\frac{a_p}{1+a_pt} = \sum_{k\geq 1}(-1)^{k-1}a_p^k t^{k-1}$ and summing over $p$ turns this into

$$E'(t) = E(t)\sum_{k\geq 1}(-1)^{k-1}p_kt^{k-1}, \qquad p_k := \sum_{p=1}^{r} a_p^k$$

Comparing coefficients of $t^{k-1}$ on both sides gives $ke_k = \sum_{m=1}^{k}(-1)^{m-1}e_{k-m}\,p_m$, which on multiplying by $(-1)^{k-1}$ and transposing is exactly (3.5) with $e_i$ in place of $c_i$. Solving the recursion in low degrees, $p_1 = e_1$, then $p_2 = e_1p_1 - 2e_2 = e_1^2 - 2e_2$, then $p_3 = e_1p_2 - e_2p_1 + 3e_3 = e_1^3 - 3e_1e_2 + 3e_3$.

The power sums determine the Chern character, since interchanging the two finite-in-each-degree summations gives

$$\sum_{p=1}^{r} e^{a_p} = \sum_{k\geq 0}\frac{1}{k!}\sum_{p=1}^{r} a_p^k = r + \sum_{k\geq 1}\frac{p_k}{k!}$$

Each $p_k$ is symmetric, and (3.5) exhibits it as an isobaric polynomial of weight $k$ in $e_1, \ldots, e_k$; by the uniqueness in [5, Symmetric Power Series in the Chern Roots] that polynomial is the one proposition 3.3.4 substitutes into. Substituting $c_i = c_i(E)$ therefore gives $\mathrm{ch}_k(E) = \frac{1}{k!}p_k$ with $p_k$ satisfying (3.5), and the displayed expansion follows from the three computed cases. Finally $\mathrm{ch}_0(E) = p_0 = r$ and $\mathrm{ch}_1(E) = p_1 = c_1$. For $r = 1$ the series is $e^{a_1}$ and $e_1 = a_1$, so $\widetilde{\mathrm{ch}}(y_1) = e^{y_1}$ and $\mathrm{ch}(L) = e^{c_1(L)} = \sum_{k\geq 0} c_1(L)^k/k!$, a finite sum by lemma 3.3.2, as we sought to show.

The denominators do not cancel. Already the factor $\frac{1}{2}$ in $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$ survives on projective space, where $c_1^2 - 2c_2$ is an odd multiple of the point class for suitable bundles, so $\mathrm{ch}_2$ is not the image of any operator on $A_*(X)$ with integer coefficients. The next example exhibits the smallest such case.

**Example 3.3.7: The Chern Character of a Line Bundle on $\mathbb{P}^n$**

Let $h = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, so that $h^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ for $0 \leq k \leq n$ by example 2.4.7, and $h^{n+1} = 0$ as an operator by lemma 3.3.2. For the line bundle $\mathcal{O}_{\mathbb{P}^n}(d)$ one has $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,h$ by proposition 2.4.4(3), so

$$\mathrm{ch}\big(\mathcal{O}_{\mathbb{P}^n}(d)\big) = e^{dh} = 1 + dh + \frac{d^2}{2}h^2 + \cdots + \frac{d^n}{n!}h^n$$

the series terminating because $h^{n+1} = 0$. Capping with the fundamental class on $\mathbb{P}^2$,

$$\mathrm{ch}\big(\mathcal{O}_{\mathbb{P}^2}(d)\big) \cap [\mathbb{P}^2] = [\mathbb{P}^2] + d\,[\mathbb{P}^1] + \frac{d^2}{2}\,[\mathrm{pt}]$$

For odd $d$ the last coefficient is not an integer, so this class lies in $A_0(\mathbb{P}^2)_{\mathbb{Q}}$ and in no subgroup coming from $A_0(\mathbb{P}^2)$. Rational coefficients are not a convenience here; without them the Chern character of $\mathcal{O}_{\mathbb{P}^2}(1)$ does not exist.

The point of the exponential is the identity $e^{a+b} = e^a e^b$, which converts both operations on bundles into operations on operators. Both of the proofs below compare two bundles at once, so they need a single splitting map that works for several bundles simultaneously.

**Theorem 3.3.8: Additivity and Multiplicativity of the Chern Character**

Let $X$ be an algebraic scheme and let $E, F$ be vector bundles on $X$.

1. If $0 \to E' \to E \to E'' \to 0$ is an exact sequence of vector bundles on $X$, then
$$\mathrm{ch}(E) = \mathrm{ch}(E') + \mathrm{ch}(E'')$$
as operators on $A_*(X)_{\mathbb{Q}}$. In particular $\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F)$.
2. $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E)\,\mathrm{ch}(F)$, the product on the right being composition of operators.

***Proof*:**
Write $r = \mathrm{rank}E$, $r' = \mathrm{rank}E'$, $r'' = \mathrm{rank}E''$ and $s = \mathrm{rank}F$; in part (1) the ranks satisfy $r = r' + r''$.

1. *Additivity.* By theorem 3.2.14 choose a flat proper $f\colon X' \to X$ with $f^*$ injective that is a splitting map for both $E'$ and $E''$, with Chern roots $a_1, \dots, a_{r'}$ and $b_1, \dots, b_{r''}$ respectively.

   The same $f$ splits $E$. Pulling the exact sequence back gives $0 \to f^*E' \xrightarrow{\iota} f^*E \xrightarrow{\pi} f^*E'' \to 0$, again exact since pullback of vector bundles is exact. Let $0 = E'_0 \subset \cdots \subset E'_{r'} = f^*E'$ and $0 = E''_0 \subset \cdots \subset E''_{r''} = f^*E''$ be the two filtrations, with line-bundle quotients $L_p$ and $M_q$, and define
$$H_p = \iota(E'_p) \quad (0 \le p \le r'), \qquad H_{r'+q} = \pi^{-1}\big(E''_q\big) \quad (1 \le q \le r'')$$
   This is a filtration of $f^*E$ by subbundles: the first stretch is the filtration of the subbundle $f^*E'$, the second consists of preimages of subbundles under a surjection of bundles. Its successive quotients are $L_1, \dots, L_{r'}$ followed by $H_{r'+q}/H_{r'+q-1} \cong E''_q/E''_{q-1} = M_q$, the isomorphism holding because $\pi$ carries $H_{r'+q}$ onto $E''_q$ with kernel $f^*E' = H_{r'}$ contained in $H_{r'+q-1}$. So $f$ is a splitting map for $E$ with Chern roots $a_1, \dots, a_{r'}, b_1, \dots, b_{r''}$.

   Now apply proposition 3.3.4(4) three times. For every $\alpha \in A_*(X)_{\mathbb{Q}}$,
$$f^*\big(\mathrm{ch}(E) \cap \alpha\big) = \Big(\sum_{p=1}^{r'} e^{a_p} + \sum_{q=1}^{r''} e^{b_q}\Big) \cap f^*\alpha = f^*\big(\mathrm{ch}(E') \cap \alpha\big) + f^*\big(\mathrm{ch}(E'') \cap \alpha\big)$$
   the first equality reading the roots of $E$ off the spliced filtration and the second applying the proposition to $E'$ and to $E''$ separately. Since $f^*$ is injective on $A_*(X)_{\mathbb{Q}}$, this gives $\mathrm{ch}(E) \cap \alpha = \mathrm{ch}(E') \cap \alpha + \mathrm{ch}(E'') \cap \alpha$ for every $\alpha$, which is the asserted identity of operators. The direct sum is the case of a split exact sequence.

2. *Multiplicativity.* First a statement about filtrations: if a bundle $A$ has a filtration with line-bundle quotients $N_1, \dots, N_a$ and $B$ one with quotients $P_1, \dots, P_b$, then $A \otimes B$ has a filtration with quotients the $N_i \otimes P_j$. Induct on $a$. For $a = 1$ the bundle $A = N_1$ is a line bundle, and tensoring the filtration $B_j$ of $B$ by it gives a filtration of $N_1 \otimes B$ with quotients $N_1 \otimes P_j$, since tensoring by a line bundle is exact and carries subbundles to subbundles. For $a > 1$, tensoring $0 \to A_{a-1} \to A \to N_a \to 0$ with the locally free $B$ keeps it exact; the subbundle $A_{a-1} \otimes B$ has the required filtration by induction, and taking preimages of the filtration of $N_a \otimes B$ under the surjection $A \otimes B \to N_a \otimes B$ extends it, exactly as in part (1).

By theorem 3.2.14 choose a flat proper $g\colon X'' \to X$ with $g^*$ injective that is a splitting map for both $E$ and $F$, with Chern roots $w_1, \dots, w_r$ and $u_1, \dots, u_s$. Applying the previous paragraph to $g^*E \otimes g^*F = g^*(E \otimes F)$ produces a filtration whose quotients are the $rs$ tensor products of line bundles from the two filtrations, and the first Chern class of such a tensor product is $w_p + u_j$ by proposition 2.4.4(3). So $g$ is a splitting map for $E \otimes F$ with Chern roots the operators $w_p + u_j$. By proposition 3.3.4(4) and the formal identity $e^{w+u} = e^w e^u$,

$$g^*\big(\mathrm{ch}(E \otimes F) \cap \alpha\big) = \Big(\sum_{p,j} e^{w_p + u_j}\Big) \cap g^*\alpha = \Big(\sum_p e^{w_p}\Big)\Big(\sum_j e^{u_j}\Big) \cap g^*\alpha$$

and the right-hand side is $g^*\big(\mathrm{ch}(E)\,\mathrm{ch}(F) \cap \alpha\big)$, again by proposition 3.3.4(4) applied twice, the two operators commuting by proposition 3.3.4(1). Injectivity of $g^*$ gives $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E)\,\mathrm{ch}(F)$, completing the proof.

The theorem says that ch converts the two ways of building bundles into the two operations of a commutative ring: extensions become sums, tensor products become composites. That is the structure of the Grothendieck group of vector bundles, in which an exact sequence is declared to be a sum and the tensor product is the multiplication, and the Chern character is the map out of it that the Riemann-Roch theorem later in this book computes with. A reader who has met Chern classes topologically [9] will recognize the axioms driving the argument, the Whitney formula and naturality being the same there; the topological Chern character satisfies the same two identities. The construction here is independent of that one, valid over any algebraically closed field and on singular schemes, and landing in rational Chow groups rather than in cohomology.

Additivity comes at a price. The total Chern class is multiplicative, so ch cannot also be, and the class that Riemann-Roch attaches to the tangent bundle, correcting the Chern character of a bundle to its Euler characteristic, has to be multiplicative. The Todd class is that multiplicative partner. The specific power series below is not derivable from anything in this chapter; it is the one that makes Riemann-Roch true, and the first evidence for it is the computation at the end of this section, where the top Todd component of the tangent bundle of $\mathbb{P}^n$ integrates to $1 = \chi(\mathcal{O}_{\mathbb{P}^n})$.

**Definition 3.3.9: The Todd Class**

Write $Q(t) = \dfrac{t}{1 - e^{-t}} \in \mathbb{Q}[[t]]$, which is a power series because $1 - e^{-t} = t\big(1 - \frac{t}{2} + \frac{t^2}{6} - \cdots\big)$ and the second factor has constant term 1, hence is invertible. For a vector bundle $E$ of rank $r$ on an algebraic scheme $X$, the *Todd class* of $E$ is the operator on $A_*(X)_{\mathbb{Q}}$ obtained from the symmetric power series

$$\mathrm{td}(a_1, \dots, a_r) = \prod_{p=1}^{r} \frac{a_p}{1 - e^{-a_p}}$$

by proposition 3.3.4, written $\mathrm{td}(E)$, with $\mathrm{td}_k(E)$ its component lowering dimension by $k$. When $X$ is smooth, $\mathrm{td}(X)$ abbreviates $\mathrm{td}(T_X)$ for the tangent bundle $T_X$.

**Proposition 3.3.10: Components of the Todd Class**

Let $E$ be a vector bundle on $X$ and write $c_i = c_i(E)$. Then

$$\mathrm{td}(E) = 1 + \tfrac{1}{2}c_1 + \tfrac{1}{12}\big(c_1^2 + c_2\big) + \tfrac{1}{24}c_1c_2 + \cdots$$

In particular $\mathrm{td}_0(E) = 1$, so $\mathrm{td}(E)$ is an invertible operator on $A_*(X)_{\mathbb{Q}}$, and for a line bundle $L$,

$$\mathrm{td}(L) = Q\big(c_1(L)\big) = 1 + \tfrac{1}{2}c_1(L) + \tfrac{1}{12}c_1(L)^2 + \cdots$$

***Proof*:**
Expand $Q$ first. From $1 - e^{-t} = t\big(1 - \frac{t}{2} + \frac{t^2}{6} - \frac{t^3}{24} + \cdots\big)$ one gets $Q(t) = (1-u)^{-1}$ with $u = \frac{t}{2} - \frac{t^2}{6} + \frac{t^3}{24} - \cdots$, so $Q(t) = 1 + u + u^2 + u^3 + \cdots$. Collecting terms,

$$Q(t) = 1 + \tfrac{t}{2} + \Big(\tfrac{1}{4} - \tfrac{1}{6}\Big)t^2 + \Big(\tfrac{1}{8} - \tfrac{1}{6} + \tfrac{1}{24}\Big)t^3 + \cdots = 1 + \tfrac{t}{2} + \tfrac{t^2}{12} + 0\cdot t^3 + \cdots$$

Now write $q_1 = \frac{1}{2}$, $q_2 = \frac{1}{12}$, $q_3 = 0$ and expand $\prod_{p=1}^{r}\big(1 + q_1a_p + q_2a_p^2 + q_3a_p^3 + \cdots\big)$ in degrees at most three, using the power sums $p_k = \sum_p a_p^k$ of proposition 3.3.6. In degree one the only contribution is $q_1p_1 = \frac{1}{2}e_1$. In degree two the contributions are $q_2p_2$ from a single factor and $q_1^2e_2$ from two distinct factors, giving

$$\tfrac{1}{12}\big(e_1^2 - 2e_2\big) + \tfrac{1}{4}e_2 = \tfrac{1}{12}\big(e_1^2 + e_2\big)$$

In degree three the contributions are $q_3p_3 = 0$, then $q_1q_2\sum_{p\neq q}a_p^2a_q$ from two distinct factors, then $q_1^3e_3$ from three distinct factors. Since $\sum_{p\neq q}a_p^2a_q = p_1p_2 - p_3 = e_1e_2 - 3e_3$, the degree-three term is

$$\tfrac{1}{24}\big(e_1e_2 - 3e_3\big) + \tfrac{1}{8}e_3 = \tfrac{1}{24}e_1e_2$$

Substituting $e_i = c_i(E)$ through proposition 3.3.4 gives the stated expansion. Invertibility follows from $\mathrm{td}(E) = \mathrm{id} + N$ with $N = \sum_{k\geq 1}\mathrm{td}_k(E)$ a sum of dimension-lowering operators, so $N^{\dim X+1} = 0$ by lemma 3.3.2 and $\mathrm{td}(E)^{-1} = \sum_{k\geq 0}(-N)^k$ is a finite sum. For $r = 1$ the series is $Q(a_1)$ with $e_1 = a_1$, so $\mathrm{td}(L) = Q(c_1(L))$, as we sought to show.

The first component $\frac{1}{2}c_1$ already forces rational coefficients, exactly as $\mathrm{ch}_2$ did, and the degree-three coefficient of $Q$ vanishes because $Q(t) - \frac{t}{2}$ is even, so $Q$ has no odd-degree terms beyond the first. The next example computes the class in the smallest case, where a single Chern class controls everything.

**Example 3.3.11: The Todd Class of a Line Bundle on $\mathbb{P}^2$**
With $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ as in example 3.3.7 and $h^3 = 0$, proposition 3.3.10 gives

$$\mathrm{td}\big(\mathcal{O}_{\mathbb{P}^2}(d)\big) = 1 + \frac{d}{2}h + \frac{d^2}{12}h^2$$

so that, capping with the fundamental class, $\mathrm{td}(\mathcal{O}_{\mathbb{P}^2}(d)) \cap [\mathbb{P}^2] = [\mathbb{P}^2] + \frac{d}{2}[\mathbb{P}^1] + \frac{d^2}{12}[\mathrm{pt}]$. For $d = 1$ the zero-dimensional component is $\frac{1}{12}[\mathrm{pt}]$, of degree $\frac{1}{12}$ under $\int_{\mathbb{P}^2}$. Individual Todd components carry denominators that no bundle removes; it is only the full expression appearing in Riemann-Roch, a Todd class multiplied by a Chern character, that is forced to have integral degree.

**Proposition 3.3.12: Multiplicativity of the Todd Class**

Let $0 \to E' \to E \to E'' \to 0$ be an exact sequence of vector bundles on an algebraic scheme $X$. Then
$$\mathrm{td}(E) = \mathrm{td}(E')\,\mathrm{td}(E'')$$
as operators on $A_*(X)_{\mathbb{Q}}$. In particular $\mathrm{td}(E \oplus F) = \mathrm{td}(E)\,\mathrm{td}(F)$, and $\mathrm{td}(E \oplus \mathcal{O}_X^{\oplus m}) = \mathrm{td}(E)$.

***Proof*:**
Choose, by theorem 3.2.14, a flat proper $f \colon X' \to X$ with $f^*$ injective splitting both $E'$ and $E''$, with roots $a_1, \ldots, a_{r'}$ and $b_1, \ldots, b_{r''}$. Part (1) of the proof of theorem 3.3.8 shows that $f$ then splits $E$, with roots the union of the two lists. Applying proposition 3.3.4(4) three times, to $E$, to $E'$ and to $E''$, and using the factorization of the defining product over the union of the two lists,
$$f^*\big(\mathrm{td}(E) \cap \alpha\big) = \Big(\prod_p Q(a_p) \prod_q Q(b_q)\Big) \cap f^*\alpha = f^*\big(\mathrm{td}(E')\,\mathrm{td}(E'') \cap \alpha\big)$$
for every $\alpha \in A_*(X)_{\mathbb{Q}}$, the two operators on the right commuting by proposition 3.3.4(1). Injectivity of $f^*$ gives $\mathrm{td}(E) = \mathrm{td}(E')\mathrm{td}(E'')$. The direct sum is the split case, and a trivial bundle contributes $Q(0)^m = 1$ because $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, completing the proof.

Additivity of ch and multiplicativity of td are what make both computable: any bundle presented by an exact sequence with known ends has both classes determined. The tangent bundle of projective space is presented in exactly that way, and the computation below is the one Riemann-Roch is checked against.

**Example 3.3.13: The Tangent Bundle of $\mathbb{P}^n$**

Example 3.2.17 has already imported the Euler sequence from [2, The Euler Sequence], dualized it to
$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$
and read off $c(T_{\mathbb{P}^n}) = (\mathrm{id} + H)^{n+1}$. What is new here is the Chern character and Todd class the same sequence yields. Here $\mathbb{P}^n$ is projective $n$-space with its standard twisting sheaf, so the convention of box 0.0.3 does not intervene and no further dualization is needed. Write $h = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$.

*The Chern character.* By theorem 3.3.8(1), $\mathrm{ch}(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}) = \mathrm{ch}(\mathcal{O}_{\mathbb{P}^n}) + \mathrm{ch}(T_{\mathbb{P}^n})$, and the left side is $(n+1)e^h$ while $\mathrm{ch}(\mathcal{O}_{\mathbb{P}^n}) = 1$. Hence
$$\mathrm{ch}(T_{\mathbb{P}^n}) = (n+1)e^h - 1$$

*The Todd class.* By proposition 3.3.12 applied to the same sequence, $\mathrm{td}(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}) = \mathrm{td}(\mathcal{O}_{\mathbb{P}^n})\,\mathrm{td}(T_{\mathbb{P}^n}) = \mathrm{td}(T_{\mathbb{P}^n})$, so
$$\mathrm{td}(T_{\mathbb{P}^n}) = Q(h)^{\,n+1}$$
with $Q$ the series of definition 3.3.9. Both classes are computed from the single operator $h$, whose powers are known: $h^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ and $h^{n+1} = 0$.

*The case $n = 1$.* Here $h^2 = 0$, so $\mathrm{ch}(T_{\mathbb{P}^1}) = 2(1 + h) - 1 = 1 + 2h$ and $\mathrm{td}(T_{\mathbb{P}^1}) = (1 + \frac{h}{2})^2 = 1 + h$. The first agrees with the identification $T_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(2)$, whose Chern character is $e^{2h} = 1 + 2h$ by example 3.3.7. The dimension-lowering component of $\mathrm{td}(T_{\mathbb{P}^1})$ caps to $h \cap [\mathbb{P}^1] = [\mathrm{pt}]$, of degree $\int_{\mathbb{P}^1} [\mathrm{pt}] = 1$.

*The case $n = 2$.* Here $h^3 = 0$ and

$$\mathrm{ch}(T_{\mathbb{P}^2}) = 3\Big(1 + h + \tfrac{h^2}{2}\Big) - 1 = 2 + 3h + \tfrac{3}{2}h^2$$

which cross-checks against proposition 3.3.6: the Whitney formula applied to the Euler sequence gives $c(T_{\mathbb{P}^2}) = (1+h)^3 = 1 + 3h + 3h^2$, so $c_1 = 3h$ and $c_2 = 3h^2$, and then $\mathrm{ch}_0 = 2 = \mathrm{rank} T_{\mathbb{P}^2}$, $\mathrm{ch}_1 = c_1 = 3h$, and $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2) = \frac{1}{2}(9h^2 - 6h^2) = \frac{3}{2}h^2$. For the Todd class,

$$\mathrm{td}(T_{\mathbb{P}^2}) = \Big(1 + \tfrac{h}{2} + \tfrac{h^2}{12}\Big)^3 = 1 + \tfrac{3}{2}h + \Big(3 \cdot \tfrac{1}{12} + 3 \cdot \tfrac{1}{4}\Big)h^2 = 1 + \tfrac{3}{2}h + h^2$$

agreeing with $\mathrm{td}_2 = \frac{1}{12}(c_1^2 + c_2) = \frac{1}{12}(9+3)h^2 = h^2$ from proposition 3.3.10. Capping with the fundamental class,

$$\mathrm{ch}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2] = 2[\mathbb{P}^2] + 3[\mathbb{P}^1] + \tfrac{3}{2}[\mathrm{pt}], \qquad \mathrm{td}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2] = [\mathbb{P}^2] + \tfrac{3}{2}[\mathbb{P}^1] + [\mathrm{pt}]$$

in $A_*(\mathbb{P}^2)_{\mathbb{Q}}$.

*The check.* In both cases the zero-dimensional component of $\mathrm{td}(T_{\mathbb{P}^n}) \cap [\mathbb{P}^n]$ has degree 1 under $\int_{\mathbb{P}^n}$, and $\chi(\mathcal{O}_{\mathbb{P}^n}) = 1$ by [2, Twisted Sheaf Cohomology]. The agreement of these two numbers is the simplest instance of the Riemann-Roch theorem proved later in this book, which computes, for a vector bundle $E$ on a smooth projective $X$, the integer $\chi(E) = \int_X \mathrm{ch}(E)\,\mathrm{td}(X) \cap [X]$; and it is the reason $t/(1 - e^{-t})$, rather than some other multiplicative series, is the one attached to the tangent bundle.

**3.3.14 Remark: Operators Now, Ring Elements Later** Both classes have been defined as operators on $A_*(X)_{\mathbb{Q}}$, and every identity above is an identity of operators, composition serving as multiplication. This is the only option available so far, since $A_*(X)$ carries no product; chapter 4 and the chapter after it supply one for smooth $X$, at which point $A^*(X)_{\mathbb{Q}}$ becomes a graded ring, capping with the fundamental class turns each of these operators into a class, and $\mathrm{ch}(E)$ and $\mathrm{td}(E)$ are elements of that ring. No statement in this section changes when that happens: an identity of operators on $A_*(X)_{\mathbb{Q}}$ specializes to an identity of classes by applying it to $[X]$.

## Exercise 3.3.1

1. Let $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ and let $L = \mathcal{O}_{\mathbb{P}^2}(d)$. Compute $\mathrm{ch}(L) \cap [\mathbb{P}^2]$ and $\mathrm{td}(L) \cap [\mathbb{P}^2]$ in $A_*(\mathbb{P}^2)_{\mathbb{Q}}$, and compute the degree $\int_{\mathbb{P}^2}$ of the zero-dimensional component of $\mathrm{ch}(L)\,\mathrm{td}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2]$. Determine for which $d$ this last number is an integer.

2. Show that $\mathrm{ch}_k(E^\vee) = (-1)^k \mathrm{ch}_k(E)$ for every vector bundle $E$ and every $k$. (Dualize a filtration by subbundles and use proposition 2.4.4(3).) Deduce that $\mathrm{ch}(E \otimes E^\vee)$ has $\mathrm{ch}_1 = 0$ for every $E$.

3. Let $E$ have rank $r$ and let $\det E = \Lambda^r E$. Show that $c_1(\det E) = c_1(E)$ and hence $\mathrm{ch}(\det E) = e^{c_1(E)}$. (Use that the top exterior power of a filtered bundle is the tensor product of the line-bundle quotients.)

4. Compute $\mathrm{ch}(T_{\mathbb{P}^3})$ and $\mathrm{td}(T_{\mathbb{P}^3})$ as polynomials in $h = c_1(\mathcal{O}_{\mathbb{P}^3}(1))$, and verify that $\int_{\mathbb{P}^3}$ of the zero-dimensional component of $\mathrm{td}(T_{\mathbb{P}^3}) \cap [\mathbb{P}^3]$ equals 1. Check the degree-two and degree-three Todd components against proposition 3.3.10 using $c(T_{\mathbb{P}^3}) = (1+h)^4$.
5. Let $S$ be a smooth projective surface and $E$ a vector bundle of rank $r$ on $S$. Write out the zero-dimensional component of $\mathrm{ch}(E)\,\mathrm{td}(T_S) \cap [S]$ in terms of $c_1(E), c_2(E), c_1(T_S), c_2(T_S)$, and specialize to $E = \mathcal{O}_S$ and to $E$ a line bundle.
6. Show that $\mathrm{ch}(E \oplus \mathcal{O}_X^{\oplus m}) = \mathrm{ch}(E) + m$ and $\mathrm{td}(E \oplus \mathcal{O}_X^{\oplus m}) = \mathrm{td}(E)$, and explain why no bundle of positive rank can have $\mathrm{ch}(E) = 0$.

## 3.4 Segre Classes of Cones and Subschemes

The Segre operators of section 3.1 came out of one recipe: projectivize, cap with powers of the tautological class, push down. Local triviality was never used in that recipe. What it used was a graded sheaf of algebras on $X$ whose relative Proj is proper over $X$ and carries a canonical line bundle, and a vector bundle supplies one such algebra among many. This section runs the recipe on the general input, a *cone*, and then on the case that matters here: the normal cone of a closed subscheme $Z \subseteq X$. The resulting class $s(Z, X)$ is an invariant of the embedding defined with no smoothness, no regularity and no dimension hypothesis, and it is the object that chapter 4 will deform $X$ onto.

The algebra that produces a cone is required to look like a polynomial algebra in the two respects that matter for Proj : it starts in degree zero at $\mathcal{O}_X$, and it is generated by its degree-one part, which is coherent.

**Definition 3.4.1: Cone**

Let $X$ be an algebraic scheme. A *cone* over $X$ is the relative spectrum

$$C = \mathrm{Spec}\left(S^\bullet\right) \xrightarrow{p} X$$

of a sheaf $S^\bullet = \bigoplus_{n \geq 0} S_n$ of graded quasi-coherent $\mathcal{O}_X$-algebras ([2, graded quasi-coherent algebra]) such that

1. $S_0 = \mathcal{O}_X$;
2. $S_1$ is coherent;
3. $S^\bullet$ is generated as an $\mathcal{O}_X$-algebra by $S_1$.

The relative spectrum is the construction of [2, Relative Spectrum], glued from the affine pieces $\mathrm{Spec}\,\Gamma(U, S^\bullet)$ over an affine open cover of $X$, exactly as the relative Proj of [2, Relative Proj] is. Writing $S_+ = \bigoplus_{n \geq 1} S_n$, the surjection $S^\bullet \to S^\bullet/S_+ = \mathcal{O}_X$ defines a closed immersion $X \hookrightarrow C$, the *vertex* of the cone.

Conditions (2) and (3) force every $S_n$ to be coherent, so $p$ is affine and of finite type; the grading is what distinguishes a cone from an arbitrary affine morphism, and it is the grading that supplies the vertex and, below, the compactification. A cone is not required to be flat over $X$, nor to have fibres of constant dimension, and both failures are the point: the cones that arise from geometry are singular, jump in dimension along strata, and carry exactly the information those defects encode.

**Example 3.4.2: Bundles, the Zero Cone, and the Affine Cone Over a Projective Variety**

1. *A vector bundle.* Let $E$ be a vector bundle of rank $e$ on $X$ with sheaf of sections $\mathcal{E}$. The functions on the total space of $E$ are the symmetric algebra on the dual, so $S^\bullet = \mathrm{Sym}^\bullet(\mathcal{E}^\vee)$ has $\mathrm{Spec}\,(S^\bullet) = E$, and conditions (1) to (3) hold with $S_1 = \mathcal{E}^\vee$ locally free of rank $e$. The vertex $X \hookrightarrow E$ is the zero section. The dual is exactly what makes $\mathrm{Proj}\,(\mathrm{Sym}^\bullet \mathcal{E}^\vee)$ the bundle of *lines* in $E$ used throughout this book (box 0.0.3), and not the bundle of rank-one quotients of $\mathcal{E}$ that [2, the projective bundle] writes as $\mathbb{P}(\mathcal{E})$.

2. *The zero cone.* Taking $S^\bullet = \mathcal{O}_X$, so that $S_+ = 0$, gives $C = X$ with its vertex the identity. The degenerate case earns its place: it is the normal cone of $X$ in itself.

3. *An affine cone.* Let $Y = \mathrm{Proj}\,(R) \subseteq \mathbb{P}^n$ be a closed subscheme, where $R = \overline{k}[x_0, \ldots, x_n]/I$ for a homogeneous ideal $I$, and take $X = \mathrm{Spec}\,\overline{k}$. Then $R$ itself satisfies (1) to (3), and $C = \mathrm{Spec}\,(R)$ is the affine cone over $Y$: the union of the lines of $\mathbb{A}^{n+1}$ through the origin lying over the points of $Y$, together with the origin, which is the vertex. For $I = 0$ this is $\mathbb{A}^{n+1}$ itself, the trivial bundle of rank $n+1$ over a point, recovering item (1).

A cone is affine over $X$, so $p$ is proper only in the degenerate case where $S^\bullet$ is coherent as an $\mathcal{O}_X$-module and $p$ is therefore finite; the zero cone of example 3.4.2(2), with $p = \mathrm{id}$, is such a case. The cones that carry geometry are not: a vector bundle of positive rank has fibres $\mathbb{A}^e$, and an affine scheme of positive dimension over a field is not proper. So $p_*$ is unavailable in general, and pushing a cycle from $C$ down to $X$ is not an option. The remedy is the one projective space applies to affine space: compactify by adding a hyperplane at infinity. Adjoining one variable in degree one does this, and does it in a way that keeps the answer a relative Proj, so the canonical line bundle comes along for free.

**Definition 3.4.3: Projective Cone and Projective Completion**

Let $C = \mathrm{Spec}\,(S^\bullet)$ be a cone over $X$. Adjoin a variable $z$ in degree one, giving the graded $\mathcal{O}_X$-algebra $S^\bullet[z]$ with $\big(S^\bullet[z]\big)_n = \bigoplus_{j=0}^n S_j\, z^{\,n-j}$, which again satisfies the conditions of definition 3.4.1. The *projective cone* of $C$ and the *projective completion* of $C$ are the relative Proj schemes

$$\mathbb{P}(C) = \mathrm{Proj}\,\big(S^\bullet\big), \qquad \mathbb{P}(C \oplus 1) = \mathrm{Proj}\,\big(S^\bullet[z]\big)$$

Both are proper over $X$ ([2, Relative Proj]). Write $q\colon \mathbb{P}(C \oplus 1) \to X$ for the structure morphism and $\mathcal{O}(1) = \mathcal{O}_{\mathbb{P}(C\oplus 1)}(1)$ for the canonical line bundle.

Three identifications justify the name, and each is read straight off the graded algebra. First, on the locus where $z$ is invertible, $D_+(z) = \mathrm{Spec}\,\big((S^\bullet[z])_{(z)}\big)$, and the assignment $s/z^n \mapsto s$ for $s \in S_n$ is an isomorphism of $\mathcal{O}_X$-algebras $(S^\bullet[z])_{(z)} \cong S^\bullet$; so the open subscheme of $\mathbb{P}(C \oplus 1)$ where $z \neq 0$ is $C$ itself. Second, $z$ is a nonzerodivisor in $S^\bullet[z]$ and a global section of $\mathcal{O}(1)$, and its zero scheme is $\mathrm{Proj}\,\big(S^\bullet[z]/(z)\big) = \mathbb{P}(C)$; so $\mathbb{P}(C)$ is an effective Cartier divisor ([2, Effective Cartier Divisor]) on $\mathbb{P}(C \oplus 1)$ with complement $C$, the hyperplane at infinity. Third, the surjection $S^\bullet[z] \to \mathcal{O}_X[z]$ killing $S_+$ realizes $X = \mathrm{Proj}\,(\mathcal{O}_X[z])$ as a closed subscheme of $\mathbb{P}(C \oplus 1)$ contained in the open piece $C$, where it is the vertex of definition 3.4.1.

So $\mathbb{P}(C \oplus 1)$ is $C$ with $\mathbb{P}(C)$ glued on at infinity, and it is proper over $X$ whereas $C$, outside the degenerate case just noted, is not. That properness is why it is introduced: it makes $q_*$ available (definition 1.2.1 and theorem 1.2.3), and with $\mathcal{O}(1)$ in hand the construction of section 3.1 can be run verbatim.

The model case is the one the terminology is borrowed from. Take $X = \operatorname{Spec} \overline{k}$ and $S^\bullet = \overline{k}[t_1, \ldots, t_n]$, so that $C = \mathbb{A}^n$ is the trivial bundle of rank $n$ over a point. Then $S^\bullet[z] = \overline{k}[t_1, \ldots, t_n, z]$ and $\mathbb{P}(C \oplus 1) = \mathbb{P}^n$, with $\mathbb{P}(C) = \mathbb{P}^{n-1}$ the hyperplane $z = 0$ at infinity, the open complement $D_+(z)$ the standard affine chart $\mathbb{A}^n$, and the vertex the origin of that chart. Compactifying a cone is compactifying affine space by a hyperplane, carried out fibrewise over $X$ and allowing the fibres to be as singular as they like.

**Definition 3.4.4: Segre Class of a Cone**

Let $C$ be a cone over an algebraic scheme $X$, with projective completion $q\colon \mathbb{P}(C \oplus 1) \to X$ and canonical line bundle $\mathcal{O}(1)$. The *Segre class* of $C$ is the cycle class

$$s(C) \;=\; q_*\Big( \sum_{i \geq 0} c_1\big(\mathcal{O}(1)\big)^i \cap \big[\mathbb{P}(C \oplus 1)\big] \Big) \;\in\; A_*(X)$$

where $[\mathbb{P}(C \oplus 1)]$ is the fundamental cycle of definition 1.1.2 and $c_1(\mathcal{O}(1)) \cap -$ is the operator of definition 2.4.1. Write $s(C)_k$ for the component of $s(C)$ in $A_k(X)$.

The sum is finite. Each cap lowers dimension by one, so the term indexed by $i$ lies in $A_{d-i}\big(\mathbb{P}(C \oplus 1)\big)$ for $d = \dim \mathbb{P}(C \oplus 1)$, and this group vanishes once $i > d$ because there are no cycles of negative dimension. Two features of the definition deserve emphasis. The Segre class of a cone is a *class*, not an operator: the capping has already happened, against the fundamental cycle of the compactification. And it is not homogeneous; $s(C)$ has components in several dimensions at once, and the interesting content is usually spread across them.

For the model cone $C = \mathbb{A}^n$ over $X = \operatorname{Spec} \overline{k}$ the class can be read off. The completion is $\mathbb{P}^n$, and $q_*\big(c_1(\mathcal{O}(1))^i \cap [\mathbb{P}^n]\big)$ lies in $A_{n-i}(\operatorname{Spec} \overline{k})$, which vanishes unless $i = n$. For $i = n$, example 2.4.7 gives $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathrm{pt}]$, whose pushforward to the point is the fundamental class. Hence $s(\mathbb{A}^n) = [\operatorname{Spec} \overline{k}]$: the trivial bundle has trivial Segre class, as it must.

Before the definition can be trusted it has to be checked against the case it generalizes. A vector bundle is a cone, and its Segre class as a cone had better be its total Segre operator evaluated on the fundamental class of the base. The bundle has to have positive rank for that comparison to mean anything: the operators $s_i(E)$ of section 3.1 are manufactured from $\mathbb{P}(E)$, which is empty for the zero bundle, so neither $s(E)$ nor $c(E)$ is defined there. The rank-zero cone is the zero cone of example 3.4.2(2), whose Segre class is computed directly below.

**Proposition 3.4.5: The Segre Class of a Vector Bundle as a Cone**

Let $E$ be a vector bundle of rank $e \geq 1$ on an algebraic scheme $X$, and let $C$ be its total space regarded as a cone (example 3.4.2(1)). Then

$$s(C) \;=\; s(E) \cap [X]$$

where $s(E) = \sum_{j \geq 0} s_j(E)$ is the total Segre operator of section 3.1.

***Proof*:**
*Step 1: the projective completion is a projective bundle.* With $S^\bullet = \mathrm{Sym}^\bullet(\mathcal{E}^\vee)$ one has $S^\bullet[z] = \mathrm{Sym}^\bullet\big(\mathcal{E}^\vee \oplus \mathcal{O}_X\big) = \mathrm{Sym}^\bullet\big((\mathcal{E} \oplus \mathcal{O}_X)^\vee\big)$, the variable $z$ being the coordinate dual to the added trivial summand. Hence $\mathbb{P}(C \oplus 1) = \mathbb{P}(E \oplus 1)$ is the bundle of lines in the rank-$(e+1)$ bundle $E \oplus 1$, its fibres are copies of $\mathbb{P}^e$, and its canonical bundle $\mathcal{O}(1)$ is the one whose first Chern class is the tautological class of section 3.1. In particular $q$ is proper and flat of relative dimension $e$ ([2, A Projective Bundle is Proper and Flat]).

*Step 2: the fundamental cycle is a flat pullback.* Write $n = \dim X$ and let $X_1, \ldots, X_r$ be the $n$-dimensional irreducible components of $X$, so that the fundamental cycle is $[X] = \sum_\lambda m_\lambda [X_\lambda]$ with $m_\lambda$ the length of $\mathcal{O}_{X,X_\lambda}$. Each preimage $q^{-1}(X_\lambda)$ is a projective bundle over the variety $X_\lambda$, hence integral, so $q^*[X_\lambda] = [q^{-1}(X_\lambda)]$ by proposition 1.3.5(2); and the length multiplicity of $q^{-1}(X_\lambda)$ in $[\mathbb{P}(E \oplus 1)]$ is again $m_\lambda$, because the local homomorphism $\mathcal{O}_{X,X_\lambda} \to \mathcal{O}_{\mathbb{P}(E\oplus 1),\, q^{-1}(X_\lambda)}$ is flat with fibre the function field of a projective space, so the two lengths agree. Summing over $\lambda$,

$$q^*[X] = \big[\mathbb{P}(E \oplus 1)\big]$$

*Step 3: the terms with $i \geq e$.* Definition 3.1.3 defines the Segre operators of a bundle $F$ of rank $f$ by $s_j(F) \cap \alpha = \pi_*\big(\xi^{\,f-1+j} \cap \pi^*\alpha\big)$, where $\pi\colon \mathbb{P}(F) \to X$ has relative dimension $f-1$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(F)}(1))$. Applied to $F = E \oplus 1$, of rank $e+1$, this reads

$$s_j(E \oplus 1) \cap \alpha = q_*\big(c_1(\mathcal{O}(1))^{\,e+j} \cap q^*\alpha\big), \qquad j \geq 0$$

so by Step 2 the terms of definition 3.4.4 indexed by $i = e + j$ with $j \geq 0$ sum to $\sum_{j\geq 0} s_j(E \oplus 1) \cap [X] = s(E \oplus 1) \cap [X]$.

*Step 4: the terms with $i < e$ vanish.* Flat pullback raises dimension by $e$ and each cap lowers it by one, so $q_*\big(c_1(\mathcal{O}(1))^i \cap q^*[X]\big)$ lies in $A_{n+e-i}(X)$. For $i < e$ this index exceeds $n = \dim X$, and $X$ carries no subvariety of dimension greater than its own, so the group is zero and every such term vanishes.

*Step 5: dropping the trivial summand.* This is the only step that uses $e \geq 1$: both $E$ and $E \oplus 1$ then have positive rank, so both have invertible total Segre operators (lemma 3.2.2) and Chern operators to compare (definition 3.2.3). The split exact sequence $0 \to E \to E \oplus 1 \to \mathcal{O}_X \to 0$ and the Whitney formula (theorem 3.2.15) give $c(E \oplus 1) = c(E)\, c(\mathcal{O}_X)$. A line bundle has no Chern classes above the rank (proposition 3.2.6), so $c(\mathcal{O}_X) = \mathrm{id} + c_1(\mathcal{O}_X)$, and $c_1(\mathcal{O}_X) = 0$ by example 2.4.3; hence $c(E \oplus 1) = c(E)$ and, inverting, $s(E \oplus 1) = s(E)$. Combining with Steps 3 and 4,

$$s(C) = s(E \oplus 1) \cap [X] = s(E) \cap [X]$$

as we sought to show.

The proposition licenses the notation: for the cone of a vector bundle the class $s(C)$ of definition 3.4.4 and the operator $s(E)$ of section 3.1 determine one another, and writing $s$ for both loses nothing. What the proposition also shows is where the new content will sit. On a bundle, the compactification $\mathbb{P}(E \oplus 1)$ is flat over $X$ with fibres of constant dimension, the Segre operators exist, and $s(C)$ is a formal consequence of them. On a general cone none of that is available, and $s(C)$ does not come from any operator; it is a class computed once, by pushing forward from a compactification that may be singular and may jump in dimension over $X$.

The cone this was built for measures how a closed subscheme sits inside an ambient scheme. When

$Z$ is a smooth subvariety of a smooth variety, the correct linear model is the normal bundle, and its rank is the codimension. When either smoothness hypothesis fails there is no bundle, but the ideal sheaf still has an associated graded algebra, and that algebra is always a cone.

**Definition 3.4.6: Normal Cone of a Closed Subscheme**

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$. The graded sheaf of algebras $\bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$ has degree-zero part $\mathcal{O}_X/\mathcal{I} = \mathcal{O}_Z$, coherent degree-one part $\mathcal{I}/\mathcal{I}^2$, and is generated in degree one, so it satisfies the conditions of definition 3.4.1. The *normal cone* of $Z$ in $X$ is the cone over $Z$

$$C_Z X \;=\; \operatorname{Spec}\Big(\bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}\Big) \;\longrightarrow\; Z$$

On an affine open $U = \operatorname{Spec} A$ of $X$ with $Z \cap U = \operatorname{Spec} A/I$, this is $\operatorname{Spec}\big(\bigoplus_{n\geq 0} I^n/I^{n+1}\big)$.

The normal cone is intrinsic to the pair $(Z, X)$ and depends on the scheme structure of $Z$, not just on its underlying set: the ideal $\mathcal{I}$ is the input, and thickening $Z$ changes it. When $Z$ is a single reduced point $P$ of a variety $X$, the normal cone is the classical tangent cone $\operatorname{Spec}\big(\operatorname{gr}_{\mathfrak{m}} \mathcal{O}_{X,P}\big)$, the cone of limiting directions of branches of $X$ through $P$, and it is where the local singularity of $X$ at $P$ becomes visible.

**Example 3.4.7: The Tangent Cone at a Node**

Assume $\operatorname{char} \overline{k} \neq 2$ and let $X = \operatorname{Spec} \overline{k}[x,y]/(y^2 - x^2 - x^3) \subseteq \mathbb{A}^2$ be the affine nodal cubic, with $P$ the origin and $\mathfrak{m} = (x, y)$ its maximal ideal.[a] The associated graded ring of $\overline{k}[x,y]$ for the filtration by powers of $(x, y)$ is again $\overline{k}[x,y]$, a domain, so initial forms multiply: the initial form of $gh$ is the product of those of $g$ and $h$. Consequently the initial forms of the elements of the principal ideal $(y^2 - x^2 - x^3)$ are generated by the initial form of its generator, namely $y^2 - x^2$. Hence

$$\bigoplus_{n\geq 0} \mathfrak{m}^n/\mathfrak{m}^{n+1} \;\cong\; \overline{k}[x,y]/(y^2 - x^2)$$

and $C_P X = \operatorname{Spec} \overline{k}[x,y]/(y^2 - x^2)$ is the union of the two lines $y = x$ and $y = -x$: precisely the two branches of the node, straightened to their tangent directions. The normal cone is one-dimensional, matching $\dim X = 1$, but it is reducible where $X$ is irreducible, and it is not a vector bundle over the point $P$ (a rank-one bundle over a point is a single line). The failure is exactly the singularity.

[a] The hypothesis on the characteristic cannot be dropped. In characteristic 2 the substitution $u = y + x$ turns the equation into $u^2 = x^3$, so the curve is a cuspidal cubic with a single branch, and $y^2 - x^2 = (y + x)^2$ makes the ring computed below the coordinate ring of a single non-reduced double line, which is irreducible; the contrast drawn at the end of the example then fails.

The normal cone is a cone over $Z$, so it has a Segre class, and that class is what records how $Z$ sits inside $X$ when no bundle is available to record it.

**Definition 3.4.8: Segre Class of a Closed Subscheme**

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme. The *Segre class of $Z$ in $X$* is the Segre class of the normal cone,

$$s(Z, X) \;=\; s\big(C_Z X\big) \;\in\; A_*(Z)$$

Two points about the target group. The class lives on $Z$, not on $X$: like the intersection classes of definition 2.2.1, it is recorded where the geometry happens, and one pushes forward along $Z \hookrightarrow X$ only when the cruder class in $A_*(X)$ is wanted. And no hypothesis whatsoever was imposed on the pair $(Z, X)$: $X$ may be singular, reducible, or nonreduced, $Z$ may be any closed subscheme including $X$ itself, and $s(Z, X)$ is defined regardless. That generality is what makes the class useful and also what makes it hard to compute; the two computations below are the cases where it can be read off.

The extreme case is immediate. Taking $Z = X$ makes $\mathcal{I} = 0$, so the normal cone is the zero cone of example 3.4.2(2), with projective completion $\operatorname{Proj}(\mathcal{O}_X[z]) = X$ and with $\mathcal{O}(1)$ free on the generator $z$, hence trivial. Every term of definition 3.4.4 beyond the first therefore vanishes, since $c_1(\mathcal{O}_X) = 0$ by example 2.4.3, and $s(X, X) = [X]$. A subscheme sitting inside itself has no normal directions, and its Segre class is its own fundamental cycle.

**3.4.9 Foreshadowing: Why the Normal Cone** The normal cone is not introduced here for its own sake. Chapter 4 constructs a family over $\mathbb{P}^1$ whose general fibre is the embedding $Z \hookrightarrow X$ and whose special fibre is the embedding of $Z$ as the vertex of $C_Z X$; intersecting with $Z$ inside $X$ is then replaced by intersecting with the vertex inside the cone, where the answer can be read off. When $C_Z X$ is a vector bundle that replacement produces the Gysin map of a regular embedding; when it is not, the Segre class $s(Z, X)$ is what survives, and it is the term that appears in the excess intersection formula.

The first computable case is the one where the normal cone degenerates to a bundle. The condition on the ideal is the classical one: it should be generated, locally, by the right number of equations imposing independent conditions.

Regular embeddings and their normal bundles are scheme theory, and they belong to, and are defined in, *Algebraic Geometry*: a closed immersion whose ideal sheaf is locally generated by a regular sequence of length $d$ is a *regular embedding of codimension $d$* ([2, Regular Embedding]), its conormal sheaf $\mathcal{I}/\mathcal{I}^2$ is then locally free of rank $d$ ([2, The Conormal Sheaf of a Regular Embedding]), and its *normal bundle* is the rank-$d$ vector bundle

$$N_Z X \;=\; \underline{\operatorname{Spec}}_Z\Big(\operatorname{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)\Big) \;\longrightarrow\; Z$$

with sheaf of sections $(\mathcal{I}/\mathcal{I}^2)^\vee$ ([2, Conormal Sheaf and Normal Bundle]). What this book adds is the cycle-level consequence below.

The definition is local on $Z$, and the number $d$ is forced: the rank of $\mathcal{I}/\mathcal{I}^2$ is a local invariant, so a regular embedding has a well-defined codimension on each connected component of $Z$. The two extremes are familiar. The origin in $\mathbb{A}^n$, whose ideal $(x_1, \ldots, x_n)$ is a regular sequence, is a regular embedding of codimension $n$ with trivial normal bundle of rank $n$; an effective Cartier divisor, whose ideal is locally generated by a single nonzerodivisor, is the codimension-one case, with normal bundle a line bundle. What makes the notion useful here is that the whole associated graded algebra, not its degree-one part, becomes a symmetric algebra.

**Lemma 3.4.10: The Associated Graded Algebra of a Regular Sequence**

Let $A$ be a ring and let $a_1, \dots, a_d$ be a regular sequence generating an ideal $I \subseteq A$. Then the homomorphism of graded $A/I$-algebras

$$(A/I)[T_1, \dots, T_d] \longrightarrow \bigoplus_{n \geq 0} I^n/I^{n+1}, \qquad T_j \longmapsto a_j \bmod I^2$$

is an isomorphism. In particular $I/I^2$ is a free $A/I$-module of rank $d$ on the classes of $a_1, \dots, a_d$, and the canonical surjection $\mathrm{Sym}^\bullet_{A/I}(I/I^2) \to \bigoplus_{n \geq 0} I^n/I^{n+1}$ is an isomorphism.

***Proof***:
This is [3, Associated Graded Ring of a Regular Sequence], proved there by induction on the length of the sequence. It is pure commutative algebra and is not re-derived here.

**Proposition 3.4.11: Segre Class of a Regular Embedding**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, with normal bundle $N = N_Z X$. Then the normal cone is the normal bundle,

$$C_Z X \;=\; N_Z X$$

and the Segre class of $Z$ in $X$ is

$$s(Z, X) \;=\; s(N) \cap [Z] \;=\; c(N)^{-1} \cap [Z]$$

where $s(N)$ and $c(N)$ are the total Segre and Chern operators of sections 3.1 and 3.2.

***Proof***:
Work on an affine open $U = \operatorname{Spec} A$ of $X$ on which $I = \mathcal{I}(U)$ is generated by a regular sequence $a_1, \dots, a_d$. There is always a canonical surjection of graded $A/I$-algebras

$$\mathrm{Sym}^\bullet_{A/I}(I/I^2) \longrightarrow \bigoplus_{n \geq 0} I^n/I^{n+1}$$

sending a product of degree-one classes to the class of the corresponding product in $I^n$; it is surjective because the target is generated in degree one. Lemma 3.4.10 says that it is injective as well, and that $I/I^2$ is free of rank $d$ on the classes of the $a_j$, so it is an isomorphism over $U$.

The surjection above is canonical, so the local isomorphisms are compatible on overlaps and glue to an isomorphism of graded $\mathcal{O}_Z$-algebras $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \xrightarrow{\sim} \bigoplus_{n \geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$. Taking relative spectra and comparing definition 3.4.6 with [2, Regular Embedding] gives $C_Z X = N_Z X$.

The normal cone is therefore the cone of a vector bundle of rank $d \geq 1$ over $Z$, so proposition 3.4.5 applies and gives $s(Z, X) = s(C_Z X) = s(N) \cap [Z]$. Finally $s(N) = c(N)^{-1}$, the Chern operators of definition 3.2.3 being defined by inverting the Segre series, completing the proof.

The normal cone is functorial in the subscheme, and for a regular embedding it sits inside a pulled-back normal bundle. Both facts are needed by chapter 4, and both are statements about cones as such, so they belong here rather than there.

**Lemma 3.4.12: The Normal Cone of a Closed Subscheme**

Let $Z, V \subseteq X$ be closed subschemes of an algebraic scheme and let $W = V \cap Z$ be their scheme-theoretic intersection. Then $W$ is the scheme-theoretic preimage of $Z$ under $V \hookrightarrow X$, and there is a canonical closed immersion

$$C_W V \hookrightarrow C_Z X$$

lying over the inclusion $W \hookrightarrow Z$.

***Proof*:**
The ideal sheaf of $W$ in $V$ is $\mathcal{I}_W = \mathcal{I}\mathcal{O}_V$, which is exactly the statement that $W$ is the scheme-theoretic preimage of $Z$ under the closed immersion $V \hookrightarrow X$. Consequently $\mathcal{I}^n\mathcal{O}_V = \mathcal{I}_W^n$ for every $n$, so the homomorphism of graded $\mathcal{O}_W$-algebras

$$\Big(\bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}\Big) \otimes_{\mathcal{O}_Z} \mathcal{O}_W \longrightarrow \bigoplus_{n\geq 0} \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$$

is surjective in each degree, the image of $\mathcal{I}^n \otimes \mathcal{O}_V \to \mathcal{O}_V$ being $\mathcal{I}_W^n$. On an affine open this is a surjection of graded rings, so by [2, Relative Spectrum](1) it induces a closed immersion of relative spectra; the local closed immersions are induced by a globally defined homomorphism of sheaves of algebras, so they glue. This exhibits $C_W V$ as a closed subscheme of the relative spectrum of the left side, which is $C_Z X \times_Z W$, again by the affine description of the relative spectrum. Composing with the closed immersion $C_Z X \times_Z W \hookrightarrow C_Z X$, the base change of $W \hookrightarrow Z$, gives the asserted closed immersion, completing the proof.

**Lemma 3.4.13: The Normal Cone of a Preimage Embeds in the Pulled-Back Normal Bundle**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with ideal sheaf $\mathcal{I}$ and normal bundle $N = N_Z X$ ([2, Regular Embedding]). Let $g\colon X' \to X$ be a morphism of algebraic schemes, let $Z' = g^{-1}(Z)$ be the scheme-theoretic preimage, and let $h\colon Z' \to Z$ be the induced morphism. Write $N' = h^*N$, a vector bundle of rank $d$ on $Z'$. Then there is a canonical closed immersion of cones over $Z'$

$$j\colon C_{Z'}X' \hookrightarrow N'$$

Moreover:

1. $j$ is an isomorphism whenever $Z' \hookrightarrow X'$ is itself a regular embedding of codimension $d$;
2. for $g = \mathrm{id}_X$ it is the identification $C_Z X = N$ of proposition 3.4.11.

***Proof*:**
Write $\mathcal{I}' \subseteq \mathcal{O}_{X'}$ for the ideal sheaf of $Z'$.

*Step 1: the conormal surjection.* By definition of the scheme-theoretic preimage, $\mathcal{I}'$ is the image of the map $g^*\mathcal{I} \to \mathcal{O}_{X'}$; that is, $\mathcal{I}'$ is generated by the pullbacks of local sections of $\mathcal{I}$. Composing $g^*\mathcal{I} \to \mathcal{I}'$ with the quotient $\mathcal{I}' \to \mathcal{I}'/\mathcal{I}'^2$ gives an $\mathcal{O}_{X'}$-linear surjection onto $\mathcal{I}'/\mathcal{I}'^2$. It kills the image of $g^*(\mathcal{I}^2)$, a product of two local sections of $\mathcal{I}$ pulling back to a product of two local sections of $\mathcal{I}'$. Since $g^*$ is right exact, $g^*(\mathcal{I}/\mathcal{I}^2)$ is the cokernel of $g^*(\mathcal{I}^2) \to g^*\mathcal{I}$, so the surjection factors through it; and $g^*(\mathcal{I}/\mathcal{I}^2)$ is annihilated by $\mathcal{I}'$, hence is the $\mathcal{O}_{Z'}$-module $h^*(\mathcal{I}/\mathcal{I}^2)$. This gives a surjection of $\mathcal{O}_{Z'}$-modules

$$h^*\big(\mathcal{I}/\mathcal{I}^2\big) \twoheadrightarrow \mathcal{I}'/\mathcal{I}'^2$$

*Step 2: the surjection of graded algebras.* The symmetric algebra functor preserves surjections, so Step 1 gives a surjection $\mathrm{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \to \mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ of graded $\mathcal{O}_{Z'}$-algebras. The graded algebra $\bigoplus_{n\geq 0} \mathcal{I}'^n/\mathcal{I}'^{n+1}$ of definition 3.4.6 is generated in degree one over $\mathcal{O}_{Z'}$, so the canonical map from $\mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ onto it is surjective as well. Composing,

$$\mathrm{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \twoheadrightarrow \bigoplus_{n\geq 0} \mathcal{I}'^n/\mathcal{I}'^{n+1}$$

is a surjection of graded quasi-coherent $\mathcal{O}_{Z'}$-algebras.

*Step 3: taking relative spectra.* On an affine open $U \subseteq Z'$ the displayed surjection is a surjection of rings $\Gamma(U, \mathrm{Sym}^\bullet h^*(\mathcal{I}/\mathcal{I}^2)) \to \Gamma(U, \bigoplus \mathcal{I}'^n/\mathcal{I}'^{n+1})$, which induces a closed immersion of affine schemes in the opposite direction; these are compatible on overlaps because the surjection is a map of sheaves, so they glue to a closed immersion over $Z'$ by [2, Relative Spectrum]. The source of that closed immersion is $C_{Z'}X'$ by definition 3.4.6. Its target is, by [2, Regular Embedding] and example 3.4.2(1), the total space of the vector bundle on $Z'$ whose sheaf of sections is $\big(h^*(\mathcal{I}/\mathcal{I}^2)\big)^\vee = h^*\big((\mathcal{I}/\mathcal{I}^2)^\vee\big)$, that is $N' = h^*N$; regularity of $i$ enters here, and only here, through the local freeness of $\mathcal{I}/\mathcal{I}^2$ of rank $d$ (lemma 3.4.10), which makes $h^*(\mathcal{I}/\mathcal{I}^2)$ locally free of rank $d$ on $Z'$. The immersion is a morphism of cones because the algebra surjection is graded.

*Step 4: part (1).* Suppose $Z' \hookrightarrow X'$ is a regular embedding of codimension $d$. Then $\mathcal{I}'/\mathcal{I}'^2$ is locally free of rank $d$ and the canonical map $\mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2) \to \bigoplus_n \mathcal{I}'^n/\mathcal{I}'^{n+1}$ is an isomorphism, both by lemma 3.4.10 applied to $X'$. So $j$ is the morphism induced by the surjection $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ of locally free $\mathcal{O}_{Z'}$-modules of the same rank $d$. Such a surjection is an isomorphism: the target is locally free, hence projective, so the surjection splits and the source decomposes as the target plus the kernel; the kernel is then a direct summand of a locally free module of finite rank, hence itself locally free, and comparing ranks locally makes that rank zero, so the kernel vanishes. Therefore $j$ is an isomorphism of cones.

*Step 5: part (2).* For $g = \mathrm{id}_X$ one has $\mathcal{I}' = \mathcal{I}$, $Z' = Z$, $h = \mathrm{id}_Z$, and the surjection of Step 2 is the canonical map $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \to \bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$. That map is the isomorphism exhibited in the proof of proposition 3.4.11, so $j$ is the identification $C_Z X = N$ recorded there, completing the proof.

Both immersions above are induced by surjections of graded algebras out of $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$, and when they are both available they agree. This is used whenever a normal cone is read inside a normal bundle.

**Lemma 3.4.14: The Two Immersions Agree**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d$ with normal bundle $N$, and let $V \subseteq X$ be a closed subscheme with $W = V \cap Z$. Then the composite

$$C_W V \hookrightarrow N|_W \hookrightarrow N$$

of the immersion $j$ of lemma 3.4.13 (for $g\colon V \hookrightarrow X$) with the restriction of $N$ to $W$ coincides with the immersion $C_W V \hookrightarrow C_Z X = N$ of lemma 3.4.12.

***Proof*:**
Both immersions are the map on relative Spec induced by a surjection of graded $\mathcal{O}_W$-algebras onto $\bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$, so it is enough to see that the two surjections coincide. The immersion of lemma 3.4.12 is induced by

$$\Big(\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}\Big) \otimes_{\mathcal{O}_Z} \mathcal{O}_W \twoheadrightarrow \bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$$

and $j$ is induced by $\operatorname{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \twoheadrightarrow \bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$. Because $i$ is a regular embedding, proposition 3.4.11 identifies $\operatorname{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$ with $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$ as graded $\mathcal{O}_Z$-algebras, and that identification is the identity in degree one. Pulling it back along $h$ turns the second surjection into the first, since both are determined by their degree-one component, which in each case is the map $\mathcal{I}/\mathcal{I}^2 \otimes \mathcal{O}_W \to \mathcal{I}_W/\mathcal{I}_W^2$ induced by $\mathcal{I}\mathcal{O}_V = \mathcal{I}_W$, as we sought to show.

For a regular embedding the Segre class carries no information beyond the normal bundle. The hypothesis $d \geq 1$ costs nothing: a regular embedding of codimension zero has zero ideal sheaf, so it is $Z = X$, and that case was settled directly above by the zero cone. The inverse $c(N)^{-1}$ is computed by the terminating geometric series $\mathrm{id} - c_1(N) + \big(c_1(N)^2 - c_2(N)\big) - \cdots$, terminating because every $c_j(N)$ with $j \geq 1$ lowers dimension and $A_k(Z) = 0$ for $k < 0$. The content of $s(Z, X)$ is therefore entirely in the irregular case: it is the invariant that continues to exist, and to enter the intersection formulas, when the ideal of $Z$ needs more equations than the codimension allows.

The codimension-one case is the one already visible in chapter 2, and it recovers the tower of self-intersections of a divisor.

**Example 3.4.15: An Effective Cartier Divisor and a Plane Conic**
Let $D \subseteq X$ be an effective Cartier divisor ([2, Effective Cartier Divisor]), so its ideal sheaf $\mathcal{I} = \mathcal{O}_X(-D)$ is invertible and locally generated by a single nonzerodivisor. This is a regular embedding of codimension one, with conormal sheaf $\mathcal{I}/\mathcal{I}^2 = \mathcal{O}_X(-D)|_D$ and normal bundle the line bundle $N_D X = \mathcal{O}_X(D)|_D$. A line bundle $L$ has $c(L) = \mathrm{id} + c_1(L)$, so $c(L)^{-1} = \sum_{i \geq 0}(-1)^i c_1(L)^i$, and proposition 3.4.11 gives

$$s(D, X) = \sum_{i \geq 0}(-1)^i\, c_1\big(\mathcal{O}_X(D)|_D\big)^i \cap [D] = [D] - c_1\big(\mathcal{O}_X(D)|_D\big) \cap [D] + \cdots \tag{3.6}$$

the series terminating in dimension zero.
Take $X = \mathbb{P}^2$ and $D$ a smooth conic, so $D \cong \mathbb{P}^1$ and $A_*(D) = \mathbb{Z}\,[D] \oplus \mathbb{Z}\,[\mathrm{pt}]$ by theorem 1.5.2; only the first two terms of (3.6) can be nonzero. The conic is cut out by a quadratic form, so $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(2)$ (example 2.1.4). To evaluate the second term, push forward along the

closed immersion $\iota\colon D \hookrightarrow \mathbb{P}^2$ and use the projection formula proposition 2.4.4(4), additivity proposition 2.4.4(3) for $\mathcal{O}(2) = \mathcal{O}(1) \otimes \mathcal{O}(1)$, the class $\iota_*[D] = 2\,[\mathbb{P}^1]$ of a conic (example 1.5.4), and example 2.4.7:

$$\iota_*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^2}(2)|_D\big) \cap [D]\Big) = c_1\big(\mathcal{O}_{\mathbb{P}^2}(2)\big) \cap \iota_*[D] = 2\,c_1\big(\mathcal{O}_{\mathbb{P}^2}(1)\big) \cap 2\,[\mathbb{P}^1] = 4\,[\mathrm{pt}]$$

Both $A_0(D)$ and $A_0(\mathbb{P}^2)$ are infinite cyclic on the class of a point and $\iota_*$ carries generator to generator, so $c_1(\mathcal{O}_{\mathbb{P}^2}(2)|_D) \cap [D] = 4\,[\mathrm{pt}]$ already in $A_0(D)$. Hence

$$s(D, \mathbb{P}^2) \;=\; [D] - 4\,[\mathrm{pt}]$$

The coefficient 4 is the self-intersection number of the conic, the degree $2 \cdot 2$ predicted by Bézout, and it enters the Segre class with a sign. For a divisor, then, $s(D, X)$ is a bookkeeping device for the alternating tower of self-intersections of $D$, and it contains exactly the information the line bundle $\mathcal{O}_X(D)|_D$ contains.

The irregular case is where the Segre class stops being a repackaging. The next computation is a closed subscheme whose normal cone is not a bundle, and the class detects the multiplicity of a singularity.

**Example 3.4.16: The Vertex of the Quadric Cone**

Let $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy - z^2)$ be the affine quadric cone of example 2.1.2, a normal surface with an isolated singularity at the vertex, and let $Z = \{P\}$ be the reduced vertex, with ideal $\mathfrak{m} = (x, y, z)$.

*The normal cone.* Write $A = \overline{k}[x, y, z]/(xy - z^2)$. The defining polynomial is homogeneous, so $A$ is a graded ring with $A_n$ its degree-$n$ piece and $\mathfrak{m}^n = \bigoplus_{m \geq n} A_m$; therefore $\mathfrak{m}^n/\mathfrak{m}^{n+1} = A_n$ and $\bigoplus_{n \geq 0} \mathfrak{m}^n/\mathfrak{m}^{n+1} = A$. The quadric cone is its own tangent cone at the vertex, and $C_P X = \operatorname{Spec} A = X$. It is not a vector bundle over the point $P$: a rank-$r$ bundle over a point is $\mathbb{A}^r$, whereas $X$ is singular at the origin. Equivalently, the embedding $\{P\} \hookrightarrow X$ is not regular, since $\mathfrak{m}$ needs three generators while $\dim X = 2$.

*The projective completion.* Adjoining $w$ in degree one gives $A[w] = \overline{k}[x, y, z, w]/(xy - z^2)$, so

$$\mathbb{P}(C_P X \oplus 1) \;=\; \operatorname{Proj}\big(\overline{k}[x, y, z, w]/(xy - z^2)\big) \;=\; Q \subseteq \mathbb{P}^3$$

the quadric surface $\{xy = z^2\}$, which is integral because $xy - z^2$ is irreducible, and $\mathcal{O}(1) = \mathcal{O}_{\mathbb{P}^3}(1)|_Q$ by the surjection $\overline{k}[x, y, z, w] \to A[w]$.

*The class.* Since $Z$ is a single point, $A_k(Z) = 0$ for $k \neq 0$, so only the term of definition 3.4.4 landing in $A_0$ survives, namely $i = \dim Q = 2$:

$$s(P, X) \;=\; \Big(\int_Q c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)|_Q\big)^2 \cap [Q]\Big)\,[P]$$

Let $\iota\colon Q \hookrightarrow \mathbb{P}^3$ be the inclusion. The quadric is the divisor of the section $xy - z^2$ of $\mathcal{O}_{\mathbb{P}^3}(2)$, so $\iota_*[Q] = 2\,[\mathbb{P}^2]$ in $A_2(\mathbb{P}^3)$ by the class map (proposition 2.1.3 and example 2.1.4). By the projection formula proposition 2.4.4(4) and the fact that capping a linear-subspace class with $c_1(\mathcal{O}_{\mathbb{P}^3}(1))$ drops its dimension by one (example 2.4.7),

$$\iota_*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)|_Q\big)^2 \cap [Q]\Big) = c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)\big)^2 \cap 2\,[\mathbb{P}^2] = 2\,[\mathrm{pt}]$$

Degrees are preserved by proper pushforward (definition 1.2.4), so the integral equals 2 and

$$s(P, X) \;=\; 2\,[P]$$

*The contrast.* Take instead the origin $P' \in \mathbb{A}^2$, where $\mathfrak{m} = (x, y)$ is a regular sequence and $\{P'\} \hookrightarrow \mathbb{A}^2$ is a regular embedding of codimension two. Proposition 3.4.11 gives $s(P', \mathbb{A}^2) = c(N)^{-1} \cap [P']$, and every $c_j(N) \cap [P']$ with $j \geq 1$ lies in $A_{-j}(P') = 0$, so $s(P', \mathbb{A}^2) = [P']$. At a smooth point of a surface the Segre class of the point is the point itself; at the vertex of the quadric cone it is twice the point. The integer it produces is the multiplicity of the singularity.

## Exercise 3.4.1

1. Let $X$ be an algebraic scheme and let $C = X$ be the zero cone of example 3.4.2(2), that is $S^\bullet = \mathcal{O}_X$. Show that $\mathbb{P}(C) = \emptyset$, that $\mathbb{P}(C \oplus 1) = X$ with $q$ the identity, and that $\mathcal{O}(1)$ is the trivial bundle. Deduce $s(X, X) = [X]$, and then explain why proposition 3.4.11 cannot be used to reach the same answer, even though $X \hookrightarrow X$ is a regular embedding of codimension zero.

2. Let $Z = \mathbb{P}^1 \subseteq \mathbb{P}^3$ be a line, cut out by two of the homogeneous coordinates. Show that this is a regular embedding of codimension two with conormal sheaf $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, hence with normal bundle $N = \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(1)$, and compute $s(\mathbb{P}^1, \mathbb{P}^3)$.

3. Let $D \subseteq \mathbb{P}^2$ be a smooth plane curve of degree $d$. Show that $s(D, \mathbb{P}^2) = [D] - \beta$ for a class $\beta \in A_0(D)$ of degree $d^2$. Then explain why, for $d \geq 3$, one may *not* conclude $\beta = d^2\,[\mathrm{pt}]$, whereas for $d \leq 2$ one may.

4. Let $f \in \overline{k}[x, y, z]$ be a homogeneous polynomial of degree $d$ defining a smooth plane curve $Y \subseteq \mathbb{P}^2$, let $X = \operatorname{Spec} \overline{k}[x, y, z]/(f)$ be the affine cone over $Y$, and let $P \in X$ be the vertex. Compute $s(P, X)$, generalizing example 3.4.16.

5. Let $X = \mathbb{A}^1$ and let $Z = \operatorname{Spec} \overline{k}[x]/(x^2)$ be the double point at the origin, with $Z_{\mathrm{red}} = \{P\}$ the reduced point. Compute both $s(Z, X)$ and $s(Z_{\mathrm{red}}, X)$, and say what the comparison shows about the dependence of the Segre class on the scheme structure of $Z$.

*4*

# *Deformation to the Normal Cone*

Classical enumerative geometry intersected two cycles by moving them until they met properly and trusting that the count did not change. Making that rigorous requires a moving lemma, and moving lemmas are hard, restrictive, and unavailable on a singular ambient scheme. The construction of this chapter avoids the manoeuvre entirely: instead of moving the cycles, it degenerates the embedding.

Given a closed subscheme $Z \subseteq X$, there is a family over $\mathbb{P}^1$ whose fibre away from $\infty$ is the given embedding $Z \subseteq X$ and whose fibre at $\infty$ is the embedding of $Z$ as the zero section of its normal cone (definition 3.4.6). Nothing has been perturbed; the ambient geometry has been deformed to a cone, where it is linear along the fibres and where intersecting is a matter of restricting to a zero section. Specializing a cycle to $\infty$ therefore converts an arbitrary intersection problem into one on a cone, and when the embedding is regular the cone is a vector bundle ([2, Regular Embedding]) and the specialization can be inverted against theorem 1.4.6 to land back on $Z$.

That composite is the Gysin map, and it is the last piece of machinery the intersection product needs. This chapter builds the deformation space, proves that specializing is well defined on rational equivalence, and assembles the Gysin map for a regular embedding together with the refined version that chapter 5 will apply to the diagonal.

## 4.1 The Deformation to the Normal Cone

This section builds the family and establishes the three facts everything later rests on: it is flat over $\mathbb{P}^1$, its fibre away from $\infty$ is the given $X$, and its fibre at $\infty$ is the normal cone.

Throughout, $Z \subseteq X$ is a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$. Fix a coordinate $t$ on $\mathbb{P}^1$, so that $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$ is the affine line with coordinate $t$, and write $U = \mathbb{P}^1 \setminus \{0\}$ for the complementary chart, an affine line with coordinate $u = t^{-1}$ vanishing exactly at $\infty$. Since $X$ is of finite type over $\overline{k}$ (box 0.0.2), the product $X \times \mathbb{P}^1$ is Noetherian, so the blow-up construction of

[2, Blow-up] applies to it.

The object to be built is a scheme over $\mathbb{P}^1$ whose fibre over every point of $\mathbb{A}^1$ is $X$ and whose fibre over $\infty$ is $C_Z X$. The mechanism that produces such a family is visible in the local algebra. A function $a$ in the ideal $\mathcal{I}$ vanishes on $Z$; dividing it by the parameter $u$ rescales the normal directions to $Z$ by the factor $u^{-1}$, and as $u$ tends to 0 the rescaling magnifies an arbitrarily small neighbourhood of $Z$ until only the behaviour of $a$ to leading order in $\mathcal{I}$ survives. Adjoining all the functions $au^{-1}$ with $a \in \mathcal{I}$ to $\mathcal{O}_X[u]$ therefore builds a family whose limit at $u = 0$ is manufactured from the successive quotients $\mathcal{I}^n/\mathcal{I}^{n+1}$, which is what $C_Z X$ is manufactured from (definition 3.4.6).

Adjoining $\mathcal{I}u^{-1}$ is exactly what makes the ideal $\mathcal{I} + (u)$ of $Z \times \{\infty\}$ principal, generated by $u$; and the universal solution to that require is the blow-up ([2, Basic Properties of Blow-ups], part (4)). The construction accordingly starts from a blow-up, and the rescaled algebra above is recovered as one of its charts.

**Definition 4.1.1: The Deformation Space**

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, and let $\mathcal{J} \subseteq \mathcal{O}_{X\times\mathbb{P}^1}$ be the ideal sheaf of the closed subscheme $Z \times \{\infty\}$. The *deformation space* of $Z$ in $X$ is the blow-up

$$M_Z X \;=\; \mathrm{Bl}_{Z\times\{\infty\}}(X \times \mathbb{P}^1) \;=\; \mathrm{Proj}\left(\bigoplus_{n\geq 0} \mathcal{J}^n\right) \xrightarrow{\;p\;} X \times \mathbb{P}^1$$

of [2, Blow-up]. Write $\pi\colon M_Z X \to \mathbb{P}^1$ for the composite of $p$ with the projection $X \times \mathbb{P}^1 \to \mathbb{P}^1$.

Nothing has been achieved yet: $M_Z X$ is a blow-up like any other, and its fibre over $\infty$ is larger than the normal cone. The next lemma computes the part of $M_Z X$ lying over the chart $U$, splitting it into the piece that carries the cone and a piece that does not. The splitting is the reason the deformation is defined on an open subscheme of $M_Z X$ rather than on all of it.

**Lemma 4.1.2: The Chart at Infinity**

Let $\mathcal{R} \subseteq \mathcal{O}_X[u, u^{-1}]$ be the sheaf of $\mathcal{O}_X[u]$-subalgebras generated by $\mathcal{I}u^{-1}$, the *extended Rees algebra* of $\mathcal{I}$; grading by powers of $u$,

$$\mathcal{R} \;=\; \bigoplus_{m\geq 0} \mathcal{O}_X\, u^m \;\oplus\; \bigoplus_{n\geq 1} \mathcal{I}^n u^{-n} \tag{4.1}$$

Write $D_+(u) \subseteq p^{-1}(X \times U)$ for the open subscheme attached to the degree-one section $u$ of $\bigoplus_n \mathcal{J}^n$. Then:

1. $D_+(u)$ is the relative spectrum $\mathrm{Spec}\,(\mathcal{R})$ over $X \times U$;
2. its closed complement $p^{-1}(X \times U) \setminus D_+(u)$ is canonically isomorphic to $\mathrm{Bl}_Z X$, carried by $p$ into $X \times \{\infty\}$ by the blow-down morphism.

The closed subscheme $\mathrm{Bl}_Z X$ so obtained is closed in $M_Z X$.

***Proof*:**
*Step 1: the ideal and its powers.* Work on an affine open $\operatorname{Spec} A \subseteq X$ with $\mathcal{I}(\operatorname{Spec} A) = I$, and set $B = A[u]$, so that $\operatorname{Spec} A \times U = \operatorname{Spec} B$ and the ideal of $Z \times \{\infty\}$ there is $J = IB + uB$. Expanding the product of the two ideals gives

$$J^n \;=\; \sum_{j=0}^{n} I^j B \cdot u^{n-j}$$

Two consequences are read off this expansion. Every term with $j < n$ is divisible by $u$, so

$$J^n + uB \;=\; I^n B + uB \tag{4.2}$$

And for $n \geq 1$,

$$J^n \cap uB \;=\; u\, J^{n-1} \tag{4.3}$$

Indeed, write $x = \sum_{j=0}^{n} a_j u^{n-j}$ with $a_j \in I^j B$. The terms with $j < n$ all lie in $uJ^{n-1}$, since $I^j B\, u^{n-1-j} \subseteq J^{n-1}$. The remaining term $a_n \in I^n B$ is a polynomial in $u$ with coefficients in $I^n$, so it lies in $uB$ precisely when its constant coefficient vanishes, that is when $a_n \in uI^n B \subseteq uJ^{n-1}$. The reverse inclusion is immediate from $uJ^{n-1} \subseteq J^n$.

*Step 2: the chart.* Write $S = \bigoplus_{n \geq 0} J^n$, so that $p^{-1}(\operatorname{Spec} A \times U) = \operatorname{Proj}(S)$ by definition 4.1.1 and $D_+(u) = \operatorname{Spec}\left(S_{(u)}\right)$ is the affine chart of the relative Proj attached to the degree-one element $u \in J = S_1$ ([2, Relative Proj]). Because $u$ is a nonzerodivisor on $B$, the assignment $x/u^n \mapsto xu^{-n}$ is a well-defined injection of $S_{(u)}$ into $A[u, u^{-1}]$, and its image is $\bigcup_{n \geq 0} J^n u^{-n}$, which by Step 1 is the $B$-subalgebra $R$ generated by $Iu^{-1}$. To see the grading (4.1), note $R = \sum_{n \geq 0} I^n B\, u^{-n}$, so the coefficient of $u^m$ in $R$ is $\sum_{n \geq 0,\, n+m \geq 0} I^n$, which is $A$ for $m \geq 0$ and $I^{-m}$ for $m < 0$. Hence $D_+(u) = \operatorname{Spec}(R)$ as a scheme over $\operatorname{Spec} B$, which is (1).

*Step 3: the complement.* Reduction modulo $u$ gives a surjection of graded $B$-algebras $S \to \overline{S}$, where $\overline{S}_n = (J^n + uB)/uB$; by (4.2) this is the image of $I^n B$ in $B/uB = A$, namely $I^n$. So $\overline{S} = \bigoplus_{n \geq 0} I^n$ is the Rees algebra of $I$, its relative Proj is $\operatorname{Bl}_Z X$ over $\operatorname{Spec} A$ by [2, Blow-up], and the surjection exhibits it as a closed subscheme of $\operatorname{Proj}(S)$ lying over $V(u) = \operatorname{Spec} A \times \{\infty\}$, with structure morphism to $\operatorname{Spec} A$ the blow-down. Since $u$ maps to 0 in $\overline{S}_1$, this closed subscheme is disjoint from $D_+(u)$.
For the reverse inclusion, the closed complement of $D_+(u)$ is $\operatorname{Proj}(S/uS)$, where $uS$ is the homogeneous ideal generated by $u \in S_1$, whose degree-$n$ part is $uJ^{n-1}$ for $n \geq 1$ and 0 in degree 0. By (4.3) the ideal $uJ^{n-1}$ is exactly the kernel of $J^n \to I^n$, so $S/uS$ and $\overline{S}$ agree in every positive degree and differ only in degree 0, where $A[u] \to A$ kills $u$. In $S/uS$ the degree-zero element $u$ annihilates every positive degree, so for a homogeneous $f \in S_1$ the image of $u$ in the localization $(S/uS)_{(f)}$ is $uf/f = 0$; hence $(S/uS)_{(f)} = \overline{S}_{(f)}$ for every such $f$, and the two relative Proj schemes coincide. The closed complement of $D_+(u)$ is therefore $\operatorname{Bl}_Z X$, which is (2).

*Step 4: globalization.* The algebra $\mathcal{R}$ and the surjection of Step 3 are defined without choices, so the local identifications agree on overlaps of an affine cover of $X$ and glue. For the final assertion, $\operatorname{Bl}_Z X$ is closed in the open subscheme $p^{-1}(X \times U)$ and is contained in $p^{-1}(X \times \{\infty\})$, which is closed in $M_Z X$ and contained in $p^{-1}(X \times U)$; writing $\operatorname{Bl}_Z X = C \cap p^{-1}(X \times U)$ with $C$ closed in $M_Z X$, one gets $\operatorname{Bl}_Z X = C \cap p^{-1}(X \times \{\infty\})$, an intersection of two closed subsets and hence closed in $M_Z X$, completing the proof.

The lemma is what makes the construction computable: over the chart at infinity the deformation is not a Proj at all but an affine scheme over $X \times U$, given by the single algebra (4.1), written $R$ when $X = \operatorname{Spec} A$ is affine, and its reduction modulo $u$ is the associated graded algebra of $\mathcal{I}$. The smallest instance already shows the mechanism.

**Example 4.1.3: A Point on a Smooth Curve**
Let $X$ be a variety of dimension one and $P \in X$ a closed point at which $X$ is smooth, and take $Z = \{P\}$ with its reduced structure. The local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring; choosing a uniformizer $s$ makes $\mathcal{I}$ locally generated by the single nonzerodivisor $s$, so $Z \hookrightarrow X$ is a regular embedding of codimension one ([2, Regular Embedding]) and $\operatorname{gr}_{\mathfrak{m}_P} \mathcal{O}_{X,P} \cong \overline{k}[\overline{s}]$ is a polynomial ring in one variable. Hence $C_P X = \mathbb{A}^1$: the deformation replaces the curve by its tangent line at $P$ and the point by the origin of that line.
The computation is explicit for $X = \mathbb{A}^1 = \operatorname{Spec} \overline{k}[x]$ and $P$ the origin, where $I = (x)$. The algebra of (4.1) is $R = \overline{k}[x, u][xu^{-1}]$, and setting $v = xu^{-1}$ gives $x = uv$, so $R$ is generated by $u$ and $v$ alone. These two are algebraically independent over $\overline{k}$: they lie in $\overline{k}(x, u)$, which has transcendence degree two, and they generate it, so they form a transcendence basis. Hence

$$R \;\cong\; \overline{k}[u, v], \qquad \operatorname{Spec}(R) = \mathbb{A}^2 \longrightarrow \mathbb{A}^1_u$$

is the trivial family with fibre $\mathbb{A}^1$. Over $u = c \neq 0$ the fibre is identified with $X$ by $x = cv$, and over $u = 0$ it is $\operatorname{Spec}\big(R/uR\big) = \operatorname{Spec} \overline{k}[v]$, which is $C_P X$. The family is the rescaling $x \mapsto x/u$ of the line, and the closed subscheme $\{v = 0\}$ is the point $P$ travelling along it.

The lemma isolates a piece of $M_Z X$ over $\infty$, namely $\operatorname{Bl}_Z X$, which the rescaling picture does not see and which is discarded.

**Definition 4.1.4: The Deformation to the Normal Cone**

With $Z \subseteq X$ and $M_Z X$ as in definition 4.1.1, and with $\operatorname{Bl}_Z X \subseteq M_Z X$ the closed subscheme of lemma 4.1.2(2), the *deformation to the normal cone* of $Z$ in $X$ is the open subscheme

$$M_Z^\circ X \;=\; M_Z X \setminus \operatorname{Bl}_Z X$$

together with the restriction of $\pi\colon M_Z X \to \mathbb{P}^1$, again written $\pi$, and the restriction of $p$, whose composite with the projection $X \times \mathbb{P}^1 \to X$ is written $\rho\colon M_Z^\circ X \to X$.

Everything the chapter needs is contained in the behaviour of $\pi$ over the two charts, and lemma 4.1.2 has already computed one of them. Collecting that computation with the elementary behaviour over $\mathbb{A}^1$ gives the properties that section 4.2 and section 4.3 consume.

**Theorem 4.1.5: The Fibres of the Deformation**

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, with $\pi\colon M^{\circ}_Z X \to \mathbb{P}^1$ as in definition 4.1.4.

1. Over $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$ the blow-down restricts to an isomorphism $\pi^{-1}(\mathbb{A}^1) \xrightarrow{\sim} X \times \mathbb{A}^1$ commuting with the projections to $\mathbb{A}^1$.
2. The fibre over $\infty$ is the normal cone, $\pi^{-1}(\infty) = C_Z X$ of definition 3.4.6, and it is an effective Cartier divisor on $M^{\circ}_Z X$, cut out by the function $u$.
3. $\pi$ is flat.
4. There is a closed immersion $Z \times \mathbb{P}^1 \hookrightarrow M^{\circ}_Z X$ over $\mathbb{P}^1$ which over $\mathbb{A}^1$ is the inclusion $Z \times \mathbb{A}^1 \subseteq X \times \mathbb{A}^1$ of (1), and over $\infty$ is the vertex $Z \hookrightarrow C_Z X$ of definition 3.4.1.

***Proof*:**

*Step 1: the chart over* $\mathbb{A}^1$. The centre $Z \times \{\infty\}$ is disjoint from the open set $X \times \mathbb{A}^1$, so [2, Basic Properties of Blow-ups](2) makes $p$ an isomorphism $p^{-1}(X \times \mathbb{A}^1) \to X \times \mathbb{A}^1$. By lemma 4.1.2(2) the deleted subscheme $\mathrm{Bl}_Z X$ lies over $X \times \{\infty\}$, so it does not meet $p^{-1}(X \times \mathbb{A}^1)$, and therefore $\pi^{-1}(\mathbb{A}^1) = p^{-1}(X \times \mathbb{A}^1) \cong X \times \mathbb{A}^1$ compatibly with the projections to $\mathbb{A}^1$. This is (1).

*Step 2: the fibre at infinity.* Over the other chart, lemma 4.1.2 gives $\pi^{-1}(U) = \operatorname{Spec}(\mathcal{R})$ with $\mathcal{R}$ the extended Rees algebra, so the scheme-theoretic fibre over $\infty = \{u = 0\}$ is $\operatorname{Spec}\big(\mathcal{R}/u\mathcal{R}\big)$. Compute the quotient degree by degree in the grading (4.1). Writing $\mathcal{R}_m$ for the coefficient of $u^m$, so that $\mathcal{R}_m = \mathcal{O}_X$ for $m \geq 0$ and $\mathcal{R}_{-n} = \mathcal{I}^n$ for $n \geq 1$, the coefficient of $u^m$ in $u\mathcal{R}$ is $\mathcal{R}_{m-1}$. Hence $(\mathcal{R}/u\mathcal{R})_m$ vanishes for $m \geq 1$, equals $\mathcal{O}_X/\mathcal{I} = \mathcal{O}_Z$ for $m = 0$, and equals $\mathcal{I}^n/\mathcal{I}^{n+1}$ for $m = -n$ with $n \geq 1$. Therefore

$$\mathcal{R}/u\mathcal{R} \;\cong\; \bigoplus_{n \geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$$

which is the graded algebra of definition 3.4.6, so $\pi^{-1}(\infty) = C_Z X$.

*Step 3: the fibre at infinity is Cartier.* The element $u$ acts invertibly on $\mathcal{O}_X[u, u^{-1}]$ and hence injectively on the subsheaf $\mathcal{R}$, so $u$ is a nonzerodivisor and $\pi^{-1}(\infty) \subseteq \pi^{-1}(U)$ is cut out by a single nonzerodivisor. On the complementary open $\pi^{-1}(\mathbb{A}^1)$ the fibre is empty, cut out by the unit 1. So $\pi^{-1}(\infty)$ is locally principal, generated at each point by a nonzerodivisor, which is an effective Cartier divisor ([2, Effective Cartier Divisor]). With Step 2 this is (2).

*Step 4: flatness.* Flatness may be checked locally on source and target, so it suffices to treat the two charts separately. Over $\mathbb{A}^1$ the morphism is the projection $X \times \mathbb{A}^1 \to \mathbb{A}^1$ by Step 1, which is the base change of $X \to \operatorname{Spec} \overline{k}$ along $\mathbb{A}^1 \to \operatorname{Spec} \overline{k}$; every scheme is flat over a field and flatness is stable under base change, so this projection is flat. Over $U$ the morphism is $\operatorname{Spec}(\mathcal{R}) \to U = \operatorname{Spec} \overline{k}[u]$. The ring $\overline{k}[u]$ is a principal ideal domain, over which a module is flat exactly when it is torsion-free. On an affine open $\operatorname{Spec} A \subseteq X$ one has $\mathcal{R}(\operatorname{Spec} A) = R \subseteq A[u, u^{-1}]$, and $A[u, u^{-1}]$ is torsion-free over $\overline{k}[u]$: given $f \in \overline{k}[u]$ nonzero of degree $d$ with leading coefficient $c \in \overline{k}^*$, and a nonzero $x = \sum_{i \leq M} a_i u^i$ with $a_M \neq 0$, the coefficient of $u^{M+d}$ in $fx$ is $c\,a_M \neq 0$, so $fx \neq 0$. A submodule of a torsion-free module is torsion-free, so $R$ is torsion-free and hence flat over $\overline{k}[u]$. This is (3).

*Step 5: the constant subscheme.* Let $\mathfrak{q} \subseteq \mathcal{R}$ be the ideal generated by $\mathcal{I}u^{-1}$; it contains $\mathcal{I}$, since

$\mathcal{I} = (\mathcal{I}u^{-1}) \cdot u$ and $u \in \mathcal{R}$. Its coefficient of $u^m$ is $\mathcal{I} \cdot \mathcal{R}_{m+1}$, which for $m \geq 0$ is $\mathcal{I}$ and for $m = -n < 0$ is $\mathcal{I} \cdot \mathcal{I}^{n-1} = \mathcal{I}^n = \mathcal{R}_{-n}$. So

$$\mathcal{R}/\mathfrak{q} \;\cong\; \bigoplus_{m\geq 0} (\mathcal{O}_X/\mathcal{I})\, u^m \;=\; \mathcal{O}_Z[u]$$

and $\mathfrak{q}$ defines a closed immersion $Z \times U \hookrightarrow \operatorname{Spec}(\mathcal{R})$ over $X \times U$. Inverting $u$ makes $u^{-1}$ a unit, so $\mathfrak{q}$ generates $\mathcal{I}$ there and the immersion restricts over $U \setminus \{\infty\}$ to $Z \times (U \setminus \{\infty\}) \subseteq X \times (U \setminus \{\infty\})$; it therefore glues with the inclusion $Z \times \mathbb{A}^1 \subseteq X \times \mathbb{A}^1$ of Step 1 to a closed immersion $Z \times \mathbb{P}^1 \hookrightarrow M_Z^{\circ} X$ over $\mathbb{P}^1$. Modulo $u$ the ideal $\mathfrak{q}$ becomes the ideal generated by the degree-one part $\mathcal{I}/\mathcal{I}^2$ of $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$, whose quotient is $\mathcal{O}_Z$; by definition 3.4.1 that is the vertex of $C_Z X$. So the immersion restricts over $\infty$ to the vertex, which is (4), completing the proof.

The theorem produces a single scheme over $\mathbb{P}^1$ carrying two embeddings at once. Away from $\infty$ the pair is the given $Z \subseteq X$, unchanged; at $\infty$ it is the vertex inside $C_Z X$. What moves in the family is the ambient scheme, not the subscheme: by (4) the subscheme is the constant family $Z \times \mathbb{P}^1$, and the ambient scheme degenerates around it until it becomes a cone with $Z$ as its vertex.

The morphism $\rho\colon M_Z^{\circ} X \to X$ records how each fibre maps back to the original scheme. Over $\mathbb{A}^1$ it is the projection $X \times \mathbb{A}^1 \to X$ by Step 1 of the proof, and over $\infty$ it is the composite $C_Z X \to Z \hookrightarrow X$, since the structure map $\mathcal{O}_X \to \mathcal{R}$ becomes $\mathcal{O}_X \to \mathcal{O}_Z$ in degree zero after reducing modulo $u$. One property the family does not have is properness: its fibre over $\infty$ is affine over $Z$ (definition 3.4.1), and an affine scheme of positive dimension over a field is not proper, so $\pi$ is not a proper morphism as soon as $C_Z X$ has positive-dimensional fibres. Cycles will therefore be moved across the family by restriction to the Cartier divisor $\pi^{-1}(\infty)$ rather than by pushforward.

**4.1.6 Remark: Why the Blow-Up Is Not Enough** The subscheme deleted in definition 4.1.4 is not a technicality. Lemma 4.1.2 splits the part of $M_Z X$ lying over the chart $U$ into $D_+(u)$ and $\mathrm{Bl}_Z X$, and the fibre over $\infty$ lies entirely in that part; its intersection with $D_+(u)$ is $C_Z X$ by Step 2 of the proof of theorem 4.1.5, while $\mathrm{Bl}_Z X$ maps into $X \times \{\infty\}$ and so lies in the fibre. The special fibre of the unmodified blow-up is therefore the disjoint union of $C_Z X$ and $\mathrm{Bl}_Z X$ as a set, and only the first piece records the normal directions along $Z$; the second is a copy of $X$ modified along $Z$, carrying no information the construction is after. Deleting it costs the properness of the blow-down $p$ over $X \times \mathbb{P}^1$ and gives a special fibre equal to $C_Z X$ on the nose.

The two extremes of the construction are the case where $C_Z X$ is a vector bundle and the case where it is not. The first is the case the Gysin map of section 4.3 will need, and codimension one is where it is easiest to see.

**Example 4.1.7: An Effective Cartier Divisor and a Line in the Plane**
Let $D \subseteq X$ be an effective Cartier divisor ([2, Effective Cartier Divisor]). Its ideal sheaf is invertible and locally generated by one nonzerodivisor, so $D \hookrightarrow X$ is a regular embedding of codimension one with normal bundle $N_D X = \mathcal{O}_X(D)|_D$ (example 3.4.15), and $C_D X = N_D X$ by proposition 3.4.11. By theorem 4.1.5 the deformation therefore degenerates $D \subseteq X$ into $D$ sitting inside the total space of a line bundle on itself, as the zero section, which is the vertex of that cone (example 3.4.2(1)). Take $X = \mathbb{P}^2$ and $D = L$ a line. Then $\mathcal{O}_{\mathbb{P}^2}(L) \cong \mathcal{O}_{\mathbb{P}^2}(1)$ by example 2.1.4, and its restriction to $L \cong \mathbb{P}^1$ is $\mathcal{O}_{\mathbb{P}^1}(1)$ (example 2.2.3), so $C_L \mathbb{P}^2$ is the total space of $\mathcal{O}_{\mathbb{P}^1}(1)$ over $L$. That total space is $\mathbb{P}^2$ with a point removed: linear projection away from a point $P \notin L$ exhibits $\mathbb{P}^2 \setminus \{P\}$ as a line

bundle over $L$ whose zero section is $L$ itself, and the degree of that bundle is the self-intersection $L \cdot L = 1$ of example 2.2.3, so it is $\mathcal{O}_{\mathbb{P}^1}(1)$. (The proof of theorem 1.5.2 notes the projection but uses only that it is an affine bundle, which is all its own argument needs.) So the family degenerates the pair $L \subseteq \mathbb{P}^2$ into the pair $L \subseteq \mathbb{P}^2 \setminus \{P\}$, with $L$ the zero section of a line bundle. On the special fibre, intersecting a cycle with $L$ has become intersecting it with the zero section of a line bundle, and flat pullback along the projection of that bundle is an isomorphism on Chow groups (theorem 1.4.6).

When the embedding is not regular the cone is not a bundle, and the deformation exhibits the singularity of the pair rather than a linear model of it. The node is the case already computed in example 3.4.7, and the whole family can be written down.

**Example 4.1.8: The Node and Its Tangent Cone**

Assume char$\overline{k} \neq 2$ and let $X = \operatorname{Spec} A$ with $A = \overline{k}[x, y]/(y^2 - x^2 - x^3)$ be the affine nodal cubic, $P$ the origin, $Z = \{P\}$ and $I = (x, y)$. Set $v = xu^{-1}$ and $w = yu^{-1}$ in $A[u, u^{-1}]$, so that $x = uv$ and $y = uw$; the algebra $R = A[u][Iu^{-1}]$ of (4.1) is generated by $u, v, w$. Substituting into the defining relation gives $u^2w^2 = u^2v^2 + u^3v^3$, and $u$ is a nonzerodivisor on the domain $A[u, u^{-1}]$, so

$$w^2 \; = \; v^2 + uv^3 \tag{4.4}$$

*The relation is the only one.* The ring $A$ is free over $\overline{k}[x]$ on $1, y$, because its defining relation is monic of degree two in $y$; hence $A[u, u^{-1}]$ is free over $\overline{k}[x, u, u^{-1}]$ on $1, y$. Using (4.4) to reduce powers of $w$, the ring $R$ is generated as a $\overline{k}[u, v]$-module by 1 and $w$. Those two are independent: if $f(u, v) + g(u, v)w = 0$, then, reading off the coefficients of the basis $1, y$, one gets $f(u, xu^{-1}) = 0$ and $g(u, xu^{-1}) = 0$ in $\overline{k}[x, u, u^{-1}]$, and $u$ and $xu^{-1}$ are algebraically independent over $\overline{k}$ (they generate $\overline{k}(x, u)$, of transcendence degree two), so $f = g = 0$. Therefore

$$R \; \cong \; \overline{k}[u, v, w]/\big(w^2 - v^2 - uv^3\big)$$

*The fibres.* Over $u = c \neq 0$ the fibre is $\operatorname{Spec} \overline{k}[v, w]/(w^2 - v^2 - cv^3)$, carried isomorphically onto $X$ by $x = cv$, $y = cw$, as substituting confirms. Over $u = 0$ the fibre is $\operatorname{Spec} \overline{k}[v, w]/(w^2 - v^2)$, the union of the two lines $w = \pm v$: the tangent cone of example 3.4.7, in agreement with theorem 4.1.5(2). The family straightens the two branches of the node into their tangent lines while keeping the singular point fixed, and the cubic term $uv^3$, which distinguishes the nodal cubic from a pair of lines, is exactly the term that the rescaling drives to zero.

## Exercise 4.1.1

1. Compute the deformation in the two degenerate cases. First take $Z = X$, so that $\mathcal{I} = 0$: identify $\mathcal{R}$, show that $M_X^\circ X = X \times \mathbb{P}^1$ with $\pi$ the projection, and check that the fibre over $\infty$ is the zero cone of example 3.4.2(2). Then take $Z = \emptyset$, so that $\mathcal{I} = \mathcal{O}_X$: identify $\mathcal{R}$, show that $M_\emptyset^\circ X = X \times \mathbb{A}^1$, and reconcile the empty fibre over $\infty$ with definition 3.4.6.
2. Let $X = \mathbb{A}^n$ and let $Z = \{0\}$ be the origin, with $I = (x_1, \ldots, x_n)$. Show that the algebra of (4.1) is a polynomial ring $R \cong \overline{k}[u, v_1, \ldots, v_n]$ with $v_i = x_iu^{-1}$, so that $\pi^{-1}(U) \cong \mathbb{A}^{n+1} \to \mathbb{A}^1_u$ is the trivial family. Identify the isomorphism of the fibre over $u = c \neq 0$ with $\mathbb{A}^n$, identify $C_0\mathbb{A}^n$, and locate the closed subscheme $Z \times U$ of theorem 4.1.5(4) inside $\mathbb{A}^{n+1}$.

3. Let $X = \mathbb{A}^1 = \operatorname{Spec} \overline{k}[x]$ and let $Z = \operatorname{Spec} \overline{k}[x]/(x^2)$ be the double point at the origin, so $I = (x^2)$. Show that $\operatorname{Spec}(\mathcal{R})$ is the affine quadric cone of example 2.1.2, and identify the fibre over $\infty$. Comment on what the computation shows about the smoothness of the deformation space.
4. Let $D \subseteq \mathbb{P}^2$ be a smooth conic. Using example 3.4.15 and proposition 3.4.11, identify $C_D\mathbb{P}^2$ as the total space of a line bundle on $D \cong \mathbb{P}^1$, and determine which line bundle it is.
5. Let $X = \operatorname{Spec} \overline{k}[x, y]/(y^2 - x^3)$ be the cuspidal cubic and $P$ the origin. Compute $C_P X$ and its fundamental cycle $[C_P X]$ in the sense of definition 1.1.2, and contrast the answer with the node of example 4.1.8.

## 4.2 Specialization of Cycles

A cycle on $X$ is spread over $\mathbb{A}^1$, its closure is followed to the fibre at infinity, and what that closure cuts on the fibre is a cycle on the normal cone. The resulting operation $\sigma\colon A_k(X) \to A_k(C_Z X)$ is the specialization homomorphism. Three things have to be proved about it: that it is well defined on rational equivalence, that on a subvariety $V$ it returns the normal cone of $V \cap Z$ in $V$, and that it is compatible with proper pushforward and flat pullback.

Throughout, $Z \subseteq X$ is a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I}$, and $\pi\colon M_Z^\circ X \to \mathbb{P}^1$ is the projection of definition 4.1.4, which by theorem 4.1.5(1) to (3) is flat with fibres

$$\pi^{-1}(\mathbb{A}^1) \;=\; X \times \mathbb{A}^1, \qquad \pi^{-1}(\infty) \;=\; C_Z X$$

Write $j\colon X \times \mathbb{A}^1 \hookrightarrow M_Z^\circ X$ for the open immersion of the first fibre, $i_\infty\colon C_Z X \hookrightarrow M_Z^\circ X$ for the closed immersion of the second, and $q\colon X \times \mathbb{A}^1 \to X$ for the projection. On $\mathbb{P}^1$ fix the coordinate $u$ vanishing to order one at $\infty$, so that $\mathbb{P}^1 \setminus \{0\} = \operatorname{Spec} \overline{k}[u]$ and $\infty$ is the closed subscheme $\{u = 0\}$; on the complementary chart $\mathbb{A}^1$ the coordinate is $t = u^{-1}$.

The fibre at infinity is cut out by a single equation (theorem 4.1.5(2)), which is what makes definition 2.2.1 applicable. One further fact is needed before anything can be intersected: the line bundle carrying that equation is trivial along the fibre itself. It looks like an accident of the construction, and it is what will make the whole operation independent of choices.

**Lemma 4.2.1: The Line Bundle of the Fibre at Infinity**

Let $D_\infty$ be the effective Cartier divisor of theorem 4.1.5(2), whose support is the fibre $\pi^{-1}(\infty) = C_Z X$. Then

$$\mathcal{O}_{M_Z^\circ X}\big(D_\infty\big) \;=\; \pi^*\mathcal{O}_{\mathbb{P}^1}(\infty), \qquad \mathcal{O}_{M_Z^\circ X}\big(D_\infty\big)\big|_{C_Z X} \;\cong\; \mathcal{O}_{C_Z X}$$

so the line bundle of $D_\infty$ restricts to the trivial bundle on the fibre it cuts out.

***Proof*:**
On $\mathbb{P}^1$ the point $\infty$ is an effective Cartier divisor ([2, Effective Cartier Divisor]): it is cut out by the nonzerodivisor $u$ on $\operatorname{Spec} \overline{k}[u]$ and by the unit 1 on the complementary chart $\mathbb{A}^1$, and its line bundle is $\mathcal{O}_{\mathbb{P}^1}(\infty)$.

By theorem 4.1.5(2) the divisor $D_\infty$ is cut out by $\pi^{\#}u$ over $\pi^{-1}(\mathbb{P}^1 \setminus \{0\})$, and over $\pi^{-1}(\mathbb{A}^1)$ it is

empty, cut out by the unit 1. Its local data $\{(\pi^{-1}(\mathbb{A}^1),\, 1),\ (\pi^{-1}(\mathbb{P}^1 \setminus \{0\}),\, \pi^{\#}u)\}$ are therefore the pullbacks along $\pi$ of the local data just recorded for $\infty$ on $\mathbb{P}^1$, so $D_\infty = \pi^*(\infty)$ and the associated line bundle is $\pi^*\mathcal{O}_{\mathbb{P}^1}(\infty)$.
For the restriction, the composite $\pi \circ i_\infty \colon C_Z X \to \mathbb{P}^1$ factors through the closed subscheme $\{\infty\} = \operatorname{Spec} \overline{k}$, since $i_\infty$ is the inclusion of the fibre over $\infty$, so

$$\mathcal{O}_{M_Z^\circ X}(D_\infty)\big|_{C_Z X} \;=\; i_\infty^* \pi^* \mathcal{O}_{\mathbb{P}^1}(\infty) \;=\; \big(\pi \circ i_\infty\big)^* \mathcal{O}_{\mathbb{P}^1}(\infty)$$

is the pullback of $\mathcal{O}_{\mathbb{P}^1}(\infty)|_{\{\infty\}}$ along the structure morphism $C_Z X \to \operatorname{Spec} \overline{k}$. A line bundle on the spectrum of a field is free of rank one, so that pullback is trivial, completing the proof.

The first consequence of the triviality is what the definition of $\sigma$ needs: the operator $D_\infty \cdot -$ annihilates every class that already lives on the fibre at infinity. Geometrically, a cycle sitting inside the special fibre is not cut further by the equation of that fibre.

**Lemma 4.2.2: Vanishing on Classes Supported at Infinity**

For every $\gamma \in A_{k+1}(C_Z X)$,

$$D_\infty \cdot (i_\infty)_* \gamma \;=\; 0 \quad \text{in } A_k(C_Z X)$$

***Proof*:**
By proposition 2.2.4 the operator $D_\infty \cdot - \colon A_{k+1}(M_Z^\circ X) \to A_k(|D_\infty|) = A_k(C_Z X)$ is a homomorphism independent of all choices, so it may be evaluated on any cycle representing the class. Represent $\gamma$ by a cycle $\sum_l n_l [T_l]$ with each $T_l \subseteq C_Z X$ a $(k+1)$-dimensional subvariety; then $(i_\infty)_* \gamma$ is represented by the same cycle, read on $M_Z^\circ X$, and every $T_l$ is contained in $|D_\infty| = C_Z X$. The second clause of definition 2.2.1 is therefore in force on each term, and

$$D_\infty \cdot [T_l] \;=\; c\Big(\mathcal{O}_{M_Z^\circ X}(D_\infty)\big|_{T_l}\Big) \;\in\; A_k(T_l)$$

By lemma 4.2.1 the restricted bundle is trivial, and the class map of proposition 2.1.3 is a group homomorphism, so it sends the trivial bundle, the identity element of $\operatorname{Pic}(T_l)$, to zero. Hence $D_\infty \cdot [T_l] = 0$ for every $l$, and summing gives $D_\infty \cdot (i_\infty)_* \gamma = 0$, as we sought to show.

With these two facts the specialization map can be built. The recipe has three moves: thicken a class on $X$ into a class on $X \times \mathbb{A}^1$, extend it across the family, and cut with the fibre at infinity. The middle move is the only one that involves a choice, since a class on the open part of $M_Z^\circ X$ has many extensions; the localization sequence measures the ambiguity exactly, and lemma 4.2.2 kills it.

**Definition 4.2.3: Specialization of Cycles**

Let $Z \subseteq X$ be a closed subscheme and $\alpha \in A_k(X)$. A *lift* of $\alpha$ is a class $\beta \in A_{k+1}(M_Z^\circ X)$ with

$$j^*\beta \;=\; q^*\alpha \quad \text{in } A_{k+1}\big(X \times \mathbb{A}^1\big)$$

The *closure lift* attached to a cycle $\sum_l n_l[V_l]$ representing $\alpha$ is the class of

$$\overline{\alpha \times \mathbb{A}^1} \;=\; \sum_l n_l\big[\overline{V_l \times \mathbb{A}^1}\big]$$

the closures being taken in $M_Z^\circ X$. The *specialization* of $\alpha$ is

$$\sigma(\alpha) \;=\; D_\infty \cdot \beta \;\in\; A_k\big(C_Z X\big)$$

for any lift $\beta$, the intersection being that of definition 2.2.1 with the Cartier divisor of lemma 4.2.1, whose support is $C_Z X$.

The definition presupposes two things, that a lift exists and that the answer does not depend on which one is chosen, and both come from the localization sequence for the closed fibre $C_Z X$ inside $M_Z^\circ X$. It also presupposes that the recipe respects rational equivalence on $X$, which is where flat pullback does its work.

**Theorem 4.2.4: The Specialization Homomorphism**

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme. Every class $\alpha \in A_k(X)$ admits a lift, and $D_\infty \cdot \beta$ is the same for all lifts $\beta$ of $\alpha$. The assignment of definition 4.2.3 is therefore a well-defined homomorphism

$$\sigma\colon A_k(X) \longrightarrow A_k\big(C_Z X\big)$$

for every $k$.

***Proof*:**
*Step 1: existence of a lift.* The fibre $C_Z X$ is closed in $M_Z^\circ X$ with open complement $X \times \mathbb{A}^1$, so theorem 1.4.1 gives an exact sequence

$$A_{k+1}\big(C_Z X\big) \xrightarrow{(i_\infty)_*} A_{k+1}\big(M_Z^\circ X\big) \xrightarrow{j^*} A_{k+1}\big(X \times \mathbb{A}^1\big) \longrightarrow 0 \tag{4.5}$$

Surjectivity of $j^*$ produces a lift of $q^*\alpha$, so lifts exist.

*Step 2: independence of the lift.* Let $\beta$ and $\beta'$ both lift $\alpha$. Then $j^*(\beta - \beta') = 0$, so by exactness of (4.5) there is $\gamma \in A_{k+1}(C_Z X)$ with $\beta - \beta' = (i_\infty)_*\gamma$. Since $D_\infty \cdot -$ is a homomorphism on $A_{k+1}(M_Z^\circ X)$ (proposition 2.2.4),

$$D_\infty \cdot \beta - D_\infty \cdot \beta' \;=\; D_\infty \cdot (i_\infty)_*\gamma \;=\; 0$$

by lemma 4.2.2. So the value depends only on $\alpha$.

*Step 3: additivity.* Flat pullback $q^*$ and open restriction $j^*$ are homomorphisms (proposition 1.3.7), so the sum of a lift of $\alpha$ and a lift of $\alpha'$ is a lift of $\alpha + \alpha'$; and $D_\infty \cdot -$ is a homomorphism. Hence $\sigma(\alpha + \alpha') = \sigma(\alpha) + \sigma(\alpha')$, and $\sigma$ is a homomorphism $A_k(X) \to A_k(C_Z X)$, completing the proof.

Nothing in the argument required $\alpha$ to be represented by any particular kind of cycle, and that is the point of routing the construction through Chow groups rather than through cycles: rational equivalence on $X$ is absorbed by $q^*$ before the family is entered, and the ambiguity created by the passage to $M_Z^\circ X$ is absorbed by lemma 4.2.2 after it. In practice, one lift is more useful than the others. Given a cycle $\sum_l n_l[V_l]$ representing $\alpha$, the flat pullback $q^*[V_l] = [V_l \times \mathbb{A}^1]$ (example 1.3.4(2)) extends across the family by taking closures in $M_Z^\circ X$, and the closure lift of definition 4.2.3 restricts along the open immersion $j$ to $q^*\alpha$ by example 1.3.4(1), so it is indeed a lift. Every one of its components dominates $\mathbb{P}^1$, hence lies in no fibre, and in particular none is contained in $|D_\infty|$; the first clause of definition 2.2.1 therefore governs every term. This is the representative used in every computation below.

**Example 4.2.5: Specializing at a Point of the Projective Line**

Let $X = \mathbb{P}^1$ and let $Z = \{P\}$ be a reduced closed point. The ideal $\mathcal{I}$ is generated at $P$ by a local coordinate, so $\mathcal{I}/\mathcal{I}^2$ is one-dimensional over $\overline{k}$ and every $\mathcal{I}^n/\mathcal{I}^{n+1}$ is one-dimensional; the normal cone $C_P\mathbb{P}^1$ of definition 3.4.6 is the affine line $\mathbb{A}^1$ over the point $P$. The deformation space is concrete here: $X \times \mathbb{P}^1 = \mathbb{P}^1 \times \mathbb{P}^1$, the centre is the single point $(P, \infty)$, and $M_Z X$ is the blow-up of that surface at that point, with exceptional divisor a copy of $\mathbb{P}^1$. Removing $\mathrm{Bl}_P\mathbb{P}^1 = \mathbb{P}^1$, the strict transform of $\mathbb{P}^1 \times \{\infty\}$, leaves $M_Z^\circ X$, whose fibre over $\infty$ is the exceptional $\mathbb{P}^1$ with one point deleted, namely $\mathbb{A}^1 = C_P\mathbb{P}^1$.

Two computations are available, one in each dimension. In dimension one, $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$ by theorem 1.5.2, and the closure of $\mathbb{P}^1 \times \mathbb{A}^1$ is all of $M_Z^\circ X$, an integral surface, so the first clause of definition 2.2.1 gives $\sigma[\mathbb{P}^1] = [\mathrm{div}(\pi^\# u)]$. The zero scheme of $\pi^\# u$ is the fibre over $\infty$, which is the reduced curve $\mathbb{A}^1$, so by definition 1.1.1 the order of $\pi^\# u$ along it is the length of the local ring of a reduced curve at its generic point, namely one. Hence

$$\sigma\big[\mathbb{P}^1\big] \;=\; \big[\mathbb{A}^1\big] \;=\; \big[C_P\mathbb{P}^1\big] \;\in\; A_1\big(C_P\mathbb{P}^1\big) = \mathbb{Z}$$

and $\sigma$ is an isomorphism in this dimension. In dimension zero the behaviour is the opposite. Let $Q \neq P$ be a closed point. The closure of $\{Q\} \times \mathbb{A}^1$ in $M_Z X$ is the strict transform of $\{Q\} \times \mathbb{P}^1$, which misses the centre $(P, \infty)$ entirely and therefore meets the fibre over $\infty$ in the single point $(Q, \infty)$; that point lies on the strict transform of $\mathbb{P}^1 \times \{\infty\}$, which was deleted. So the closure lift of $[Q]$ in $M_Z^\circ X$ does not meet $|D_\infty|$ at all, and definition 2.2.1 records the intersection class on the empty intersection:

$$\sigma[Q] \;=\; 0 \;\in\; A_0\big(C_P\mathbb{P}^1\big)$$

consistent with $A_0(\mathbb{A}^1) = 0$ (proposition 1.5.1). Since $A_0(\mathbb{P}^1) = \mathbb{Z}$ by theorem 1.5.2, specialization is not injective. A cycle disjoint from $Z$ escapes to infinity as the family degenerates, and the operation records nothing of it.

The example computed $\sigma$ on the class of a subvariety by taking the closure of its cylinder and reading the fibre at infinity, and the answer was in both cases a normal cone: the normal cone of $Z$ in $\mathbb{P}^1$ in the first computation, and the empty cone $C_\emptyset\{Q\}$ in the second. That this always happens is the main theorem of the section, and it is what makes $\sigma$ computable. Proving it requires knowing that the deformation space of a closed subscheme sits inside the deformation space of the ambient one, which is the content of the next two results.

The first is a statement about normal cones alone. If $V \subseteq X$ is a closed subscheme, the normal cone of $V \cap Z$ in $V$ has an intrinsic map to the normal cone of $Z$ in $X$, obtained by restricting the ideal $\mathcal{I}$ to $V$.

The second is the corresponding statement one level up, for the whole family rather than its special fibre. It is stated for an arbitrary morphism because the two compatibilities at the end of the section need the proper and the flat cases, while the theorem below needs the case of a closed immersion.

**Lemma 4.2.6: Functoriality of the Deformation Space**

Let $f\colon X' \to X$ be a morphism of algebraic schemes, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$ its scheme-theoretic preimage. Write $\pi'\colon M^\circ_{Z'}X' \to \mathbb{P}^1$ for the projection of the corresponding deformation space. Then there is a unique morphism $\widetilde{f}\colon M_{Z'}X' \to M_Z X$ over $f \times \mathrm{id}_{\mathbb{P}^1}$, and it satisfies

$$\widetilde{f}^{-1}\big(M^\circ_Z X\big) \;=\; M^\circ_{Z'}X'$$

so that it restricts to a morphism $\widetilde{f}\colon M^\circ_{Z'}X' \to M^\circ_Z X$. Moreover:

1. $\pi \circ \widetilde{f} = \pi'$; over $\mathbb{A}^1$ the morphism $\widetilde{f}$ restricts to $f \times \mathrm{id}_{\mathbb{A}^1}$, and over $\infty$ it restricts to the morphism of normal cones $C(f)\colon C_{Z'}X' \to C_Z X$ induced by the identity $\mathcal{I}\mathcal{O}_{X'} = \mathcal{I}'$.
2. If $f$ is proper, then $\widetilde{f}$ and $C(f)$ are proper.
3. If $f$ is flat of relative dimension $n$, then $\widetilde{f}$ and $C(f)$ are flat of relative dimension $n$, and the square formed by $\widetilde{f}$, $f \times \mathrm{id}_{\mathbb{P}^1}$ and the two blow-up morphisms is Cartesian.

***Proof*:**
Write $g = f \times \mathrm{id}_{\mathbb{P}^1}$, let $\mathcal{J} \subseteq \mathcal{O}_{X\times\mathbb{P}^1}$ be the ideal sheaf of $Z \times \{\infty\}$, and let $\mathcal{J}'$ be the ideal sheaf of $Z' \times \{\infty\}$ in $X' \times \mathbb{P}^1$.

*Step 1: the ideals correspond.* Scheme-theoretically $Z \times \{\infty\} = (Z \times \mathbb{P}^1) \cap (X \times \{\infty\})$, so $\mathcal{J}$ is the sum of the ideal sheaves of the two factors, and the same holds upstairs. The inverse image ideal of a sum is the sum of the inverse image ideals; the ideal of $Z \times \mathbb{P}^1$ pulls back to that of $Z' \times \mathbb{P}^1$ because $\mathcal{I}\mathcal{O}_{X'} = \mathcal{I}'$ by definition of the scheme-theoretic preimage, and the ideal of $X \times \{\infty\}$, generated by $u$, pulls back to the ideal of $X' \times \{\infty\}$, generated by the same $u$. Hence

$$\mathcal{J}\mathcal{O}_{X'\times\mathbb{P}^1} \;=\; \mathcal{J}'$$

*Step 2: existence and uniqueness.* On $M_{Z'}X'$ the ideal $\mathcal{J}'\mathcal{O}_{M_{Z'}X'}$ is invertible by [2, Basic Properties of Blow-ups](3), and by Step 1 it is the inverse image ideal of $\mathcal{J}$ under the composite $M_{Z'}X' \to X' \times \mathbb{P}^1 \xrightarrow{g} X \times \mathbb{P}^1$. The universal property, [2, Basic Properties of Blow-ups](4), therefore factors that composite uniquely through the blow-down $p\colon M_Z X \to X \times \mathbb{P}^1$, and the factorization is the asserted $\widetilde{f}$. Uniqueness of the factorization says precisely that $\widetilde{f}$ is the only morphism $M_{Z'}X' \to M_Z X$ lying over $g$. Since $\pi$ is the second projection composed with $p$, and $\widetilde{f}$ lies over $g$, which is the identity in the second factor, $\pi \circ \widetilde{f} = \pi'$.

*Step 3: the two open subschemes correspond.* The assertion is local on $X$ and on $\mathbb{P}^1$. Over $\mathbb{A}^1$ the blow-ups are isomorphisms, since the centres lie over $\infty$ ([2, Basic Properties of Blow-ups](2)), and both deformation spaces contain the whole preimage of $\mathbb{A}^1$; there $\widetilde{f}^{-1}(X \times \mathbb{A}^1) = X' \times \mathbb{A}^1$ because $\pi \circ \widetilde{f} = \pi'$. Over $\mathbb{P}^1 \setminus \{0\} = \operatorname{Spec} \overline{k}[u]$, take an affine open $U = \operatorname{Spec} A$ of $X$ with

$\mathcal{I}|_U$ given by an ideal $I \subseteq A$, and put $J = IA[u] + (u) \subseteq A[u]$, the ideal of $Z \times \{\infty\}$ over $U \times (\mathbb{P}^1 \setminus \{0\})$, so that by [2, Blow-up] the deformation space is there the relative Proj of the Rees algebra $\bigoplus_{n \geq 0} J^n$, in which $u$ is an element of degree one. By lemma 4.1.2 the open subscheme $M_Z^\circ X = M_Z X \setminus \mathrm{Bl}_Z X$ is over $U \times (\mathbb{P}^1 \setminus \{0\})$ exactly the basic open $D_+(u)$ of that relative Proj, with coordinate ring the extended Rees algebra $A[u][I/u]$, its closed complement there being $\mathrm{Bl}_Z X$.
The same description applies upstairs over any affine open $U' = \operatorname{Spec} A'$ of $f^{-1}(U)$, with $I' = IA'$ and $J' = I'A'[u] + (u)$, giving $M_{Z'}^\circ X' = D_+(u)$ there. By Step 1, $J^n A'[u] = J'^n$ for every $n$, so the graded homomorphism $\bigoplus_n J^n \otimes_{A[u]} A'[u] \to \bigoplus_n J'^n$ is surjective in every degree; it therefore induces a morphism of relative Proj schemes defined on all of $M_{Z'}X'$, and that morphism lies over $g$; by the uniqueness of Step 2 it is $\widetilde{f}$. A morphism of Proj schemes induced by a graded homomorphism $\varphi$ pulls the basic open $D_+(x)$ back to $D_+(\varphi(x))$, and $\varphi(u) = u$, so $\widetilde{f}^{-1}(D_+(u)) = D_+(u)$. Hence $\widetilde{f}^{-1}(M_Z^\circ X) = M_{Z'}^\circ X'$.

The three numbered assertions remain.

1. *The restrictions to the two fibres.* Over $\mathbb{A}^1$ both blow-ups are isomorphisms onto $X \times \mathbb{A}^1$ and $X' \times \mathbb{A}^1$, and $\widetilde{f}$ lies over $g$, so it restricts there to $f \times \mathrm{id}_{\mathbb{A}^1}$. Over $\infty$, the chart of Step 3 has coordinate ring $A[u][I/u]$, and the fibre over $\infty$ is cut out by $u$; on that chart the identification of the fibre with the normal cone given by theorem 4.1.5(2) reads
$$A[u][I/u]/(u) \;\cong\; \bigoplus_{n \geq 0} I^n/I^{n+1}$$
carrying the class of $a/u$, for $a \in I$, to the class of $a$ in degree one. The morphism $\widetilde{f}$ is induced by $A[u] \to A'[u]$ and sends $a/u$ to $f^\#(a)/u$, so on fibres it induces the homomorphism of graded algebras determined in degree one by $\bar{a} \mapsto \overline{f^\# a}$. That is the homomorphism defining $C(f)$.

2. *Properness.* Suppose $f$ is proper. The blow-up $M_{Z'}X' \to X' \times \mathbb{P}^1$ is projective, hence proper, by [2, Basic Properties of Blow-ups](1), and $g$ is proper as a base change of $f$, so the composite $M_{Z'}X' \to X \times \mathbb{P}^1$ is proper. Factor $\widetilde{f}$ as
$$M_{Z'}X' \xrightarrow{(\mathrm{id}, \widetilde{f})} M_{Z'}X' \times_{X \times \mathbb{P}^1} M_Z X \xrightarrow{\mathrm{pr}_2} M_Z X$$
The first map is the graph of $\widetilde{f}$ over $X \times \mathbb{P}^1$, a closed immersion because $M_Z X \to X \times \mathbb{P}^1$ is separated, and the second is a base change of the proper morphism $M_{Z'}X' \to X \times \mathbb{P}^1$, hence proper. So $\widetilde{f}$ is proper. Restricting over the open subscheme $M_Z^\circ X$, whose preimage is $M_{Z'}^\circ X'$ by Step 3, preserves properness, and $C(f)$ is the base change of $\widetilde{f}$ along the closed immersion $C_Z X \hookrightarrow M_Z^\circ X$, hence proper.

3. *Flatness.* Suppose $f$ is flat of relative dimension $n$; then so is $g$, being a base change of $f$. Flatness gives $\mathcal{J}^n \mathcal{O}_{X' \times \mathbb{P}^1} = g^* \mathcal{J}^n$ for every $n$, so the Rees algebra of $\mathcal{J}'$ is the pullback $g^*$ of the Rees algebra of $\mathcal{J}$, and [2, Relative Proj Commutes with Base Change] identifies
$$M_{Z'}X' \;\cong\; M_Z X \times_{X \times \mathbb{P}^1} \left(X' \times \mathbb{P}^1\right)$$
over $X' \times \mathbb{P}^1$. Under this identification the second projection is a morphism over $g$, hence equals $\widetilde{f}$ by the uniqueness of Step 2, and the square is Cartesian. So $\widetilde{f}$ is a base change of $g$, therefore flat with the same fibres, that is, flat of relative dimension $n$. The same holds after restricting to the two open subschemes, which correspond by Step 3, and $C(f)$ is again a base change of $\widetilde{f}$, so it too is flat of relative dimension $n$, completing the proof.

The computation of $\sigma$ on a subvariety can now be carried out. The statement is the one the construction was designed to produce: specializing a subvariety returns the normal cone of its intersection with $Z$, which is the linearized model of how $V$ meets $Z$.

**Theorem 4.2.7: Specialization of a Subvariety**

Let $Z \subseteq X$ be a closed subscheme, let $V \subseteq X$ be a $k$-dimensional subvariety, and let $W = V \cap Z$ be the scheme-theoretic intersection. Then

$$\sigma[V] \;=\; \big[C_W V\big] \;\in\; A_k\big(C_Z X\big)$$

the fundamental cycle of the normal cone $C_W V$, regarded as a closed subscheme of $C_Z X$ through lemma 3.4.12. In particular $C_W V$ is pure of dimension $k$.

***Proof*:**

*Step 1: the closure lift.* Let $\widetilde{V}$ be the closure of $V \times \mathbb{A}^1$ in $M_Z^\circ X$, with its reduced structure, a subvariety of dimension $k+1$. Since $V \times \mathbb{A}^1$ is closed in $X \times \mathbb{A}^1$, the open restriction of $[\widetilde{V}]$ is $j^*[\widetilde{V}] = [V \times \mathbb{A}^1] = q^*[V]$ by example 1.3.4, so $[\widetilde{V}]$ is a lift of $[V]$ and $\sigma[V] = D_\infty \cdot [\widetilde{V}]$.

*Step 2: the smaller deformation space.* Apply lemma 4.2.6 to the closed immersion $\iota\colon V \hookrightarrow X$, whose scheme-theoretic preimage of $Z$ is $W$ by lemma 3.4.12. It supplies a proper morphism

$$\widetilde{\iota}\colon M_W^\circ V \longrightarrow M_Z^\circ X$$

commuting with the projections to $\mathbb{P}^1$ and restricting over $\mathbb{A}^1$ to the closed immersion $V \times \mathbb{A}^1 \hookrightarrow X \times \mathbb{A}^1$. Over $\infty$ it restricts to the morphism $C(\iota)\colon C_W V \to C_Z X$, which is induced by the graded homomorphism attached to $\mathcal{I}\mathcal{O}_V = \mathcal{I}_W$ and is therefore the closed immersion of lemma 3.4.12.

The scheme $M_W V$ is integral. Indeed $V \times \mathbb{P}^1$ is a variety, and on an affine open $\operatorname{Spec} B \subseteq V \times \mathbb{P}^1$ the ideal $\mathfrak{b}$ of $W \times \{\infty\}$ is either the unit ideal, in which case the corresponding piece of the blow-up is $\operatorname{Spec} B$, or a nonzero proper ideal, in which case the charts of $\operatorname{Proj} \bigoplus_n \mathfrak{b}^n$ have coordinate rings $B[\mathfrak{b}/b]$ for $0 \neq b \in \mathfrak{b}$, subrings of the function field $k(V \times \mathbb{P}^1)$ containing $B$. Every chart is thus the spectrum of a domain with fraction field $k(V)(t)$, so $M_W V$ is reduced and irreducible, and so is its open subscheme $M_W^\circ V$. The latter has dimension $k+1$, since it contains $V \times \mathbb{A}^1$ as a nonempty, hence dense, open subscheme.

*Step 3: the image and the degree.* The morphism $\widetilde{\iota}$ is proper, hence closed, and $V \times \mathbb{A}^1$ is dense in $M_W^\circ V$, so the image is the closure of $\widetilde{\iota}(V \times \mathbb{A}^1) = V \times \mathbb{A}^1$, which is $\widetilde{V}$. Over $\mathbb{A}^1$ the morphism is an isomorphism onto $V \times \mathbb{A}^1$, which is dense in both $M_W^\circ V$ and $\widetilde{V}$, so the two schemes have the same function field and $\deg\big(M_W^\circ V/\widetilde{V}\big) = 1$. By definition 1.2.1,

$$\widetilde{\iota}_*\big[M_W^\circ V\big] \;=\; \big[\widetilde{V}\big]$$

*Step 4: cutting the smaller family.* The projection $\pi'\colon M_W^\circ V \to \mathbb{P}^1$ is flat by theorem 4.1.5(3), applied to the closed subscheme $W \subseteq V$. Each nonempty fibre is cut out locally by one equation, so every component has dimension at least $k$ by Krull's principal ideal theorem, and each fibre is a proper closed subset of the integral $M_W^\circ V$, so every component has dimension at most $k$. Thus $\pi'$ is flat of relative dimension $k$ in the sense of definition 1.3.1, and its fibre over $\infty$, which is $C_W V$, is pure of dimension $k$.

Since $\pi \circ \widetilde{\iota} = \pi'$, the pullback Cartier divisor is $\widetilde{\iota}^* D_\infty = \pi'^*(\infty)$, cut out by $\pi'^{\#}u$; it is defined because $M^\circ_W V$ is dominant over $\mathbb{P}^1$, its restriction over $\mathbb{A}^1$ being the nonempty $V \times \mathbb{A}^1$, and so is not contained in $|D_\infty|$. The first clause of definition 2.2.1 therefore gives

$$\widetilde{\iota}^* D_\infty \cdot \left[M^\circ_W V\right] \;=\; \left[\operatorname{div}(\pi'^{\#}u)\right] \;=\; \sum_T \operatorname{ord}_T\left(\pi'^{\#}u\right)[T]$$

the sum over the codimension-one subvarieties $T$ of $M^\circ_W V$. By definition 1.1.1 the coefficient $\operatorname{ord}_T(\pi'^{\#}u)$ is the length of $\mathcal{O}_{M^\circ_W V,T}/(\pi'^{\#}u)$, which is the local ring at $T$ of the zero scheme of $\pi'^{\#}u$, that is, of the fibre $C_W V$. The coefficient vanishes unless $T \subseteq C_W V$, and the $k$-dimensional subvarieties of $C_W V$ are exactly its irreducible components, by the purity just established. Comparing with the fundamental cycle of definition 1.1.2,

$$\widetilde{\iota}^* D_\infty \cdot \left[M^\circ_W V\right] \;=\; \left[C_W V\right]$$

*Step 5: descent to $C_Z X$.* Combining the previous steps with the projection formula proposition 2.3.3(1) for the proper morphism $\widetilde{\iota}$,

$$\sigma[V] \;=\; D_\infty \cdot \widetilde{\iota}_*\left[M^\circ_W V\right] \;=\; \widetilde{\iota}_*\Big(\widetilde{\iota}^* D_\infty \cdot \left[M^\circ_W V\right]\Big) \;=\; \widetilde{\iota}_*\left[C_W V\right]$$

as classes in $A_k(C_Z X)$. By Step 2 the morphism $\widetilde{\iota}$ restricts over $\infty$ to the closed immersion of lemma 3.4.12, and a closed immersion carries a fundamental cycle to the fundamental cycle of its image, every degree in definition 1.2.1 being 1. Hence $\widetilde{\iota}_*[C_W V] = [C_W V]$ read inside $C_Z X$, as we sought to show.

The theorem is what turns $\sigma$ from a construction into a computation. It also explains the two features of example 4.2.5: for $V = \mathbb{P}^1$ the intersection $V \cap Z$ is $Z$ itself and the answer is $[C_P\mathbb{P}^1]$, while for $V = \{Q\}$ with $Q \notin Z$ the intersection is empty, the normal cone of the empty subscheme is empty, and the answer is 0. Specialization does not preserve the geometry of $V$, only its normal geometry along $Z$: the class $[C_W V]$ remembers how $V$ meets $Z$ and forgets everything about $V$ away from $Z$. The multiplicities carry information: the fundamental cycle of definition 1.1.2 weights each component of $C_W V$ by the length of the local ring there, so a component along which the cone is non-reduced contributes with multiplicity greater than one. In the example below the cone is reduced and every multiplicity is one, the two components recording the two tangent directions of a node; the cuspidal cubic of exercise 4.2.1 (3) is the opposite case, a single tangent line doubled. And the purity statement is not an extra hypothesis but a by-product: the normal cone of any closed subscheme in a variety is pure of the dimension of that variety, because it is a fibre of a flat family whose other fibres are the variety itself.

**Example 4.2.8: Specializing the Nodal Cubic to its Tangent Cone**

Assume $\operatorname{char}\overline{k} \neq 2$, let $X = \operatorname{Spec}\overline{k}[x,y]/(y^2 - x^2 - x^3)$ be the affine nodal cubic, and let $Z = \{P\}$ be the reduced origin. The curve $X$ is a variety of dimension one, so $A_1(X) = \mathbb{Z}[X]$ by example 1.1.6(3). Taking $V = X$ in theorem 4.2.7 gives $W = X \cap Z = Z$ and

$$\sigma[X] \;=\; \left[C_P X\right]$$

and example 3.4.7 computed the normal cone: $C_P X = \operatorname{Spec}\overline{k}[x,y]/(y^2 - x^2)$, the union of the two lines $L_+ = \{y = x\}$ and $L_- = \{y = -x\}$. The polynomial $y^2 - x^2 = (y-x)(y+x)$ is squarefree, so the cone is reduced and each component carries multiplicity one:

$$\sigma[X] \;=\; [L_+] + [L_-] \;\in\; A_1(C_P X)$$

which is free of rank two on the two lines, again by example 1.1.6(3). The specialization has replaced an irreducible curve by a reducible cone, one line for each branch of the node, and the two summands are exactly the two tangent directions along which $X$ approaches $P$. The class of $X$ in $A_1(X) = \mathbb{Z}$ carried no information about the singularity; its specialization, in a group of rank two, carries the branch structure.

What remains is to check that $\sigma$ is compatible with the two functorialities of chapter 1. Both statements are forced by the shape of the construction: $\sigma$ was built from the flat pullback $q^*$, the localization sequence, and the divisor operator, and each of the three commutes with proper pushforward and with flat pullback. The compatibilities are what allow $\sigma$, and hence the Gysin map of section 4.3, to be computed by moving to a cover or restricting to an open set.

**Proposition 4.2.9: Compatibility with Proper Pushforward**

Let $f\colon X' \to X$ be a proper morphism, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$. Write $\sigma'$ for the specialization homomorphism of $Z' \subseteq X'$ and $C(f)\colon C_{Z'}X' \to C_Z X$ for the induced proper morphism of normal cones. Then for every $\alpha' \in A_k(X')$,

$$\sigma(f_*\alpha') \;=\; C(f)_*(\sigma'(\alpha')) \quad \text{in } A_k(C_Z X)$$

***Proof*:**
Let $\widetilde{f}\colon M^\circ_{Z'}X' \to M^\circ_Z X$ be the proper morphism of lemma 4.2.6, and let $\beta'$ be the closure lift of $\alpha'$, so that $j'^*\beta' = q'^*\alpha'$ and every component of $\beta'$ dominates $\mathbb{P}^1$.

*Step 1: the pushed-forward lift.* Since $\pi \circ \widetilde{f} = \pi'$, the square with horizontal maps $\widetilde{f}$ and $f \times \mathrm{id}_{\mathbb{A}^1}$ and vertical maps the two open immersions $j', j$ is Cartesian: the preimage $\widetilde{f}^{-1}(X \times \mathbb{A}^1)$ is $\pi'^{-1}(\mathbb{A}^1) = X' \times \mathbb{A}^1$, and $\widetilde{f}$ restricts there to $f \times \mathrm{id}_{\mathbb{A}^1}$ by lemma 4.2.6(1). As $j$ is flat of relative dimension 0 and $\widetilde{f}$ is proper, proposition 1.3.6 gives

$$j^*\widetilde{f}_*\beta' \;=\; \left(f \times \mathrm{id}_{\mathbb{A}^1}\right)_* j'^*\beta' \;=\; \left(f \times \mathrm{id}_{\mathbb{A}^1}\right)_* q'^*\alpha'$$

The square with horizontal maps $f \times \mathrm{id}_{\mathbb{A}^1}$ and $f$ and vertical maps $q'$ and $q$ is Cartesian as well, since $X' \times \mathbb{A}^1 = X' \times_X (X \times \mathbb{A}^1)$, so a second application of proposition 1.3.6, now with $q$ flat of relative dimension 1 and $f$ proper, gives $(f \times \mathrm{id}_{\mathbb{A}^1})_* q'^*\alpha' = q^* f_*\alpha'$. Hence $\widetilde{f}_*\beta'$ is a lift of $f_*\alpha'$.

*Step 2: the projection formula.* Every component of $\beta'$ dominates $\mathbb{P}^1$, and $\pi \circ \widetilde{f} = \pi'$, so no component of the support of $\beta'$ maps into $|D_\infty| = C_Z X$; the pullback divisor $\widetilde{f}^*D_\infty$ is therefore defined, and it equals $\pi'^*(\infty) = D'_\infty$. By proposition 2.3.3(1),

$$D_\infty \cdot \widetilde{f}_*\beta' \;=\; \widetilde{f}_*(D'_\infty \cdot \beta')$$

as classes in $A_k(C_Z X)$. The left side is $\sigma(f_*\alpha')$ by Step 1 and theorem 4.2.4, and $D'_\infty \cdot \beta' = \sigma'(\alpha')$ is a class on $C_{Z'}X'$, on which $\widetilde{f}_*$ acts as $C(f)_*$ by lemma 4.2.6(1). Therefore $\sigma(f_*\alpha') = C(f)_*(\sigma'(\alpha'))$, completing the proof.

**Proposition 4.2.10: Compatibility with Flat Pullback**

Let $f\colon X' \to X$ be flat of relative dimension $n$, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$. Then the induced morphism $C(f)\colon C_{Z'}X' \to C_Z X$ is flat of relative dimension $n$, and for every $\alpha \in A_k(X)$,

$$\sigma'\big(f^*\alpha\big) \;=\; C(f)^*\big(\sigma(\alpha)\big) \quad \text{in } A_{k+n}\big(C_{Z'}X'\big)$$

***Proof*:**

Flatness of $C(f)$ is lemma 4.2.6(3), which also makes $\widetilde{f}\colon M^\circ_{Z'}X' \to M^\circ_Z X$ flat of relative dimension $n$ with $\widetilde{f}^{-1}(C_Z X) = C_{Z'}X'$ and $\widetilde{f}^{-1}(X \times \mathbb{A}^1) = X' \times \mathbb{A}^1$.

*Step 1: the pulled-back lift.* Let $\beta$ be the closure lift of $\alpha$. Contravariant functoriality of flat pullback (proposition 1.3.5(3)) applied to the two equal composites $j \circ (f \times \mathrm{id}_{\mathbb{A}^1}) = \widetilde{f} \circ j'$ and $q \circ (f \times \mathrm{id}_{\mathbb{A}^1}) = f \circ q'$ gives

$$j'^*\widetilde{f}^*\beta \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)^* j^*\beta \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)^* q^*\alpha \;=\; q'^* f^*\alpha$$

so $\widetilde{f}^*\beta$ is a lift of $f^*\alpha$. Its components are components of flat preimages of the components of $\beta$, each of which dominates $\mathbb{P}^1$, so they dominate $\mathbb{P}^1$ as well and none lies in $|D'_\infty|$.

*Step 2: the projection formula.* The pullback Cartier divisor is $\widetilde{f}^*D_\infty = \pi'^*(\infty) = D'_\infty$, again because $\pi \circ \widetilde{f} = \pi'$. By proposition 2.3.3(2),

$$\widetilde{f}^*\big(D_\infty \cdot \beta\big) \;=\; D'_\infty \cdot \widetilde{f}^*\beta$$

as classes in $A_{k+n}(\widetilde{f}^{-1}(C_Z X)) \;=\; A_{k+n}(C_{Z'}X')$. The right side is $\sigma'(f^*\alpha)$ by Step 1 and theorem 4.2.4. On the left, $D_\infty \cdot \beta = \sigma(\alpha)$ is a class on $C_Z X$, and the restriction of $\widetilde{f}^*$ to classes on $C_Z X$ is the flat pullback along the base change $C(f)$, since flat pullback is computed on preimages and $\widetilde{f}^{-1}(C_Z X) = C_{Z'}X'$. Hence $\sigma'(f^*\alpha) = C(f)^*(\sigma(\alpha))$, as we sought to show.

The two propositions say that specialization is a natural transformation, covariant for proper morphisms and contravariant for flat ones, between the Chow groups of $X$ and those of the normal cone. The hypothesis common to both, $Z' = f^{-1}(Z)$, is not a restriction imposed for convenience: the normal cone of $Z'$ in $X'$ maps to that of $Z$ in $X$ precisely because the ideal of $Z$ pulls back to the ideal of $Z'$, and without that identity there is no morphism $C(f)$ to state the compatibility with.

## Exercise 4.2.1

1. Take $Z = X$. Exercise 4.1.1 (1) identifies the deformation space as $M^\circ_X X = X \times \mathbb{P}^1$ with special fibre $C_X X = X$. Deduce from theorem 4.2.7 that $\sigma\colon A_k(X) \to A_k(X)$ is the identity.
2. Let $X = \mathbb{P}^2$ and let $Z = L$ be a line, an effective Cartier divisor on $X$. Using example 3.4.15, identify $C_L\mathbb{P}^2$ with the total space of the line bundle $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$ on $L \cong \mathbb{P}^1$, and compute $A_k(C_L\mathbb{P}^2)$ for all $k$ by theorem 1.4.6. Then compute $\sigma[\mathbb{P}^2]$, $\sigma[L']$ for a line $L' \neq L$, and $\sigma[Q]$ for a closed point $Q$, and say in which dimensions $\sigma$ is an isomorphism.
3. Let $X = \operatorname{Spec} \overline{k}[x,y]/(y^2 - x^3)$ be the affine cuspidal cubic and $Z = \{P\}$ the reduced origin. Taking the cone $C_P X = \operatorname{Spec} \overline{k}[x,y]/(y^2)$ and its fundamental cycle $2\,[L]$ from exercise 4.1.1 (5),

compute $\sigma[X]$ as a cycle. Compare the answer with example 4.2.8 and say what distinguishes a cusp from a node in this computation. Does the answer depend on char$\overline{k}$?

4. Let $\iota\colon Z \hookrightarrow C_Z X$ be the vertex of the normal cone (definition 3.4.1) and $i_Z\colon Z \hookrightarrow X$ the inclusion. Show that $\sigma \circ (i_Z)_* = \iota_*$ as homomorphisms $A_k(Z) \to A_k(C_Z X)$. (Hint: apply theorem 4.2.7 to a subvariety $V \subseteq Z$ and identify $C_V V$.)
5. Show that if $V \subseteq X$ is a subvariety with $V \cap Z = \emptyset$, then $\sigma[V] = 0$. Deduce that for $X = \mathbb{P}^n$ with $n \geq 1$ and $Z = \{P\}$ a closed point, $\sigma$ vanishes on all of $A_0(\mathbb{P}^n)$, and check that this is consistent with a direct computation of $A_0(C_P\mathbb{P}^n)$.

## 4.3 The Gysin Map for a Regular Embedding

The specialization homomorphism of section 4.2 accepts an arbitrary closed subscheme, and its target is the Chow group of a cone that is usually not a bundle. This section takes the case in which it is. For a regular embedding the normal cone is the normal bundle (proposition 3.4.11), so specialization lands in $A_k(N)$, and flat pullback along $N \to Z$ is an isomorphism (theorem 1.4.6) that can be inverted. The composite lowers dimension by the codimension and lands on $Z$.

Two things have to be established beyond that one-line recipe. The first is that the recipe computes what a reader expects when the intersection is proper: the cycle $[V \cap Z]$, with the multiplicities the scheme structure supplies. The second is that the construction survives base change. For any morphism $X' \to X$ the normal cone of the preimage of $Z$ still sits inside the pulled-back normal bundle, and the same recipe produces a homomorphism landing on that preimage rather than on $Z$. It is this refined form, not the plain one, that chapter 5 applies to the diagonal.

Both rest on the following comparison between the normal cone of a pullback and the pullback of the normal bundle. Regularity of $i$ is what makes the comparison available: it is exactly the hypothesis under which the conormal sheaf is locally free, so that the symmetric algebra it generates is a bundle into which the associated graded algebra downstairs maps.

The lemma says that regularity of $i$ propagates one step: it is not inherited by $Z' \hookrightarrow X'$, which can be badly irregular, but the failure is confined inside a bundle of the correct rank $d$. The cone $C_{Z'}X'$ may be singular, may be reducible, and may have components of several dimensions, and none of that prevents it from being a closed subscheme of a rank-$d$ bundle on $Z'$. That containment is what lets the same recipe be applied over $X'$ as over $X$, and the discrepancy between $C_{Z'}X'$ and all of $N'$ is what the excess intersection formula of chapter 5 measures.

With $g = \mathrm{id}_X$ the lemma reduces to the case already in hand, and the construction can be stated.

**Definition 4.3.1: The Gysin Map of a Regular Embedding**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 0$ with normal bundle $\pi\colon N \to Z$. For $d = 0$ the ideal sheaf is zero, $i$ is an open and closed immersion, $N$ is the zero bundle over $Z$, and the formula below reads as flat pullback along $i$; the construction is stated for $d \geq 1$ below and the degenerate case is recorded here so that statements about codimension-zero embeddings, which occur in chapter 5, have a definition to refer to. By proposition 3.4.11 the normal cone $C_Z X$ is $N$, so the specialization homomorphism of definition 4.2.3 is a homomorphism $\sigma\colon A_k(X) \to A_k(N)$ (theorem 4.2.4). The *Gysin map* of $i$ is

$$i^! \;=\; (\pi^*)^{-1} \circ \sigma \;:\; A_k(X) \longrightarrow A_{k-d}(Z)$$

where $\pi^*\colon A_{k-d}(Z) \to A_k(N)$ is flat pullback along $\pi$.

The inverse in the definition exists because $N$ is a vector bundle of rank $d$, hence an affine bundle of relative dimension $d$ (definition 1.4.5), so $\pi^*$ is an isomorphism by theorem 1.4.6. The codimension-zero case is excluded because it is empty of content: an ideal generated locally by the empty regular sequence is the zero ideal, so a regular embedding of codimension zero is $Z = X$.

The smallest case can be run through at once. Take $X = \mathbb{A}^1$ and $Z = \{0\}$, whose ideal is generated by the single nonzerodivisor $t$, so that $i$ is a regular embedding of codimension one and $N$ is a one-dimensional vector space over the point $Z$, with total space $\mathbb{A}^1$. By theorem 4.2.7, $\sigma[\mathbb{A}^1] = [C_Z\mathbb{A}^1] = [N]$, and $\pi^*[Z] = [N]$ by proposition 1.3.5(2), so $i^![\mathbb{A}^1] = [Z]$ in $A_0(Z) = \mathbb{Z}$: the origin with multiplicity one.

**Proposition 4.3.2: The Gysin Map is a Homomorphism of Degree $-d$**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$. Then:

1. $i^!\colon A_k(X) \to A_{k-d}(Z)$ is a well-defined homomorphism of abelian groups for every $k$, and it is the unique map satisfying $\pi^* \circ i^! = \sigma$;
2. $i^! = 0$ on $A_k(X)$ for $k < d$.

***Proof*:**

1. A vector bundle of rank $d$ is an affine bundle of relative dimension $d$ in the sense of definition 1.4.5, a local trivialization of the bundle being in particular a local product decomposition with $\mathbb{A}^d$. Theorem 1.4.6 therefore makes $\pi^*\colon A_{k-d}(Z) \to A_k(N)$ an isomorphism for every $k$. The specialization homomorphism $\sigma\colon A_k(X) \to A_k(N)$ is a homomorphism defined on rational equivalence classes by theorem 4.2.4, and $C_Z X = N$ by proposition 3.4.11, so the composite $(\pi^*)^{-1} \circ \sigma$ is a homomorphism $A_k(X) \to A_{k-d}(Z)$. If $\tau\colon A_k(X) \to A_{k-d}(Z)$ satisfies $\pi^* \circ \tau = \sigma$, then $\pi^*\tau = \pi^* i^!$, and injectivity of $\pi^*$ gives $\tau = i^!$.
2. For $k < d$ the target $A_{k-d}(Z)$ is a Chow group in negative dimension. No scheme has a subvariety of negative dimension, so $Z_{k-d}(Z) = 0$ and hence $A_{k-d}(Z) = 0$, forcing $i^! = 0$ there, as we sought to show.

The characterization $\pi^* \circ i^! = \sigma$ is how the map is used in practice: rather than inverting $\pi^*$, one produces a candidate class on $Z$ and checks that its pullback to $N$ is the specialization. The degree $-d$ matches the naive expectation that intersecting a $k$-dimensional cycle with a codimension-$d$ subscheme leaves something of dimension $k-d$. What the construction adds to that expectation is that the answer is defined for every class, with no hypothesis on how $Z$ and a representing cycle meet, and that it is a rational-equivalence invariant.

The first thing to verify is that the machinery reproduces the naive answer when the naive answer makes sense. The condition for that is exactly regularity of the induced embedding.

**Proposition 4.3.3: The Gysin Map of a Regular Intersection**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, let $V \subseteq X$ be a $k$-dimensional subvariety, and let $W = V \cap Z$ be the scheme-theoretic intersection, that is the preimage of $Z$ under the inclusion $V \hookrightarrow X$. If $W \hookrightarrow V$ is a regular embedding of codimension $d$, then $W$ is pure of dimension $k-d$ and

$$i^![V] \;=\; \iota_*[W] \;\in\; A_{k-d}(Z)$$

where $[W]$ is the fundamental cycle of definition 1.1.2 and $\iota\colon W \hookrightarrow Z$ is the inclusion.

***Proof*:**
Apply lemma 3.4.13 to the closed immersion $g\colon V \hookrightarrow X$, whose preimage of $Z$ is $W$: it produces a closed immersion $j\colon C_W V \hookrightarrow N|_W$ of cones over $W$, and by part (1) of that lemma the hypothesis on $W \hookrightarrow V$ makes $j$ an isomorphism. Thus $C_W V = N|_W$, the restriction of $N$ to $W$, viewed as a closed subscheme of $C_Z X = N$.

By theorem 4.2.7, $\sigma[V] = [C_W V]$ in $A_k(N)$, the fundamental cycle of the normal cone of $V$ along $W$ read inside $C_Z X$. The previous paragraph identifies that cycle as $[N|_W]$. Now $\pi_W : N|_W \to W$ is flat of relative dimension $d$, being a vector bundle of rank $d$, so proposition 1.3.5(2) gives $\pi_W^*[W] = [N|_W]$; and flat pullback of a cycle is computed componentwise from scheme-theoretic preimages (definition 1.3.3), so the cycle $\pi_W^*[W]$ on $N|_W$ is the cycle $\pi^*\iota_*[W]$ on $N$. Hence

$$\pi^*\big(\iota_*[W]\big) = [N|_W] = \sigma[V]$$

and proposition 4.3.2(1) gives $i^![V] = \iota_*[W]$.

Finally, $\sigma[V]$ lies in $A_k(N)$, so $[N|_W]$ is a $k$-cycle; since $\pi_W$ raises dimension by exactly $d$ on each component, every irreducible component of $W$ has dimension $k-d$, completing the proof.

The hypothesis is the scheme-theoretic form of "$V$ meets $Z$ properly and without excess tangency". It is not a condition on dimensions alone: a subvariety can meet $Z$ in the expected dimension $k-d$ and still fail it, if the intersection is not cut out by $d$ equations forming a regular sequence, and then $i^![V]$ carries multiplicities the set $V \cap Z$ does not display. When the hypothesis does hold, the Gysin map returns the fundamental cycle of the intersection scheme, so the multiplicities are the lengths that scheme records and nothing more.

**Example 4.3.4: Linear Subspaces of Projective Space**

Fix homogeneous coordinates $x_0, \dots, x_n$ on $\mathbb{P}^n$ and let $Z = \{x_0 = \cdots = x_{d-1} = 0\} \cong \mathbb{P}^{n-d}$ with $1 \le d \le n$. On each standard affine chart $\{x_r \neq 0\}$ meeting $Z$, the ideal of $Z$ is generated by the $d$ ratios $x_0/x_r, \dots, x_{d-1}/x_r$, which form a regular sequence in the polynomial ring of that chart: each is a nonzerodivisor modulo its predecessors, the quotients being polynomial rings in the remaining coordinates. So $i\colon Z \hookrightarrow \mathbb{P}^n$ is a regular embedding of codimension $d$ and $i^!\colon A_m(\mathbb{P}^n) \to A_{m-d}(Z)$ is defined.

By theorem 1.5.2, $A_m(\mathbb{P}^n) = \mathbb{Z}\, h_m$ with $h_m$ the class of a linear subspace $\mathbb{P}^m \subseteq \mathbb{P}^n$, and likewise $A_{m-d}(Z) = \mathbb{Z}\, h^Z_{m-d}$ for $Z \cong \mathbb{P}^{n-d}$. Take $m \ge d$ and represent $h_m$ by $L = \{x_{m+1} = \cdots = x_n = 0\}$, the span of the coordinate points $e_0, \dots, e_m$. Then

$$L \cap Z \;=\; \{x_0 = \cdots = x_{d-1} = 0\} \cap L$$

is the span of $e_d, \dots, e_m$, a linear $\mathbb{P}^{m-d}$, and its ideal in $L \cong \mathbb{P}^m$ is generated by the restrictions of $x_0, \dots, x_{d-1}$, again a regular sequence of length $d$ on every chart of $L$ meeting it. So $L \cap Z \hookrightarrow L$ is a regular embedding of codimension $d$, and proposition 4.3.3 gives

$$i^!(h_m) \;=\; \left[\mathbb{P}^{m-d}\right] \;=\; h^Z_{m-d}$$

the intersection scheme being reduced and irreducible, so that its fundamental cycle carries multiplicity one. For $m < d$ the target $A_{m-d}(Z)$ vanishes and $i^!(h_m) = 0$ by proposition 4.3.2(2). The Gysin map of a linear subspace is therefore the shift $h_m \mapsto h^Z_{m-d}$, carrying generator to generator in every degree where the target is nonzero.

The computation is the expected one, and its value is that no general position was invoked to get it: the representative $L$ was chosen for convenience, and proposition 4.3.2 guarantees that any other representative of $h_m$, including one contained in $Z$ or tangent to it, returns the same class. That is the difference between this construction and a count of intersection points.

The codimension-one case can be compared with the operator built by hand in chapter 2, and the two agree wherever the first clause of definition 2.2.1 applies.

**Proposition 4.3.5: Agreement with Intersection with a Divisor**

Let $D$ be an effective Cartier divisor on an algebraic scheme $X$ with $D \neq X$, and let $i\colon D \hookrightarrow X$ be the inclusion, a regular embedding of codimension one with normal bundle $N = \mathcal{O}_X(D)|_D$ (example 3.4.15). Let $V \subseteq X$ be a $k$-dimensional subvariety with $V \not\subseteq |D|$, and let $W = V \cap D$ be the scheme-theoretic intersection, a closed subscheme with support $V \cap |D|$. Then $W$ is pure of dimension $k-1$, the fundamental cycle of $W$ and the intersection class of definition 2.2.1 agree,

$$[W] \;=\; D \cdot [V] \;\in\; A_{k-1}\big(V \cap |D|\big)$$

already as cycles and not as classes, and the Gysin class of $V$ is their pushforward,

$$i^![V] \;=\; \iota_*\big(D \cdot [V]\big) \;\in\; A_{k-1}(D)$$

where $\iota\colon W \hookrightarrow D$ is the inclusion.

***Proof*:**
Recall that $W = V \cap D$ is the closed subscheme of $V$ cut out by the ideal $\mathcal{I} \cdot \mathcal{O}_V$, where $\mathcal{I}$ is the ideal sheaf of $D$.

*Step 1: the hypothesis of proposition 4.3.3 holds.* Since $D$ is an effective Cartier divisor, $\mathcal{I}$ is locally generated by a single nonzerodivisor $f \in \mathcal{O}_X(U)$. As $V$ is integral and $V \not\subseteq |D|$, the restriction $f|_V$ is a nonzero element of the domain $\mathcal{O}_V(U \cap V)$, hence a nonzerodivisor, and it generates $\mathcal{I} \cdot \mathcal{O}_V$ there. So $W \hookrightarrow V$ is a regular embedding of codimension one, and proposition 4.3.3 gives $i^![V] = \iota_*[W]$ in $A_{k-1}(D)$, with $W$ pure of dimension $k-1$.

*Step 2: the fundamental cycle of $W$ is the divisor of the section.* The intersection class of definition 2.2.1 is computed by its first clause, $V$ not being contained in $|D|$: the local equations of $D$ restrict to a rational section $s_V$ of $\mathcal{O}_X(D)|_V$, given on $U \cap V$ by $f|_V$, and $D \cdot [V] = [\operatorname{div}(s_V)] = \sum_T \operatorname{ord}_T(s_V)\,[T]$, the sum over codimension-one subvarieties $T \subseteq V$. Fix such a $T$ and let $A = \mathcal{O}_{V,T}$, a one-dimensional local domain. The local ring of $W$ at the generic point of $T$ is $A/(f|_V)$, so the coefficient of $[T]$ in the fundamental cycle $[W]$ of definition 1.1.2 is $\operatorname{length}_A\big(A/(f|_V)\big)$, which is $\operatorname{ord}_T(s_V)$ by definition 1.1.1. The coefficient is zero exactly when $f|_V$ is a unit in $A$, that is when $T$ is not a component of $W$, so the two cycles have the same components as well as the same multiplicities. Hence $[W] = [\operatorname{div}(s_V)]$ as cycles on $V \cap D$, so $[W] = D \cdot [V]$ in $A_{k-1}(V \cap |D|)$; pushing that identity forward along $\iota$ and combining it with Step 1 gives $i^![V] = \iota_*[W] = \iota_*\big(D \cdot [V]\big)$ in $A_{k-1}(D)$, completing the proof.

The proposition is the consistency check in codimension one: the operator $D \cdot -$, built in chapter 2 out of local equations and orders of vanishing, is the specialization construction in disguise. The restriction $V \not\subseteq |D|$ is exactly the first clause of definition 2.2.1; the second clause, where $V$ lies inside the support and $D \cdot [V]$ is computed from the restricted line bundle, corresponds on the Gysin side to a cycle supported on the zero section of $N$, and evaluating it needs the self-intersection formula of chapter 5.

Everything so far records the answer on $Z$. The construction gives more: it records it on the part of $Z$ that the cycle meets, and it does so for a cycle living on any scheme over $X$, not only on $X$ itself. Lemma 3.4.13 is what makes this possible, since it supplies a bundle of rank $d$ on the preimage of $Z$ regardless of how singular that preimage is.

**Definition 4.3.6: The Refined Gysin Homomorphism**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle $N$, and let $g\colon X' \to X$ be a morphism of algebraic schemes. Form the fibre square

$$\begin{tikzcd} Z' \arrow[r, "i'"] \arrow[d, "h"'] & X' \arrow[d, "g"] \\ Z \arrow[r, "i"] & X \end{tikzcd}$$

with $Z' = g^{-1}(Z)$ and $i'$ the induced closed immersion, and write $\pi'\colon N' = h^*N \to Z'$. Let $\sigma'\colon A_k(X') \to A_k(C_{Z'}X')$ be the specialization homomorphism of definition 4.2.3 for the pair $(Z', X')$, well defined by theorem 4.2.4, and let $j\colon C_{Z'}X' \hookrightarrow N'$ be the closed immersion of lemma 3.4.13. The *refined Gysin homomorphism* of $i$ along $g$ is

$$i^! \;=\; (\pi'^*)^{-1} \circ j_* \circ \sigma' \;:\; A_k(X') \longrightarrow A_{k-d}(Z')$$

where $j_*$ is proper pushforward along the closed immersion $j$ and $\pi'^*\colon A_{k-d}(Z') \to A_k(N')$ is the isomorphism of theorem 1.4.6.

The notation deliberately reuses $i^!$: the map depends on $g$, but the source and target groups record which $g$ is meant, and no ambiguity arises in practice. Each of the three maps is defined for the reasons already in place. Specialization for $(Z', X')$ needs no hypothesis on $Z'$, which is why the construction tolerates a preimage that is singular or of the wrong dimension; $j_*$ is defined because a closed immersion is proper, and proper pushforward preserves the dimension of a cycle (definition 1.2.1); and $\pi'^*$ is invertible because $N'$ is a bundle of rank $d$ on $Z'$, by definition 1.4.5 and theorem 1.4.6 as before. The only step that used regularity of $i$ is lemma 3.4.13, and it used it to guarantee that the rank is $d$ on the nose.

**Proposition 4.3.7: The Refined Map Extends the Gysin Map**

In the situation of definition 4.3.6 with $X' = X$ and $g = \mathrm{id}_X$, the refined Gysin homomorphism is the Gysin map $i^!$ of definition 4.3.1.

***Proof*:**
With $g = \mathrm{id}_X$ one has $Z' = Z$, $h = \mathrm{id}_Z$ and $N' = N$. By lemma 3.4.13(2) the closed immersion $j$ is the identification $C_ZX = N$ of proposition 3.4.11, so $j_*$ is the identity on $A_k(N)$ and $\sigma' = \sigma$. The composite $(\pi^*)^{-1} \circ \mathrm{id} \circ \sigma$ is definition 4.3.1, as claimed.

**Example 4.3.8: Two Lines in the Plane**

Let $X = \mathbb{P}^2$ and $Z = L = \{x_0 = 0\}$, an effective Cartier divisor and hence a regular embedding of codimension one with $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$ (example 3.4.15).
*A transverse line.* Take $g\colon X' = L' \hookrightarrow \mathbb{P}^2$ with $L' = \{x_1 = 0\}$. Then $Z' = g^{-1}(L)$ is cut out on $L'$ by $x_0$, so $Z' = \{p\}$ with $p = [0:0:1]$, a reduced point, and $Z' \hookrightarrow L'$ is a regular embedding of codimension one. By lemma 3.4.13(1) the immersion $j$ is an isomorphism, so $C_{Z'}X' = N' = N|_p$, a one-dimensional vector space over $p$. Specialization gives $\sigma'[L'] = [C_{Z'}L'] = [N']$ by theorem 4.2.7,

and $\pi'^*[p] = [N']$ by proposition 1.3.5(2), so

$$i^![L'] \;=\; [p] \;\in\; A_0(L \cap L') = \mathbb{Z}$$

The class is recorded on the single point $L \cap L'$, not on $L$, and since $\{p\}$ is the scheme-theoretic intersection $L' \cap L$, proposition 4.3.5 identifies it there with the intersection class $L \cdot [L'] \in A_0(L' \cap |L|)$. Pushing it forward along $\{p\} \hookrightarrow L$ returns the class of a point on $L$, which by the same proposition is the plain Gysin class of $L'$ in $A_0(L)$.

*The same line.* Take instead $g = i$, so $X' = L$ and $Z' = L$. The normal cone of $L$ in itself is the zero cone of example 3.4.2(2), so $C_{Z'}X' = L$ and $j$ is the vertex of $N' = N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$, that is the zero section $\zeta\colon L \to N$. Hence

$$i^![L] \;=\; (\pi^*)^{-1}\big(\zeta_*[L]\big) \;\in\; A_0(L)$$

Nothing in this section evaluates the right-hand side. What it must equal is known from chapter 2: example 2.2.3 computes $L \cdot [L] = [q]$, the class of one point, so the class of the zero section in $A_1(N)$ has to be $\pi^*[q]$. Identifying it is the self-intersection formula, proved in chapter 5.

The two halves of the example are the two regimes the refined map is built to handle uniformly. In the first, the preimage $Z'$ has the expected dimension and the answer is a cycle on it. In the second, $Z'$ is far too large, the cone degenerates onto the zero section, and the answer is carried entirely by the normal bundle. No hypothesis distinguishes the cases in definition 4.3.6; the difference shows up only in how much of $N'$ the cone $C_{Z'}X'$ fills.

**4.3.9 Foreshadowing: What the Gysin Map Is Not Yet** The Gysin map is a homomorphism of Chow groups, not a product. Three things it is expected to satisfy are not proved here and are the business of chapter 5: that $i^!$ applied to the diagonal $X \hookrightarrow X \times X$ of a smooth variety turns $A_*(X)$ into a commutative ring; that $i^!$ of a class supported on $Z$ is computed by capping with the top Chern class of $N$, which is the self-intersection formula and the missing step in example 4.3.8; and that two refined Gysin maps for two regular embeddings commute, which is what makes the resulting product associative. Everything in the present section is input to those proofs rather than a consequence of them.

The third item is settled before this chapter ends, in section 4.4; the first two belong to chapter 5.

The properties chapter 5 consumes are the two functorialities. They say that the refined Gysin map is a morphism of the theory over $X$: it commutes with pushing a cycle forward along a proper $X$-morphism and with pulling it back along a flat one, in each case through the induced morphism of preimages.

**Theorem 4.3.10: Compatibility of the Refined Gysin Homomorphism**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$. Let $X' \to X$ and $X'' \to X$ be morphisms of algebraic schemes and $p\colon X'' \to X'$ a morphism over $X$; write $Z' \subseteq X'$ and $Z'' \subseteq X''$ for the preimages of $Z$ and $q\colon Z'' \to Z'$ for the morphism induced by $p$.

1. *Proper pushforward.* If $p$ is proper then so is $q$, and for every $\alpha \in A_k(X'')$,
$$i^!\big(p_*\alpha\big) \;=\; q_*\big(i^!\alpha\big) \quad \text{in } A_{k-d}(Z')$$
2. *Flat pullback.* If $p$ is flat of relative dimension $n$ then so is $q$, and for every $\beta \in A_k(X')$,
$$i^!\big(p^*\beta\big) \;=\; q^*\big(i^!\beta\big) \quad \text{in } A_{k+n-d}(Z'')$$

***Proof*:**
Write $C' = C_{Z'}X'$ and $C'' = C_{Z''}X''$, let $h'\colon Z' \to Z$ and $h''\colon Z'' \to Z$ be the structure morphisms, and set $N' = h'^*N$ and $N'' = h''^*N$ with projections $\pi'$ and $\pi''$. Let $j'\colon C' \hookrightarrow N'$ and $j''\colon C'' \hookrightarrow N''$ be the closed immersions of lemma 3.4.13.

Since $X'' \to X$ factors through $p$, the preimage of $Z$ in $X''$ is $Z'' = p^{-1}(Z')$, so $q$ is the base change of $p$ along $Z' \hookrightarrow X'$; properness and flatness of relative dimension $n$ are both stable under base change, which proves the first assertion of each part. From $h'' = h' \circ q$ one gets $N'' = q^*N'$, so

$$\begin{tikzcd} N'' \arrow[r, "\bar{q}"] \arrow[d, "\pi''"'] & N' \arrow[d, "\pi'"] \\ Z'' \arrow[r, "q"] & Z' \end{tikzcd}$$

is a fibre square, called the *bundle square* below, with $\bar{q}$ the projection. The ideal sheaves satisfy $\mathcal{I}'\mathcal{O}_{X''} = \mathcal{I}''$ because $Z'' = p^{-1}(Z')$, so $p$ induces a morphism of graded algebras and hence a morphism of cones $\bar{p}\colon C'' \to C'$ over $q$; and the *cone square*

$$\begin{tikzcd} C'' \arrow[r, "j''"] \arrow[d, "\bar{p}"'] & N'' \arrow[d, "\bar{q}"] \\ C' \arrow[r, "j'"] & N' \end{tikzcd}$$

commutes, since $j'$ and $j''$ are induced by the canonical surjections of lemma 3.4.13 and those surjections are compatible with pullback along $p$, being built from the single conormal surjection of $\mathcal{I}$ on $X$. Concretely, both composites in the corresponding square of graded algebras are maps out of $\mathrm{Sym}^\bullet\big(h'^*(\mathcal{I}/\mathcal{I}^2)\big)$, hence are determined by their degree-one parts, and in degree one both send the class of a local section $a$ of $\mathcal{I}$ to the class of its pullback in $\mathcal{I}''/\mathcal{I}''^2$, because $\mathcal{I}\mathcal{O}_{X''} = \mathcal{I}''$.

1. Let $p$ be proper. Then $\bar{q}$ is proper, being the base change of $q$ along $\pi'$, and $\bar{p}$ is proper as well: it is the composite $C'' \hookrightarrow N'' \to N'$ of a closed immersion and a proper morphism, corestricted to the closed subscheme $C' \subseteq N'$. The compatibility of specialization with proper pushforward (proposition 4.2.9) gives
$$\sigma'\big(p_*\alpha\big) = \bar{p}_*\big(\sigma''\alpha\big)$$

Pushing forward along $j'$ and using the cone square together with functoriality of proper pushforward (definition 1.2.1),

$$j'_*\sigma'\big(p_*\alpha\big) = j'_*\bar{p}_*\big(\sigma''\alpha\big) = \bar{q}_*j''_*\big(\sigma''\alpha\big)$$

It remains to move $\bar{q}_*$ past the inverse of $\pi'^*$. The bundle square has $q$ proper and $\pi'$ flat, so proposition 1.3.6 gives $\pi'^*q_* = \bar{q}_*\pi''^*$. Applying this to $\gamma = (\pi''^*)^{-1}j''_*\sigma''\alpha = i^!\alpha$,

$$\pi'^*\big(q_*(i^!\alpha)\big) = \bar{q}_*\big(\pi''^*\gamma\big) = \bar{q}_*j''_*\big(\sigma''\alpha\big) = j'_*\sigma'\big(p_*\alpha\big) = \pi'^*\big(i^!(p_*\alpha)\big)$$

Since $\pi'^*$ is injective, $q_*(i^!\alpha) = i^!(p_*\alpha)$.

2. Let $p$ be flat of relative dimension $n$. Flatness gives $\mathcal{I}'^m\mathcal{O}_{X''} = p^*\mathcal{I}'^m$ for every $m$ and makes the formation of the quotients $\mathcal{I}'^m/\mathcal{I}'^{m+1}$ commute with pullback, so $C'' = C' \times_{Z'} Z''$ and the induced $\bar{p}\colon C'' \to C'$ is flat of relative dimension $n$. Consequently the cone square is a fibre square as well, both vertical maps being base change along $Z'' \to Z'$. The compatibility of specialization with flat pullback (proposition 4.2.10) gives

$$\sigma''\big(p^*\beta\big) = \bar{p}^*\big(\sigma'\beta\big)$$

Applying $j''_*$ and using proposition 1.3.6 for the cone square, whose horizontal maps $j', j''$ are proper and whose vertical maps $\bar{p}, \bar{q}$ are flat,

$$j''_*\sigma''\big(p^*\beta\big) = j''_*\bar{p}^*\big(\sigma'\beta\big) = \bar{q}^*j'_*\big(\sigma'\beta\big)$$

Finally $\pi' \circ \bar{q} = q \circ \pi''$ is a composite of flat morphisms in two ways, so proposition 1.3.5(3) gives $\bar{q}^*\pi'^* = \pi''^*q^*$. Applying this to $\delta = (\pi'^*)^{-1}j'_*\sigma'\beta = i^!\beta$,

$$\pi''^*\big(q^*(i^!\beta)\big) = \bar{q}^*\big(\pi'^*\delta\big) = \bar{q}^*j'_*\big(\sigma'\beta\big) = j''_*\sigma''\big(p^*\beta\big) = \pi''^*\big(i^!(p^*\beta)\big)$$

and injectivity of $\pi''^*$ gives $q^*(i^!\beta) = i^!(p^*\beta)$, completing the proof.

Part (1) is what makes the refined map a refinement rather than a separate construction: a class computed on a small piece of $X$ and then pushed forward agrees with the class computed on $X$ directly. Part (2) says that the map commutes with restriction to an open subscheme, that being flat pullback of relative dimension zero: $(i^!\alpha)|_{Z\cap U} = i^!(\alpha|_U)$. It does *not* say that $i^!\alpha$ can be reassembled from its restrictions to an open cover, and it could not: Chow groups are not a Zariski sheaf, and $A_0(\mathbb{P}^1) = \mathbb{Z}$ restricts to zero on each of the two standard charts (theorem 1.5.2, proposition 1.5.1). What the compatibility does give is the useful one-way statement of exercise 4.3.1 (4): a class vanishing on an open set containing $Z$ has vanishing Gysin image. Both are also the form in which chapter 5 uses the map: the projection formula and the compatibility of the intersection product with proper pushforward are read off them.

The first corollary is the statement that the plain Gysin map of a subvariety is always the pushforward of a class living on the intersection.

**Corollary 4.3.11: The Gysin Class of a Subvariety Lives on the Intersection**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq X$ be a $k$-dimensional subvariety, with inclusion $\iota_V\colon V \hookrightarrow X$ and $W = V \cap Z = \iota_V^{-1}(Z)$. Then

$$i^![V] \;=\; \iota_*\big(i_V^![V]\big)$$

where $i_V^!\colon A_k(V) \to A_{k-d}(W)$ is the refined Gysin homomorphism along $\iota_V$ and $\iota\colon W \hookrightarrow Z$ is the inclusion.

***Proof*:**
Apply theorem 4.3.10(1) with $X' = X$, $X'' = V$ and $p = \iota_V$, which is proper as a closed immersion. The preimage of $Z$ in $V$ is $W$ and the induced morphism $q$ is $\iota$. By definition 1.2.1 the class $[V] \in A_k(V)$ has $p_*[V] = [V] \in A_k(X)$, the extension $k(V) \subseteq k(V)$ having degree one. On the left-hand side the refined Gysin homomorphism along $\mathrm{id}_X$ is the Gysin map of definition 4.3.1 by proposition 4.3.7, so the identity reads $i^![V] = \iota_*(i_V^![V])$, as we sought to show.

The class $i^![V] \in A_{k-d}(Z)$ discards information: it records a cycle on all of $Z$, whereas the geometry produced it on $V \cap Z$, which may be much smaller. The refined class $i_V^![V] \in A_{k-d}(V \cap Z)$ retains that localization, and it is the object every localized intersection number is a degree of. Proposition 4.3.3 is the special case in which the refined class is the fundamental cycle $[V \cap Z]$ itself, and example 4.3.8 exhibits both a case where it is and a case where it is not.

## Exercise 4.3.1

1. Let $Z = \{x_2 = x_3 = 0\} \subseteq \mathbb{P}^3$, a line. Show that $Z \hookrightarrow \mathbb{P}^3$ is a regular embedding of codimension two, and compute $i^!\colon A_m(\mathbb{P}^3) \to A_{m-2}(Z)$ on the generator of each group, for $m = 0, 1, 2, 3$.

2. Let $X$ be a variety of dimension $n$ and let $D \subseteq X$ be an effective Cartier divisor with $D \neq X$, with inclusion $i$. Show that $i^![X] = [D]$, the fundamental cycle of the scheme $D$. Then take $X = \mathbb{P}^2$ and $D$ a smooth plane cubic, and compute the image of $i^![\mathbb{P}^2]$ in $A_1(\mathbb{P}^2)$.

3. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq X$ be a $k$-dimensional subvariety with $V \cap Z = \emptyset$. Show that $i^![V] = 0$. Then let $Z = \{0\}$ be the origin in $X = \mathbb{A}^n$ with $n \geq 1$, verify that this is a regular embedding of codimension $n$, and determine $i^!$ on every group $A_k(\mathbb{A}^n)$.

4. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $u\colon U \hookrightarrow X$ be an open immersion. Show that the preimage of $Z$ in $U$ is $Z \cap U$ and that $i^!\big(u^*\alpha\big) = \big(i^!\alpha\big)\big|_{Z\cap U}$ for every $\alpha \in A_k(X)$. Deduce that if $Z \subseteq U$ and $\alpha$ restricts to zero on $U$, then $i^!\alpha = 0$.

5. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq Z$ be a $k$-dimensional subvariety of $Z$. Show that $\sigma[V] = \zeta_*[V]$, where $\zeta\colon Z \to N$ is the zero section, and hence that $i^![V] = (\pi^*)^{-1}\zeta_*[V]$. Now take $X = \mathbb{A}^2$, $Z = \{y = 0\}$ and $V = Z$, and evaluate both $i^![Z]$ and $D \cdot [Z]$ of definition 2.2.1, checking that they agree.

## 4.4 Commutativity, Composition, and Sections

The Gysin map is built; the algebra it obeys is not yet available. Three identities are used downstream, and all three are properties of the construction of section 4.3 rather than of anything later, so they belong here.

They are not independent, and the dependency runs in a direction worth naming before starting. Commutativity of two Gysin maps looks the hardest and is in fact a consequence of the composition rule: two regular embeddings determine a single regular embedding of the product, that embedding factors through the two intermediate stages in two ways, and the composition rule applied to both factorizations equates the two orders. The work is therefore concentrated in composition, and the section is arranged accordingly. Composition in turn needs a commutativity statement of its own, but only the case of a *divisor*, and that case the construction hands over directly: specialization is intersection with a Cartier divisor, so two of them commute because divisors do.

**Lemma 4.4.1: The Product of Two Regular Embeddings**

Let $i\colon Z \hookrightarrow P$ and $i'\colon Z' \hookrightarrow P'$ be regular embeddings of codimensions $d$ and $d'$. Then

$$i \times i' \;:\; Z \times Z' \;\hookrightarrow\; P \times P'$$

is a regular embedding of codimension $d + d'$, and it factors in two ways as a composite of regular embeddings,

$$Z \times Z' \xhookrightarrow{\mathrm{id}_Z \times i'} Z \times P' \xhookrightarrow{i \times \mathrm{id}_{P'}} P \times P', \qquad Z \times Z' \xhookrightarrow{i \times \mathrm{id}_{Z'}} P \times Z' \xhookrightarrow{\mathrm{id}_P \times i'} P \times P'$$

the four factors being regular of codimensions $d'$, $d$, $d$ and $d'$ respectively.

***Proof*:**
*Step 1: each factor is regular.* The morphism $i \times \mathrm{id}_{P'}$ is the base change of $i$ along the projection $\mathrm{pr}_1\colon P \times P' \to P$, which is flat, and the scheme-theoretic preimage of $Z$ under that projection is $Z \times P'$. A regular embedding pulls back along a flat morphism to a regular embedding of the same codimension ([2, Regular Embeddings Pull Back Along Flat Morphisms]), so $i \times \mathrm{id}_{P'}$ is regular of codimension $d$. The same argument along $\mathrm{pr}_2$ makes $\mathrm{id}_P \times i'$ and $\mathrm{id}_Z \times i'$ regular of codimension $d'$, the latter as the base change of $i'$ along $Z \times P' \to P'$; and $i \times \mathrm{id}_{Z'}$ is regular of codimension $d$, being the base change of $i$ along the flat projection $P \times Z' \to P$. That accounts for all four factors named in the statement.

*Step 2: the composite is regular.* Work on affine opens $\operatorname{Spec} A \subseteq P$ and $\operatorname{Spec} A' \subseteq P'$ over which the ideals of $Z$ and $Z'$ are generated by regular sequences $f_1, \ldots, f_d$ and $g_1, \ldots, g_{d'}$. On $\operatorname{Spec}(A \otimes_{\overline{k}} A')$ the ideal of $Z \times Z'$ is generated by the images of the $f$'s and the $g$'s together, and that concatenation is a regular sequence. Indeed $A \to A \otimes_{\overline{k}} A'$ is flat, so the $f$'s remain a regular sequence after it by the argument of Step 1; and modulo them the ring is $(A/(f)) \otimes_{\overline{k}} A'$, which is a flat base change of $A'$, so the $g$'s are regular there by the same argument with the two factors exchanged. Hence $i \times i'$ is a regular embedding of codimension $d + d'$.

The two factorizations are immediate from the definitions of the four morphisms, and Step 1 identified each factor as a regular embedding of the stated codimension, completing the proof.

The first identity says that the Gysin map of a section undoes the flat pullback along the morphism it is a section of. It is the only one of the three that can be proved directly from the proper-intersection computation already available.

**Theorem 4.4.2: Gysin Maps of Sections**

Let $\pi\colon P \to S$ be smooth of relative dimension $e \geq 1$ and let $s\colon S \to P$ be a section, so that $\pi \circ s = \mathrm{id}_S$. Then $s$ is a regular embedding of codimension $e$ and

$$s^! \circ \pi^* \;=\; \mathrm{id} \quad \text{on } A_k(S) \text{ for every } k$$

***Proof*:**
That $s$ is a regular embedding of codimension $e$ is [2, A Section of a Smooth Morphism Is a Regular Embedding]. Both $s^!$ and $\pi^*$ are homomorphisms, so it suffices to prove the identity on the class of a $k$-dimensional subvariety $V \subseteq S$.

*Step 1: the preimage and its components.* The morphism $\pi$ is flat of relative dimension $e$ by [2, Smooth Morphisms Are Flat], so $\pi^*$ is defined and $\pi^*[V] = [P_V]$ for $P_V = \pi^{-1}(V)$ the scheme-theoretic preimage (definition 1.3.3). The projection $\pi_V\colon P_V \to V$ is smooth of relative dimension $e$, being the base change of $\pi$ along $V \hookrightarrow S$ ([2, Stability of Smooth Morphisms](1)), hence flat by the same citation as above; since its base $V$ is a variety, proposition 1.3.2 applies to $\pi_V$ and $P_V$ is pure of dimension $k+e$. Because $\pi \circ s = \mathrm{id}_S$, the section restricts to a section $s_V\colon V \to P_V$ of $\pi_V$, and indeed $s^{-1}(P_V) = (\pi \circ s)^{-1}(V) = V$ scheme-theoretically, so $s(S) \cap P_V = s_V(V)$.

Flatness of $\pi_V$ makes every irreducible component of $P_V$ dominate $V$, by the going-down argument already used in the proof of proposition 1.3.2. So the generic point of each component lies in the generic fibre, which is smooth over the field $k(V)$ and therefore regular ([2, Smooth Implies Regular]); its local rings are consequently domains. Hence $P_V$ is generically reduced along every component, so $[P_V] = \sum_j [V_j]$ with all multiplicities one (definition 1.1.2), and no two components share a generic point of the generic fibre. The subvariety $s_V(V)$ dominates $V$, so its generic point lies in the generic fibre and therefore in exactly one component, which is moreover the unique component containing $s_V(V)$; call it $V_1$, and let $V_2, \ldots, V_r$ be the others.

*Step 2: the other components miss the section altogether.* Let $j \geq 2$ and put $W_j = V_j \cap s(S) = V_j \cap s_V(V)$, using Step 1's identification of $s(S) \cap P_V$. Suppose $W_j \neq \emptyset$. Applying theorem 4.2.7 to the subvariety $V_j \subseteq P_V$, of dimension $k+e$, its normal cone $C_{W_j}V_j$ is pure of dimension $k+e$, and lemma 3.4.12 embeds it in $C_{s_V(V)}P_V$ over the inclusion $W_j \hookrightarrow s_V(V)$. Now $s_V(V) \hookrightarrow P_V$ is a section of the smooth morphism $\pi_V$, hence a regular embedding of codimension $e$ ([2, A Section of a Smooth Morphism Is a Regular Embedding]), so that cone is the normal bundle $N_V$, of rank $e$ over $s_V(V) \cong V$ (proposition 3.4.11). The part of $N_V$ lying over $W_j$ has dimension $\dim W_j + e$, so purity forces $\dim W_j = k$. But $W_j$ is then a $k$-dimensional closed subscheme of the $k$-dimensional variety $s_V(V)$, hence all of it, so $V_j \supseteq s_V(V)$, contradicting the uniqueness of $V_1$ in Step 1. Therefore $W_j = \emptyset$ for every $j \geq 2$, and in particular $s^![V_j] = 0$, the Chow groups of the empty scheme vanishing.

*Step 3: the remaining component.* Step 2 has a consequence worth isolating, because the naive statement it replaces is false: an irreducible component of a smooth $V$-scheme need not itself be smooth over $V$, so one may not argue that $V_1 \to V$ is smooth because $V_1$ is the component through the section. What is true, and all that is needed, is that

$$U \;=\; P_V \setminus \bigcup_{j\geq 2} V_j$$

is an open subscheme of $P_V$ containing $s_V(V)$, by Step 2, and $U \subseteq V_1$. Being a regular embedding of a given codimension is a local condition along the subscheme ([2, Regular Embedding]), and $s_V(V) \hookrightarrow P_V$ is regular of codimension $e$ as recorded in Step 2. Restricting to the open $U$ and using that $U$ is also open in $V_1$, the embedding $s_V(V) \hookrightarrow V_1$ is regular of codimension $e$.

The scheme-theoretic intersection $V_1 \cap s(S)$ is $s_V(V)$: this is a scheme-theoretic and not a set-theoretic claim, so it needs an argument rather than a comparison of points. By Step 1, $s^{-1}(P_V) = V$ scheme-theoretically; $s^{-1}(V_1)$ is a closed subscheme of $V$ because the ideal sheaf of $V_1$ contains that of $P_V$; it has full support since $V_1 \supseteq s_V(V)$; and $V$ is reduced, so $s^{-1}(V_1) = V$.

Proposition 4.3.3 therefore applies to the regular embedding $s$ of codimension $e$ and the $(k+e)$-dimensional subvariety $V_1$, and gives

$$s^![V_1] \;=\; \iota_*\big[s_V(V)\big] \;\in\; A_k\big(s(S)\big)$$

Identifying $s(S)$ with $S$ through the isomorphism $s$, the right side is $[V]$.

Summing Steps 2 and 3 gives $s^!\big(\pi^*[V]\big) = s^![P_V] = [V]$, as we sought to show.

The remaining two identities concern a pair of Gysin maps rather than a single one, and both rest on one observation about definition 4.3.6 that is worth isolating first. The refined map is built from the specialization of the pair $(Z', X')$, the cone $C_{Z'}X'$, and the bundle $N'$ into which that cone embeds. Nothing else about the ambient $X$ enters. So an embedding may be enlarged, or crossed with a further factor, without changing the resulting homomorphism, provided the preimage and the normal bundle come along unchanged.

**Lemma 4.4.3: The Refined Gysin Map Does Not See the Ambient Scheme**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with ideal sheaf $\mathcal{I}$, and let $\varphi\colon \widetilde{X} \to X$ be a morphism whose preimage $\widetilde{Z} = \varphi^{-1}(Z)$, with ideal sheaf $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$, is again a regular embedding $\widetilde{i}\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ *of the same codimension* $d$. Then for every morphism $g\colon X' \to \widetilde{X}$, writing $Z' = g^{-1}(\widetilde{Z}) = (\varphi \circ g)^{-1}(Z)$,

$$\widetilde{i}^!_g \;=\; i^!_{\varphi\circ g} \;:\; A_k(X') \longrightarrow A_{k-d}(Z')$$

***Proof*:**
First, the hypothesis on codimensions already forces the conormal sheaf to be the pullback. The canonical map

$$\varphi_Z^*\big(\mathcal{I}/\mathcal{I}^2\big) \;\longrightarrow\; \widetilde{\mathcal{I}}/\widetilde{\mathcal{I}}^2, \qquad \varphi_Z\colon \widetilde{Z} \to Z,$$

is surjective, since $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$ is generated by the image of $\mathcal{I}$. Both sides are locally free of rank $d$ ([2, The Conormal Sheaf of a Regular Embedding]), the source as the pullback of a rank-$d$ bundle and the target because $\widetilde{i}$ is regular of codimension $d$. A surjection of locally free sheaves of equal rank is an isomorphism, its kernel being locally free of rank zero. So the canonical map is an isomorphism, and we may use it below.

The two closed subschemes $g^{-1}(\widetilde{Z})$ and $(\varphi \circ g)^{-1}(Z)$ of $X'$ coincide because $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$ and pullback of ideals is transitive, so the target group is the same and both homomorphisms are the composite of definition 4.3.6 for one and the same pair $(Z', X')$. In particular they use the same specialization $\sigma'$ and the same cone $C_{Z'}X'$, neither of which mentions the ambient scheme at all.

What must be compared is the bundle and the embedding of the cone in it. Write $h\colon Z' \to Z$ and $\widetilde{h}\colon Z' \to \widetilde{Z}$ for the structure morphisms, so $h = \varphi_Z \circ \widetilde{h}$. The hypothesis gives a canonical isomorphism $\widetilde{h}^*(\widetilde{\mathcal{I}}/\widetilde{\mathcal{I}}^2) \cong \widetilde{h}^*\varphi_Z^*(\mathcal{I}/\mathcal{I}^2) = h^*(\mathcal{I}/\mathcal{I}^2)$, that is $\widetilde{N}' \cong N'$ as bundles on $Z'$, and hence $\pi'$ is the same projection in both constructions.

It remains to see that the closed immersion $j$ of lemma 3.4.13 is carried to the other by this isomorphism. That immersion is induced by a surjection of graded $\mathcal{O}_{Z'}$-algebras out of $\mathrm{Sym}^\bullet$ of the pullback conormal sheaf, and such a morphism is determined by its degree-one part. For $i$ the degree-one part sends the class of a local section $\alpha$ of $\mathcal{I}$ to the class of $\alpha\mathcal{O}_{X'}$ in $\mathcal{I}'/\mathcal{I}'^2$; for $\widetilde{i}$ it sends the class of a local section of $\widetilde{\mathcal{I}}$ to the class of its extension to $X'$. The hypothesis isomorphism identifies the class of $\alpha$ with the class of $\alpha\mathcal{O}_{\widetilde{X}}$, and $(\alpha\mathcal{O}_{\widetilde{X}})\mathcal{O}_{X'} = \alpha\mathcal{O}_{X'}$ by transitivity of extension of ideals along $X' \to \widetilde{X} \to X$. So the two degree-one parts correspond, the two immersions $j$ agree, and the three factors $(\pi'^*)^{-1}$, $j_*$, $\sigma'$ agree one by one.

The hypothesis is satisfied whenever $\varphi$ is flat, by [2, Regular Embeddings Pull Back Along Flat Morphisms], and that is the only case used below: $\varphi$ will be a projection off a product.

Three lemmas stand between here and the composition rule. The first is the only place a genuine commutativity statement is needed, and it is needed only for a divisor, where the construction delivers it directly: specialization *is* intersection with a Cartier divisor (definition 4.2.3), so two such intersections commute by theorem 2.3.2.

**Lemma 4.4.4: The Gysin Map Commutes with Intersection by a Divisor**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, let $g\colon X' \to X$ be a morphism with $Z' = g^{-1}(Z)$, and let $D$ be a Cartier divisor on $X'$. Then for every $\alpha \in A_k(X')$,

$$i^!\big(D \cdot \alpha\big) \;=\; D \cdot \big(i^!\alpha\big) \quad \text{in } A_{k-d-1}(Z')$$

both sides being read through corollary 2.4.5, on $X'$ for the left and on the closed subscheme $Z'$ for the right.

***Proof*:**
Write $M = M^\circ_{Z'}X'$ for the deformation space of definition 4.1.4, with blow-down $\rho\colon M \to X'$, fibre $D_\infty = C_{Z'}X'$ over $\infty$, and $\kappa\colon C_{Z'}X' \hookrightarrow N'$, $\pi'\colon N' \to Z'$ as in definition 4.3.6, so that $i^! = (\pi'^*)^{-1} \circ \kappa_* \circ \sigma'$. Throughout, a Cartier divisor is intersected against classes on closed subschemes of the ambient through corollary 2.4.5; the relevant restrictions are in general *not* Cartier on those subschemes, and it is to avoid needing them to be that the corollary was stated.

*Step 0: $\rho^*D$ is a Cartier divisor on $M$.* The blow-down is not flat, its fibre at infinity being a cone, so this needs an argument. Over $\mathbb{A}^1$ the map is the projection $X' \times \mathbb{A}^1 \to X'$, which is flat, and flat pullback of a Cartier divisor is Cartier. Over the chart at infinity, lemma 4.1.2 and (4.1) present $\mathcal{O}_M$ as the extended Rees algebra, a graded subsheaf of $\mathcal{O}_{X'}[u, u^{-1}]$. A local equation $f$ of $D$ is a non-zero divisor on $\mathcal{O}_{X'}$, hence on the free module $\mathcal{O}_{X'}[u, u^{-1}]$, hence on any subsheaf of it; so $\rho^\# f$ is a non-zero divisor and $\rho^*D$ is Cartier.

*Step 1: specialization.* Let $\beta \in A_{k+1}(M)$ be a lift of $\alpha$ in the sense of definition 4.2.3, so that $\beta$ restricts to $q^*\alpha$ on $X' \times \mathbb{A}^1$. Then $\rho^*D \cdot \beta$ is a lift of $D \cdot \alpha$: restriction along that open immersion is flat pullback of relative dimension zero, proposition 2.3.3(2) applies with no hypothesis on supports, and $\rho^*D$ restricts to $q^*D$ there by theorem 4.1.5(1). Hence, by theorem 4.2.4 and by theorem 2.3.2 applied on $M$ to the Cartier divisors $D_\infty$ and $\rho^*D$, the first by theorem 4.1.5(2) and the second by Step 0,

$$\sigma'\big(D \cdot \alpha\big) \;=\; D_\infty \cdot \big(\rho^*D \cdot \beta\big) \;=\; \rho^*D \cdot \big(D_\infty \cdot \beta\big) \;=\; \rho^*D \cdot \sigma'(\alpha)$$

an identity of classes on the cone, the refined form of theorem 2.3.2 placing both sides in the Chow group of $|D_\infty| \cap |\rho^*D| \cap \operatorname{supp}\beta$.

*Step 2: pushforward along the cone.* On $N'$ put $D_{N'} = \pi'^*D$, the flat pullback along $\pi'$ of $D$ restricted to $Z'$; it is a Cartier divisor there, $\pi'$ being flat. Its restriction to the cone and the restriction of $\rho^*D$ agree, both being computed from the line bundle $\mathcal{O}_{X'}(D)$ pulled back to the cone with its canonical section. A closed immersion carries a cycle to itself, so corollary 2.4.5, applied to $C_{Z'}X' \subseteq N'$, gives

$$\kappa_*\big(\rho^*D \cdot \gamma\big) \;=\; D_{N'} \cdot \kappa_*\gamma$$

for every class $\gamma$ on the cone, with no hypothesis on its support. This is the step at which the projection formula would otherwise be invoked, and proposition 2.3.3(1) could not be: its hypothesis is that no component of the support maps into $|D|$, which $\sigma'(\alpha)$ need not satisfy.

*Step 3: the bundle inverse.* Since $\pi'$ is flat, proposition 2.3.3(2) gives $\pi'^*\big(D \cdot \delta\big) = D_{N'} \cdot \pi'^*\delta$ for every $\delta \in A_*(Z')$, again with no support hypothesis. Applying this to $\delta = i^!\alpha$ and using Steps 1 and 2,

$$\pi'^*\Big(D \cdot i^!\alpha\Big) = D_{N'} \cdot \pi'^*\big(i^!\alpha\big) = D_{N'} \cdot \kappa_*\sigma'(\alpha) = \kappa_*\sigma'\big(D \cdot \alpha\big) = \pi'^*\Big(i^!\big(D \cdot \alpha\big)\Big)$$

and $\pi'^*$ is injective by theorem 1.4.6, which gives the identity.

The second lemma settles the composition rule when the outer embedding is the zero section of a vector bundle. This is the case in which the normal bundle of the composite genuinely splits, and it is where the conormal sequence of a chain does its work.

**Lemma 4.4.5: Composition Through a Zero Section**

Let $E$ be a vector bundle of rank $e \geq 1$ on an algebraic scheme $Y$ with zero section $\zeta \colon Y \hookrightarrow E$, and let $i \colon Z \hookrightarrow Y$ be a regular embedding of codimension $d \geq 1$. Let $g \colon S \to Y$ be a morphism, $E' = g^*E$ with projection $\pi' \colon E' \to S$, and let $\mathrm{pr} \colon E' \to E$ be the induced morphism. Then for every $\alpha \in A_k(E')$,

$$(\zeta \circ i)^!_{\mathrm{pr}}(\alpha) \;=\; i^!_g\Big(\zeta^!_{\mathrm{pr}}(\alpha)\Big) \quad \text{in } A_{k-d-e}\big(g^{-1}(Z)\big)$$

***Proof*:**
Write $S_Z = g^{-1}(Z)$. The preimage of $Y$ under pr is the zero section of $E'$, which we identify with $S$; the preimage of $Z$ is $S_Z$. Both refined maps are therefore defined and land as stated. That $\zeta^!_{\mathrm{pr}} = (\pi'^*)^{-1}$ is not merely the observation that the normal cone of a zero section is the bundle itself, which is the case $\mathcal{I} = 0$ of (4.6) below: that identifies the cone and the bundle

and makes the immersion of lemma 3.4.13 an identity, but leaves $\zeta^!$ equal to $(\pi'^*)^{-1} \circ \sigma'$, and specialization is emphatically not the identity in general (example 4.2.5). What settles it is that the preimage of $Y$ under pr is the zero section of $E'$, a regular embedding of codimension $e$, so lemma 4.4.3 identifies $\zeta^!_{\mathrm{pr}}$ with the refined Gysin map of that zero section along id, which is its plain Gysin map (proposition 4.3.7); and $\pi' \colon E' \to S$ is smooth of relative dimension $e$ with the zero section as a section, so theorem 4.4.2 gives $\zeta^! \circ \pi'^* = \mathrm{id}$, that is $\zeta^! = (\pi'^*)^{-1}$.

*Step 1: the cone of the composite splits off the bundle.* Let $\mathcal{I}$ be the ideal sheaf of $S_Z$ in $S$ and let $\mathcal{S}^\bullet = \mathrm{Sym}^\bullet(\mathcal{E}^\vee)$ present $E' = \mathrm{Spec}_S \mathcal{S}^\bullet$, so that $\mathcal{S}^0 = \mathcal{O}_S$ and $\mathcal{S}^1 = \mathcal{E}^\vee$. The ideal $\mathcal{J}$ of $S_Z$ in $E'$ is generated by $\mathcal{I}$ in degree zero together with all of $\mathcal{S}^1$: a function vanishes on the zero section over $S_Z$ exactly when its degree-zero part lies in $\mathcal{I}$. Consequently

$$\mathcal{J}^n \;=\; \bigoplus_{q \le n} \mathcal{I}^{n-q}\,\mathcal{S}^q \;\oplus\; \bigoplus_{q > n} \mathcal{S}^q, \qquad \mathcal{J}^n/\mathcal{J}^{n+1} \;=\; \bigoplus_{p+q=n} \big(\mathcal{I}^p/\mathcal{I}^{p+1}\big) \otimes \mathcal{S}^q$$

so that the total graded algebra factors, $\bigoplus_n \mathcal{J}^n/\mathcal{J}^{n+1} = \big(\bigoplus_p \mathcal{I}^p/\mathcal{I}^{p+1}\big) \otimes_{\mathcal{O}_{S_Z}} \mathcal{S}^\bullet$. Taking Spec and using definition 3.4.6 this reads

$$C_{S_Z}E' \;\cong\; C_{S_Z}S \times_{S_Z} E'|_{S_Z} \tag{4.6}$$

Applied with $S = Y$ and $g = \mathrm{id}$ it identifies the normal bundle of the composite as $N_Z E = N_Z Y \oplus i^* E$, which is the splitting of [2, The Conormal Sequence of a Chain of Regular Embeddings] for the chain $Z \subseteq Y \subseteq E$; the splitting is the one induced by the bundle projection.

*Step 2: reduction.* By theorem 1.4.6 every class on $E'$ is $\pi'^*\beta$ for a unique $\beta \in A_{k-e}(S)$, and by linearity we may take $\beta = [V]$ for $V \subseteq S$ a subvariety. Both sides of the assertion commute with proper pushforward along $V \hookrightarrow S$ and its base change $V \times_S E' \hookrightarrow E'$, by theorem 4.3.10(1), so we may replace $S$ by $V$. Then $S$ is a variety, $\alpha = [E']$, and $\zeta^!\alpha = [S]$.

*Step 3: comparison of fundamental cycles.* Write $h \colon S_Z \to Z$ for the induced morphism, $r \colon h^*N_Z Y \to S_Z$ and $q \colon h^* N_Z E \to h^* N_Z Y$ for the projections, the latter using the splitting of Step 1, so that $r \circ q$ is the projection of $h^* N_Z E$. Both $S$ and $E'$ are varieties, so theorem 4.2.7 gives $\sigma[S] = [C_{S_Z}S]$ and $\sigma[E'] = [C_{S_Z}E']$. By (4.6) the second cone is the preimage of the first under $q$, so $q^*[C_{S_Z}S] = [C_{S_Z}E']$ by definition 1.3.3. Now definition 4.3.6 reads the two Gysin classes as

$$r^*\big(i^![S]\big) = [C_{S_Z}S], \qquad (r \circ q)^*\big((\zeta i)^![E']\big) = [C_{S_Z}E']$$

Combining the three displays, $(r \circ q)^*\big((\zeta i)^![E']\big) = q^* r^*\big(i^![S]\big) = (r \circ q)^*\big(i^![S]\big)$ by proposition 1.3.5(3), and $(r \circ q)^*$ is injective by theorem 1.4.6. Hence $(\zeta i)^![E'] = i^![S] = i^!\zeta^![E']$, which by Step 2 is the assertion in general.

The third lemma is the only input that is not about Gysin maps at all. It says that cutting a class on $X \times \mathbb{P}^1$ by a fibre gives an answer that does not depend on which fibre.

**Lemma 4.4.6: Cutting by a Fibre of $X \times \mathbb{P}^1$ Is Independent of the Fibre**

Let $X$ be an algebraic scheme and for $t \in \mathbb{P}^1$ a closed point let $D_t$ denote the Cartier divisor $X \times \{t\}$ on $X \times \mathbb{P}^1$. Then for every $\beta \in A_k(X \times \mathbb{P}^1)$ the class

$$D_t \cdot \beta \;\in\; A_{k-1}\big(X \times \{t\}\big) \;=\; A_{k-1}(X)$$

is the same for all $t$, the identification being the projection $X \times \{t\} \to X$.

***Proof*:**
Write $X \times \mathbb{P}^1 = \mathbb{P}(\mathcal{O}_X^{\oplus 2})$ with projection $p$ and tautological class $\xi$; for the trivial bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ is pulled back from $\mathbb{P}^1$, and $D_t$ is the divisor of the section of it cutting out $t$, so $\mathcal{O}(D_t) = \mathcal{O}_{\mathbb{P}(E)}(1)$ for every $t$. By theorem 3.1.9 with $r = 1$ there are unique classes $\alpha_0 \in A_{k-1}(X)$ and $\alpha_1 \in A_k(X)$ with

$$\beta \;=\; p^*\alpha_0 \;+\; \xi \cap p^*\alpha_1$$

It suffices to evaluate $D_t\cdot$ on each summand. For the first, take $\alpha_0 = [V]$ with $V \subseteq X$ a subvariety; then $p^*[V] = [V \times \mathbb{P}^1]$, which is not contained in $|D_t|$, so definition 2.2.1(1) computes $D_t \cdot p^*[V] = [V \times \{t\}]$, the divisor of the restricted section, and under $X \times \{t\} \cong X$ this is $[V]$. For the second, $D_t \cdot (\xi \cap p^*\alpha_1) = \xi^2 \cap p^*\alpha_1$ once pushed into $A_{k-2}(X \times \mathbb{P}^1)$, and $\xi^2 = 0$ by theorem 3.2.9, the bundle being trivial. To descend that vanishing to the refined class on $X \times \{t\}$, note that the first computation identifies the pushforward $\iota_{t*} \colon A_m(X) = A_m(X \times \{t\}) \to A_m(X \times \mathbb{P}^1)$ with $\gamma \mapsto \xi \cap p^*\gamma$, and that map is injective by the uniqueness clause of theorem 3.1.9, being the restriction of the isomorphism $\Psi$ to the summand $\alpha_0 = 0$. Hence the refined class vanishes as well, and $D_t \cdot \beta = \alpha_0$ for every $t$, as we sought to show.

**Theorem 4.4.7: Composition of Gysin Maps**

Let $i \colon Z \hookrightarrow Y$ and $j \colon Y \hookrightarrow P$ be regular embeddings of codimensions $d \geq 1$ and $e \geq 1$. Then $j \circ i$ is a regular embedding of codimension $d + e$, and for every morphism $a \colon W \to P$, writing $W_Y = a^{-1}(Y)$, $W_Z = a^{-1}(Z)$ and $a' \colon W_Y \to Y$ for the induced morphism,

$$(j \circ i)^!_a \;=\; i^!_{a'} \circ j^!_a \;:\; A_k(W) \longrightarrow A_{k-d-e}(W_Z)$$

***Proof*:**
That $j \circ i$ is a regular embedding of codimension $d + e$ is the first assertion of [2, The Conormal Sequence of a Chain of Regular Embeddings]. Both sides commute with proper pushforward (theorem 4.3.10(1)), and $A_k(W)$ is generated by the classes of $k$-dimensional subvarieties, so it suffices to prove the identity for $W$ a variety and the class $[W]$.

*The two deformations.* Let $M = M^{\circ}_Y P$ and $M' = M^{\circ}_{W_Y} W$ be the deformation spaces of definition 4.1.4, with projections to $\mathbb{P}^1$ written $\pi$ and $\pi'$. The morphism $a$ induces $\widetilde{a} \colon M' \to M$ over $\mathbb{P}^1$ by lemma 4.2.6, whose hypothesis $W_Y = a^{-1}(Y)$ holds by definition. Let

$$\chi \;:\; Z \times \mathbb{P}^1 \xhookrightarrow{i \times \mathrm{id}} Y \times \mathbb{P}^1 \hookrightarrow M$$

be the composite of $i \times \mathrm{id}$ with the closed immersion of theorem 4.1.5(4). Its first factor is a regular embedding of codimension $d$, being the base change of $i$ along the flat projection

$Y \times \mathbb{P}^1 \to Y$ ([2, Regular Embeddings Pull Back Along Flat Morphisms]); its second is a regular embedding of codimension $e$. This last point needs the charts rather than the fibres: $\pi^{-1}(\mathbb{A}^1)$ is open, and there the immersion is $j \times \mathrm{id}$, but $\pi^{-1}(\infty)$ is closed, so regularity of the immersion into it says nothing about a neighbourhood. On the complementary chart $\pi^{-1}(\mathbb{P}^1 \setminus \{0\})$, lemma 4.1.2 presents the coordinate ring as an extended Rees algebra $\mathcal{R}$, in which the ideal of $Y \times \mathbb{P}^1$ is generated by $g_1/u, \ldots, g_e/u$ for $g_1, \ldots, g_e$ a local regular sequence cutting out $Y$ in $P$. Here $u$ is a non-zero divisor on $\mathcal{R}$ and $\mathcal{R}/u\mathcal{R} = \mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$, in which the images of the $g_m/u$ are the coordinates and so form a regular sequence; hence $u, g_1/u, \ldots, g_e/u$ is a regular sequence, and permuting a regular sequence in a Noetherian local ring leaves it regular, so $g_1/u, \ldots, g_e/u$ is one. So $\chi$ is a regular embedding of codimension $d + e$, again by the chain result cited above, and $\widetilde{a}^{-1}(Z \times \mathbb{P}^1) = W_Z \times \mathbb{P}^1$.

*Cutting by a fibre.* For a closed point $t \in \mathbb{P}^1$ let $D_t$ be the fibre $\pi'^{-1}(t)$ on $M'$. It is an effective Cartier divisor: for $t = \infty$ by theorem 4.1.5(2), and for $t \neq \infty$ because $M'$ is a variety, as the proof of theorem 4.2.7 records, so that any nonzero local equation is a non-zero divisor. Theorem 4.1.5(1),(2) identify the fibres. Over $\mathbb{A}^1$ the fibre is $W \times \{t\}$, which is reduced, so definition 2.2.1(1) computes $D_t \cdot [M']$ as its fundamental cycle with multiplicity one; over $\infty$ the same intersection is by definition the specialization, which theorem 4.2.7 evaluates. So

$$D_t \cdot [M'] \;=\; [M'_t] \;=\; \begin{cases} [W] & t \neq \infty \\ [C_{W_Y} W] & t = \infty \end{cases} \tag{4.7}$$

Now apply $\chi^!$ along $\widetilde{a}$. The scheme $M'_t$ maps to $M_t$, the inclusion $M_t \hookrightarrow M$ pulls $Z \times \mathbb{P}^1$ back to $Z \times \{t\}$, and the induced embedding is $j \circ i$ for $t \neq \infty$ and $\widetilde{\jmath} \circ i$ for $t = \infty$, where $\widetilde{\jmath}$ is the zero section of $N_Y P$; both are regular of codimension $d + e$. So lemma 4.4.3, applied with $\varphi$ the inclusion of the fibre, gives

$$\chi^!\big(D_t \cdot [M']\big) \;=\; \begin{cases} (j \circ i)^!\,[W] & t \neq \infty \\ (\widetilde{\jmath} \circ i)^!\,[C_{W_Y} W] & t = \infty \end{cases} \tag{4.8}$$

*The value at infinity.* The cone $C_{W_Y} W$ is a closed subscheme of the pullback bundle $(a')^* N_Y P$ over $W_Y$ (lemma 3.4.13). Lemma 4.4.5, stated for an arbitrary class on that bundle, therefore applies with $E = N_Y P$ and $S = W_Y$ to the pushforward of $[C_{W_Y} W]$; and theorem 4.3.10(1), for the closed immersion $C_{W_Y} W \hookrightarrow (a')^* N_Y P$, says that computing before or after that pushforward gives the same answer, the induced morphism on the preimages of $Z$ being the identity of $W_Z$ because the vertex of $C_{W_Y} W$ is $W_Y$, the cone algebra being generated in degree one. This gives

$$(\widetilde{\jmath} \circ i)^!\,[C_{W_Y} W] \;=\; i^!\Big(\widetilde{\jmath}^!\,[C_{W_Y} W]\Big) \;=\; i^!_{a'}\Big(j^!_a[W]\Big)$$

the last equality because $\widetilde{\jmath}^!$ is the inverse of the bundle pullback and $\sigma[W] = [C_{W_Y} W]$, which is exactly how definition 4.3.6 computes $j^!_a[W]$.

*Independence of $t$.* By lemma 4.4.4, applied to the regular embedding $\chi$ and the divisor $D_t$, whose restriction to $W_Z \times \mathbb{P}^1$ is the Cartier divisor $W_Z \times \{t\}$,

$$\chi^!\big(D_t \cdot [M']\big) \;=\; D_t \cdot \big(\chi^![M']\big)$$

and $\chi^![M']$ is a class on $W_Z \times \mathbb{P}^1$ not depending on $t$, so lemma 4.4.6 makes the right-hand side independent of $t$.

One point of bookkeeping remains. Lemma 4.4.4 records its identity in $A_*(W_Z \times \mathbb{P}^1)$, whereas (4.8) produces classes on $W_Z \times \{t\}$; the two are related by the pushforward $\iota_{t*}$ along the closed immersion $W_Z \times \{t\} \hookrightarrow W_Z \times \mathbb{P}^1$, through theorem 4.3.10(1) applied to $M'_t \hookrightarrow M'$. That costs nothing, because the proof of lemma 4.4.6 identifies $\iota_{t*}$ with $\gamma \mapsto \xi \cap p^*\gamma$, which is one and the same map for every $t$ and is injective. Comparing the two lines of (4.8) at $t \neq \infty$ and at $t = \infty$, their images under $\iota_{t*}$ agree, hence so do the classes themselves, and $(j \circ i)^![W] = i^!_{a'}\big(j^!_a[W]\big)$, as we sought to show.

Commutativity now follows, and the route is the one announced at the start of the section: the two embeddings are made into a single embedding of a product, which factors in two ways.

**Corollary 4.4.8: Commutativity of Gysin Maps**

Let $i\colon Z \hookrightarrow P$ and $i'\colon Z' \hookrightarrow P'$ be regular embeddings of codimensions $d$ and $d'$, let $W$ be an algebraic scheme, and let $a\colon W \to P$ and $b\colon W \to P'$ be morphisms. Put $W_1 = a^{-1}(Z)$, $W_2 = b^{-1}(Z')$ and $W_{12} = W_1 \cap W_2$. Then for every $\alpha \in A_k(W)$,

$$i^!_{a|_{W_2}}\big(i'^!_b\alpha\big) \;=\; i'^!_{b|_{W_1}}\big(i^!_a\alpha\big) \quad \text{in } A_{k-d-d'}(W_{12})$$

***Proof*:**
Let $c = (a,b)\colon W \to P \times P'$. Lemma 4.4.1 makes $i \times i'$ a regular embedding of codimension $d + d'$ and exhibits its two factorizations through $Z \times P'$ and through $P \times Z'$, all four factors being regular embeddings. The preimages under $c$ are

$$c^{-1}(Z \times P') = W_1, \qquad c^{-1}(P \times Z') = W_2, \qquad c^{-1}(Z \times Z') = W_{12},$$

because the ideal of $Z \times P'$ is $\mathcal{I}_Z\mathcal{O}_{P\times P'}$ and $\mathrm{pr}_1 \circ c = a$, and symmetrically. The third follows from the first two: the ideal of $Z \times Z'$ is the sum $\mathcal{I}_Z\mathcal{O}_{P\times P'} + \mathcal{I}_{Z'}\mathcal{O}_{P\times P'}$, so $Z \times Z'$ is the scheme-theoretic intersection of $Z \times P'$ and $P \times Z'$, and extension of ideals along $c$ carries a sum of ideals to the sum of the extensions.

Each of the four factors is the preimage of $i$ or of $i'$ along a projection off a product. Such a projection is the base change of a structure morphism to $\operatorname{Spec}\overline{k}$, hence flat, so lemma 4.4.3 applies to each factor and identifies its refined Gysin map with that of $i$ or of $i'$ along the corresponding composite. Explicitly, with $\varphi = \mathrm{pr}_1$ one gets $(i \times \mathrm{id}_{P'})^!_c = i^!_a$ and $(i \times \mathrm{id}_{Z'})^!_{c''} = i^!_{a|_{W_2}}$ for $c''\colon W_2 \to P \times Z'$, and with $\varphi = \mathrm{pr}_2$ one gets $(\mathrm{id}_P \times i')^!_c = i'^!_b$ and $(\mathrm{id}_Z \times i')^!_{c'} = i'^!_{b|_{W_1}}$ for $c'\colon W_1 \to Z \times P'$. Here $c'$ and $c''$ are the morphisms induced by $c$ on the preimages $W_1$ and $W_2$, so that $\mathrm{pr}_2 \circ c' = b|_{W_1}$ and $\mathrm{pr}_1 \circ c'' = a|_{W_2}$; they are exactly the morphisms that theorem 4.4.7 calls $a'$, for the two factorizations respectively.

Apply theorem 4.4.7 to each factorization in turn, with $c$ as the morphism to $P \times P'$. The first gives $(i \times i')^!_c = i'^!_{b|_{W_1}} \circ i^!_a$ and the second gives $(i \times i')^!_c = i^!_{a|_{W_2}} \circ i'^!_b$. The left sides agree, so the right sides do, which is the assertion.

## Exercise 4.4.1

1. Theorem 4.4.2 assumes $e \geq 1$, and so does definition 4.3.6. Show that for $e = 0$ a section $s$ of a smooth $\pi\colon P \to S$ of relative dimension zero is an open and closed immersion, so that

the only sensible reading of $s^!$ is flat pullback along $s$, and that with that reading $s^! \circ \pi^* = \mathrm{id}$ follows from proposition 1.3.5(3) with no deformation at all.

2. In lemma 4.4.1, show that the normal bundle of $i \times i'$ is $\mathrm{pr}_1^* N \oplus \mathrm{pr}_2^* N'$, and that the two factorizations exhibit the two summands as the normal bundles of the two intermediate stages. Which of the two summands does lemma 4.4.3 match to $N$ when applied with $\varphi = \mathrm{pr}_1$?

3. Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d$ and take $P' = P$ and $i' = i$, but keep two distinct morphisms $a, b\colon W \to P$. Write out what corollary 4.4.8 asserts, and identify the regular embedding of $P \times P$ to which lemma 4.4.1 applies. Then show that for $a = b$ the identity degenerates to a tautology, and explain why the corollary is in no case a statement about $i^!$ commuting with itself as an endomorphism.

4. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d$ and $\varphi\colon U \hookrightarrow X$ an open immersion with $Z \subseteq U$. Check that $\varphi$ satisfies the hypotheses of lemma 4.4.3, and deduce that the refined Gysin map of $i$ may be computed in any open neighbourhood of $Z$.

5. In theorem 4.4.7 verify directly, from the definition of a regular embedding, that a local regular sequence for $Z$ in $Y$ lifted to $P$ and appended to one for $Y$ in $P$ generates the ideal of $Z$ in $P$, and that the concatenation is regular. Then write out the sequence $0 \to N_Z Y \to N_Z P \to N_Y P|_Z \to 0$ of [2, The Conormal Sequence of a Chain of Regular Embeddings] for $Z = \{0\} \subseteq Y = \mathbb{A}^1 \subseteq P = \mathbb{A}^2$, identifying all three bundles and the splitting.

# 5

# *Intersection Products*

The refined Gysin map of chapter 4 intersects a cycle with a regularly embedded subscheme. It is not yet a product: it takes a cycle on the ambient scheme and a regular embedding, not two cycles. The step that converts it into one is a piece of bookkeeping that costs nothing and gives everything.

For $X$ smooth, the diagonal $\delta\colon X \hookrightarrow X \times X$ is a regular embedding, and two cycles $\alpha, \beta$ on $X$ determine a single cycle $\alpha \times \beta$ on $X \times X$. Applying $\delta^!$ to it lands back on $X$ and, as this chapter shows, the result is bilinear, commutative, associative, and has $[X]$ as a unit. That is a ring, and it is the Chow ring $A^*(X)$: the object the classical geometers were computing in when they counted, and which *Algebraic Geometry* [2] could construct only on a surface, by hand.

Nothing here needs a moving lemma. The cycles are never perturbed; the diagonal is what gets degenerated, and chapter 4 already did that work. The moving lemma is stated in its place, both because it is true and because seeing what it would have been used for is the clearest way to see why the deformation replaced it.

The chapter closes by paying two debts. The self-intersection formula computes $i^*i_*$ as capping with the top Chern class, which recovers the $E^2 = -1$ of a blown-up surface as a special case of a general theorem; and the agreement theorem shows that on a smooth projective surface this product is the one built cohomologically upstream, so the surface theory that book developed by hand is subsumed rather than paralleled.

## 5.1 The Intersection Product

The first ingredient is a way of assembling a cycle on $X$ and a cycle on $Y$ into a single cycle on $X \times Y$. On cycles the rule is forced, $[V] \times [W] = [V \times W]$, and the content is that it survives rational equivalence. The mechanism that makes it survive is flat pullback along the two projections, which is already known to preserve rational equivalence (proposition 1.3.7), so the whole construction is arranged to be a composite of operations that do.

**Lemma 5.1.1: Projections of a Product Are Flat**

Let $X$ and $Y$ be algebraic schemes with $Y$ pure of dimension $l$, and let $p\colon X \times Y \to X$ be the projection. Then $p$ is flat of relative dimension $l$, and for every $k$-dimensional subvariety $V \subseteq X$ the scheme-theoretic preimage is $p^{-1}(V) = V \times Y$, so that

$$p^*[V] \;=\; [V \times Y]$$

the fundamental cycle of $V \times Y$, which is pure of dimension $k + l$.

***Proof*:**
The product is taken over $\operatorname{Spec} \overline{k}$, so $p$ is the base change of the structure morphism $Y \to \operatorname{Spec} \overline{k}$ along $X \to \operatorname{Spec} \overline{k}$. Every module over a field is free, hence flat, so $Y \to \operatorname{Spec} \overline{k}$ is flat, and flatness is stable under base change; thus $p$ is flat. The fibre of $p$ over a point $x \in X$ is $Y \times_{\operatorname{Spec} \overline{k}} \operatorname{Spec} k(x)$, which is pure of dimension $l$, extension of the ground field preserving the dimension of a scheme of finite type. So $p$ is flat of relative dimension $l$ in the sense of definition 1.3.3.

The scheme-theoretic preimage of $V$ is $p^{-1}(V) = (X \times Y) \times_X V = V \times Y$, and definition 1.3.3 computes $p^*[V]$ as the fundamental cycle of that preimage. For the dimension, restrict $p$ to $V$: the induced morphism $V \times Y \to V$ is again flat with all fibres of pure dimension $l$, and its base $V$ is a variety, so proposition 1.3.2 applies and every irreducible component of $V \times Y$ has dimension $k + l$, as we sought to show.

The lemma supplies both the cycle and the guarantee that it has the expected dimension, so the exterior product can be defined by the geometric formula and its bookkeeping checked afterwards.

**Definition 5.1.2: The Exterior Product of Cycles**

Let $X$ and $Y$ be algebraic schemes. For a $k$-dimensional subvariety $V \subseteq X$ and an $l$-dimensional subvariety $W \subseteq Y$, the product $V \times W$ is a closed subscheme of $X \times Y$, pure of dimension $k + l$ by lemma 5.1.1. The *exterior product* of $[V]$ and $[W]$ is its fundamental cycle (definition 1.1.2),

$$[V] \times [W] \;=\; [V \times W] \;\in\; Z_{k+l}(X \times Y)$$

Extending bilinearly gives the exterior product of cycles

$$\times \;:\; Z_k(X) \times Z_l(Y) \longrightarrow Z_{k+l}(X \times Y)$$

**Example 5.1.3: The Two Rulings of a Quadric Surface**

Take $X = Y = \mathbb{P}^1$. By theorem 1.5.2, $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$ and $A_0(\mathbb{P}^1) = \mathbb{Z}\,[\mathrm{pt}]$, the second generated by the class of any closed point. Fix closed points $p, q \in \mathbb{P}^1$. The four exterior products of generators are

$$[\mathbb{P}^1] \times [\mathbb{P}^1] = [\mathbb{P}^1 \times \mathbb{P}^1], \qquad [\mathbb{P}^1] \times [q] = [\mathbb{P}^1 \times \{q\}], \qquad [p] \times [\mathbb{P}^1] = [\{p\} \times \mathbb{P}^1], \qquad [p] \times [q] = [(p, q)]$$

each product of varieties being a variety here: $\mathbb{P}^1 \times \mathbb{P}^1$ is covered by four copies of $\mathbb{A}^2$, any two of which meet, so it is irreducible, and it is reduced because those charts are.

The geometry is visible through the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, $([a:b],[c:d]) \mapsto [ac:ad:bc:bd]$, whose image is the quadric surface $\{x_0x_3 = x_1x_2\}$. The middle two classes are the classes of the two families of lines on that quadric: $\mathbb{P}^1 \times \{q\}$ and $\{p\} \times \mathbb{P}^1$ each map to a line, and two members of the same family are disjoint while members of opposite families meet in one point. The first class is the fundamental class of the surface and the last is the class of a point.
Not every subvariety of a product is a product. The diagonal $\Delta = \{([a:b],[a:b])\} \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ is a curve meeting both families, so it is neither of the form $V \times \mathbb{P}^1$ nor of the form $\mathbb{P}^1 \times W$; its class is nonetheless a sum of exterior products, $[\Delta] = [\mathbb{P}^1] \times [q] + [p] \times [\mathbb{P}^1]$, which exercise 5.1.1 (2) verifies by exhibiting a rational function.

Rational equivalence is not visible in definition 5.1.2, which is written on cycles. The route to it is to recognize the exterior product as a composite of a flat pullback and a proper pushforward, one factor at a time.

**Proposition 5.1.4: The Exterior Product Descends to Rational Equivalence**

Let $X$ and $Y$ be algebraic schemes.

1. *Flat descriptions.* Let $W \subseteq Y$ be an $l$-dimensional subvariety, let $p_W : X \times W \to X$ be the projection and $\iota_W : X \times W \hookrightarrow X \times Y$ the closed immersion. Then for every $\alpha \in Z_k(X)$,
$$\alpha \times [W] \;=\; (\iota_W)_*\big(p_W^*\alpha\big)$$
Symmetrically, if $V \subseteq X$ is a $k$-dimensional subvariety, $q_V : V \times Y \to Y$ the projection and $\iota_V : V \times Y \hookrightarrow X \times Y$ the closed immersion, then $[V] \times \beta = (\iota_V)_*(q_V^*\beta)$ for every $\beta \in Z_l(Y)$.

2. *Descent.* The exterior product carries $\mathrm{Rat}_k(X) \times Z_l(Y)$ and $Z_k(X) \times \mathrm{Rat}_l(Y)$ into $\mathrm{Rat}_{k+l}(X \times Y)$, so it induces a bilinear map
$$\times \;:\; A_k(X) \times A_l(Y) \longrightarrow A_{k+l}(X \times Y)$$

3. *Closed immersions.* If $f : X' \hookrightarrow X$ and $g : Y' \hookrightarrow Y$ are closed immersions of algebraic schemes, then $(f \times g)_*(\alpha \times \beta) = f_*\alpha \times g_*\beta$ for all $\alpha \in A_k(X')$ and $\beta \in A_l(Y')$.

***Proof*:**

1. Both sides are additive in $\alpha$, so it suffices to take $\alpha = [V]$ for a $k$-dimensional subvariety $V \subseteq X$. The subscheme $W$ is a variety, hence pure of dimension $l$, so lemma 5.1.1 applies to $p_W$ and gives $p_W^*[V] = [V \times W]$, the fundamental cycle of $V \times W$ computed inside $X \times W$. Its components are the components of $V \times W$, each carrying the same length whether $V \times W$ is regarded in $X \times W$ or in $X \times Y$, and each maps isomorphically to itself under the closed immersion $\iota_W$; so $(\iota_W)_*[V \times W] = [V \times W]$ by definition 1.2.1, the field extension in each degree being trivial. This is $[V] \times [W]$. The symmetric statement is the same argument with the roles of the two factors exchanged.

2. Let $\alpha \in \mathrm{Rat}_k(X)$ and $\beta \in Z_l(Y)$. Writing $\beta = \sum_j n_j[W_j]$ and using bilinearity, it is enough to show $\alpha \times [W] \in \mathrm{Rat}_{k+l}(X \times Y)$ for a single $l$-dimensional subvariety $W$. By part (1) this class is $(\iota_W)_*(p_W^*\alpha)$. Flat pullback carries cycles rationally equivalent to zero

to cycles rationally equivalent to zero (proposition 1.3.7), and so does proper pushforward along the closed immersion $\iota_W$ (theorem 1.2.3); hence $\alpha \times [W] \in \mathrm{Rat}_{k+l}(X \times Y)$. The symmetric half is identical, using the second formula of part (1) and additivity in the first variable.

For classes, let $\alpha \sim \alpha'$ in $Z_k(X)$ and $\beta \sim \beta'$ in $Z_l(Y)$. Then

$$\alpha \times \beta - \alpha' \times \beta' \;=\; (\alpha - \alpha') \times \beta \;+\; \alpha' \times (\beta - \beta')$$

and both terms on the right lie in $\mathrm{Rat}_{k+l}(X \times Y)$ by the previous paragraph. So the product of the classes is well defined, and it is bilinear because it is bilinear on cycles.

3. Both sides are bilinear, so it suffices to check on $\alpha = [V']$ and $\beta = [W']$ for subvarieties $V' \subseteq X'$ and $W' \subseteq Y'$. A closed immersion identifies $V'$ with its image $f(V')$ and $W'$ with $g(W')$, so $V' \times W'$ and $f(V') \times g(W')$ are the same closed subscheme of $X \times Y$; their fundamental cycles therefore agree, and each component of $V' \times W'$ maps isomorphically under $f \times g$. Hence $(f \times g)_*[V' \times W'] = [f(V') \times g(W')] = f_*[V'] \times g_*[W']$ by definition 1.2.1, completing the proof.

The proof is short because the exterior product was built out of the two functorialities of chapter 1 rather than defined by an independent construction, and that is the only reason it respects rational equivalence: nothing about a product of schemes is used beyond the flatness of its projections. Part (3) is the compatibility that lets a product computed on small closed subschemes be compared with the same product computed on the ambient scheme, and it is what the refined form of the intersection product below rests on.

What the exterior product does not do is produce a class on $X$. A class on $X \times X$ is returned to $X$ by pulling back along the diagonal, and chapter 4 supplies the pullback provided the diagonal is a regular embedding. The diagonal morphism $\delta\colon X \to X \times X$ exists for every scheme, and it is a closed immersion exactly when $X$ is separated, that being the definition of separatedness; every variety appearing from here on is separated, and the hypothesis is carried in the statements that use it.

The first step is to identify the conormal sheaf of the diagonal. It requires no hypothesis on $X$ beyond separatedness, and it is what converts smoothness into the regularity of $\delta$.

**Lemma 5.1.5: The Conormal Sheaf of the Diagonal**

Let $X$ be an algebraic scheme, let $\delta\colon X \to X \times X$ be the diagonal and let $\mathcal{J} \subseteq \mathcal{O}_{X \times X}$ be the ideal sheaf of the closed subscheme $\delta(X)$. Regard $\mathcal{J}/\mathcal{J}^2$ as a sheaf of $\mathcal{O}_X$-modules through the isomorphism $\delta\colon X \to \delta(X)$. Then there is a canonical isomorphism

$$\mathcal{J}/\mathcal{J}^2 \;\cong\; \Omega_X$$

of $\mathcal{O}_X$-modules, under which the class of $1 \otimes a - a \otimes 1$ corresponds to $da$.

***Proof*:**
There is nothing to prove: this is how the sheaf of differentials is defined upstream. The cotangent sheaf is $\Omega_{X/Y} := \Delta^*(\mathscr{I}/\mathscr{I}^2)$ for $\mathscr{I}$ the ideal sheaf of the diagonal, with universal derivation $d(b) = 1 \otimes b - b \otimes 1$ modulo $\mathscr{I}^2$ ([2, Sheaf of Differentials]); taking $Y = \operatorname{Spec} \overline{k}$ and using that $\delta$ is a closed immersion gives the statement. The affine content, that $I/I^2$ with this $d$ satisfies the universal property of $\Omega_{B/A}$, is [2, Differentials via the Diagonal], as we sought to show.

The identification uses no smoothness, and it is the reason the sheaf of differentials can be constructed from the diagonal in the first place: $\Omega_X$ is the conormal sheaf of $\delta$, so it records the first infinitesimal neighbourhood of the diagonal and nothing more. What smoothness adds is that $\Omega_X$ is then locally free of the right rank, and local freeness of the conormal sheaf is the visible half of being a regular embedding. The other half, that the ideal is generated by a regular sequence and not by the right number of elements, is where the local rings of $X \times X$ enter.

**Theorem 5.1.6: The Diagonal of a Smooth Variety Is a Regular Embedding**

Let $X$ be a separated smooth variety of dimension $n$ over $\overline{k}$ and let $\delta\colon X \to X \times X$ be the diagonal. Then $\delta$ is a regular embedding of codimension $n$ ([2, Regular Embedding]), and its normal bundle is the tangent bundle of $X$,

$$N_\delta \;=\; T_X$$

***Proof*:**
Write $\Delta = \delta(X)$ and let $\mathcal{J}$ be its ideal sheaf.

*Step 1: the diagonal is closed of codimension $n$.* Separatedness of $X$ says exactly that $\delta$ is a closed immersion, and $\delta$ is an isomorphism onto $\Delta$, so $\Delta$ is a closed subscheme of $X \times X$ isomorphic to $X$ and of dimension $n$. Smoothness is stable under products and dimensions add, so $X \times X$ is smooth of dimension $2n$ ([2, Stability of Smooth Morphisms](3)); moreover $X \times X$ is pure of dimension $2n$ by lemma 5.1.1 applied with $V = X$. Its local rings are therefore regular, of dimension $2n$ at every closed point. The codimension of $\Delta$ in $X \times X$ is $2n - n = n$.

*Step 2: the ideal is generated by $n$ elements along the diagonal.* Fix an affine open $U = \operatorname{Spec} A$ of $X$ and keep the notation of lemma 5.1.5, so that $U \times U = \operatorname{Spec} B$ with $B = A \otimes_{\overline{k}} A$, and $J = \ker \mu$ is the ideal of $\delta(U)$. By lemma 5.1.5 there is an isomorphism $\Omega_A \cong J/J^2$ of $A$-modules, and smoothness of $X$ makes $\Omega_A$ locally free of rank $n$.

The differentials $da$ generate $\Omega_A$, so after shrinking $U$ around a chosen closed point there are $t_1, \dots, t_n \in A$ whose differentials generate $\Omega_A$: pick elements whose differentials span the fibre of $\Omega_A$ at the point, which has dimension $n$, and apply Nakayama together with coherence to spread the generation to a neighbourhood. The resulting surjection $A^n \to \Omega_A$ onto a locally free module of rank $n$ is an isomorphism, since after localizing at any prime it becomes a surjective endomorphism of a finitely generated module over a commutative ring and every such endomorphism is injective. So $dt_1, \dots, dt_n$ is a basis of $\Omega_A$.

Transporting through lemma 5.1.5, the classes of

$$u_i \;=\; t_i \otimes 1 - 1 \otimes t_i, \qquad i = 1, \dots, n$$

form a basis of $J/J^2$ over $A$. In particular $J = (u_1, \dots, u_n) + J^2$. Let $z \in \delta(U)$ and put $R = \mathcal{O}_{X\times X,z}$ with maximal ideal $\mathfrak{m}$. Localizing the last equality gives $JR = (u_1, \dots, u_n)R + J^2R$, and $J \subseteq \mathfrak{m}$ forces $J^2R \subseteq \mathfrak{m}\,JR$; so the finitely generated $R$-module $N = JR/(u_1, \dots, u_n)R$ satisfies $N = \mathfrak{m}N$ and vanishes by Nakayama. Hence

$$JR \;=\; (u_1, \dots, u_n)R \quad \text{for every } z \in \delta(U)$$

The coherent sheaf $\mathcal{J}/(u_1, \dots, u_n)\mathcal{O}_{U\times U}$ therefore has vanishing stalk at every point of $\delta(U)$, so its support is a closed subset of $U \times U$ disjoint from $\delta(U)$, and on the open complement, an open neighbourhood of $\delta(U)$ in $X \times X$, the ideal sheaf of $\Delta$ is generated by $u_1, \dots, u_n$.

*Step 3: the generators form a regular sequence.* Let $z \in \delta(U)$ be a closed point, corresponding to the closed point $x \in U$, and keep $R = \mathcal{O}_{X\times X,z}$. By Step 1 the ring $R$ is regular local of dimension $2n$, and

$$R/JR \;=\; \mathcal{O}_{\Delta,z} \;=\; \mathcal{O}_{X,x}$$

is regular local of dimension $n$, again by smoothness of $X$. Regularity makes the embedding dimensions equal to the Krull dimensions, so $\dim_{\overline{k}} \mathfrak{m}/\mathfrak{m}^2 = 2n$ while the quotient $\mathfrak{m}/(\mathfrak{m}^2 + JR)$, which is the corresponding space for $R/JR$, has dimension $n$. The image of $JR$ in $\mathfrak{m}/\mathfrak{m}^2$ is therefore $n$-dimensional, and it is spanned by the images of $u_1, \dots, u_n$ by Step 2; those $n$ images are consequently linearly independent in $\mathfrak{m}/\mathfrak{m}^2$, that is, $u_1, \dots, u_n$ is part of a regular system of parameters of $R$. A regular system of parameters of a regular local ring is a regular sequence, and so is any part of one: extend $u_1, \dots, u_n$ to a regular system of parameters $u_1, \dots, u_{2n}$ of $R$ and put $R_i = R/(u_1, \dots, u_i)$, whose maximal ideal is generated by the images of $u_{i+1}, \dots, u_{2n}$. Its embedding dimension is therefore at most $2n - i$, while its dimension is at least $2n - i$, since killing one element drops the dimension by at most one; the two agree, so $R_i$ is regular local and hence a domain, and the image of $u_{i+1}$ in it is nonzero, since otherwise $\mathfrak{m}R_i$ would be generated by $2n - i - 1$ elements. Each $u_{i+1}$ is thus a nonzerodivisor on $R_i$, that is, $u_1, \dots, u_n$ is a regular sequence in $R$.

This was proved at closed points, and it propagates to the rest of $\Delta$ by localization. If $z' \in \Delta$ is arbitrary, its closure meets the closed points of $X \times X$, since closed points are dense in a closed subset of a scheme of finite type over a field; choosing a closed point $z$ in that closure, $\mathcal{O}_{X\times X,z'}$ is a localization of $\mathcal{O}_{X\times X,z}$ at a prime containing $J$. Localization is exact, so a nonzerodivisor stays a nonzerodivisor and a regular sequence stays one, the quotient remaining nonzero because $z' \in \Delta$. Thus every point of $\Delta$ has a neighbourhood in $X \times X$ on which $\mathcal{J}$ is generated by $n$ sections whose images in the local rings along $\Delta$ form a regular sequence, which is the local condition of [2, Regular Embedding]. Since $\Delta$ has codimension $n$ by Step 1, $\delta$ is a regular embedding of codimension $n$.

*Step 4: the normal bundle.* The normal bundle of a regular embedding is the vector bundle whose sheaf of sections is the dual of the conormal sheaf ([2, Conormal Sheaf and Normal Bundle]), and lemma 5.1.5 identifies that conormal sheaf $\mathcal{J}/\mathcal{J}^2$ with $\Omega_X$. The dual of $\Omega_X$ is the tangent sheaf ([2, Tangent Sheaf]), whose associated bundle is $T_X$. Hence $N_\delta = T_X$, completing the proof.

Two features of the statement are worth separating. The codimension is $n$, which is what makes $\delta^!$ lower dimension by exactly $n$ and so makes the product of a $k$-cycle and an $l$-cycle a cycle of dimension $k + l - n$, the dimension a transverse intersection would have. The normal bundle is $T_X$, which is where the geometry of $X$ enters the product: every formula below in which a Chern class of the tangent bundle appears is tracing back to this identification, and the self-intersection formula of section 5.3 is the first of them. Smoothness is used twice and cannot be weakened to either half alone: the local rings of $X \times X$ must be regular for Step 3, and $\Omega_X$ must be locally free for Step 2.

**Example 5.1.7: The Diagonal of Affine Space**

Let $X = \mathbb{A}^n = \operatorname{Spec} \overline{k}[t_1, \dots, t_n]$. Then $X \times X = \operatorname{Spec} \overline{k}[t_1, \dots, t_n, s_1, \dots, s_n] = \mathbb{A}^{2n}$, and the ideal of the diagonal is $J = (t_1 - s_1, \dots, t_n - s_n)$, so the generators of theorem 5.1.6 are visible globally: $u_i = t_i - s_i$. That they form a regular sequence needs no local algebra here, since $\overline{k}[t, s] = \overline{k}[t_1, \dots, t_n, u_1, \dots, u_n]$ is again a polynomial ring and each successive quotient $\overline{k}[t, u]/(u_1, \dots, u_i)$ is a polynomial ring, hence a domain, in which $u_{i+1}$ is a nonzero element and therefore a nonzerodivisor. The conormal sheaf is free on the classes of the $u_i$, matching the basis $dt_1, \dots, dt_n$ of $\Omega_{\mathbb{A}^n}$ under lemma 5.1.5, and $N_\delta$ is the trivial bundle of rank $n$, which is $T_{\mathbb{A}^n}$.

The Chow groups are as small as possible: $A_k(\mathbb{A}^m) = 0$ for $k < m$ and $A_m(\mathbb{A}^m) = \mathbb{Z}[\mathbb{A}^m]$ by proposition 1.5.1. So the only product that can be nonzero is $[\mathbb{A}^n] \cdot [\mathbb{A}^n]$, and it is computed by the exterior product $[\mathbb{A}^n] \times [\mathbb{A}^n] = [\mathbb{A}^{2n}]$ followed by $\delta^!$. The intersection $\mathbb{A}^{2n} \cap \Delta$ is $\Delta$ itself, and $\Delta \hookrightarrow \mathbb{A}^{2n}$ is a regular embedding of codimension $n$, so proposition 4.3.3 gives $\delta^![\mathbb{A}^{2n}] = [\Delta]$, that is

$$[\mathbb{A}^n] \cdot [\mathbb{A}^n] \;=\; [\mathbb{A}^n]$$

the fundamental class reproducing itself.

The exterior product turns a pair of classes on $X$ into one class on $X \times X$, and the diagonal is a regular embedding, so the Gysin map of definition 4.3.1 returns a class on $X$.

**Definition 5.1.8: The Intersection Product**

Let $X$ be a separated smooth variety of dimension $n$ over $\overline{k}$, with diagonal $\delta\colon X \hookrightarrow X \times X$, a regular embedding of codimension $n$ by theorem 5.1.6. For $\alpha \in A_k(X)$ and $\beta \in A_l(X)$ the *intersection product* is

$$\alpha \cdot \beta \;=\; \delta^!\big(\alpha \times \beta\big) \;\in\; A_{k+l-n}(X)$$

where $\alpha \times \beta \in A_{k+l}(X \times X)$ is the exterior product of proposition 5.1.4 and $\delta^!\colon A_{k+l}(X \times X) \to A_{k+l-n}(X)$ is the Gysin map of definition 4.3.1.

The definition is the whole of the construction, and every property of the product is a property of one of its two inputs. It is defined for arbitrary classes, with no hypothesis on how representing cycles meet, because the Gysin map has none; and it depends only on the rational equivalence classes of the two factors, because both the exterior product and the Gysin map do. The dimension bookkeeping is the expected one: $k + l - n$ is the dimension of a transverse intersection of a $k$-dimensional and an $l$-dimensional subvariety of an $n$-dimensional variety, so in codimension the product adds. That grading, and the ring axioms it grades, are established in section 5.2; until then the product is exactly what definition 5.1.8 says it is.

Bilinearity and the vanishing below the expected dimension are inherited from the two inputs, and are recorded here for the computations that use them.

**Proposition 5.1.9: Bilinearity and Vanishing in Low Dimension**

Let $X$ be a separated smooth variety of dimension $n$.

1. The product $A_k(X) \times A_l(X) \to A_{k+l-n}(X)$ is bilinear.
2. If $k + l < n$ then $\alpha \cdot \beta = 0$ for all $\alpha \in A_k(X)$ and $\beta \in A_l(X)$.

***Proof*:**

1. The exterior product is bilinear by proposition 5.1.4(2) and $\delta^!$ is a homomorphism of abelian groups by proposition 4.3.2(1), so the composite is bilinear in each variable.
2. The class $\alpha \times \beta$ lies in $A_{k+l}(X \times X)$ with $k + l < n$, and the codimension of $\delta$ is $n$, so proposition 4.3.2(2) gives $\delta^!(\alpha \times \beta) = 0$. Equivalently the target group $A_{k+l-n}(X)$ is a Chow group in negative dimension and vanishes, as we sought to show.

The product of definition 5.1.8 records its answer on all of $X$, and that discards information the construction has available. If $\alpha$ is carried by a closed subscheme $V$ and $\beta$ by a closed subscheme $W$, the intersection ought to be carried by $V \cap W$, and the refined Gysin homomorphism of definition 4.3.6 delivers exactly that, because the preimage of $V \times W$ under the diagonal is the scheme-theoretic intersection. Indeed the ideal sheaf of $V \times W$ in $X \times X$ is generated by the pullbacks of $\mathcal{I}_V$ and $\mathcal{I}_W$ along the two projections, and composing either projection with $\delta$ is the identity, so that ideal pulls back to $\mathcal{I}_V + \mathcal{I}_W$, the ideal of $V \cap W$.

**Definition 5.1.10: The Refined Intersection Product**

Let $X$ be a separated smooth variety of dimension $n$ and let $V, W \subseteq X$ be closed subschemes, with $g\colon V \times W \to X \times X$ the closed immersion. Since $\delta^{-1}(V \times W) = V \cap W$, the refined Gysin homomorphism of definition 4.3.6 along $g$ is a homomorphism

$$\delta^! \;:\; A_m(V \times W) \longrightarrow A_{m-n}(V \cap W)$$

For $\alpha \in A_k(V)$ and $\beta \in A_l(W)$ the *refined intersection product* is

$$\alpha \cdot_X \beta \;=\; \delta^!\big(\alpha \times \beta\big) \;\in\; A_{k+l-n}(V \cap W)$$

**Proposition 5.1.11: The Refined Product Determines the Product**

Let $X$ be a separated smooth variety of dimension $n$, let $V, W \subseteq X$ be closed subschemes with inclusions $i_V$ and $i_W$, and let $\iota\colon V \cap W \hookrightarrow X$ be the inclusion. For all $\alpha \in A_k(V)$ and $\beta \in A_l(W)$,

$$\iota_*\big(\alpha \cdot_X \beta\big) \;=\; \big(i_{V*}\alpha\big) \cdot \big(i_{W*}\beta\big) \;\in\; A_{k+l-n}(X)$$

***Proof*:**
Apply theorem 4.3.10(1) to the regular embedding $\delta$ with $X' = X \times X$, $X'' = V \times W$ and $p = g$ the closed immersion, which is proper. The preimage of $\delta(X)$ in $X'$ is $\delta(X)$ itself and the preimage in $X''$ is $V \cap W$, so the induced morphism $q$ of preimages is $\iota$, read through the isomorphism $\delta\colon X \to \delta(X)$. The theorem gives

$$\delta^!\big(g_*\gamma\big) \;=\; \iota_*\big(\delta^!\gamma\big)$$

for $\gamma \in A_{k+l}(V \times W)$, where the left-hand $\delta^!$ is the refined map along $\mathrm{id}_{X \times X}$, which is the Gysin map of definition 4.3.1 by proposition 4.3.7. Take $\gamma = \alpha \times \beta$. By proposition 5.1.4(3) applied to the closed immersions $i_V$ and $i_W$,

$$g_*(\alpha \times \beta) \;=\; i_{V*}\alpha \times i_{W*}\beta$$

so the left-hand side is $\delta^!\big(i_{V*}\alpha \times i_{W*}\beta\big)$, which is $(i_{V*}\alpha) \cdot (i_{W*}\beta)$ by definition 5.1.8, and the right-hand side is $\iota_*(\alpha \cdot_X \beta)$ by definition 5.1.10, as we sought to show.

The pattern is the one recorded in corollary 4.3.11 for a single subvariety: the class computed on the ambient scheme is the pushforward of a class computed on the intersection. The refined product is the finer object, and every localized intersection number in the classical sense is a degree taken of it; the product of definition 5.1.8 is its pushforward to $X$.

**Example 5.1.12: Two Lines in the Projective Plane**

Let $X = \mathbb{P}^2$, which is a smooth variety of dimension 2, and let $L = \{x_0 = 0\}$ and $L' = \{x_1 = 0\}$ be two distinct lines, meeting in the single reduced point $P = [0:0:1]$.

*The exterior product.* By definition 5.1.2, $[L] \times [L'] = [L \times L']$, and $L \times L' \cong \mathbb{P}^1 \times \mathbb{P}^1$ is a variety of dimension 2 by the argument of example 5.1.3, so the exterior product is the class of a single subvariety of $\mathbb{P}^2 \times \mathbb{P}^2$ with multiplicity one.

*The Gysin pullback.* The intersection of $L \times L'$ with the diagonal is $\delta^{-1}(L \times L') = L \cap L' = \{P\}$, a reduced point. In the affine chart of $L \times L'$ around $(P, P)$, with coordinates $y$ on $L \setminus \{x_2 = 0\}$ and $z$ on $L' \setminus \{x_2 = 0\}$, that point is the origin of $\mathbb{A}^2$, cut out by the regular sequence $y, z$; so $\{P\} \hookrightarrow L \times L'$ is a regular embedding of codimension $2 = \operatorname{codim}(\delta)$. Proposition 4.3.3 therefore applies to $\delta$ and the subvariety $L \times L'$ and gives

$$[L] \cdot [L'] \;=\; \delta^{!}\big[L \times L'\big] \;=\; [P] \;\in\; A_0(\mathbb{P}^2)$$

The refined form of definition 5.1.10 records a class on $L \cap L' = \{P\}$, and that class is the generator of $A_0(\{P\}) = \mathbb{Z}$: by proposition 5.1.11 its image in $A_0(\mathbb{P}^2)$ is $[P]$, and the pushforward $A_0(\{P\}) \to A_0(\mathbb{P}^2)$ carries generator to generator.

*The self-intersection.* By theorem 1.5.2 the two lines have the same class, $[L] = [L'] = h_1$ in $A_1(\mathbb{P}^2)$. Since the product is defined on classes, the computation just made also evaluates

$$[L] \cdot [L] \;=\; h_1 \cdot h_1 \;=\; [P]$$

No cycle was perturbed to get this, and no representative of $h_1$ was required to meet $L$ properly: the two factors were replaced by equal classes, which proposition 5.1.4(2) and proposition 4.3.2(1) permit. The refined product, by contrast, does depend on the representatives: computed on $L$ against $L$ it lives on $A_0(L)$ rather than on $A_0(\{P\})$, and evaluating it there is the business of section 5.3.

## Exercise 5.1.1

1. Let $P = \operatorname{Spec} \overline{k}$ and let $X$ be an algebraic scheme. Show that $A_0(P) = \mathbb{Z}[P]$, and that under the canonical isomorphism $X \times P \cong X$ the exterior product of proposition 5.1.4 sends $(\alpha, m[P])$ to $m\alpha$. Then show that the exterior product of cycles is associative: for algebraic schemes $X, Y, T$ and cycles $\alpha, \beta, \gamma$ on them, $(\alpha \times \beta) \times \gamma = \alpha \times (\beta \times \gamma)$ under the identification $(X \times Y) \times T = X \times (Y \times T)$.

2. Let $S = \mathbb{P}^1 \times \mathbb{P}^1$ with coordinates $([a:b],[c:d])$, let $\Delta \subseteq S$ be the diagonal, and let $F = \mathbb{P}^1 \times \{[1:0]\}$ and $G = \{[0:1]\} \times \mathbb{P}^1$. Show that

$$[\Delta] \;=\; [\mathbb{P}^1] \times [\mathrm{pt}] + [\mathrm{pt}] \times [\mathbb{P}^1] \quad \text{in } A_1(S)$$

by computing the divisor of the rational function $f = (ad - bc)/(ad)$ on $S$ chart by chart.

3. Let $X = \operatorname{Spec} \overline{k}[x,y]/(y^2 - x^3)$, a curve with a cusp at the origin $p$. Using lemma 5.1.5 and the presentation $\Omega_X = \big(\mathcal{O}_X\,dx \oplus \mathcal{O}_X\,dy\big)/\big(d(y^2 - x^3)\big)$, show that the conormal sheaf of the diagonal of $X$ is not locally free of rank one, and deduce that $\delta\colon X \to X \times X$ is not a regular embedding of codimension one. Where does the proof of theorem 5.1.6 break down for this $X$?

4. Let $X$ be a separated smooth variety of dimension $n$ and let $V, W \subseteq X$ be closed subschemes with $V \cap W = \emptyset$. Show that $[V] \cdot [W] = 0$ in $A_*(X)$, where $[V]$ and $[W]$ denote the

images in $A_*(X)$ of the fundamental cycles. Give a second proof of the vanishing when $\dim V + \dim W < n$ that does not use disjointness.

5. Fix $0 \leq a, b \leq n$ with $a + b \geq n$, and let $L = \{x_{a+1} = \cdots = x_n = 0\}$ and $M = \{x_0 = \cdots = x_{n-b-1} = 0\}$ be coordinate subspaces of $\mathbb{P}^n$, so that $L \cong \mathbb{P}^a$, $M \cong \mathbb{P}^b$ and $L \cap M \cong \mathbb{P}^{a+b-n}$. Show that $L \cap M \hookrightarrow L \times M$ is a regular embedding of codimension $n$, and deduce from proposition 4.3.3 that $h_a \cdot h_b = h_{a+b-n}$ in $A_*(\mathbb{P}^n)$, where $h_m$ is the class of a linear $\mathbb{P}^m$ as in example 4.3.4.

## 5.2 The Chow Ring

Section 5.1 produced a biadditive pairing on the Chow groups of a smooth variety. This section proves that the pairing obeys the three laws that make it a ring, and it indexes the groups by codimension so that the degrees add rather than subtract. The ring is then made contravariant: a morphism of smooth varieties pulls classes back, and so does a local complete intersection morphism between arbitrary algebraic schemes, which is the form Grothendieck-Riemann-Roch will need.

Two ingredients are imported rather than built. One is a pair of statements about regular embeddings that belong to scheme theory; the other is the commutativity of two Gysin maps, which chapter 4 flagged and did not prove. Both are isolated below, so that everything resting on them is visible at a glance.

**5.2.1 Convention: Subscripts on Refined Gysin Maps** When several refinements of one Gysin map occur in the same formula, the morphism along which the map is refined is written as a subscript: $i^!_g$ is the refined homomorphism of definition 4.3.6 for the morphism $g$, and an unadorned $i^!$ is the Gysin map of definition 4.3.1.

One reading has to be fixed first. Definition 4.3.1 is stated in codimension $d \geq 1$, whereas a regular embedding of codimension 0 is an open and closed immersion, its ideal sheaf vanishing locally. For such an $i\colon Z \hookrightarrow P$ both $i^!$ and its refinements $i^!_g$ mean flat pullback throughout this section, along $i$ and along the induced immersion $g^{-1}(Z) \hookrightarrow P'$ for $g\colon P' \to P$; this is the reading definition 5.2.11 uses again. When the case arises from a diagonal or a graph the smooth variety in question has dimension 0, hence is $\operatorname{Spec} \overline{k}$: the embedding is then the identity under the canonical identification of $W \times \operatorname{Spec} \overline{k}$ with $W$, its flat pullback is the identity on Chow groups, and every statement below in which that happens holds by inspection. The arguments are written for the remaining case.

A cycle class on a smooth $n$-dimensional variety carries two equally good indices, its dimension and its codimension, and the product adds the second while subtracting the first from $2n$. A ring wants its grading to be additive, so the codimension index is the one to record.

**Definition 5.2.2: The Chow Ring**

Let $X$ be a smooth variety of dimension $n$. For $0 \le p \le n$ write

$$A^p(X) \;=\; A_{n-p}(X)$$

and call $A^p(X)$ the group of cycle classes of *codimension* $p$. The graded abelian group

$$A^*(X) \;=\; \bigoplus_{p=0}^{n} A^p(X)$$

equipped with the intersection product $\alpha \cdot \beta = \delta^!(\alpha \times \beta)$ of section 5.1, is the *Chow ring* of $X$. That the product makes it a ring is theorem 5.2.7.

The two indices are interchangeable data, and both are used: a statement about the geometry of cycles is usually cleaner in dimension, a statement about the ring is cleaner in codimension. The class $[X]$ lies in $A^0(X)$, a divisor class lies in $A^1(X)$, and the class of a point on a complete $X$ lies in $A^n(X)$.

Projective space is the first ring that can be written down, and it can be written down before the ring axioms are available, because each product in it is a single application of proposition 4.3.3.

**Example 5.2.3: The Chow Ring of Projective Space**

Let $X = \mathbb{P}^n$ with homogeneous coordinates $x_0, \dots, x_n$. By theorem 1.5.2, $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ with $h_k$ the class of a linear subspace $\mathbb{P}^k$, so $A^p(\mathbb{P}^n) = \mathbb{Z}\, h_{n-p}$ is infinite cyclic for $0 \le p \le n$ and zero otherwise.

Fix $p, q \ge 0$ with $p + q \le n$ and take the linear subspaces

$$L = \{x_0 = \cdots = x_{p-1} = 0\}, \qquad L' = \{x_p = \cdots = x_{p+q-1} = 0\}$$

of dimensions $n-p$ and $n-q$, so that $[L] = h_{n-p}$ and $[L'] = h_{n-q}$. Their product is $\delta^!\big([L]\times[L']\big) = \delta^![L\times L']$, and proposition 4.3.3 evaluates it once the scheme-theoretic intersection $W = (L\times L')\cap\Delta$ is known to be regularly embedded in $L \times L'$ with codimension $n$. Set-theoretically $W$ consists of the pairs $(z, z)$ with $z \in L \cap L'$, so $W \cong L \cap L' = \{x_0 = \cdots = x_{p+q-1} = 0\} \cong \mathbb{P}^{n-p-q}$.

Work on the chart $\{x_n \neq 0\}$ of each factor, which meets $W$ because the coordinate point $e_n$ lies in $L \cap L'$. Writing $u_i = x_i/x_n$ for the affine coordinates on the first factor and $v_i = x_i/x_n$ for those on the second, the chart of $L \times L'$ is the affine space with coordinate ring

$$\overline{k}\big[u_p, \dots, u_{n-1};\ v_0, \dots, v_{p-1},\ v_{p+q}, \dots, v_{n-1}\big]$$

the coordinates $u_0, \dots, u_{p-1}$ vanishing on $L$ and $v_p, \dots, v_{p+q-1}$ on $L'$. The diagonal is cut out on $\mathbb{P}^n \times \mathbb{P}^n$ in this chart by the $n$ equations $u_i = v_i$ with $0 \le i \le n-1$, and restricting them to $L \times L'$ leaves

$$v_0,\ \dots,\ v_{p-1}, \qquad u_p,\ \dots,\ u_{p+q-1}, \qquad u_i - v_i \ \ (p+q \le i \le n-1)$$

a list of $n$ linear forms in distinct variables. Each is a nonzerodivisor modulo its predecessors, every successive quotient being a polynomial ring, so the list is a regular sequence and $W \hookrightarrow L \times L'$ is a regular embedding of codimension $n$; the quotient is the polynomial ring on the remaining $n-p-q$ variables, so $W$ is reduced and irreducible with fundamental cycle $[\mathbb{P}^{n-p-q}]$. The same computation runs on every chart meeting $W$. Proposition 4.3.3 therefore gives

$$h_{n-p} \cdot h_{n-q} \;=\; h_{n-p-q}$$

For $p+q > n$ the product lands in $A^{p+q}(\mathbb{P}^n) = A_{n-p-q}(\mathbb{P}^n) = 0$ and is zero, by proposition 4.3.2(2). Writing $h = h_{n-1}$ for the class of a hyperplane, induction on $p$ gives $h^p = h_{n-p}$, and

$$A^*(\mathbb{P}^n) \;\cong\; \mathbb{Z}[h]/(h^{n+1})$$

once the ring axioms below license the power notation.

No general position argument entered the computation. The two subspaces were chosen in special position relative to the coordinates but in general position relative to each other, and proposition 4.3.2 guarantees that any other pair of representatives, including two equal ones, returns the same product. What the example does not show is that the answer depends on the two classes through a ring structure, and that is the content of the rest of the section.

The first import is scheme-theoretic. Both of its statements concern regular embeddings and smooth morphisms rather than cycles, so both belong to the scheme theory upstream, where they are now proved. Neither is re-derived here.

**Lemma 5.2.4: Sections, Flat Base Change, and Graphs**

1. Let $\pi\colon P \to S$ be a smooth morphism of relative dimension $e \geq 0$ and $s\colon S \to P$ a section of it. Then $s$ is a regular embedding of codimension $e$ with normal bundle $s^*T_{P/S}$.
2. Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d$ and $g\colon P' \to P$ a flat morphism. Then $g^{-1}(Z) \hookrightarrow P'$ is a regular embedding of codimension $d$, with normal bundle the pullback of the normal bundle of $i$.
3. Let $X$ be a smooth variety of dimension $n \geq 0$, let $V$ be an algebraic scheme, and let $f\colon V \to X$ be a morphism. Then the scheme-theoretic preimage of the diagonal under $f \times \mathrm{id}_X\colon V \times X \to X \times X$ is the graph $\Gamma_f$, the first projection restricts to an isomorphism $\Gamma_f \to V$, and $\gamma_f\colon V \to V \times X$ is a regular embedding of codimension $n$ with normal bundle $f^*T_X$.

***Proof*:**
Parts (1) and (2) are upstream results and are cited rather than re-derived, both being statements about regular embeddings as such: part (1) is [2, A Section of a Smooth Morphism Is a Regular Embedding] and part (2) is [2, Regular Embeddings Pull Back Along Flat Morphisms]. Part (3) follows from part (1). A $T$-valued point of $(f \times \mathrm{id}_X)^{-1}(\Delta)$ is a pair $(t_V, t_X)$ of $T$-points of $V$ and $X$ with $f \circ t_V = t_X$, so the functor of points of the preimage is that of $V$, and the isomorphism is induced by the first projection, with inverse $\gamma_f$. The projection $\mathrm{pr}_1\colon V \times X \to V$ is the base change of $X \to \operatorname{Spec} \overline{k}$, hence smooth of relative dimension $n$, and $\gamma_f$ is a section of it, so part (1) applies. Its relative tangent bundle is $\mathrm{pr}_2^* T_X$, whose pullback along $\gamma_f$ is $f^*T_X$, completing the proof.

Part (3) extends the diagonal computation of section 5.1 from the identity morphism to an arbitrary one, and it is the reason the intersection product can be transported along a map. Taking $V = X$ and $f = \mathrm{id}_X$ returns the diagonal.

The second import is about Gysin maps. That two regular embeddings can be intersected against in either order is what makes the product associative, and that the Gysin map of a composite of two regular embeddings factors is what makes the pullback along a local complete intersection morphism well defined. Both are proved in section 4.4, as corollary 4.4.8 and theorem 4.4.7, and are used here without further comment; the section identity theorem 4.4.2 of the same section is the third.

This theorem and parts (1) and (2) of lemma 5.2.4 are the only results the arguments below use whose proofs are cited rather than given. On the theorem rest the associativity of the intersection product, the identity of theorem 5.2.10(1) between intersecting with the graph and pulling back, and with it the projection formula, the multiplicativity and functoriality of pullback, and the independence of the local complete intersection pullback from its factorization. On the lemma rests everything that passes through a graph: its part (3) is deduced from part (1), so the unit law, definition 5.2.8 and all four parts of theorem 5.2.10 rest on it, and Step 1 of theorem 5.2.12(1) uses parts (1) and (2) directly. Commutativity of the product itself and the grading rest on neither.

The last preparation is bookkeeping. A Gysin map applied to an exterior product in which one factor is inert should return that factor untouched.

**Lemma 5.2.5: Gysin Maps and Exterior Products**

Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$, let $W$ be an algebraic scheme, and let $\mathrm{pr}\colon P \times W \to P$ and $\mathrm{pr}'\colon W \times P \to P$ be the projections, whose scheme-theoretic preimages of $Z$ are $Z \times W$ and $W \times Z$. Then for all $\xi \in A_k(P)$ and $\gamma \in A_l(W)$,

$$i^!_{\mathrm{pr}}(\xi \times \gamma) = (i^! \xi) \times \gamma, \qquad i^!_{\mathrm{pr}'}(\gamma \times \xi) = \gamma \times (i^! \xi)$$

***Proof*:**
Both sides are additive in $\gamma$, so it suffices to treat $\gamma = [W']$ for a subvariety $W' \subseteq W$ of dimension $l$, with inclusion $\iota\colon W' \hookrightarrow W$. Write $u = \mathrm{id}_P \times \iota\colon P \times W' \to P \times W$, a closed immersion and hence proper, and $\rho\colon P \times W' \to P$ for the projection. As $W'$ is a variety of dimension $l$ over $\overline{k}$, the structure morphism $W' \to \operatorname{Spec} \overline{k}$ is flat with $l$-dimensional fibre, so its base change $\rho$ is flat of relative dimension $l$; and $\rho = \mathrm{pr} \circ u$, so $u$ is a morphism over $P$.

*Step 1: the exterior product as a pullback followed by a pushforward.* For a subvariety $V \subseteq P$ the preimage $\rho^{-1}(V)$ is $V \times W'$, so $\rho^*[V] = [V \times W']$ by definition 1.3.3, and $u_*[V \times W'] = [V \times W']$ by definition 1.2.1, the immersion $u$ carrying $V \times W'$ isomorphically onto its image. Hence $\xi \times [W'] = u_*(\rho^* \xi)$ for every $\xi$. The same computation on $Z$ gives $q^* \eta = \eta \times [W']$ for the projection $q\colon Z \times W' \to Z$ and every $\eta \in A_*(Z)$.

*Step 2: the two compatibilities.* Theorem 4.3.10(2), applied with $X' = P$ and the identity, $X'' = P \times W'$ and $\rho$, and with $\rho$ itself as the morphism over $P$, gives

$$i^!_\rho(\rho^* \xi) \;=\; q^*(i^! \xi) \;=\; (i^! \xi) \times [W']$$

where proposition 4.3.7 was used to read the refined map along the identity as $i^!$. Theorem 4.3.10(1), applied with $X' = P \times W$ and pr, $X'' = P \times W'$ and $\rho$, and with the proper morphism $u$ over $P$, gives

$$i^!_{\mathrm{pr}}(u_* \eta) \;=\; q'_*(i^!_\rho \eta)$$

for the closed immersion $q' \colon Z \times W' \hookrightarrow Z \times W$ induced by $u$. Combining the two with Step 1, and using $q'_*\big(\zeta \times [W']\big) = \zeta \times [W']$ by definition 1.2.1 again, yields the first identity. Exchanging the two factors throughout gives the second, as we sought to show.

One comparison is used repeatedly below: when the preimage of $Z$ is as well behaved as $Z$ itself, the refined map is the plain one.

**Proposition 5.2.6: Refined Gysin Maps with a Regular Preimage**

Let $i \colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$ and $g \colon P' \to P$ a morphism of algebraic schemes such that $Z' = g^{-1}(Z)$ is regularly embedded in $P'$ with codimension $d$. Then the refined Gysin homomorphism $i^!_g$ of definition 4.3.6 is the Gysin map $(i')^!$ of definition 4.3.1 for $i' \colon Z' \hookrightarrow P'$.

***Proof*:**
Write $C = C_{Z'}P'$ and $N' = h^*N$ as in definition 4.3.6, and let $j \colon C \hookrightarrow N'$ be the immersion of lemma 3.4.13. By part (1) of that lemma $j$ is an isomorphism, and it is a morphism of cones over $Z'$, so the projections satisfy $\pi_{N'} \circ j = \pi_C$. Since $i'$ is a regular embedding, proposition 3.4.11 identifies $C$ with the normal bundle of $i'$, and definition 4.3.1 reads $(i')^! = (\pi_C^*)^{-1} \circ \sigma'$. Flat pullback along the isomorphism $j$ is inverse to proper pushforward along it, each carrying a subvariety to its preimage or image with multiplicity one, so $j_*\pi_C^* = j_*j^*\pi_{N'}^* = \pi_{N'}^*$ by proposition 1.3.5(3). Hence $(\pi_{N'}^*)^{-1} \circ j_* = (\pi_C^*)^{-1}$, and composing with $\sigma'$ turns definition 4.3.6 into definition 4.3.1, as we sought to show.

**Theorem 5.2.7: The Chow Ring of a Smooth Variety**

Let $X$ be a smooth variety of dimension $n$, with diagonal $\delta \colon X \hookrightarrow X \times X$. Then, for all $\alpha, \beta, \gamma \in A^*(X)$:

1. the product is biadditive and $A^p(X) \cdot A^q(X) \subseteq A^{p+q}(X)$;
2. $\alpha \cdot \beta = \beta \cdot \alpha$;
3. $(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma)$;
4. $[X] \cdot \alpha = \alpha$.

Thus $A^*(X)$ is a commutative graded ring with unit $[X]$.

***Proof*:**

1. Biadditivity is proposition 5.1.9(1). For the degrees, let $\alpha \in A^p(X) = A_{n-p}(X)$ and $\beta \in A^q(X) = A_{n-q}(X)$. Their exterior product lies in $A_{2n-p-q}(X \times X)$, and $\delta$ is a regular embedding of codimension $n$ (theorem 5.1.6), so $\delta^!$ lowers dimension by $n$ by proposition 4.3.2(1). The product therefore lies in $A_{n-p-q}(X) = A^{p+q}(X)$.
2. Let $\tau \colon X \times X \to X \times X$ exchange the two factors. It carries a product of subvarieties $V \times W$ isomorphically onto $W \times V$, so $\tau_*(\alpha \times \beta) = \beta \times \alpha$ by definition 1.2.1, the exchange inducing an isomorphism of function fields. Since $\tau$ is an involution fixing the diagonal pointwise,

$\tau^{-1}(\Delta) = \Delta$, which is regularly embedded in $X \times X$ with codimension $n$; proposition 5.2.6 therefore identifies $\delta^!_\tau$ with $\delta^!$. Applying theorem 4.3.10(1) with $X' = X \times X$ and the identity, $X'' = X \times X$ and $\tau$, and with $\tau$ itself as the morphism over $X \times X$, which is proper because it is an isomorphism, and noting that the induced morphism of preimages is $\tau|_\Delta = \mathrm{id}_\Delta$,

$$\delta^!(\tau_*\xi) \;=\; \delta^!_\tau(\xi) \;=\; \delta^!(\xi)$$

for every $\xi \in A_*(X \times X)$. With $\xi = \alpha \times \beta$ this reads $\beta \cdot \alpha = \alpha \cdot \beta$.

3. Write $\Xi = \alpha \times \beta \times \gamma \in A_*(X \times X \times X)$, which is unambiguous because the exterior product is associative on cycles, and let $\mathrm{pr}_{12}, \mathrm{pr}_{23} \colon X \times X \times X \to X \times X$ be the two projections. Their preimages of the diagonal are $W_1 = \Delta \times X$ and $W_2 = X \times \Delta$, each identified with $X \times X$ by the two projections that are not constrained; under those identifications $\mathrm{pr}_{23}|_{W_1}$ and $\mathrm{pr}_{12}|_{W_2}$ are the identity of $X \times X$, so the refined maps along them are $\delta^!$ by proposition 4.3.7. By lemma 5.2.5, applied to $i = \delta$ with $W = X$ on each side in turn,

$$(\alpha \cdot \beta) \times \gamma = \delta^!_{\mathrm{pr}_{12}}(\Xi), \qquad \alpha \times (\beta \cdot \gamma) = \delta^!_{\mathrm{pr}_{23}}(\Xi)$$

Applying $\delta^!$ to each and using corollary 4.4.8 with $i = i' = \delta$, $W = X \times X \times X$ and the two projections,

$$(\alpha \cdot \beta) \cdot \gamma = \delta^!\big(\delta^!_{\mathrm{pr}_{12}}\Xi\big) = \delta^!\big(\delta^!_{\mathrm{pr}_{23}}\Xi\big) = \alpha \cdot (\beta \cdot \gamma)$$

both sides being computed on $W_{12} = (\Delta \times X) \cap (X \times \Delta)$, the small diagonal, which the first projection identifies with $X$.

4. By additivity it suffices to take $\alpha = [V]$ for a $k$-dimensional subvariety $V \subseteq X$, and by part (2) it suffices to evaluate $[V] \cdot [X] = \delta^![V \times X]$, the exterior product of the two fundamental cycles being the fundamental cycle of the variety $V \times X$ of dimension $k + n$ (definition 5.1.2). Apply lemma 5.2.4(3) to the inclusion $f \colon V \hookrightarrow X$: the scheme-theoretic intersection $(V \times X) \cap \Delta$, which is the preimage of $\Delta$ under $f \times \mathrm{id}_X$, is the graph $\Gamma_f$, isomorphic to $V$ and regularly embedded in $V \times X$ with codimension $n$. Proposition 4.3.3 therefore gives $\delta^![V \times X] = \iota_*[\Gamma_f] = [V]$ in $A_k(\Delta) = A_k(X)$, where $\iota$ is the inclusion of the graph in the diagonal, which the projections identify with the inclusion of $V$ in $X$, completing the proof.

The ring exists, and its two ends are already familiar: $A^0(X) = \mathbb{Z}\,[X]$ because a variety has only itself as a subvariety of top dimension, and $A^1(X) = A_{n-1}(X)$ is the group of divisor classes on which chapter 2 built the operator $D \cdot -$ by hand. The operators of section 3.2 can now be recorded as elements: for a vector bundle $E$ on $X$ the class $c_i(E) \cap [X]$ lies in $A^i(X)$, so Chern classes become members of a ring rather than endomorphisms of a graded group, and an identity of operators may be read as an identity in $A^*(X)$ by evaluating both sides on $[X]$.

The grading also explains an absence. On $\mathbb{P}^3$ two distinct lines meeting in a point have classes in $A^2$, so their product lies in $A^4(\mathbb{P}^3) = 0$: the ring records no trace of the point in which they meet. The computation is not defective. The intersection is not proper, one of the two lines can be moved off the other within its rational equivalence class, and the invariant answer to “how much do they meet” is zero.

A morphism $f \colon X \to Y$ of smooth varieties does not act on cycles in either direction in any naive way, but its graph is a regular embedding by lemma 5.2.4(3), and intersecting with the graph is a pullback.

**Definition 5.2.8: Pullback Along a Morphism of Smooth Varieties**

Let $f\colon X \to Y$ be a morphism of smooth varieties, of dimensions $m$ and $n$, and let $\gamma_f\colon X \to X \times Y$ be its graph, a regular embedding of codimension $n$ by lemma 5.2.4(3); for $n = 0$, where $Y = \operatorname{Spec} \overline{k}$ and $\gamma_f$ is an isomorphism, $\gamma_f^!$ is flat pullback along it. The *pullback* along $f$ is

$$f^*\colon A_l(Y) \longrightarrow A_{l+m-n}(X), \qquad f^*\beta \;=\; \gamma_f^!\big([X] \times \beta\big)$$

Equivalently $f^*\beta = \gamma_f^!\big(\mathrm{pr}_Y^*\beta\big)$, since $\mathrm{pr}_Y\colon X \times Y \to Y$ is flat of relative dimension $m$ with $\mathrm{pr}_Y^*\beta = [X] \times \beta$.

In codimension the map preserves degree: $\beta \in A^q(Y)$ means $l = n - q$, and then $l + m - n = m - q$, so $f^*\beta \in A^q(X)$. The definition is the one the geometry suggests. To pull a cycle on $Y$ back to $X$, place it on $X \times Y$, where it is inert in the first factor, and intersect with the graph, which is the locus where the second coordinate is the image of the first.

**Example 5.2.9: Pullback to a Linear Subspace**

Let $i\colon \mathbb{P}^m \hookrightarrow \mathbb{P}^n$ be the linear subspace $\{x_0 = \cdots = x_{n-m-1} = 0\}$ with $1 \le m < n$, and write $h$ and $h'$ for the hyperplane classes of $\mathbb{P}^n$ and $\mathbb{P}^m$. On the chart where $x_n \neq 0$ in both factors, with affine coordinates $u_j$ on $\mathbb{P}^m$ for $n - m \le j \le n - 1$ and $v_j$ on $\mathbb{P}^n$ for $0 \le j \le n-1$, the graph $\Gamma_i = \{(z,z) \big| z \in \mathbb{P}^m\}$ is cut out of $\mathbb{P}^m \times \mathbb{P}^n$ by the $n$ equations

$$v_j \;\; (j < n - m), \qquad v_j - u_j \;\; (j \ge n - m)$$

Fix $1 \le p \le m$ and represent $h_{n-p}$ by $L = \{x_{n-m} = \cdots = x_{n-m+p-1} = 0\}$, a linear subspace of dimension $n - p$ meeting $\mathbb{P}^m$ in the linear subspace $\mathbb{P}^{m-p}$. Then $[\mathbb{P}^m] \times h_{n-p}$ is represented by the subvariety $\mathbb{P}^m \times L$, of dimension $m + n - p$, whose intersection with $\Gamma_i$ is $\{(z,z) \big| z \in \mathbb{P}^{m-p}\}$. On the chart above, the coordinates $v_{n-m}, \ldots, v_{n-m+p-1}$ vanish on $L$, so restricting the $n$ equations of the graph to $\mathbb{P}^m \times L$ leaves

$$v_j \;\; (j < n - m), \qquad u_j \;\; (n - m \le j < n - m + p), \qquad v_j - u_j \;\; (j \ge n - m + p)$$

again $n$ coordinates and coordinate differences in distinct variables, hence a regular sequence exactly as in example 5.2.3. The intersection is therefore regularly embedded of codimension $n$, and reduced and irreducible, so proposition 4.3.3 gives

$$i^*h_{n-p} \;=\; \big[\mathbb{P}^{m-p}\big] \;=\; h'_{m-p}$$

For $p > m$ the target $A_{m-p}(\mathbb{P}^m)$ vanishes and the pullback is zero. The restriction of the class of a linear subspace is the class of a linear subspace of the same codimension, which is what a subspace meeting $\mathbb{P}^m$ properly should give; in the notation of example 5.2.3 this reads $i^*(h^p) = (h')^p$ for $p \le m$.

**Theorem 5.2.10: Properties of Pullback**

Let $f\colon X \to Y$ be a morphism of smooth varieties, of dimensions $m$ and $n$.

1. For $\alpha \in A_*(X)$ and $\beta \in A_*(Y)$,
$$\gamma_f^!\big(\alpha \times \beta\big) \;=\; \alpha \cdot f^*\beta$$
and in particular $f^*[Y] = [X]$.
2. If $f$ is proper then $f_*\big(\alpha \cdot f^*\beta\big) = f_*\alpha \cdot \beta$ for all $\alpha \in A_*(X)$ and $\beta \in A_*(Y)$.
3. $f^*\colon A^*(Y) \to A^*(X)$ is a homomorphism of graded rings.
4. $\mathrm{id}_X^* = \mathrm{id}$, and $(g \circ f)^* = f^* \circ g^*$ for a further morphism $g\colon Y \to Z$ of smooth varieties.

***Proof*:**
Throughout, $\Gamma_f \subseteq X \times Y$ is the graph and $\gamma_f\colon X \to X \times Y$ the regular embedding of codimension $n$ onto it (lemma 5.2.4(3)).

1. By lemma 5.2.5, applied to $\gamma_f$ with an inert factor $X$ on the left,
$$\alpha \times f^*\beta \;=\; \alpha \times \gamma_f^!\big([X] \times \beta\big) \;=\; (\gamma_f)^!_{\mathrm{pr}_{23}}\big(\alpha \times [X] \times \beta\big)$$
where $\mathrm{pr}_{23}\colon X \times X \times Y \to X \times Y$ is the projection, whose preimage of $\Gamma_f$ is $X \times \Gamma_f$, identified with $X \times X$ by the first two projections. Applying $\delta^!$ and invoking corollary 4.4.8 for the regular embeddings $\delta$ and $\gamma_f$ with the projections $\mathrm{pr}_{12}$ and $\mathrm{pr}_{23}$,
$$\alpha \cdot f^*\beta \;=\; \delta^!\Big((\gamma_f)^!_{\mathrm{pr}_{23}}\big(\alpha \times [X] \times \beta\big)\Big) \;=\; (\gamma_f)^!_\rho\Big(\delta^!_{\mathrm{pr}_{12}}\big(\alpha \times [X] \times \beta\big)\Big)$$
On the left the outer map is refined along $\mathrm{pr}_{12}|_{X\times\Gamma_f}$, which is the identity of $X \times X$ under the identification above, so it is $\delta^!$ by proposition 4.3.7. On the right the preimage of $\Delta$ under $\mathrm{pr}_{12}$ is $\Delta \times Y$, identified with $X \times Y$, and $\rho$ is the restriction of $\mathrm{pr}_{23}$ to it, which is the identity of $X \times Y$; so the outer map is $\gamma_f^!$, again by proposition 4.3.7. The inner map is evaluated by lemma 5.2.5 and the unit law theorem 5.2.7(4):
$$\delta^!_{\mathrm{pr}_{12}}\big((\alpha \times [X]) \times \beta\big) = \big(\alpha \cdot [X]\big) \times \beta = \alpha \times \beta$$
which gives the stated identity. Taking $\alpha = [X]$ and $\beta = [Y]$, the left side is $\gamma_f^![X \times Y]$; the variety $X \times Y$ meets $\Gamma_f$ in $\Gamma_f$ itself, a regular embedding of codimension $n$, so proposition 4.3.3 gives $\gamma_f^![X \times Y] = [\Gamma_f] = [X]$. Hence $[X] \cdot f^*[Y] = [X]$, and $f^*[Y] = [X]$ by the unit law.
2. Let $f$ be proper. On cycles, $(f \times \mathrm{id}_Y)_*\big(\alpha \times \beta\big) = f_*\alpha \times \beta$: for subvarieties $V \subseteq X$ and $W \subseteq Y$ the image of $V \times W$ is $f(V) \times W$, and the field extension $k(f(V) \times W) \subseteq k(V \times W)$ is obtained from $k(f(V)) \subseteq k(V)$ by tensoring with $k(W)$ over $\overline{k}$ and passing to fraction fields, so the two degrees agree and definition 1.2.1 applies. Now $(f \times \mathrm{id}_Y)^{-1}(\Delta_Y) = \Gamma_f$ by lemma 5.2.4(3), and $\Gamma_f$ is regularly embedded of codimension $n$, so proposition 5.2.6 identifies $(\delta_Y)^!_{f\times\mathrm{id}}$ with $\gamma_f^!$. Applying theorem 4.3.10(1) with $X' = Y \times Y$ and the identity,

$X'' = X \times Y$ and $f \times \mathrm{id}_Y$, and with $f \times \mathrm{id}_Y$ as the proper morphism over $Y \times Y$, whose induced morphism of preimages is $f$ itself,

$$f_*\alpha \cdot \beta \;=\; \delta_Y^!\big((f \times \mathrm{id}_Y)_*(\alpha \times \beta)\big) \;=\; f_*\Big(\gamma_f^!\big(\alpha \times \beta\big)\Big) \;=\; f_*\big(\alpha \cdot f^*\beta\big)$$

the last step by part (1).

3. Degrees were checked after definition 5.2.8, additivity is clear from the definition, and $f^*[Y] = [X]$ is part (1). For multiplicativity, let $\beta, \beta' \in A_*(Y)$ and put $\Xi = [X] \times \beta \times \beta' \in A_*(X \times Y \times Y)$. By lemma 5.2.5,

$$[X] \times \big(\beta \cdot \beta'\big) = (\delta_Y)^!_{\mathrm{pr}_{23}}(\Xi), \qquad \big(f^*\beta\big) \times \beta' = (\gamma_f)^!_{\mathrm{pr}_{12}}(\Xi)$$

with $\mathrm{pr}_{23} \colon X \times Y \times Y \to Y \times Y$ and $\mathrm{pr}_{12} \colon X \times Y \times Y \to X \times Y$. Applying $\gamma_f^!$ to the first identity gives $f^*(\beta \cdot \beta')$, the map being refined along $\mathrm{pr}_{12}|_{X \times \Delta_Y}$, which is the identity of $X \times Y$. Corollary 4.4.8, for the regular embeddings $\gamma_f$ and $\delta_Y$ with the projections $\mathrm{pr}_{12}$ and $\mathrm{pr}_{23}$, exchanges the order:

$$f^*\big(\beta \cdot \beta'\big) = \gamma_f^!\Big((\delta_Y)^!_{\mathrm{pr}_{23}}\Xi\Big) = (\delta_Y)^!_{\psi}\Big((\gamma_f)^!_{\mathrm{pr}_{12}}\Xi\Big)$$

where $\psi \colon \Gamma_f \times Y \to Y \times Y$ is the restriction of $\mathrm{pr}_{23}$. The first and third projections identify $\Gamma_f \times Y$ with $X \times Y$, and under that identification $\psi$ is $f \times \mathrm{id}_Y$, whose preimage of $\Delta_Y$ is $\Gamma_f$; so $(\delta_Y)^!_\psi = \gamma_f^!$ by proposition 5.2.6. Hence

$$f^*\big(\beta \cdot \beta'\big) = \gamma_f^!\big(f^*\beta \times \beta'\big) = f^*\beta \cdot f^*\beta'$$

the last equality by part (1) with $\alpha = f^*\beta$.

4. For $f = \mathrm{id}_X$ the graph is the diagonal, and part (1) with $\alpha = [X]$ gives $\mathrm{id}^*\beta = [X] \cdot \beta = \beta$. For the composite, let $\zeta \in A_*(Z)$ and put $\Theta = [X \times Y] \times \zeta \in A_*(X \times Y \times Z)$. By lemma 5.2.5 and $[X \times Y] = [X] \times [Y]$,

$$[X] \times g^*\zeta = [X] \times \gamma_g^!\big([Y] \times \zeta\big) = (\gamma_g)^!_{\mathrm{pr}_{23}}(\Theta)$$

with $\mathrm{pr}_{23} \colon X \times Y \times Z \to Y \times Z$, so $f^*g^*\zeta = \gamma_f^!\big((\gamma_g)^!_{\mathrm{pr}_{23}}\Theta\big)$, the outer map being read on $X \times \Gamma_g \cong X \times Y$. Corollary 4.4.8, for the regular embeddings $\gamma_f$ and $\gamma_g$ with the projections $\mathrm{pr}_{12}$ and $\mathrm{pr}_{23}$, exchanges the order, giving $(\gamma_g)^!_\rho\big((\gamma_f)^!_{\mathrm{pr}_{12}}\Theta\big)$ with $\rho \colon \Gamma_f \times Z \to Y \times Z$ the restriction of $\mathrm{pr}_{23}$. By lemma 5.2.5 and part (1),

$$(\gamma_f)^!_{\mathrm{pr}_{12}}\Theta = \gamma_f^![X \times Y] \times \zeta = [X] \times \zeta$$

Identifying $\Gamma_f \times Z$ with $X \times Z$, the morphism $\rho$ becomes $f \times \mathrm{id}_Z$, whose preimage of $\Gamma_g$ is $\{(x,z) \big| z = g(f(x))\} = \Gamma_{g \circ f}$, regularly embedded of codimension $\dim Z$ by lemma 5.2.4(3). So proposition 5.2.6 identifies $(\gamma_g)^!_\rho$ with $\gamma^!_{g \circ f}$, and

$$f^*g^*\zeta \;=\; \gamma^!_{g \circ f}\big([X] \times \zeta\big) \;=\; (g \circ f)^*\zeta$$

completing the proof.

Part (1) says that intersecting with the graph is the same as pulling back and then multiplying, which is what makes part (2) a statement about a ring rather than about a Gysin map. The projection formula is the identity used most often downstream: it converts a computation on $X$ into a computation on $Y$ whenever a class on $X$ arose by restriction, and it is how an intersection number computed on a subvariety is compared with one computed on the ambient variety.

Smoothness of the source is more than the construction needs. What was used about $X$ is that its graph in $X \times Y$ is regularly embedded, and by lemma 5.2.4(3) that holds for any algebraic scheme mapping to a smooth variety. Dropping smoothness of the target as well requires a hypothesis on the morphism instead: it must factor as a regular embedding followed by a smooth morphism.

**Definition 5.2.11: Gysin Pullback for a Local Complete Intersection Morphism**

Throughout this definition $f$ is assumed to admit a *global* factorization $f = p \circ i$ with $i$ a regular embedding and $p$ smooth. The upstream definition ([2, Local Complete Intersection Morphism]) requires such a factorization only locally on the source, which is strictly weaker and does not suffice to fix a single $i$ and $p$; the global hypothesis is Fulton's convention and is what the construction below needs. Let $f\colon X \to Y$ be a local complete intersection morphism of algebraic schemes ([2, Local Complete Intersection Morphism]) which admits a factorization on all of $X$, and fix one: $f = p \circ i$ with $i\colon X \hookrightarrow P$ a regular embedding of codimension $d$ and $p\colon P \to Y$ smooth of relative dimension $e$. The *Gysin pullback* of $f$ is

$$f^! \;=\; i^! \circ p^* \;:\; A_k(Y) \longrightarrow A_{k+e-d}(X)$$

where $p^*$ is flat pullback (definition 1.3.3) and $i^!$ is the Gysin map of definition 4.3.1, read for $d = 0$ as flat pullback along $i$, which is then an open and closed immersion. The integer $e - d$ is the *virtual relative dimension* of $f$.

The factorization on all of $X$ is a hypothesis, not a consequence. The definition just cited asks only that every point of $X$ have a neighbourhood on which $f$ factors in this way, and local factorizations need not be glued into one on all of $X$; without one there is no $p^*$ and no $i^!$ to compose, and $f^!$ is not defined. Wherever $f^!$ is written a factorization on all of $X$ is therefore part of the data. The cases used below carry an evident one: a regular embedding and a smooth morphism are their own factorizations, and a morphism of smooth varieties is factored by its graph.

**Theorem 5.2.12: The Gysin Pullback Is Well Defined**

Let $f\colon X \to Y$ be a local complete intersection morphism admitting a factorization on all of $X$.

1. $f^!$ and the virtual relative dimension $e - d$ are independent of the chosen factorization.
2. If $f$ is smooth of relative dimension $e$ then $f^!$ is flat pullback, and if $f$ is a regular embedding then $f^!$ is its Gysin map.
3. If $X$ and $Y$ are smooth varieties then every morphism $f\colon X \to Y$ meets the hypotheses above, its graph supplying a factorization on all of $X$, and $f^!$ is the pullback $f^*$ of definition 5.2.8.

***Proof*:**

1. Let $f = p \circ i = p' \circ i'$ be two factorizations, with data $(P, i, p, d, e)$ and $(P', i', p', d', e')$. Form $P'' = P \times_Y P'$ with projections $q \colon P'' \to P$ and $q' \colon P'' \to P'$, and set $p'' = p \circ q = p' \circ q'$. Being base changes of $p'$ and $p$, the morphisms $q$ and $q'$ are smooth of relative dimensions $e'$ and $e$, so $p''$ is smooth of relative dimension $e + e'$. It suffices to prove $i^! \circ p^* = (i'')^! \circ (p'')^*$ for $i'' = (i, i') \colon X \to P''$, since the same argument with the roles of the two factorizations exchanged gives $i'^! \circ p'^* = (i'')^! \circ (p'')^*$.

   *Step 1: factoring $i''$.* Write $W_1 = q^{-1}(X) = X \times_Y P'$, a closed subscheme of $P''$, and let $\tilde{i} \colon W_1 \hookrightarrow P''$ be its inclusion; since $q$ is flat, lemma 5.2.4(2) makes $\tilde{i}$ a regular embedding of codimension $d$. The projection $\bar{q} \colon W_1 \to X$ is the base change of $p'$ along $f$, hence smooth of relative dimension $e'$, and $\gamma = (\mathrm{id}_X, i') \colon X \to W_1$ is a section of it, hence a regular embedding of codimension $e'$ by lemma 5.2.4(1). As $i'' = \tilde{i} \circ \gamma$, theorem 4.4.7 gives $(i'')^! = \gamma^! \circ \tilde{i}^!$ and makes $i''$ a regular embedding of codimension $d + e'$.

   *Step 2: the comparison.* Let $\beta \in A_k(Y)$. Flat pullback is functorial by proposition 1.3.5(3), so $(p'')^*\beta = q^*\big(p^*\beta\big)$. Applying theorem 4.3.10(2) to the regular embedding $i$ with $X' = P$ and the identity, $X'' = P''$ and $q$, and with the flat $q$ as the morphism over $P$, and using proposition 5.2.6 to read the refined map along $q$ as $\tilde{i}^!$,

   $$\tilde{i}^!\big((p'')^*\beta\big) \;=\; \bar{q}^*\big(i^! p^*\beta\big)$$

   Since $\gamma$ is a section of the smooth morphism $\bar{q}$, theorem 4.4.2 gives $\gamma^! \circ \bar{q}^* = \mathrm{id}$, whence

   $$(i'')^!(p'')^*\beta \;=\; \gamma^! \bar{q}^*\big(i^! p^*\beta\big) \;=\; i^! p^*\beta$$

   *Step 3: the degree.* Step 1 exhibits $i''$ as a regular embedding of codimension $d + e'$, and the same step applied to the other factorization exhibits it as one of codimension $d' + e$. The codimension of a regular embedding is determined by the rank of its conormal sheaf, so $d + e' = d' + e$ and therefore $e - d = e' - d'$.

2. If $f$ is smooth, take $P = X$ with $i = \mathrm{id}_X$, a regular embedding of codimension zero, and $p = f$; then $f^! = \mathrm{id}^* \circ f^* = f^*$. If $f$ is a regular embedding, take $P = Y$ with $p = \mathrm{id}_Y$, smooth of relative dimension zero, and $i = f$; then $f^!$ is the Gysin map of $f$. Part (1) makes both readings legitimate.

3. Let $X$ and $Y$ be smooth of dimensions $m$ and $n$. Factor $f$ as

   $$X \xrightarrow{\gamma_f} X \times Y \xrightarrow{\mathrm{pr}_Y} Y$$

   The first map is a regular embedding of codimension $n$ by lemma 5.2.4(3), and the second is smooth of relative dimension $m$, being the base change of $X \to \operatorname{Spec} \bar{k}$. So $f$ is a local complete intersection morphism with a factorization on all of $X$, and by part (1) its Gysin pullback may be computed from this factorization: $f^!\beta = \gamma_f^!\big(\mathrm{pr}_Y^*\beta\big)$, which is definition 5.2.8, completing the proof.

The virtual relative dimension is the bookkeeping device the name suggests: for a smooth morphism it is the actual relative dimension, for a regular embedding it is minus the codimension, and for a general local complete intersection morphism it is the difference of the two, which need not be realized by any fibre. That generality is what Grothendieck-Riemann-Roch is stated in, the morphisms there being proper local complete intersection morphisms between possibly singular schemes.

One classical statement remains to be placed. The construction in this chapter never moved a cycle; the ambient embedding was degenerated instead. The classical construction did the opposite, and it rested on the following.

**Theorem 5.2.13: The Moving Lemma**

Let $X$ be a smooth quasi-projective variety of dimension $n$, let $\alpha$ be a $k$-cycle on $X$ and $\beta$ an $l$-cycle on $X$. Then there is a $k$-cycle $\alpha'$ rationally equivalent to $\alpha$ such that $\alpha'$ and $\beta$ intersect properly: every irreducible component of $|\alpha'| \cap |\beta|$ has dimension $k+l-n$. Moreover the cycle class obtained by counting the components of $|\alpha'| \cap |\beta|$ with their intersection multiplicities depends only on the classes of $\alpha$ and $\beta$.

No proof is given, and none is used. The lemma is stated because it is what the intersection product looked like before chapter 4: a product defined for cycles in general position and extended to all pairs by first moving one of them, the whole construction resting on the second assertion above, that the answer does not depend on how the cycle was moved. That second assertion is the difficult half, and it is what makes the moving lemma a theorem rather than a definition; the argument goes back to Chow, and a complete treatment is in Fulton [7].

Two costs were paid on that route and are not paid here. The lemma requires quasi-projectivity, so the classical product exists only on quasi-projective varieties, whereas theorem 5.2.7 holds on every smooth variety. And a product defined by moving is defined only up to the choice made, so every property of it, commutativity included, has to be proved compatible with all choices. The deformation construction makes no choice: $\delta^!$ is a homomorphism on classes from the outset, and the proofs above use only its functorialities.

## Exercise 5.2.1

1. Let $L$ and $L'$ be distinct lines in $\mathbb{P}^3$ meeting in a point. Compute $[L]\cdot[L']$ in $A^*(\mathbb{P}^3)$ from the grading alone, and compute $[L]\cdot[H]$ for a plane $H$ containing neither line. Then explain why the first answer is not a defect of the theory, by exhibiting a cycle rationally equivalent to $[L]$ whose support misses $L'$.

2. Let $X$ be a smooth variety and let $V, W \subseteq X$ be subvarieties with $V \cap W = \emptyset$. Show that $[V]\cdot[W] = 0$ in $A^*(X)$, working from the definition of the product rather than from the grading.

3. Let $f\colon X \to Y$ be a proper morphism of smooth varieties of the same dimension $n$, with $\deg f$ the degree of definition 1.2.1. Show that $f_*\big(f^*\beta\big) = (\deg f)\,\beta$ for every $\beta \in A^*(Y)$, and deduce that $f^*$ is injective on $A^*(Y)\otimes\mathbb{Q}$ when $\deg f \neq 0$.

4. Let $p\colon P \to Y$ be smooth of relative dimension $e$ and let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$. Show that $p\circ i$ is a local complete intersection morphism of virtual relative dimension $e-d$. Then take $Y = \operatorname{Spec}\overline{k}$, $P = \mathbb{P}^n$ and $Z = \mathbb{P}^{n-d}$ a linear subspace, and compute the image of the generator of $A_0(\operatorname{Spec}\overline{k})$ under $(p\circ i)^!$.

5. Let $i\colon \mathbb{P}^2 \hookrightarrow \mathbb{P}^3$ be a plane and let $D \subseteq \mathbb{P}^3$ be a surface of degree $r$ not containing it, so that $[D] = r\,h$ in $A^1(\mathbb{P}^3)$ by example 1.5.4. Compute $i^*[D]$ in $A^1(\mathbb{P}^2)$, and verify the projection

formula $i_*\big(i^*[D]\big) = [D] \cdot [\mathbb{P}^2]$ by computing both sides in $A^*(\mathbb{P}^3)$.

## 5.3 Self-Intersection and Excess

The Gysin map returns the fundamental cycle of the intersection scheme when the intersection is as proper as it can be (proposition 4.3.3). At the opposite extreme, when the cycle being intersected already lies inside $Z$, nothing proved so far evaluates it at all: the second half of example 4.3.8 left exactly that computation open, and for the same reason proposition 4.3.5 reached only the first clause of definition 2.2.1. This section closes the gap. The answer is the top Chern class of the normal bundle, and the statement interpolating between the two extremes, in which the intersection has the wrong dimension by a prescribed amount, is the excess intersection formula.

Everything below is stated for the Gysin map of a regular embedding, so by section 5.1 it is at the same time a statement about the product: on a smooth variety the product is $\delta^!$ of an exterior product, and $\delta$ is a regular embedding like any other. The formulas are then spent on blow-ups, which is where a subvariety is forced to meet the locus it is intersected with in the worst possible way.

Throughout, $i\colon Z \hookrightarrow X$ is a regular embedding of codimension $d \geq 1$ of algebraic schemes, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ and normal bundle $\pi\colon N \to Z$.

The reduction to a single computation is immediate from the construction. Apply definition 4.3.6 to the morphism $g = i$ itself: the preimage of $Z$ in $Z$ is $Z$, so the normal cone in question is $C_Z Z$, the zero cone of example 3.4.2(2), and the immersion of that cone into $N$ is the vertex, that is the zero section. Specialization is therefore the identity, and the whole matter is the identification of the class of the zero section inside the Chow group of the total space of the bundle.

**Lemma 5.3.1: The Class of the Zero Section**

Let $\pi\colon N \to Z$ be a vector bundle of rank $d \geq 1$ on an algebraic scheme $Z$, with zero section $\zeta\colon Z \to N$. Then for every $\alpha \in A_k(Z)$,

$$\zeta_*\alpha \;=\; \pi^*\big(c_d(N) \cap \alpha\big) \quad \text{in } A_k(N)$$

equivalently $(\pi^*)^{-1}\zeta_*\alpha = c_d(N) \cap \alpha$, the top Chern class of $N$ applied to $\alpha$.

***Proof*:**
Write $P = \mathbb{P}(N \oplus 1)$ for the projective completion of definition 3.4.3, where $N \oplus 1 = N \oplus \mathcal{O}_Z$ has rank $d+1$, with projection $p\colon P \to Z$ and tautological class $\eta = c_1(\mathcal{O}_P(1))$. Let $u\colon N \hookrightarrow P$ be the open immersion, so that $p \circ u = \pi$, and let $\bar{\zeta} = u \circ \zeta\colon Z \to P$, a section of $p$ and hence a closed immersion. Fix $\alpha \in A_k(Z)$ and put $\beta = \bar{\zeta}_*\alpha \in A_k(P)$.

*Step 1: two trivializations.* Writing $S^\bullet = \mathrm{Sym}^\bullet(N^\vee)$, the completion is $P = \mathrm{Proj}\,\big(S^\bullet[z]\big)$ with $z$ the degree-one variable dual to the trivial summand, and $N$ is the chart $D_+(z)$. On that chart the sheaf $\mathcal{O}_P(1)$ is free on the generator $z$, so

$$\mathcal{O}_P(1)\big|_N \;\cong\; \mathcal{O}_N$$

Since $\bar{\zeta}$ factors through $N$, it follows that $\bar{\zeta}^*\mathcal{O}_P(1) \cong \mathcal{O}_Z$ as well. Finally, the exact sequence $0 \to N \to N \oplus 1 \to \mathcal{O}_Z \to 0$ and theorem 3.2.15 give $c(N \oplus 1) = c(N)\,c(\mathcal{O}_Z) = c(N)$, the total Chern class of a trivial bundle being the identity (example 3.2.12(1)); in particular $c_i(N \oplus 1) = c_i(N)$ for every $i$, and this vanishes for $i > d$ by theorem 3.2.9(2).

*Step 2: $\eta$ annihilates $\beta$.* The morphism $\bar{\zeta}$ is proper, being a closed immersion, so the projection formula (proposition 2.4.4(4)) and Step 1 give

$$\eta \cap \bar{\zeta}_*\alpha \;=\; \bar{\zeta}_*\Big(c_1\big(\bar{\zeta}^*\mathcal{O}_P(1)\big) \cap \alpha\Big) \;=\; \bar{\zeta}_*\big(c_1(\mathcal{O}_Z) \cap \alpha\big) \;=\; 0$$

the last equality because $c_1$ of a trivial line bundle is the zero operator (example 2.4.3).

*Step 3: the top coefficient.* The bundle $N \oplus 1$ has rank $d+1$, so theorem 3.1.9 writes $\beta$ uniquely as

$$\beta \;=\; \sum_{j=0}^{d} \eta^j \cap p^*\alpha_j, \qquad \alpha_j \in A_{k-d+j}(Z) \tag{5.1}$$

By definition 3.1.3 applied to $N \oplus 1$, whose projective bundle has relative dimension $d$, one has $p_*\big(\eta^j \cap p^*\gamma\big) = s_{j-d}(N \oplus 1) \cap \gamma$, which vanishes for $j < d$ and is $\gamma$ for $j = d$ by proposition 3.1.5(1),(2). Applying $p_*$ to (5.1) therefore gives $p_*\beta = \alpha_d$. On the other hand $p \circ \bar{\zeta} = \mathrm{id}_Z$, so functoriality of proper pushforward (definition 1.2.1) gives $p_*\beta = \alpha$. Hence $\alpha_d = \alpha$.

*Step 4: the remaining coefficients.* Theorem 3.2.9(1) applied to $N \oplus 1$, together with the identification of its Chern classes in Step 1, reads

$$\sum_{i=0}^{d} \eta^{d+1-i} \cap p^*\big(c_i(N) \cap \gamma\big) \;=\; 0$$

for every $\gamma \in A_*(Z)$, so that $\eta^{d+1} \cap p^*\gamma = -\sum_{i=1}^{d} \eta^{d+1-i} \cap p^*(c_i(N) \cap \gamma)$. Capping (5.1) with $\eta$, using Step 2 on the left and this relation on the term $j = d$, and reindexing the resulting sum by $j = d - i$,

$$0 \;=\; \sum_{j=0}^{d-1} \eta^{j+1} \cap p^*\alpha_j \;-\; \sum_{j=0}^{d-1} \eta^{j+1} \cap p^*\big(c_{d-j}(N) \cap \alpha_d\big) \;=\; \sum_{m=1}^{d} \eta^m \cap p^*\big(\alpha_{m-1} - c_{d-m+1}(N) \cap \alpha_d\big)$$

The exponents occurring lie in $\{0, 1, \ldots, d\}$, so this is a representation of the zero class in the form of theorem 3.1.9, and uniqueness forces every coefficient to vanish: $\alpha_{m-1} = c_{d-m+1}(N) \cap \alpha_d$ for $1 \le m \le d$. Taking $m = 1$ and using Step 3,

$$\alpha_0 \;=\; c_d(N) \cap \alpha$$

*Step 5: restriction to the bundle.* The open immersion $u$ is flat of relative dimension zero. The square with $\bar{\zeta}$ and $\zeta$ over $u$ is a fibre square, $\bar{\zeta}$ factoring through $N$, so proposition 1.3.6 gives $u^*\beta = \zeta_*\alpha$. Applying $u^*$ to (5.1), every term with $j \ge 1$ contains a factor $\eta \cap -$, whose pullback is $c_1(\mathcal{O}_P(1)|_N) \cap u^*(-) = 0$ by proposition 2.4.4(4) and Step 1; and the term $j = 0$ is $u^*p^*\alpha_0 = \pi^*\alpha_0$ by proposition 1.3.5(3). Hence $\zeta_*\alpha = \pi^*\alpha_0 = \pi^*(c_d(N) \cap \alpha)$, as we sought to show.

The zero section meets every fibre of $N$ in exactly one point, and yet the class it carries is not the pullback of $\alpha$ but the pullback of $\alpha$ cut down by the top Chern class. On a trivial bundle of rank $d \ge 1$ the top Chern class vanishes and the zero section therefore carries the zero class, and the same holds whenever $N$ admits a nowhere-vanishing section: such a section spans a trivial line subbundle with locally free quotient $Q$ of rank $d-1$, so theorem 3.2.15 gives $c(N) = c(Q)$ and $c_d(N) = 0$, no bundle carrying a nonzero Chern class above its rank. Nonvanishing of $\zeta_*\alpha$ is therefore an obstruction to the existence of such a section and not a criterion for one. On $\mathbb{P}^1$ the bundle $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(1)$ has $c_2$ the zero operator, the groups $A_{k-2}(\mathbb{P}^1)$ vanishing for every $k \le 1$, and yet $\mathcal{O}_{\mathbb{P}^1}(-1)$ has no nonzero section, so every section of the sum vanishes wherever its second coordinate does. Since $\pi^*$ is an isomorphism (theorem 1.4.6), the identity is equivalent to the evaluation of $(\pi^*)^{-1}$ on classes supported on the zero section, and that is what the Gysin map of chapter 4 was waiting for.

**Theorem 5.3.2: The Self-Intersection Formula**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \ge 1$ with normal bundle $N$. Then for every $\alpha \in A_k(Z)$:

1. the refined Gysin homomorphism of $i$ along $i$ itself is the operator $c_d(N) \cap -$, that is $i^!_i\alpha = c_d(N) \cap \alpha$ in $A_{k-d}(Z)$, in the notation of box 5.2.1;
2. the Gysin map of definition 4.3.1 satisfies

$$i^!\big(i_*\alpha\big) \;=\; c_d(N) \cap \alpha \quad \text{in } A_{k-d}(Z)$$

***Proof*:**

1. Apply definition 4.3.6 with $X' = Z$ and $g = i$. The ideal sheaf of $Z$ in $Z$ is zero, so the preimage $Z' = i^{-1}(Z)$ is $Z$ itself, $h = \mathrm{id}_Z$ and $N' = h^*N = N$. The associated graded algebra of the zero ideal is $\mathcal{O}_Z$ concentrated in degree zero, so $C_{Z'}X' = C_Z Z$ is the zero cone of example 3.4.2(2), whose total space is $Z$; and the immersion $j\colon C_Z Z \hookrightarrow N$ of lemma 3.4.13 is induced by the surjection of graded algebras $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \to \mathcal{O}_Z$ killing every positive degree, which is the vertex of definition 3.4.1, that is the zero section $\zeta$. For the specialization homomorphism $\sigma'\colon A_k(Z) \to A_k(C_Z Z) = A_k(Z)$ of the pair $(Z, Z)$, theorem 4.2.7 gives $\sigma'[V] = [C_V V] = [V]$ for every $k$-dimensional subvariety $V \subseteq Z$, the preimage of $Z$ in $V$ being $V$. Such classes generate $A_k(Z)$ (definition 1.1.4), so $\sigma'$ is the identity. Therefore

$$i^!_i\alpha \;=\; (\pi^*)^{-1} j_*\sigma'\alpha \;=\; (\pi^*)^{-1}\zeta_*\alpha \;=\; c_d(N) \cap \alpha$$

   the last equality by lemma 5.3.1.

2. Apply theorem 4.3.10(1) with $X' = X$, $X'' = Z$ and $p = i$, which is proper as a closed immersion. The preimage of $Z$ in $Z$ is $Z$ and the induced morphism $q$ is $\mathrm{id}_Z$, so the identity reads $i^!(i_*\alpha) = i^!_i\alpha$, where the left-hand side is the refined map along $\mathrm{id}_X$, which is the Gysin map of definition 4.3.1 by proposition 4.3.7, and the right-hand side is the refined map along $i$ evaluated in part (1). Combining the two gives $i^!(i_*\alpha) = c_d(N) \cap \alpha$, completing the proof.

The theorem says what it means to intersect a subvariety with itself. Pushing $\alpha$ into $X$ and pulling it back records no information about how $Z$ sits inside $X$ except through the normal bundle, and the whole of that information is the single operator $c_d(N)$. Two consequences are worth separating. The composite $i^! i_*$ is not the identity, and it is not even injective: on a subvariety with trivial normal bundle it is zero. And the sign of a self-intersection number, which has no counterpart in a naive count of intersection points, is the sign of a Chern number of $N$; this is why the exceptional curve of a blow-up can meet itself a negative number of times. Since the Gysin pullback of an l.c.i. morphism restricts to the Gysin map on a regular embedding (theorem 5.2.12(2)), part (2) is at the same time a statement about the pullback of definition 5.2.11: for the l.c.i. morphism $i$ the composite $i^! \circ i_*$ is the operator $c_d(N) \cap -$.

In codimension one the theorem completes the comparison begun in proposition 4.3.5, which matched the Gysin map with the operator $D \cdot -$ of chapter 2 only for cycles not contained in the support of $D$.

**Corollary 5.3.3: The Gysin Map on a Divisor Class**

Let $D$ be an effective Cartier divisor on an algebraic scheme $X$ with $D \neq X$, and let $i\colon D \hookrightarrow X$ be the inclusion. Then for every $\alpha \in A_k(D)$,

$$i^!(i_*\alpha) \;=\; c_1\big(\mathcal{O}_X(D)|_D\big) \cap \alpha$$

In particular, for a $k$-dimensional subvariety $V \subseteq D$ the class $i^![V]$ is the image in $A_{k-1}(D)$ of the intersection class $D \cdot [V]$ computed by the second clause of definition 2.2.1.

***Proof*:**
The inclusion of an effective Cartier divisor is a regular embedding of codimension one with normal bundle $N = \mathcal{O}_X(D)|_D$ (example 3.4.15), and $c_1(N)$ is the top Chern class of a line bundle, so theorem 5.3.2(2) gives the displayed identity. For the second assertion, let $V \subseteq D$ be a $k$-dimensional subvariety with inclusion $\iota_V$ into $D$. Since $V$ is contained in $|D|$, the second clause of definition 2.2.1 computes $D \cdot [V] = c_1(\mathcal{O}_X(D)|_V) \cap [V]$ in $A_{k-1}(V)$. Pushing that class forward along $\iota_V$ and applying the projection formula (proposition 2.4.4(4)) to the proper morphism $\iota_V$, whose pullback of $\mathcal{O}_X(D)|_D$ is $\mathcal{O}_X(D)|_V$, gives $\iota_{V*}(D \cdot [V]) = c_1(\mathcal{O}_X(D)|_D) \cap [V]$, which is $i^!(i_*[V])$ by the first assertion, as we sought to show.

**Example 5.3.4: A Line Meeting Itself in the Plane**

Let $X = \mathbb{P}^2$ and $L = \{x_0 = 0\}$, an effective Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(L) \cong \mathcal{O}_{\mathbb{P}^2}(1)$ (example 2.1.4), so that $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L \cong \mathcal{O}_{\mathbb{P}^1}(1)$ (example 4.1.7). The second half of example 4.3.8 produced the class $i_i^![L] = (\pi^*)^{-1}\zeta_*[L]$, the refined Gysin map along $i$ applied to $[L]$, and could go no further. Theorem 5.3.2(1) evaluates it:

$$i_i^![L] \;=\; c_1\big(\mathcal{O}_{\mathbb{P}^1}(1)\big) \cap [L] \;=\; [\mathrm{pt}]$$

by example 2.4.7, a single point of $L$ with multiplicity one. This is the value predicted there from example 2.2.3, where $L \cdot [L] = [q]$ was computed from the restricted line bundle, and corollary 5.3.3 identifies the two computations in general rather than in this instance.
For a smooth conic $C \subseteq \mathbb{P}^2$ the same argument gives $N = \mathcal{O}_{\mathbb{P}^2}(2)|_C$, and example 3.4.15 evaluated $c_1(\mathcal{O}_{\mathbb{P}^2}(2)|_C) \cap [C] = 4\,[\mathrm{pt}]$, so $i^!(i_*[C]) = 4\,[\mathrm{pt}]$. The self-intersection number of a plane curve of degree $m$ is $m^2$, and the Gysin map records it on the curve itself rather than in $A_0(\mathbb{P}^2)$.

The examples so far are divisors, where the normal bundle is a line bundle and $c_d(N)$ is the operator of chapter 2. The construction that supplies higher codimension, and with it the geometry the formula was built for, is the blow-up: its exceptional divisor is a projectivized normal bundle, and it meets itself in every point.

**Lemma 5.3.5: The Exceptional Divisor of a Blow-up Along a Regular Centre**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with $Z \neq \emptyset$, ideal sheaf $\mathcal{I}$ and normal bundle $N$. Let $f\colon \widetilde{X} = \mathrm{Bl}_Z X \to X$ be the blow-up ([2, Blow-up]) and let $\widetilde{Z} = f^{-1}(Z)$ be its exceptional divisor, the closed subscheme cut out by $\mathcal{I}' = \mathcal{I}\mathcal{O}_{\widetilde{X}}$. Then:

1. $\widetilde{Z} = \mathbb{P}(N)$, with the restriction $g\colon \mathbb{P}(N) \to Z$ of $f$ as its bundle projection;
2. $j\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ is a regular embedding of codimension one whose conormal surjection $g^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is the universal surjection $g^*(N^\vee) \to \mathcal{O}_{\mathbb{P}(N)}(1)$ of box 3.2.5. In particular the normal bundle of $j$ is $\mathcal{O}_{\mathbb{P}(N)}(-1)$, sitting inside $g^*N$ as the tautological subbundle with quotient the universal quotient bundle $Q$ of rank $d-1$.

***Proof*:**

Realize the Rees algebra as the subalgebra $S = \bigoplus_{n\geq 0} \mathcal{I}^n t^n$ of $\mathcal{O}_X[t]$, so that $\widetilde{X} = \mathrm{Proj}\,(S)$ over $X$ by [2, Blow-up]. The algebra $S$ is generated in degree one, so the affine charts $D_+(at)$ attached to local sections $a$ of $\mathcal{I}$ cover $\widetilde{X}$; on such a chart $\widetilde{X}$ is $\mathrm{Spec}\,(S_{(at)})$, whose elements are the fractions $y/a^n$ with $y$ a local section of $\mathcal{I}^n$.

1. The homogeneous ideal $\mathcal{I}S \subseteq S$ has degree-$n$ part $\mathcal{I} \cdot \mathcal{I}^n t^n = \mathcal{I}^{n+1} t^n$, so
$$S/\mathcal{I}S \;=\; \bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$$
which is $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$ by lemma 3.4.10, glued over an affine cover as in the proof of proposition 3.4.11. Taking relative Proj of the surjection $S \to S/\mathcal{I}S$ exhibits $\mathrm{Proj}\left(\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)\right)$ as the closed subscheme of $\widetilde{X}$ cut out by the ideal generated by $\mathcal{I}$, that is by $\mathcal{I}'$, and its structure morphism to $Z$ is the restriction of $f$. Since $N$ has $(\mathcal{I}/\mathcal{I}^2)^\vee$ as its sheaf of sections ([2, Conormal Sheaf and Normal Bundle]), that relative Proj is $\mathbb{P}(N)$ in the convention of box 0.0.3.
2. *The ideal is invertible.* On the chart $D_+(at)$ the ideal $\mathcal{I}'$ is generated by the image of $a$, since every local section $y$ of $\mathcal{I}$ satisfies $y = a \cdot (y/a)$ with $y/a$ in $S_{(at)}$. That generator is a nonzerodivisor: if $a \cdot (y/a^n) = 0$ in $S_{(at)}$ with $y$ a local section of $\mathcal{I}^n$, then $(at)^m\, a\, y\, t^n = 0$ in $\mathcal{O}_X[t]$ for some $m \geq 0$, that is $a^{m+1}y = 0$, and the same relation forces $y/a^n = 0$ in $S_{(at)}$. Hence $\widetilde{Z}$ is an effective Cartier divisor and $j$ is a regular embedding of codimension one (example 3.4.15), which also recovers [2, Basic Properties of Blow-ups](3); its conormal sheaf is $\mathcal{I}'/\mathcal{I}'^2 = \mathcal{I}' \otimes \mathcal{O}_{\widetilde{Z}}$, invertible on $\widetilde{Z}$.
*Which invertible sheaf.* The ideal $\mathcal{I}'$ is generated by the image of $\mathcal{I}$, so the canonical map $g^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is surjective. Compare it on $D_+(at) \cap \widetilde{Z}$ with the universal surjection of $\mathbb{P}(N) = \mathrm{Proj}\,\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$. There $\mathcal{I}'/\mathcal{I}'^2$ is free on the class of $a$ by the previous paragraph, while $\mathcal{O}_{\mathbb{P}(N)}(1)$ is free on the image $\bar{a}$ of the degree-one element $at$. A local section $b$ of $\mathcal{I}$ satisfies $b = a \cdot (b/a)$ upstairs, and its class satisfies $\bar{b} = \bar{a} \cdot \overline{b/a}$ downstairs, the reduction $S \to S/\mathcal{I}S$ carrying $b/a$ to $\overline{b/a}$. Under the two trivializations the two surjections therefore have the same coefficient on every local section, so they agree and $\mathcal{I}'/\mathcal{I}'^2 = \mathcal{O}_{\mathbb{P}(N)}(1)$.

Dualizing gives the inclusion $\mathcal{O}_{\mathbb{P}(N)}(-1) \hookrightarrow g^*N$, which is the tautological subbundle of box 3.2.5, whose quotient $Q$ is locally free of rank $d-1$, completing the proof.

The lemma is the reason the blow-up is the standard source of examples for the self-intersection formula: the exceptional divisor is a divisor whose normal bundle is the tautological line bundle, with first Chern class $-\xi$ for the tautological class $\xi$ of section 3.1. In the smallest case the resulting number was computed by hand upstream, and the general formula reproduces it.

**Example 5.3.6: The Exceptional Curve of a Blown-up Surface**

Let $X = \mathbb{P}^2$ and let $Z = \{P\}$ with $P = [0:0:1]$. On the affine chart $\{x_2 \neq 0\}$ the ideal of $P$ is $(x_0/x_2, x_1/x_2)$, a regular sequence of length two in the polynomial ring of the chart, so $Z \hookrightarrow X$ is a regular embedding of codimension two whose normal bundle $N$ is a rank-two vector bundle over a point, that is a two-dimensional vector space. Write $f\colon \widetilde{X} = \mathrm{Bl}_P\mathbb{P}^2 \to \mathbb{P}^2$ for the blow-up, $\widetilde{Z}$ for its exceptional divisor, written $E$ in [2, Intersection Theory on the Blow-up], and $j\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ for its inclusion. By lemma 5.3.5,

$$\widetilde{Z} \;=\; \mathbb{P}(N) \;\cong\; \mathbb{P}^1, \qquad N_{\widetilde{Z}}\widetilde{X} \;=\; \mathcal{O}_{\mathbb{P}^1}(-1)$$

The self-intersection formula now evaluates the intersection of $\widetilde{Z}$ with itself. By corollary 5.3.3 and additivity of the first Chern class (proposition 2.4.4(3)),

$$j^!\big(j_*[\widetilde{Z}]\big) \;=\; c_1\big(\mathcal{O}_{\mathbb{P}^1}(-1)\big) \cap [\mathbb{P}^1] \;=\; -\,c_1\big(\mathcal{O}_{\mathbb{P}^1}(1)\big) \cap [\mathbb{P}^1] \;=\; -[\mathrm{pt}]$$

using example 2.4.7. Its degree (definition 1.2.4) is $-1$, which is the self-intersection number $E^2 = -1$ recorded upstream. The same computation applies at any point of any smooth surface, differing only in the verification that the maximal ideal at that point is generated by a regular sequence of length two; what the general formula contributes is that the number $-1$ is not an accident of the plane but the degree of the tautological line bundle on the projectivized normal space.

The negative sign is where the Gysin map departs furthest from counting points. A cycle contained in $Z$ has no intersection points with $Z$ to count that are not already in it, and the number the formula returns is a Chern number of the normal bundle, with whatever sign that bundle has. The upstream proof of $E^2 = -1$ argued from the surface pairing and the geometry of the blow-down; the argument here uses no property of surfaces and no property of blow-ups beyond lemma 5.3.5.

Between the proper case, where the answer is the fundamental cycle of $V \cap Z$, and the self-intersection case, where it is a top Chern class, lies everything else: the preimage $Z'$ has some dimension between the expected one and $\dim X'$, and the normal cone fills only part of the pulled-back normal bundle. When the preimage is itself regularly embedded the discrepancy is a vector bundle, and it is the only new input the general formula needs.

The comparison rests on the conormal sheaves. For a regular embedding of codimension $d'$ the conormal sheaf $\mathcal{I}'/\mathcal{I}'^2$ is locally free of rank $d'$ ([2, The Conormal Sheaf of a Regular Embedding]), and since $\mathcal{I}' = \mathcal{I}\mathcal{O}_{X'}$ is generated by the image of $\mathcal{I}$, the canonical map $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is a surjection of vector bundles. A surjection of vector bundles has locally free kernel, being locally split, so dualizing gives an inclusion of vector bundles with locally free cokernel; in particular $d' \leq d$.

**Definition 5.3.7: The Excess Bundle**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle $N$ and ideal sheaf $\mathcal{I}$, let $g\colon X' \to X$ be a morphism of algebraic schemes, and form the fibre square of definition 4.3.6, with $Z' = g^{-1}(Z)$, ideal sheaf $\mathcal{I}' = \mathcal{I}\mathcal{O}_{X'}$, and $h\colon Z' \to Z$. Suppose $i'\colon Z' \hookrightarrow X'$ is a regular embedding of codimension $d'$ with normal bundle $N'$. The dual of the conormal surjection $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is an inclusion of vector bundles $N' \hookrightarrow h^*N$ on $Z'$, and the *excess bundle* of the square is its cokernel

$$E \;=\; h^*N/N'$$

a vector bundle of rank $e = d - d'$ on $Z'$. When $e = 0$ the excess bundle is the zero bundle, and the operator $c_0(E)$ is then read as the identity, definition 3.2.3 being stated for bundles of positive rank.

The excess bundle measures how much of the normal directions to $Z$ fail to be normal directions to $Z'$ after base change. It vanishes exactly when $d' = d$, that is when the preimage has the expected codimension and the two normal bundles agree; at the other extreme, when $X' = Z$ and $g = i$, the subbundle $N'$ is zero and the excess bundle is all of $N$. Its rank is the number of dimensions by which the intersection is too large.

The identity that converts the excess bundle into a correction term is the following, which contains lemma 5.3.1 as the case $N' = 0$ and is deduced from it.

**Lemma 5.3.8: The Class of a Subbundle**

Let $0 \to N' \xrightarrow{j} N \to E \to 0$ be an exact sequence of vector bundles on an algebraic scheme $Z$, with $\operatorname{rank} E = e$ and $\operatorname{rank} N' = d'$, and write $\pi\colon N \to Z$ and $\pi'\colon N' \to Z$ for the projections. Then for every $\gamma \in A_m(Z)$,

$$j_*\pi'^*\gamma \;=\; \pi^*\big(c_e(E) \cap \gamma\big) \quad \text{in } A_{m+d'}(N)$$

***Proof*:**
If $e = 0$ then $j$ is an isomorphism of bundles and both sides are $\pi^*\gamma$, the operator $c_0(E)$ being the identity by the convention of definition 5.3.7. So assume $e \geq 1$, write $\pi_E\colon E \to Z$ for the projection and $\zeta_E\colon Z \to E$ for the zero section, and let $\rho\colon N \to E$ be the morphism of total spaces induced by the surjection $N \to E$.

*Step 1: the square is a fibre square with $\rho$ flat.* Both assertions are local on $Z$. Over an affine open on which $N'$, $N$ and $E$ are free, the surjection $N \to E$ splits, so there $N \cong N' \times E$ over the base with $\rho$ the second projection, which is flat of relative dimension $d'$, and the scheme-theoretic preimage $\rho^{-1}\big(\zeta_E(Z)\big)$ is $N'$ embedded by $j$. Hence $\rho$ is flat of relative dimension $d'$ and

$$\begin{tikzcd} N' \arrow[r, "j"] \arrow[d, "\pi'"'] & N \arrow[d, "\rho"] \\ Z \arrow[r, "\zeta_E"] & E \end{tikzcd}$$

is a fibre square, with proper horizontal maps and flat vertical maps.

*Step 2: transport the zero-section identity.* Proposition 1.3.6 applied to that square gives $\rho^*\zeta_{E*}\gamma = j_*\pi'^*\gamma$. By lemma 5.3.1 applied to the bundle $E$, the left-hand side is $\rho^*\pi_E^*(c_e(E)\cap\gamma)$, and $\pi_E \circ \rho = \pi$, so proposition 1.3.5(3) rewrites it as $\pi^*(c_e(E)\cap\gamma)$. Combining the two displays gives the assertion, completing the proof.

The refined Gysin map is built from a specialization followed by a pushforward into the pulled-back normal bundle, and when the preimage is regularly embedded the specialization is already the Gysin map of the preimage; the pushforward is then the subbundle inclusion, and lemma 5.3.8 evaluates it.

**Theorem 5.3.9: The Excess Intersection Formula**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle $N$, let $g\colon X' \to X$ be a morphism of algebraic schemes, and suppose the preimage $Z' = g^{-1}(Z)$ is regularly embedded in $X'$ of codimension $d' \geq 1$. Let $E$ be the excess bundle of definition 5.3.7, of rank $e = d - d'$. Then for every $\alpha \in A_k(X')$,

$$i_g^!\alpha \;=\; c_e(E)\cap i'^!\alpha \quad \text{in } A_{k-d}(Z')$$

where $i_g^!$ is the refined Gysin homomorphism of $i$ along $g$ and $i'^!$ is the Gysin map of $i'$, in the notation of box 5.2.1.

***Proof*:**
Write $\mathcal{I}$ and $\mathcal{I}'$ for the ideal sheaves of $Z$ in $X$ and of $Z'$ in $X'$, let $N' = N_{Z'}X'$ with projection $\pi'$, and let $\varpi\colon h^*N \to Z'$ be the projection of the pulled-back normal bundle.

*Step 1: the immersion of the cone is the subbundle inclusion.* By lemma 3.4.13 the closed immersion $j\colon C_{Z'}X' \hookrightarrow h^*N$ is induced by the canonical surjection of graded $\mathcal{O}_{Z'}$-algebras

$$\operatorname{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \longrightarrow \bigoplus_{n\geq 0}\mathcal{I}'^n/\mathcal{I}'^{n+1}$$

Since $i'$ is a regular embedding, lemma 3.4.10 identifies the target with $\operatorname{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ by the canonical surjection, which is the identity in degree one, and identifies $C_{Z'}X'$ with $N'$ (proposition 3.4.11). A homomorphism of graded algebras out of a symmetric algebra is determined by its degree-one part, so the displayed map is $\operatorname{Sym}^\bullet(\varphi)$ for $\varphi\colon h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ the conormal surjection. Taking relative spectra, $j\colon N' \to h^*N$ is the inclusion of vector bundles dual to $\varphi$, that is the inclusion of definition 5.3.7, with cokernel $E$ of rank $e$.

*Step 2: evaluation.* Let $\sigma'\colon A_k(X') \to A_k(C_{Z'}X') = A_k(N')$ be the specialization homomorphism of the pair $(Z', X')$. By definition 4.3.1 applied to $i'$, the Gysin map is $i'^! = (\pi'^*)^{-1}\circ\sigma'$, so $\sigma'\alpha = \pi'^*(i'^!\alpha)$. Hence, using definition 4.3.6 and then lemma 5.3.8 with $\gamma = i'^!\alpha$,

$$\varpi^*\big(i_g^!\alpha\big) \;=\; j_*\sigma'\alpha \;=\; j_*\pi'^*\big(i'^!\alpha\big) \;=\; \varpi^*\Big(c_e(E)\cap i'^!\alpha\Big)$$

The pullback $\varpi^*$ is injective, being an isomorphism by theorem 1.4.6, so $i_g^!\alpha = c_e(E)\cap i'^!\alpha$, as we sought to show.

The formula says that an intersection of the wrong dimension is computed by intersecting properly on the preimage and then correcting by the top Chern class of what was left over. When $e = 0$ the correction is the identity and the two Gysin maps agree, which is the regime of proposition 4.3.3:

the preimage has the expected codimension and the answer is read off it directly. When $e = d$, so that $d' = 0$ and $Z' = X'$, the hypothesis $d' \geq 1$ puts the case outside the statement, though only nominally: definition 5.2.11 reads the Gysin map of a codimension-zero embedding as flat pullback along it, which for $i' = \mathrm{id}_{X'}$ is the identity, so with that reading the uncorrected term is $\alpha$ itself, the correction is all of $c_d(h^*N)$, and for $g = i$ the formula returns theorem 5.3.2(1). The formula also explains the role of the normal cone in the general construction: without a regularity hypothesis on $Z'$ there is no excess bundle, and the discrepancy between $C_{Z'}X'$ and $h^*N$ is recorded by a Segre class instead.

**Example 5.3.10: A Plane Through a Line in Three-Space**

Let $X = \mathbb{P}^3$ with coordinates $x_0, \ldots, x_3$, let $Z = L = \{x_2 = x_3 = 0\}$ be a line and let $X' = H = \{x_3 = 0\}$ be a plane containing it, with $g \colon H \hookrightarrow \mathbb{P}^3$ the inclusion. By exercise 3.4.1 (2) the embedding $i$ is regular of codimension two with $N = \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 2}$. The expected dimension of $H \cap L$ is $\dim H - 2 = 0$, but the preimage is $Z' = g^{-1}(L) = L$, of dimension one.

*The excess bundle.* The line $L$ is an effective Cartier divisor on $H \cong \mathbb{P}^2$ cut out by $x_2$, so $i'$ is a regular embedding of codimension $d' = 1$ with $N' = \mathcal{O}_{\mathbb{P}^2}(1)|_L \cong \mathcal{O}_{\mathbb{P}^1}(1)$ (example 2.1.4, example 3.4.15). The excess bundle has rank $e = 2 - 1 = 1$, and the Whitney formula (theorem 3.2.15) applied to $0 \to N' \to N \to E \to 0$ gives, with $h = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$,

$$c(E) \;=\; c(N)\,c(N')^{-1} \;=\; (\mathrm{id} + h)^2(\mathrm{id} + h)^{-1} \;=\; \mathrm{id} + h$$

so $c_1(E) = h$.

*The class.* The plane $H$ is a variety not contained in $|L|$, and its scheme-theoretic intersection with $L$ is $L$, so $i'^![H] = [L]$ by proposition 4.3.5, which is the computation of exercise 4.3.1 (2). Theorem 5.3.9 then gives

$$i_g^![H] \;=\; c_1(E) \cap [L] \;=\; h \cap [L] \;=\; [\mathrm{pt}]$$

in $A_0(L)$, of degree one. The uncorrected answer $[L]$ has the wrong dimension and the wrong degree; the correction cuts it down by one dimension and returns the intersection number a plane and a line in general position would have. What the refined map adds is that the class is recorded on $L$, the locus where the two meet, and not only as a number.

Applying the formula to a blow-up produces the statement that lets Chow groups be computed after blowing up. The square in question is the defining one for the exceptional divisor, and lemma 5.3.5 has already identified all of its data.

**Theorem 5.3.11: The Blow-up Formula**

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with $Z \neq \emptyset$ and normal bundle $N$, let $f\colon \widetilde{X} = \mathrm{Bl}_Z X \to X$ be the blow-up with exceptional divisor $j\colon \widetilde{Z} = \mathbb{P}(N) \hookrightarrow \widetilde{X}$ and $g\colon \widetilde{Z} \to Z$, and let $Q$ be the universal quotient bundle of rank $d-1$ on $\mathbb{P}(N)$. Then for every $\beta \in A_k(\widetilde{X})$:

1. $i^!_f \beta = c_{d-1}(Q) \cap j^! \beta$ in $A_{k-d}(\widetilde{Z})$, where $i^!_f$ is the refined Gysin homomorphism of $i$ along $f$ and $j^!$ is the Gysin map of $j$;
2. $i^!(f_*\beta) = g_*\Big(c_{d-1}(Q) \cap j^!\beta\Big)$ in $A_{k-d}(Z)$, where the left-hand $i^!$ is the Gysin map of definition 4.3.1.

***Proof*:**

1. The exceptional divisor is the scheme-theoretic preimage $f^{-1}(Z)$, so the square formed by $j, f, i$ and $g$ is the square of definition 4.3.6 for the morphism $f$. By lemma 5.3.5(2) the preimage $\widetilde{Z}$ is regularly embedded in $\widetilde{X}$ of codimension $d' = 1$, and its conormal surjection, which is the one definition 5.3.7 dualizes, is there identified with the universal surjection; so the excess bundle of the square is the universal quotient $Q$, of rank $e = d-1$. Theorem 5.3.9 therefore gives $i^!_f \beta = c_{d-1}(Q) \cap j^! \beta$.
2. The blow-down $f$ is proper ([2, Basic Properties of Blow-ups](1)), so theorem 4.3.10(1) applies with $X' = X$, $X'' = \widetilde{X}$ and $p = f$. The preimage of $Z$ in $\widetilde{X}$ is $\widetilde{Z}$ and the induced morphism on preimages is $g$, so $i^!(f_*\beta) = g_*(i^!_f\beta)$, the left-hand side being the Gysin map of $i$ by proposition 4.3.7. Substituting part (1) completes the proof.

Part (1) computes an intersection with $Z$ upstairs, where the geometry is a divisor and therefore elementary, at the cost of one Chern class of the universal quotient bundle. Part (2) is the form used in practice: it computes the Gysin image of any class pushed down from the blow-up, and it is the identity from which the Chow groups of $\widetilde{X}$ are assembled from those of $X$ and of $\mathbb{P}(N)$. The smallest instance is a consistency check that can be run in both directions.

**Example 5.3.12: The Fundamental Class of a Blow-up**

Let $X$ be a variety of dimension $n$ and $Z \subsetneq X$ a nonempty subvariety with $Z \neq X$, regularly embedded of codimension $d \geq 1$, and take $\beta = [\widetilde{X}]$. The Rees algebra is a graded subalgebra of $\mathcal{O}_X[t]$ and hence a sheaf of domains, so $\widetilde{X}$ is integral, that is a variety; and $f$ restricts to an isomorphism over $X \setminus Z$ ([2, Basic Properties of Blow-ups](2)), so the two function fields agree and $f_*[\widetilde{X}] = [X]$ by definition 1.2.1. Also $i^![X] = [Z]$ by proposition 4.3.3, the intersection of $X$ with $Z$ being $Z$ itself, and $j^![\widetilde{X}] = [\widetilde{Z}]$ by proposition 4.3.5 applied to the subvariety $\widetilde{X}$ of itself. Substituting into theorem 5.3.11(2),

$$[Z] \;=\; g_*\Big(c_{d-1}(Q) \cap \big[\mathbb{P}(N)\big]\Big)$$

The identity can be checked directly, and the check is a use of the Segre operators rather than of anything proved in this section. The tautological sequence $0 \to \mathcal{O}_{\mathbb{P}(N)}(-1) \to g^*N \to Q \to 0$ and the Whitney formula (theorem 3.2.15) give $c(Q) = c(g^*N)\,(\mathrm{id} - \xi)^{-1}$ with $\xi = c_1(\mathcal{O}_{\mathbb{P}(N)}(1))$, the inverse being the terminating series $\sum_{m \geq 0} \xi^m$, so

$$c_{d-1}(Q) \;=\; \sum_{m=0}^{d-1} \xi^m \circ c_{d-1-m}\big(g^*N\big)$$

Capping with $[\mathbb{P}(N)] = g^*[Z]$ and pushing forward, naturality of the Chern operators for the flat morphism $g$ (proposition 3.2.10(3)) and the definition of the Segre operators (definition 3.1.3), for which $\mathbb{P}(N) \to Z$ has relative dimension $d-1$, give

$$g_*\Big(\xi^m \cap g^*\big(c_{d-1-m}(N) \cap [Z]\big)\Big) \;=\; s_{m-d+1}(N) \cap \big(c_{d-1-m}(N) \cap [Z]\big)$$

for each $m$. Every term with $m < d-1$ vanishes, the Segre operator having negative index, and the term $m = d-1$ is $s_0(N) \cap [Z] = [Z]$; summing over $m$ gives $g_*\big(c_{d-1}(Q) \cap [\mathbb{P}(N)]\big) = [Z]$, so the two computations agree.

### Exercise 5.3.1

1. Let $Y \subseteq \mathbb{P}^2$ be a smooth plane curve of degree $m$, with inclusion $i$. Show that $i\colon Y \hookrightarrow \mathbb{P}^2$ is a regular embedding of codimension one, identify its normal bundle, and compute the degree of $i^!(i_*[Y]) \in A_0(Y)$. Then explain why the answer determines $i^!(i_*[Y])$ itself when $m \leq 2$ but not when $m \geq 3$.

2. Let $Z = \{x_0 = 0\} \cong \mathbb{P}^{n-1}$ be a hyperplane in $\mathbb{P}^n$ with $n \geq 1$, with inclusion $i$. Identify the normal bundle of $i$ and compute $i^!(i_*\alpha)$ for $\alpha$ the generator $h^Z_m$ of $A_m(Z)$ for each $0 \leq m \leq n-1$. Compare the answer with the Gysin map $i^!$ of example 4.3.4 in the case $d = 1$.

3. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ whose normal bundle is trivial, $N \cong \mathcal{O}_Z^{\oplus d}$. Show that $i^! \circ i_* = 0$ on $A_k(Z)$ for every $k$. Then let $X = \mathbb{P}^1 \times \mathbb{P}^1$, let $Z = \{q\} \times \mathbb{P}^1$ for a point $q \in \mathbb{P}^1$, and verify the hypothesis, deducing that the ruling has self-intersection zero.

4. In $\mathbb{P}^4$ with coordinates $x_0, \ldots, x_4$, let $Z = \{x_3 = x_4 = 0\}$ and $X' = \{x_2 = x_4 = 0\}$ be two planes, so that $Z \cap X' = \{x_2 = x_3 = x_4 = 0\}$ is a line $L$, and let $g\colon X' \hookrightarrow \mathbb{P}^4$ be the inclusion. Show that $Z \hookrightarrow \mathbb{P}^4$ is a regular embedding of codimension two with $N = \mathcal{O}_{\mathbb{P}^2}(1)^{\oplus 2}$, identify the preimage $Z'$ of $Z$ in $X'$ and its normal bundle, compute the excess bundle, and evaluate $i^!_g[X']$. Interpret the degree of the answer.

5. Let $\widetilde{X} = \mathrm{Bl}_P\mathbb{P}^2$ as in example 5.3.6, with exceptional divisor $\widetilde{Z} \cong \mathbb{P}^1$ and the remaining notation that of theorem 5.3.11. Identify the universal quotient bundle $Q$ on $\widetilde{Z}$ and compute $c_1(Q)$. Then verify theorem 5.3.11(2) for $\beta = [\widetilde{X}]$ by computing both sides, and compute $i^!\big(f_*j_*[\widetilde{Z}]\big)$.

## 5.4 Bézout and the Agreement Theorem

Two counts were recorded on credit. Example 1.2.6 asserted that two plane curves of degrees $c$ and $d$ meet in a zero-cycle of degree $cd$, and example 1.5.4 read that assertion off an identity $h_1 \cdot h_1 = h_0$

in a ring that did not then exist. Both are settled here, in the sharp form: not only the total count, but its distribution over the points of $C \cap D$, which is what an intersection multiplicity is.

Every enumerative count in projective space reduces to one computation, the product of the classes of two linear subspaces, because theorem 1.5.2 leaves nothing else in $A_*(\mathbb{P}^n)$ to compute, and that computation was carried out in example 5.2.3. Written in dimension rather than codimension, with $h_k = [\mathbb{P}^k] \in A_k(\mathbb{P}^n)$ the class of a linear subspace of dimension $k$ and $h = h_{n-1}$ the hyperplane class, it reads

$$h_r \cdot h_s \;=\; \begin{cases} h_{r+s-n} & \text{if } r+s \geq n \\ 0 & \text{if } r+s < n \end{cases}$$

for $0 \leq r, s \leq n$, together with $h^p = h_{n-p}$ for $0 \leq p \leq n$, the vanishing $h^{n+1} = 0$, and the presentation

$$A^*(\mathbb{P}^n) \;\cong\; \mathbb{Z}[h]/(h^{n+1})$$

as a graded ring with $h$ in codimension one. Nothing else about $\mathbb{P}^n$ is needed below.

The presentation converts every question about cycles on $\mathbb{P}^n$ into a question about polynomials in one variable. A class in $A^*(\mathbb{P}^n)$ is a polynomial in $h$ of degree at most $n$; multiplying two of them and truncating above degree $n$ is the whole intersection calculus of projective space. The truncation is geometric: $h^p h^q = 0$ for $p + q > n$ says that two subvarieties whose codimensions exceed the ambient dimension can be moved apart, and rational equivalence records that they have been.

The single integer attached to a subvariety by theorem 1.5.2 now deserves a name, since it is the only input Bézout's theorem takes.

**Definition 5.4.1: Degree of a Subvariety of Projective Space**

Let $V \subseteq \mathbb{P}^n$ be a closed subvariety of dimension $k$. Since $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ is free of rank one (theorem 1.5.2), there is a unique integer $\deg V$ with

$$[V] \;=\; (\deg V)\, h_k \;\in\; A_k(\mathbb{P}^n)$$

called the *degree* of $V$. The same definition applies to a closed subscheme pure of dimension $k$, using its fundamental cycle.

The definition is a tautology until the integer is computed, and the two computations that matter are the classical ones: a hypersurface has the degree of its equation, and the degree of anything is the number of points it cuts on a complementary linear space.

**Proposition 5.4.2: Degrees of Hypersurfaces and Hyperplane Sections**

Let $H \subseteq \mathbb{P}^n$ be a hyperplane, so that $\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1)$.

1. If $V \subseteq \mathbb{P}^n$ is a closed subvariety of dimension $k$, then $H^k \cdot [V] = (\deg V)\, h_0$ in $A_0(\mathbb{P}^n)$, so $\deg V = \int_{\mathbb{P}^n} \left(H^k \cdot [V]\right)$, the iterated divisor operator of definition 2.2.1 applied $k$ times.
2. A linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$ has degree 1.
3. If $F$ is a nonzero homogeneous form of degree $d \geq 1$ on $\mathbb{P}^n$, the hypersurface $V(F)$ has degree $d$.

***Proof*:**

1. Example 2.2.5 computes $H \cdot h_j = h_{j-1}$ for $1 \leq j \leq n$. The operator $H \cdot -$ is a homomorphism of Chow groups (proposition 2.4.4(1)), so from $[V] = (\deg V)\, h_k$ one gets $H \cdot [V] = (\deg V)\, h_{k-1}$, and iterating $k$ times gives $H^k \cdot [V] = (\deg V)\, h_0$. The class $h_0$ is that of a single closed point, of degree $[\bar{k} : \bar{k}] = 1$ by definition 1.2.4, whence $\int_{\mathbb{P}^n}(H^k \cdot [V]) = \deg V$.

2. By theorem 1.5.2 the class of a linear $\mathbb{P}^k$ is the generator $h_k$ itself, so the coefficient is 1.

3. Write $Z = V(F)$ for the closed subscheme cut out by $F$, a hypersurface, and let $D$ be the associated effective Cartier divisor, with $\mathcal{O}_{\mathbb{P}^n}(D) \cong \mathcal{O}_{\mathbb{P}^n}(d)$ because $F$ is a section of $\mathcal{O}_{\mathbb{P}^n}(d)$. Applying proposition 4.3.5 with $V = \mathbb{P}^n$, which is legitimate since $\mathbb{P}^n$ is a variety not contained in $|D|$, the fundamental cycle of $Z$ is $[Z] = D \cdot [\mathbb{P}^n]$. By definition 2.4.1 this is $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) \cap [\mathbb{P}^n]$, and $\mathcal{O}_{\mathbb{P}^n}(d) \cong \mathcal{O}_{\mathbb{P}^n}(1)^{\otimes d}$, so additivity of the first Chern class (proposition 2.4.4(3)) gives

$$[Z] \;=\; d\left(c_1(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n]\right) \;=\; d\, h_{n-1}$$

the last equality being example 2.4.3(2). Hence $\deg Z = d$, completing the proof.

Part (1) is the reason the word “degree” is the same word as in the classical theory: cutting a $k$-dimensional subvariety by $k$ hyperplanes leaves a zero-cycle whose degree is the invariant, and by proposition 2.4.6 the hyperplanes need not be chosen in general position for the answer to come out right. Part (3) recovers the plane-curve count of example 1.5.4(2) as the case $n = 2$. Degrees of subvarieties that are not hypersurfaces are computed by pushing forward from a parametrization, as the next example illustrates.

**Example 5.4.3: The Twisted Cubic**

Let $\nu \colon \mathbb{P}^1 \to \mathbb{P}^3$ be the third Veronese embedding, $[s : t] \mapsto [s^3 : s^2t : st^2 : t^3]$, and let $C = \nu(\mathbb{P}^1)$ be the twisted cubic, a closed subvariety of $\mathbb{P}^3$ of dimension one. Since $\nu$ is a closed immersion, $\nu_*[\mathbb{P}^1] = [C]$ by definition 1.2.1, the extension of function fields being trivial.
Let $H = \{x_0 = 0\} \subseteq \mathbb{P}^3$. The pullback $\nu^*H$ is the Cartier divisor on $\mathbb{P}^1$ cut by $s^3$, that is $3\,[0 : 1]$, so $\nu^*H \cdot [\mathbb{P}^1] = 3\left[[0 : 1]\right]$ by definition 2.2.1. The projection formula for divisors (proposition 2.3.3(1)) applied to the proper morphism $\nu$ gives

$$H \cdot [C] \;=\; \nu_*\big(\nu^*H \cdot [\mathbb{P}^1]\big) \;=\; 3\,[P], \qquad P = \nu([0:1]) = [0:0:0:1]$$

a zero-cycle of degree 3. By proposition 5.4.2(1) with $k = 1$, $\deg C = 3$: the twisted cubic has degree three, all of it concentrated at the single point where $H$ is tangent to $C$ to order three. A general plane, by contrast, meets $C$ in three distinct points, and proposition 2.4.6 is what makes the two computations agree without either being privileged.

With degrees in hand, Bézout’s theorem is a restatement of example 5.2.3.

**Theorem 5.4.4: Bézout's Theorem**

Let $V, W \subseteq \mathbb{P}^n$ be closed subvarieties of dimensions $r$ and $s$. If $r + s \geq n$, then

$$[V] \cdot [W] \;=\; (\deg V)(\deg W)\, h_{r+s-n} \;\in\; A_{r+s-n}(\mathbb{P}^n)$$

and if $r + s < n$ then $[V] \cdot [W] = 0$. In the boundary case $r + s = n$,

$$\int_{\mathbb{P}^n} \big([V] \cdot [W]\big) \;=\; (\deg V)(\deg W)$$

***Proof*:**
By definition 5.4.1 the classes are $[V] = (\deg V)\, h_r$ and $[W] = (\deg W)\, h_s$. The intersection product is $\mathbb{Z}$-bilinear (proposition 5.1.9(1)), so

$$[V] \cdot [W] \;=\; (\deg V)(\deg W)\, (h_r \cdot h_s)$$

and example 5.2.3 evaluates $h_r \cdot h_s$ as $h_{r+s-n}$ when $r + s \geq n$ and as 0 otherwise. For $r + s = n$ the class $h_0$ is that of a single reduced closed point, of degree one by definition 1.2.4, and the degree map is a homomorphism, so $\int_{\mathbb{P}^n}([V] \cdot [W]) = (\deg V)(\deg W)$, as we sought to show.

The theorem is stated for classes, and in that form it costs nothing beyond the ring structure. What it does not yet say is where the count sits. The product $[V] \cdot [W]$ is defined as $\delta^!([V] \times [W])$, and the refined form of the Gysin map records that class on the preimage of the diagonal, that is on $V \cap W$ (corollary 4.3.11 and definition 5.1.10). Reading the refined class off requires knowing that $V \cap W$ sits inside $V \times W$ as a regular embedding, which is a hypothesis and not a theorem. The following lemma verifies it in the case where the verification is cheapest and most used: two curves on a smooth surface.

**Lemma 5.4.5: Distinct Curves on a Smooth Surface Meet Regularly**

Let $S$ be a smooth variety of dimension two over $\overline{k}$ and let $C, D \subseteq S$ be distinct one-dimensional closed subvarieties. Then:

1. $C \cap D$ is a finite set of closed points, and for $P \in C \cap D$ with local equations $f, g \in \mathcal{O}_{S,P}$ of $C$ and $D$, the local ring of the scheme $C \cap D$ at $P$ is $\mathcal{O}_{S,P}/(f, g)$, of length equal to its dimension $\dim_{\overline{k}} \mathcal{O}_{S,P}/(f, g)$ as a $\overline{k}$-vector space;
2. the closed immersion $C \cap D \hookrightarrow C \times D$ induced by the diagonal is a regular embedding of codimension two;
3. consequently

$$[C] \cdot [D] \;=\; \sum_{P \in C \cap D} \dim_{\overline{k}} \big(\mathcal{O}_{S,P}/(f, g)\big)\, [P] \;\in\; A_0(S)$$

***Proof*:**
Since $S$ is smooth its local rings are regular, hence factorial, so each of $C$ and $D$ is an effective Cartier divisor and the local equations $f, g$ exist and are nonzerodivisors.

1. $C \cap D$ is a closed subscheme of the irreducible curve $C$ not equal to $C$, because $D \neq C$ and both are irreducible of dimension one; a proper closed subscheme of a curve is finite. Scheme-theoretically $C \cap D$ is cut out on $S$ by the ideal $(f, g)$, so its local ring at $P$ is $\mathcal{O}_{S,P}/(f,g)$. That ring is Artinian local with residue field $\overline{k}$, since $\overline{k}$ is algebraically closed and $P$ is a closed point, so each factor of a composition series is a copy of $\overline{k}$ and its length equals its $\overline{k}$-dimension.

2. Fix $P \in C \cap D$ and write $\mathcal{O} = \mathcal{O}_{S,P}$, a regular local ring of dimension two with residue field $\overline{k}$, and $A = \mathcal{O}/(f)$, $B = \mathcal{O}/(g)$, the local rings of $C$ and $D$ at $P$. Let $R$ be the local ring of $C \times D$ at $(P, P)$, that is the localization of $A \otimes_{\overline{k}} B$ at the maximal ideal corresponding to $(P, P)$.
*Step 1: $R$ is Cohen-Macaulay of dimension two.* The product of two curves over $\overline{k}$ is a surface, so $\dim R = 2$. Because $\mathcal{O}$ is regular of dimension two and $f$ is a nonzerodivisor, $A$ has depth one, so $\mathfrak{m}_A$ contains a nonzerodivisor $a$; likewise $\mathfrak{m}_B$ contains a nonzerodivisor $b$. Now $B$ is free as a $\overline{k}$-module, so $A \to A \otimes_{\overline{k}} B$ is flat and $a \otimes 1$ is a nonzerodivisor, with quotient $(A/aA) \otimes_{\overline{k}} B$; and $A/aA$ is again free over $\overline{k}$, so $B \to (A/aA) \otimes_{\overline{k}} B$ is flat and $1 \otimes b$ is a nonzerodivisor on that quotient. Localizing at the maximal ideal, the pair $a \otimes 1$, $1 \otimes b$ is a regular sequence in $R$ of length two, so $\operatorname{depth} R \geq 2 = \dim R$ and $R$ is Cohen-Macaulay.
*Step 2: the diagonal cuts $R$ by a system of parameters.* The diagonal $\delta\colon S \hookrightarrow S \times S$ is a regular embedding of codimension two (theorem 5.1.6), so its ideal is generated near $(P, P)$ by two elements $\theta_1, \theta_2$. The scheme-theoretic preimage of $\delta(S)$ under $C \times D \hookrightarrow S \times S$ is $C \cap D$, whose ideal in $R$ is therefore generated by the images of $\theta_1$ and $\theta_2$. By part (1) that quotient is Artinian, so the two images form a system of parameters of the two-dimensional local ring $R$. In a Cohen-Macaulay local ring every system of parameters is a regular sequence ([4, A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence]), so $C \cap D \hookrightarrow C \times D$ is a regular embedding of codimension two.

3. The subvarieties $C$ and $D$ are integral over the algebraically closed field $\overline{k}$, so $C \times D$ is a subvariety of $S \times S$ of dimension two, with $[C] \times [D] = [C \times D]$ (definition 5.1.2). Part (2) supplies the hypothesis of proposition 4.3.3 for the regular embedding $\delta$ of codimension two and the subvariety $C \times D$, so

$$[C] \cdot [D] \;=\; \delta^![C \times D] \;=\; [C \cap D]$$

the fundamental cycle of the scheme $C \cap D$ read on $S$ through $\delta$. By definition 1.1.2 that cycle is $\sum_P \operatorname{length}(\mathcal{O}_{C \cap D, P})\,[P]$, and part (1) identifies each length with $\dim_{\overline{k}} \mathcal{O}_{S,P}/(f,g)$, completing the proof.

The lemma is the bridge from a statement about classes to a statement about points. In $\mathbb{P}^2$ every curve is a hypersurface, so its degree is the degree of its equation by proposition 5.4.2(3); the lemma settles a pair of distinct integral curves, and additivity of the local count in each argument extends it to arbitrary curves without a common component. Bézout's theorem then acquires its classical form.

**Corollary 5.4.6: Bézout's Theorem for Plane Curves**

Let $F$ and $G$ be nonzero forms on $\mathbb{P}^2$ of degrees $c$ and $d$ with no common irreducible factor, and let $C = V(F)$ and $D = V(G)$ be the plane curves they cut out, so that $C$ and $D$ have no common component; neither is assumed irreducible or reduced. Then $C \cap D$ is finite and

$$\sum_{P \in C\cap D} \dim_{\overline{k}} \left(\mathcal{O}_{\mathbb{P}^2,P}/(f_P, g_P)\right) \;=\; cd$$

where $f_P$ and $g_P$ are the local equations of $C$ and $D$ at $P$, obtained by dehomogenizing $F$ and $G$.

***Proof*:**
Factor $F = \prod_i F_i^{m_i}$ and $G = \prod_j G_j^{n_j}$ into powers of pairwise nonassociate irreducible forms; by hypothesis no $F_i$ is an associate of any $G_j$. The curves $C_i = V(F_i)$ and $D_j = V(G_j)$ are one-dimensional closed subvarieties of $\mathbb{P}^2$, of degrees $\deg F_i$ and $\deg G_j$ by proposition 5.4.2(3), and $C_i \neq D_j$ for all $i, j$. For a form $u$ and a point $P$ write $u_P \in \mathcal{O}_{\mathbb{P}^2,P}$ for the dehomogenization of $u$ at $P$, so that $f_P = F_P$ and $g_P = G_P$.

*Step 1: the count for a pair of components.* The plane $\mathbb{P}^2$ is a smooth variety of dimension two and $C_i \neq D_j$ are distinct one-dimensional closed subvarieties, so lemma 5.4.5 applies to the pair $(C_i, D_j)$: the intersection $C_i \cap D_j$ is finite and $[C_i]\cdot[D_j] = \sum_P \dim_{\overline{k}}(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P}, G_{j,P}))\,[P]$ in $A_0(\mathbb{P}^2)$. Each point is closed with residue field $\overline{k}$, so taking degrees (definition 1.2.4) and comparing with theorem 5.4.4 for $n = 2$, $r = s = 1$ gives

$$\sum_{P \in C_i\cap D_j} \dim_{\overline{k}} \left(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P}, G_{j,P})\right) \;=\; (\deg F_i)(\deg G_j)$$

Since $C \cap D = \bigcup_{i,j} C_i \cap D_j$ as sets, $C \cap D$ is finite.

*Step 2: the local count is additive in each argument.* Fix $P$ and write $\mathcal{O} = \mathcal{O}_{\mathbb{P}^2,P}$, a regular local ring of dimension two. Let $u$ be a divisor of $F$ and let $v, w$ be divisors of $G$, or the same with the roles of $F$ and $G$ exchanged; then $V(u) \cap V(vw) \subseteq C \cap D$ is finite by Step 1, so $\mathcal{O}/(u_P, (vw)_P)$ and its two companions below are Artinian as in lemma 5.4.5(1), of finite $\overline{k}$-dimension. We may assume $u_P$ and $w_P$ lie in the maximal ideal, since otherwise one of the three quotients is zero and the additivity claimed below is trivial. Being Artinian, $\mathcal{O}/(u_P, w_P)$ exhibits $u_P, w_P$ as a system of parameters of $\mathcal{O}$, hence as a regular sequence ([4, A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence]); in particular $w_P$ is a nonzerodivisor on $A = \mathcal{O}/(u_P)$. The sequence

$$0 \longrightarrow A/(v_P) \xrightarrow{\;w_P\;} A/(v_P w_P) \longrightarrow A/(w_P) \longrightarrow 0$$

is then exact: surjectivity on the right and exactness in the middle are immediate, and if $w_P h \in (v_P w_P)A$ then $w_P(h - v_P a) = 0$ in $A$ for some $a$, so $h \in (v_P)A$ because $w_P$ is a nonzerodivisor on $A$. Counting $\overline{k}$-dimensions gives

$$\dim_{\overline{k}} \mathcal{O}/(u_P, v_P w_P) \;=\; \dim_{\overline{k}} \mathcal{O}/(u_P, v_P) \;+\; \dim_{\overline{k}} \mathcal{O}/(u_P, w_P)$$

that is, additivity in the second argument, and the exchanged case gives additivity in the first.

Iterating over the two factorizations,

$$\dim_{\overline{k}}\left(\mathcal{O}_{\mathbb{P}^2,P}/(f_P,g_P)\right) \;=\; \sum_{i,j} m_i n_j \,\dim_{\overline{k}}\left(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P},G_{j,P})\right)$$

*Step 3: summation.* Sum the last display over the finitely many points of $C\cap D$; a point outside $C_i\cap D_j$ contributes nothing to the $(i,j)$ term, since there $F_{i,P}$ or $G_{j,P}$ is a unit, so Step 1 evaluates each term and

$$\sum_{P\in C\cap D}\dim_{\overline{k}}\left(\mathcal{O}_{\mathbb{P}^2,P}/(f_P,g_P)\right) \;=\; \sum_{i,j} m_i n_j (\deg F_i)(\deg G_j) \;=\; \Big(\sum_i m_i \deg F_i\Big)\Big(\sum_j n_j \deg G_j\Big) \;=\; cd$$

the last equality because $\deg F = \sum_i m_i \deg F_i$ and $\deg G = \sum_j n_j \deg G_j$, completing the proof.

This is the count promised in example 1.2.6 and the identity $h_1\cdot h_1 = h_0$ anticipated in example 1.5.4(2), now with the local distribution attached: the integer $\dim_{\overline{k}}\mathcal{O}_{\mathbb{P}^2,P}/(f_P,g_P)$ says how many of the $cd$ units of intersection are spent at $P$. Nothing in the argument assumed that the curves are irreducible, reduced, smooth, or that they meet transversally; only that they share no component, which is exactly what makes the intersection finite.

**Example 5.4.7: A Cuspidal Cubic and Its Cuspidal Tangent**

Let $C=\{y^2z=x^3\}\subseteq\mathbb{P}^2$, a cubic curve with a cusp at $P=[0:0:1]$, and let $D=\{y=0\}$, the tangent line to $C$ at the cusp. The cubic is irreducible and is not the line, so the two share no component and corollary 5.4.6 predicts a total of $3\cdot 1 = 3$.

Set-theoretically, $y=0$ together with $y^2z=x^3$ forces $x^3=0$, hence $x=0$: the only intersection point is $P$. Working in the chart $z=1$ with $\mathcal{O}=\overline{k}[x,y]_{(x,y)}$, the local equations are $f=y^2-x^3$ and $g=y$, and

$$\mathcal{O}/(f,g) \;=\; \overline{k}[x,y]_{(x,y)}/(y^2-x^3,\,y) \;\cong\; \overline{k}[x]_{(x)}/(x^3)$$

of dimension 3 over $\overline{k}$, with basis $1,x,x^2$. So all three units of intersection sit at the cusp, and $[C]\cdot[D]=3\,[P]$. Compare example 2.2.2, where a conic met its tangent line in a single point of multiplicity two: in both cases the multiplicity is the order to which the equation of the line vanishes along the curve, and here the cusp raises it from two to three.

The integer attached to $P$ in the example is not special to curves in a plane. Whenever two subvarieties of a smooth variety meet in the expected dimension, the product distributes over the components of the intersection with well-defined integer coefficients, and those coefficients are the intersection multiplicities.

**Definition 5.4.8: Proper Intersection and Intersection Multiplicity**

Let $X$ be a smooth variety of pure dimension $n$ and let $V, W \subseteq X$ be closed subvarieties of dimensions $r$ and $s$. Say that $V$ and $W$ *meet properly* if every irreducible component of $V \cap W$ has dimension exactly $m = r + s - n$. In that case $V \cap W$ is pure of dimension $m$, so $A_m(V \cap W) = Z_m(V \cap W)$ is free on the classes of its irreducible components $Z_1, \ldots, Z_t$ (example 1.1.6(3)), and the refined intersection class of definition 5.1.10 may be written uniquely as

$$[V] \cdot [W] \;=\; \sum_{j=1}^{t} i(Z_j;\, V \cdot W;\, X)\,[Z_j] \;\in\; A_m(V \cap W)$$

The integer $i(Z_j;\, V \cdot W;\, X)$ is the *intersection multiplicity* of $V$ and $W$ along $Z_j$.

The definition would be empty without the refinement: the class $[V] \cdot [W]$ read in $A_m(X)$ carries no information about where the intersection happens, and it is the localization on $V \cap W$ supplied by the refined Gysin map (definition 4.3.6, corollary 4.3.11) that makes the coefficients meaningful. The proper-intersection hypothesis is what makes the coefficients unique, since $A_m$ of an $m$-dimensional scheme is free on components; drop it and the class still exists but no longer decomposes canonically.

When the scheme-theoretic intersection is as good as its dimension suggests, the multiplicities are lengths and nothing else. Proposition 4.3.3 says that if $V \cap W \hookrightarrow V \times W$ is a regular embedding of codimension $n$, then $[V] \cdot [W] = [V \cap W]$ is the fundamental cycle, so $i(Z_j; V \cdot W; X)$ is the length of the local ring of $V \cap W$ at the generic point of $Z_j$. That is exactly what lemma 5.4.5 verified for curves on a surface, and it is the source of the numbers computed in example 5.4.7. In general the regularity hypothesis can fail, and then the multiplicity and the naive length part company. The correction is homological.

**Theorem 5.4.9: Serre's Tor Formula**

Let $X$ be a smooth variety of pure dimension $n$ and let $V, W \subseteq X$ be closed subvarieties meeting properly, with $Z$ an irreducible component of $V \cap W$ and $\mathcal{O} = \mathcal{O}_{X,Z}$ the local ring of $X$ at the generic point of $Z$. Then

$$i(Z;\, V \cdot W;\, X) \;=\; \sum_{j \geq 0} (-1)^j \operatorname{length}_{\mathcal{O}} \operatorname{Tor}_j^{\mathcal{O}}\big(\mathcal{O}/\mathcal{I}_V,\, \mathcal{O}/\mathcal{I}_W\big)$$

where $\mathcal{I}_V$ and $\mathcal{I}_W$ are the ideals of $V$ and $W$ in $\mathcal{O}$. The sum is finite, each term having finite length and all terms vanishing for $j > n$.

The proof is omitted: it runs entirely through homological algebra over regular local rings, in particular the vanishing, finite-length and additivity theorems for Tor that this book does not develop, and it is genuinely outside the scope of a treatment built on deformation to the normal cone. Serre proved the formula in *Algèbre locale, multiplicités*, and the comparison with the multiplicities of definition 5.4.8 is carried out in Fulton [7]. The formula matters here because it exhibits the multiplicity as a purely local, purely algebraic invariant of the two local rings, with no reference to the deformation that defined it.

The correction terms are invisible in the case the previous results have been using, and that is worth recording, since it says the classical count of plane curves needs no homological algebra at all.

**Proposition 5.4.10: Vanishing of the Higher Terms for Divisors**

Let $X$ be a smooth variety of pure dimension $n$ and let $C, D \subseteq X$ be closed subvarieties of dimension $n-1$ with $C \neq D$, so that $C$ and $D$ meet properly. For every irreducible component $Z$ of $C \cap D$, with $\mathcal{O} = \mathcal{O}_{X,Z}$ and local equations $f, g \in \mathcal{O}$,

$$\operatorname{Tor}_j^{\mathcal{O}}\big(\mathcal{O}/(f),\, \mathcal{O}/(g)\big) = 0 \quad \text{for } j \geq 1$$

so Serre's formula reduces to $i(Z;\, C \cdot D;\, X) = \operatorname{length}_{\mathcal{O}} \mathcal{O}/(f, g)$.

***Proof*:**
Since $X$ is smooth, $\mathcal{O}$ is a regular local ring and $C$, $D$ are effective Cartier divisors near $Z$, so $f$ and $g$ are nonzerodivisors and

$$0 \longrightarrow \mathcal{O} \xrightarrow{\ f\ } \mathcal{O} \longrightarrow \mathcal{O}/(f) \longrightarrow 0$$

is a free resolution of $\mathcal{O}/(f)$ of length one. Hence $\operatorname{Tor}_j(\mathcal{O}/(f), \mathcal{O}/(g)) = 0$ for $j \geq 2$, while $\operatorname{Tor}_1$ is the kernel of multiplication by $f$ on $\mathcal{O}/(g)$. Because $\mathcal{O}$ is regular and $g$ is a nonzerodivisor, $\mathcal{O}/(g)$ is Cohen-Macaulay, hence has no embedded associated primes ([4, A Cohen-Macaulay Ring Has No Embedded Associated Primes]), so its associated primes are the minimal primes of $(g)$, which are the primes corresponding to the components of $D$ passing through $Z$. As $C$ and $D$ share no component, $f$ lies in none of them, so multiplication by $f$ on $\mathcal{O}/(g)$ is injective and $\operatorname{Tor}_1 = 0$. The remaining term is $\operatorname{Tor}_0(\mathcal{O}/(f), \mathcal{O}/(g)) = \mathcal{O}/(f, g)$, and theorem 5.4.9 therefore reads $i(Z; C \cdot D; X) = \operatorname{length}_{\mathcal{O}} \mathcal{O}/(f, g)$, as we sought to show.

Everything so far has been internal to this book: a product, a ring, a count. The comparison that remains is external. *Algebraic Geometry* [2] builds a pairing on the divisors of a smooth projective surface without any of this machinery, setting $C \cdot D = \chi(\mathcal{O}_S) - \chi(\mathcal{O}_S(-C)) - \chi(\mathcal{O}_S(-D)) + \chi(\mathcal{O}_S(-C - D))$, and derives adjunction, the behaviour of self-intersection, and the classification of surfaces from it. Two constructions with nothing in common beyond the answer they are supposed to give must be shown to give it.

**Theorem 5.4.11: The Agreement Theorem for Surfaces**

Let $S$ be a smooth projective surface over $\overline{k}$ and let $C, D$ be Cartier divisors on $S$, with associated classes $[C] = c_1(\mathcal{O}_S(C)) \cap [S]$ and $[D] = c_1(\mathcal{O}_S(D)) \cap [S]$ in $A^1(S) = A_1(S)$. Then

$$\int_S ([C] \cdot [D]) \;=\; \int_S (C \cdot (D \cdot [S])) \;=\; C \cdot D$$

where the first is the intersection product of section 5.1, the second the iterated divisor operator of definition 2.2.1, and the third the cohomological intersection number [2, Intersection Number (Cohomological)].

***Proof*:**
Write $\langle C,D\rangle_1$, $\langle C,D\rangle_2$ and $\langle C,D\rangle_3$ for the three quantities.

*Step 1: all three are $\mathbb{Z}$-bilinear and depend only on linear equivalence.* For the first, $C \mapsto [C] = C\cdot[S]$ is additive in $C$ by additivity of the first Chern class (proposition 2.4.4(3)) and depends only on $\mathcal{O}_S(C)$ by proposition 2.4.2; the product is $\mathbb{Z}$-bilinear (proposition 5.1.9(1)) and $\int_S$ is a homomorphism (definition 1.2.4). For the second, $D \mapsto D\cdot[S]$ is additive and linear-equivalence invariant for the same two reasons, and $C\cdot -$ is a homomorphism of Chow groups additive in $C$ by proposition 2.4.4(1),(3). For the third, bilinearity and invariance under linear equivalence are [2, Properties of the Intersection Pairing](1),(3).

*Step 2: the three agree for distinct prime divisors.* Let $C$ and $D$ be distinct one-dimensional closed subvarieties of $S$, with local equations $f, g$ at a point $P$. Lemma 5.4.5(3) gives

$$\langle C,D\rangle_1 \;=\; \int_S \Big( \sum_{P\in C\cap D} \dim_{\overline{k}} \left(\mathcal{O}_{S,P}/(f,g)\right)[P]\Big) \;=\; \sum_{P\in C\cap D} \dim_{\overline{k}} \left(\mathcal{O}_{S,P}/(f,g)\right)$$

each $P$ being a closed point with residue field $\overline{k}$. For the second quantity, $S \not\subseteq |D|$ and $D \not\subseteq |C|$, so the first clause of definition 2.2.1 applies twice: $D\cdot[S] = [\mathrm{div}(s_S)] = [D]$, and $C\cdot[D] = [\mathrm{div}(f|_D)] = \sum_P \mathrm{ord}_P(f|_D)\,[P]$. By definition 1.1.1 the coefficient is $\mathrm{ord}_P(f|_D) = \mathrm{length}_{\mathcal{O}_{D,P}}\left(\mathcal{O}_{D,P}/(f|_D)\right)$, and $\mathcal{O}_{D,P}/(f|_D) = \mathcal{O}_{S,P}/(f,g)$, whose length equals its $\overline{k}$-dimension. Hence $\langle C,D\rangle_2$ is the same sum. For the third, [2, Properties of the Intersection Pairing](5) evaluates $C\cdot D$ on effective divisors with no common component as $\sum_P \dim_{\overline{k}} \mathcal{O}_{S,P}/(f,g)$, again the same sum. So the three agree on such a pair.

*Step 3: reduction of the general case to Step 2.* Fix a very ample divisor $H$ on the projective surface $S$. By Serre's theorem on global generation after an ample twist ([2, Global Section Generation of Twisted Sheaves]) there is an $m \geq 1$ such that $\mathcal{O}_S(C+mH)$, $\mathcal{O}_S(D+mH)$ and $\mathcal{O}_S(mH)$ are all generated by their global sections. Set

$$A_1 = C+mH, \quad A_2 = mH, \quad B_1 = D+mH, \quad B_2 = mH$$

so that $C = A_1 - A_2$ and $D = B_1 - B_2$ as Cartier divisors, and by Step 1 it suffices to prove the agreement for each of the four pairs $(A_i, B_j)$.
A globally generated line bundle has a nonzero global section, so each $A_i$ is linearly equivalent to an effective divisor $A_i'$; let $\Gamma_1,\ldots,\Gamma_t$ be the finitely many prime components of $A_1'$ and $A_2'$ and choose a closed point $p_a \in \Gamma_a$ for each. For a fixed $j$, the sections of $\mathcal{O}_S(B_j)$ vanishing at $p_a$ form a proper linear subspace of the finite-dimensional space $H^0(S,\mathcal{O}_S(B_j))$, since that bundle is globally generated; a vector space over the infinite field $\overline{k}$ is not a finite union of proper subspaces, so some section $\tau_j$ vanishes at no $p_a$. Its divisor $B_j' = \mathrm{div}(\tau_j)$ is effective, linearly equivalent to $B_j$, and contains none of the $\Gamma_a$.
Writing $A_i' = \sum_a m_a\Gamma_a$ and $B_j' = \sum_b n_b\Lambda_b$ as nonnegative combinations of prime divisors, no $\Gamma_a$ equals any $\Lambda_b$, so bilinearity reduces each $\langle A_i', B_j'\rangle_\bullet$ to $\sum_{a,b} m_a n_b \langle\Gamma_a,\Lambda_b\rangle_\bullet$, and every term is covered by Step 2. Invariance under linear equivalence transfers the conclusion back from $A_i', B_j'$ to $A_i, B_j$, and Step 1 assembles the four pairs into the general case, completing the proof.

The theorem is what connects the two. Every numerical statement about divisors on a smooth projective surface proved in *Algebraic Geometry* [2] out of Euler characteristics, the adjunction formula, the identification of $C^2$ with the degree of the normal bundle, the Hodge index theorem, is a statement about the product of this chapter, and conversely every general theorem proved here specializes to a surface fact that book had to obtain by hand. The surface theory is subsumed rather than paralleled, and section 5.3 already used that when it recovered $E^2 = -1$ from the self-intersection formula.

The middle term of the theorem is worth isolating. It says that on a smooth projective surface the ring product of two divisor classes computes the same integer as the iterated divisor operator built in chapter 2, so the operator calculus of that chapter, commutativity and the projection formula included, transfers verbatim to the ring. The two constructions were built from different material: one from local equations and orders of vanishing, the other from a degeneration of the diagonal. Nothing forced them to agree except that both were built to count.

## Exercise 5.4.1

1. Let $V, W \subseteq \mathbb{P}^3$ be surfaces of degrees 2 and 3 with no common component. Compute $[V] \cdot [W]$ in $A_1(\mathbb{P}^3)$ and read off the degree of the curve $V \cap W$. Then let $V_1, V_2, V_3 \subseteq \mathbb{P}^3$ be surfaces of degrees $d_1, d_2, d_3$ and compute $\int_{\mathbb{P}^3} \big([V_1] \cdot [V_2] \cdot [V_3]\big)$.
2. Let $L \cong \mathbb{P}^r$ and $M \cong \mathbb{P}^s$ be linear subspaces of $\mathbb{P}^n$ with $r + s < n$. Show that $[L] \cdot [M] = 0$, and explain why this conclusion is *not* a consequence of the observation that $L$ and $M$ can be chosen disjoint. (Consider the case $L = M$.)
3. Keep $C = \{y^2 z = x^3\} \subseteq \mathbb{P}^2$ from example 5.4.7 and take instead $D' = \{x = 0\}$, the line joining the cusp $[0 : 0 : 1]$ to the inflection point $[0 : 1 : 0]$. Compute $C \cap D'$ and the multiplicity of $[C] \cdot [D']$ at each of its points, and verify corollary 5.4.6.
4. Let $Q = \mathbb{P}^1 \times \mathbb{P}^1$, with rulings $f_1 = \{a\} \times \mathbb{P}^1$ and $f_2 = \mathbb{P}^1 \times \{b\}$ as in exercise 2.2.1 (5), where $\mathcal{O}_Q(f_1) = \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(1)$ and $\mathcal{O}_Q(f_2) = \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^1}(1)$. Using theorem 5.4.11, write down the three numbers $\int_Q([f_1] \cdot [f_2])$, $\int_Q([f_1] \cdot [f_1])$ and $\int_Q([f_2] \cdot [f_2])$. Then show that the diagonal $\Delta \subseteq Q$ satisfies $\Delta \sim f_1 + f_2$ and compute $\int_Q([\Delta] \cdot [\Delta])$.
5. Let $L \subseteq \mathbb{P}^2$ be a line. Compute $\int_{\mathbb{P}^2}([L] \cdot [L])$ from example 5.2.3, and identify precisely which conclusion of lemma 5.4.5 fails when $C = D = L$. Explain how the intersection product still returns an answer, and reconcile it with example 2.2.3.

# Part III

# Cycle Classes, Riemann-Roch, and Enumerative Geometry

The Chow ring is built, and with it the question that motivated the construction has an answer: two subvarieties of a smooth variety can be intersected, and the result is a class that behaves. What the ring does not yet do is talk to anything else. A cycle class is an algebraic invariant of an algebraic object, and the classical invariants of a complex variety are topological; until the two are compared, the theory is self-contained in the unhelpful sense.

This part makes the comparison and then spends it. Chapter 6 maps cycles to cohomology and finds that the map is far from injective, which turns a single equivalence relation into a tower of four: rational, algebraic, homological, numerical. Where the tower collapses is, in almost every case, an open problem, and two of the gaps are among the most famous questions in the subject. **??** pays the comparison forward: the Chern character carries $K$-theory into the Chow ring, and Grothendieck-Riemann-Roch measures its failure to commute with pushforward by a Todd class, recovering the classical Riemann-Roch theorems as the case of a map to a point. **??** then uses the whole apparatus for the purpose the subject was invented for, which is counting.

# 6

# Cycle Classes and Adequate Equivalence Relations

Rational equivalence was chosen in chapter 1 for its formal properties, not because it was the only candidate, and it is in fact the finest of a family. Coarsening it produces new groups, each with its own virtues: algebraic equivalence brings in families over curves and makes the quotient finitely generated; homological equivalence is whatever cohomology cannot distinguish; numerical equivalence remembers only intersection numbers. The four sit in a tower, each contained in the next, and the inclusions are the subject of this chapter.

The comparison that generates the tower is the cycle class map. For a smooth complex variety a subvariety carries a fundamental class in homology, and Poincaré duality turns it into a cohomology class; the content of **??** is that this assignment is blind to rational equivalence and respects everything the Chow ring can do. Once it exists, homological equivalence is defined as its kernel, and the question of how much information the map loses becomes precise.

It loses some. Cohomology sees a cycle only through its class, and the Hodge conjecture asks whether every cohomology class that looks algebraic is algebraic. Whether homological and numerical equivalence agree is the standard conjecture $D$. Neither is proved here or anywhere; what this chapter owes the reader is exact statements, correct attributions, and a clear account of which implications are theorems.

The chapter closes with correspondences. A cycle on $X \times Y$ acts on cycles and on cohomology by pulling back, intersecting, and pushing forward, and composing two such cycles is again the intersection product. That is the operation the theory of motives is built from, and it is the form in which [1] consumes this book.

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